Expanding the precision medicine landscape of paediatric solid tumours with *ex vivo* drug screening

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Expanding the precision medicine landscape of paediatric solid tumours with *ex vivo* drug screening

Verbreding van het precisiegeneeskundelandschap voor solide pediatrische tumoren met *ex vivo* medicijntesten
(met een samenvatting in het Nederlands)

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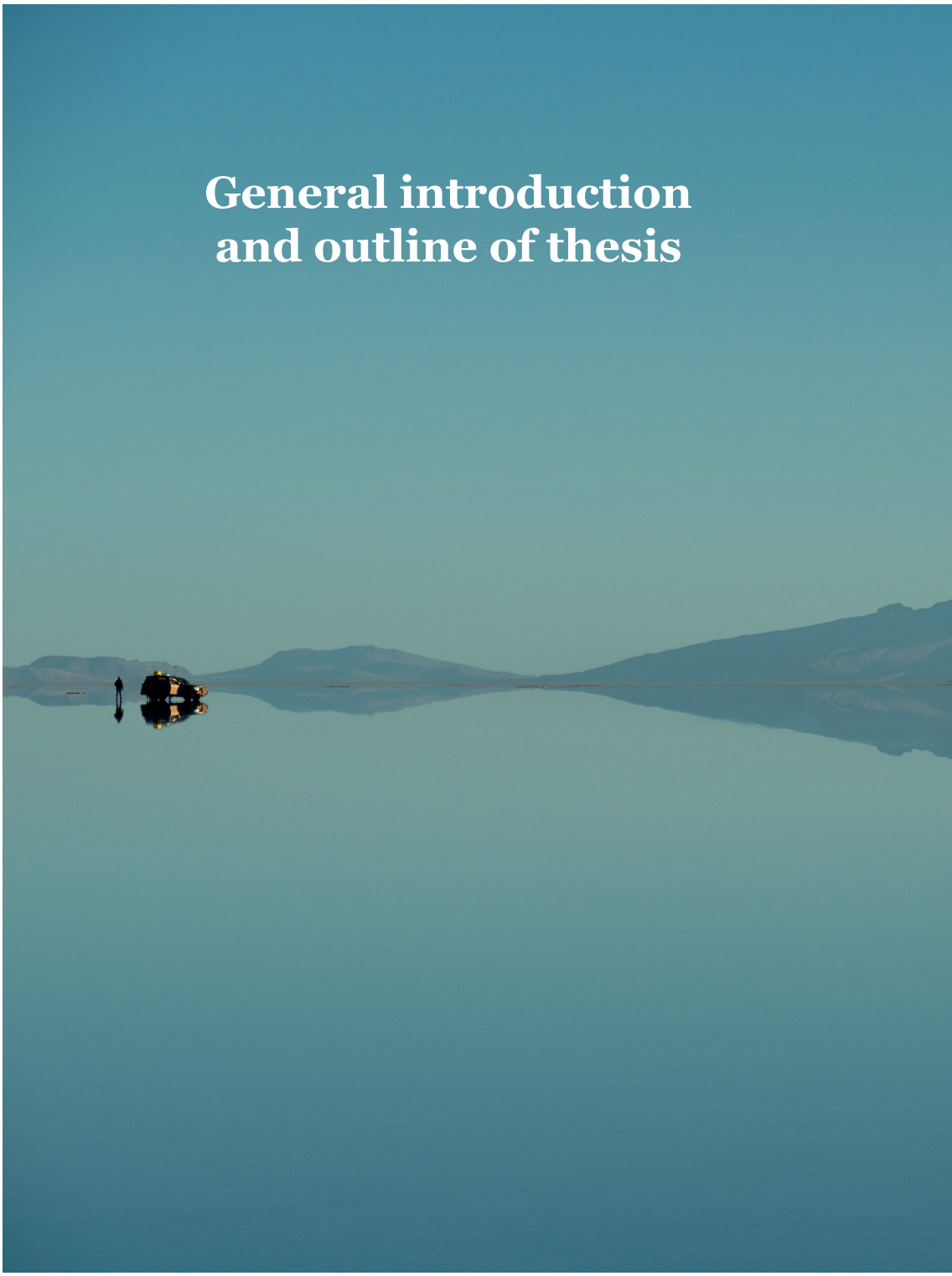
General introduction and outline of thesis



1

Pediatric cancer

Progress and remaining challenges

Each year, approximately 400,000 children in the world are newly diagnosed with cancer.[1] The highest incidence is observed in children aged 0-4 years and adolescents aged 12-18 years. Before 1970, the survival rate for children diagnosed with cancer was below 20%. Since then, survival has improved dramatically in high-income countries thanks to sustained efforts in basic, translational and clinical cancer research (Fig. 1).[2] Currently, in the Netherlands, approximately 84% of children diagnosed with cancer survive at least five years after diagnosis.[3] However, despite this remarkable improvement in survival percentage, two major challenges remain.

First, cancer remains a leading cause of disease-related death in children in developed countries and outcomes remain poor for children with high-risk disease.[4] The overall survival rate masks substantial differences between tumor types and risk groups. While five-year survival exceeds 95% for diseases such as Hodgkin lymphoma, retinoblastoma and low-grade glioma, outcomes remain poor for several high-risk malignancies, including high-risk neuroblastoma (~50%), metastatic osteosarcoma (30%), and high-grade glioma (7%).[5–8] Further improving survival for these high-risk patients is a major priority in pediatric oncology. Second, nearly 75% of survivors experience at least one treatment-related adverse event, and 40% experience a severe life-threatening or disabling event.[2,9,10] Curative treatment often requires multimodal therapy, including combinations of chemotherapy, surgery, radiotherapy, stem cell transplantation and immunotherapy. As a result, childhood cancer survivors have an increased risk of (chronic) health conditions, including cardiovascular disease, endocrine disorders, infertility, neurocognitive impairment and secondary cancers.

Improving outcomes for high-risk patients while simultaneously reducing treatment-related toxicity and preserving quality of life represents one of the central goals of pediatric oncology research. This thesis aims to contribute to this goal by investigating precision medicine approaches that can support the identification of more effective, patient-specific therapies for extracranial solid tumors with a focus on neuroblastoma.

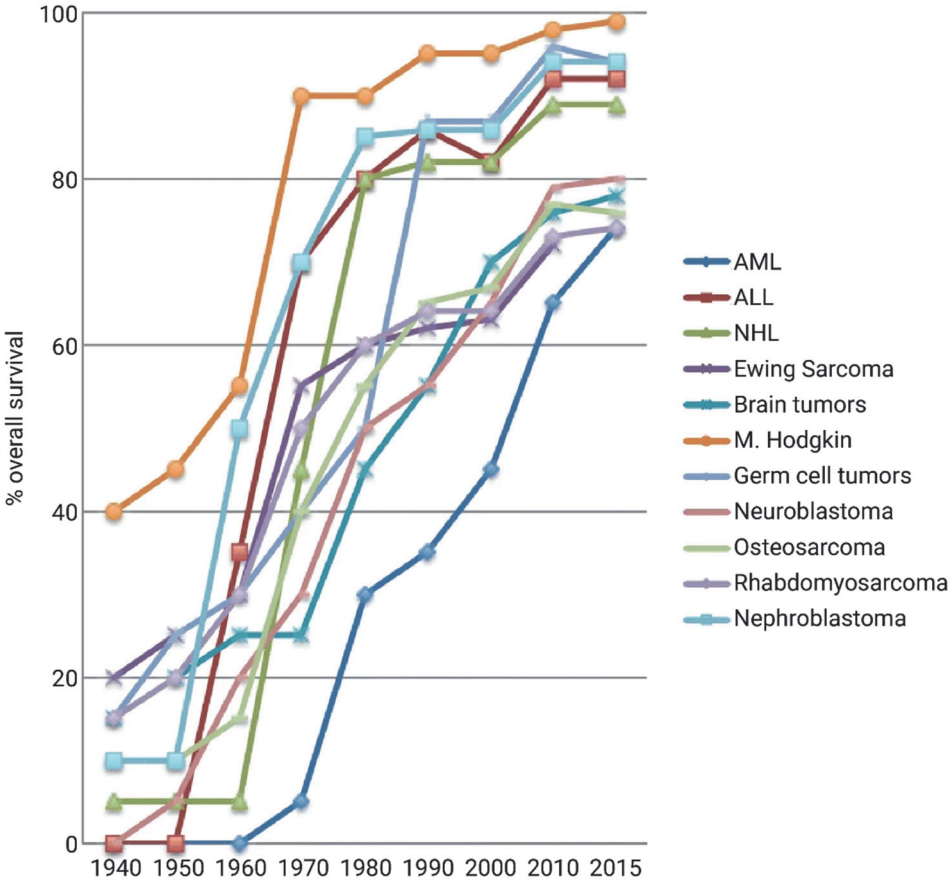


Figure 1: Improvements in pediatric cancer survival over time. *Overall survival rates for major pediatric tumor types in Germany from 1940 to 2015, by Burdach et al.[2] Two-year-overall survival is shown until 1980 and five-year overall survival thereafter. Survival has improved substantially across all pediatric malignancies over the past decades. Despite these improvements, survival gains have varied between tumor types and outcomes remain poor for several high-risk tumors.*

Extracranial solid tumors

Extracranial solid tumors arise outside the central nervous system. These tumors originate from a variety of tissues, including bone (osteosarcoma and Ewing sarcoma), skeletal muscle (rhabdomyosarcoma), connective tissue (synovial sarcoma), peripheral nerves (malignant peripheral nerve sheath tumor), liver (hepatoblastoma), and the developing sympathetic nervous system

(neuroblastoma). These cancers encompass a broad range of clinical presentations and treatment strategies, including surgery, chemotherapy, radiotherapy and targeted or immunotherapeutic approaches. Despite advances in conventional treatment, outcomes remain poor for several high-risk tumors. Neuroblastoma is the most common extracranial solid tumor and the primary disease studied throughout this thesis.

Neuroblastoma

Origin and clinical course

Neuroblastoma arises from neural crest progenitor cells of the developing sympathetic nervous system and remains challenging to cure.[6,11] Annually, approximately 25 to 30 children are diagnosed with neuroblastoma in the Netherlands.[12] Neuroblastoma predominantly affects young children, with a median age at diagnosis of 18 months.[6] Primary tumors can arise anywhere along the sympathetic nervous system, although more than half originate in the adrenal gland.[13] In those cases, patients present with abdominal pain, nausea and a swollen abdomen. Approximately 50% of patients present with metastatic disease at diagnosis, most commonly involving the bone marrow, bone, liver and skin.[14] The diagnosis is typically determined using radiological investigations including ultrasound and full-body MRI, blood and urinary analyses including catecholamine measurements, metaiodobenzylguanidine (MIBG) nuclear imaging, and tumor tissue biopsies and bone marrow aspirates and biopsies.

A defining characteristic of neuroblastoma is its remarkable difference in clinical course across different risk groups. It ranges from spontaneous regression without treatment to highly aggressive metastatic disease.[14] This heterogeneity is captured in multiple risk stratification systems, which incorporate age at diagnosis, disease stage, histopathological features and molecular characteristics, to guide therapy intensity.[15] Risk-adapted treatments maintain excellent outcomes for low- and intermediate-risk patients while minimizing treatment-related adverse effects.[14] In contrast, children with high-risk neuroblastoma require intensive multimodal treatment. This consists of induction chemotherapy, surgery, consolidation with (tandem) high-dose chemotherapy with stem cell rescue, and radiotherapy, followed by maintenance

therapy with anti-GD2 antibody-mediated immunotherapy and retinoic acid.[12,14] The introduction of the anti-GD2 antibody as maintenance treatment has significantly improved survival for children with high-risk neuroblastoma.[16] Nevertheless, over 50% of high-risk neuroblastoma patients develop relapsed and/or refractory disease.[14,17,18] Long-term cure of these patients is uncommon, highlighting the need for novel therapeutic approaches and improved methods to identify effective treatments for individual patients. The efficacy of these treatments is influenced not only by tumor-intrinsic properties, but also by the surrounding tumor microenvironment.

The tumor microenvironment

Neuroblastoma tumor cells are surrounded by the tumor microenvironment consisting of fibroblasts, endothelial cells, Schwann cells, and a variety of immune cells, including macrophages, T-cells, B-cells, natural killer cells and dendritic cells.[19–21] These non-malignant cells interact with tumor cells and contribute to tumor progression, immune evasion and response to treatment.[19] Among the components of the tumor microenvironment, immune cells are of particular interest because of their role in anti-tumor immunity. Tumors can be broadly classified as immunologically “hot” or “cold” depending on the extent of immune-cell infiltration. Neuroblastoma is generally considered a cold tumor type which uses multiple mechanisms to escape immune surveillance, including the transcriptional downregulation of MHC class I molecules, expression of immune checkpoint molecules, and the secretion of immunoregulatory mediators that suppress immune infiltration and activity.[14,20] This immunosuppressive phenotype is closely linked to tumor genotype. For example, MYCN-amplified tumors show lower immune-cell infiltration and reduced antigen presentation, illustrating how tumor-intrinsic genetic drivers shape the composition of the microenvironment.[20] Tumor infiltration by T and NK cells has been associated with improved survival, highlighting the clinical relevance of the immune microenvironment in neuroblastoma.[20] The importance of immune-tumor interactions has stimulated the development of immunomodulatory therapies to enhance anti-tumor immune responses. Building on the success of anti-GD2 antibody therapy, novel approaches including immune checkpoint blockade, antibody-

based therapies and adoptive T cell therapy such as CAR-T cells are currently being explored for neuroblastoma.[14,20,22–25] Preclinical models that incorporate immune cells are essential to further test these immunomodulatory therapies *in vitro*.[26–29]

Genetic alterations

Beyond the tumor microenvironment, neuroblastoma is characterized by a relatively small number of recurrent genomic alterations. These alterations include gene amplifications, mutations or chromosomal aberrations and can affect gene expression, protein function or cellular signaling pathways. They can provide important prognostic information to use for risk stratification or indicate sensitivity to specific therapies which can be targeted through treatments directed against the affected pathway.

The most important genomic alteration is MYCN amplification, first identified in neuroblastoma in 1983,[30,31] which occurs in approximately 20% of tumors and is associated with poor clinical outcome.[15] Consequently, MYCN status plays an important role in risk stratification. ALK mutations and amplifications occur in 6-12% and 1-2% of primary neuroblastomas, respectively.[14,32] The genomic landscape also evolves under treatment: alterations in ALK as well as the RAS–MAPK and TP53 pathways become enriched at relapse and contribute to therapy resistance. [33,34] Additional recurrent alterations involve telomere maintenance mechanisms, including ATRX alterations and TERT rearrangements.[14] In addition, specific chromosomal aberrations, including 1p deletion, 11q deletion and 17q gain, are associated with an increased risk of relapse and unfavorable outcome.[35,36]

The common genomic alterations in neuroblastoma differ markedly in their current clinical actionability. Activating ALK mutations and ALK amplifications are among the clearest therapeutically actionable alterations in neuroblastoma and have stimulated the use of ALK-directed therapies in clinical trials.[34,37] For example, lorlatinib, a third-generation ALK inhibitor that was originally developed and approved for adult ALK-positive non-small cell lung cancer, was repurposed for ALK-driven neuroblastoma. It has been evaluated in patients with relapsed or refractory disease and is currently being explored as a first-line treatment for newly diagnosed patients with ALK activation (NCT03126916).[34,38]

In contrast, although MYCN is an attractive therapeutic target, the N-Myc protein remains largely undruggable due to its intrinsically disordered structure.[39] Accordingly, current preclinical efforts focus mainly on indirect MYCN-targeting strategies like Aurora kinase inhibition.[40–42] More broadly, this illustrates a central challenge in pediatric precision oncology: the identification of a genomic driver does not necessarily translate into an effective or readily available matched therapy.

Precision medicine

Pediatric precision medicine

Precision medicine aims to tailor treatments to individual patients to maximize anti-tumor efficacy while minimizing treatment-related toxicity. Neuroblastoma, as discussed above, provides an example of using patient-specific characteristics to tailor treatment intensity according to risk. Beyond risk-adapted treatment, precision medicine uses patient-specific characteristics to guide therapeutic decision-making. In adult oncology, this has transformed clinical care through approaches such as hormone receptor-guided treatment in breast cancer.[43]

The successful implementation of precision medicine in adult oncology has stimulated similar efforts in pediatric cancer, but important biological and practical challenges remain. Adult cancer therapies cannot simply be carried over to pediatric oncology, as pediatric cancers differ fundamentally from adult cancers in their biology and clinical characteristics. Whereas adult cancers typically arise through the gradual accumulation of genetic mutations over time, pediatric cancers often originate from disruptions in normal developmental processes. Consequently, pediatric cancers generally have a substantially lower mutational burden and harbor driver alterations distinct from adult tumors.[44] These differences highlight the need to develop pediatric-specific therapeutic strategies. Furthermore, pediatric cancers are rare diseases, resulting in small patient populations that complicate clinical trial design and slow down evidence generation. In addition, the relatively small market size reduces commercial incentives for pediatric-specific drug development. Therefore, identifying patient-specific therapeutic vulnerabilities

may enable the selection of already existing treatments and improve clinical strategies, especially for pediatric cancers. These therapeutic vulnerabilities can be identified through both molecular and functional precision medicine approaches.

Molecular precision medicine

Molecular profiling through sequencing of the genome and transcriptome has improved diagnostic classification and risk stratification of pediatric tumors and has become standard of care for all children treated at the Princess Máxima Center. In addition, molecular profiling can identify therapeutically targetable molecular alterations that may guide treatment selection. This has improved treatment for selected patients, but molecular profiling alone is insufficient to identify effective therapies for all patients. Large adult precision oncology studies have demonstrated the feasibility of molecularly profiling tumors and matching patients to targeted therapies. However, therapeutic benefit remains limited, as actionable alterations are not identified in all patients and responses to matched therapies are variable.[45–52]

To evaluate the therapeutic potential of molecular precision medicine in clinical practice, several precision oncology programs have been established in which pediatric tumors undergo molecular profiling and the results are discussed in multidisciplinary molecular tumor boards to identify potential treatment options.[48,51,53–56] At the Princess Máxima Center, this approach is implemented through the nationwide iTHER program for children with relapsed, refractory or high-risk tumors. These pediatric precision oncology programs have encountered similar challenges to those observed in adult oncology studies. Although potentially actionable alterations can be identified in a substantial proportion of pediatric patients, only a subset are considered clinically relevant targets and an even smaller proportion of patients ultimately receive matched treatment.[53,54] Barriers to receiving matched treatment include rapid disease progression, disease stabilization with conventional therapy, lack of access to targeted agents and patient or family preferences. For example, the iTHER program identified actionable alterations in 82% of patients, with 53% classified as moderate- to very-high-priority targets. Despite the identification of these alterations, only 3.5% of all successfully profiled patients ultimately received matched treatment.[54]

Although patients receiving matched therapies for selected (high-priority) genomic targets have shown encouraging treatment responses, molecular profiling does not yet result in an effective matched treatment for many pediatric cancer patients.[50,53,54,56–58] The presence of a molecular alteration does not necessarily predict sensitivity to the corresponding targeted therapy.[34,59,60] An explanation might be that molecular profiling provides a snapshot of a tumor and does not fully capture the complex biological processes that ultimately determine treatment response, including epigenetic cell states, pathway activity and interactions with the tumor microenvironment.[19,21,51,61,62] These limitations have stimulated interest in complementary approaches that directly measure drug response in patient-derived models, giving rise to the field of functional precision medicine.

Functional precision medicine

Instead of identifying genetic alterations, functional precision medicine directly tests the effects of drugs or drug combinations on living tumor cells. This strategy is increasingly being explored to guide treatment decisions in both adult and pediatric oncology.[50,58,60,61,63–67] Preclinical models for functional precision medicine must meet three key requirements: they must be feasible with limited patient material, predictive of in-patient drug response and rapid enough to support real-time treatment decisions. In the INFORM study, molecular profiling results were discussed in molecular tumor boards after a mean turnaround time of approximately 25 days.[53] To allow functional profiling results to be integrated into these discussions, drug testing should be completed within a similar timeframe.

Three main types of preclinical models are used for functional precision medicine: patient-derived xenografts (PDX), tumor organoids and *ex vivo* short-term cultures of tumor material. Each offers distinct advantages and limitations for precision oncology applications, which are summarized in Figure 2 and discussed in detail below. In PDX models, patient tumor material is implanted into (immunodeficient) mice to establish *in vivo* tumor growth. PDX models are widely used due to their ability to preserve major genetic characteristics of the originating tumor and enable *in vivo* assessment of drug efficacy and tolerability.[68,69] Importantly, drug responses observed in PDX models have been

shown to reflect clinical responses in patients.[70] However, engraftment and stable expansion typically require one to eleven months, limiting their suitability for real-time clinical decision-making.[50,58,71,72] PDX models are further constrained by low engraftment rates, high costs and low scalability.[50,58] Ethical considerations further restrict their use, and regulatory agencies encourage the development of patient-derived non-animal alternatives to reduce animal testing, in line with the 3Rs framework (Replace, Reduce, Refine).[73]

Tumor organoid models are long-term 3D cultures established from patient- or PDX-derived tumor cells. Originally pioneered in Utrecht,[74,75] organoids are now widely established for neuroblastoma and other pediatric solid tumors to recapitulate the biology of solid tumors *in vitro*.[63,72,76] Organoids preserve key aspects of the 3D tumor architecture and, once established, can be expanded and screened at a large scale while requiring fewer resources than PDX models.[77] Organoid drug responses have been shown to correlate with clinical responses in patients and are currently being evaluated in prospective clinical trials in adult oncology.[60,66,78] Nevertheless, model validation is required to confirm tumor cell identity, as stromal or other non-malignant cells can overgrow long-term cultures depending on culture conditions.[79–81] Their application in precision medicine remains limited by relatively low establishment rates, loss of the tumor microenvironment and prolonged establishment timelines of four to seven months, which may exceed progression-free survival of relapsed or high-risk patients.[50,53,58,60,63,65,66,72,77,82]

Ex vivo short-term drug screening can be performed rapidly, bypassing the prolonged timelines associated with organoid and PDX establishment. Patient- or PDX-derived tumor material is dissociated directly after tumor resection or biopsy and cultured only briefly prior to drug testing. Promising results are increasingly emerging from adult oncology, where functional precision medicine approaches are currently being evaluated in prospective clinical trials.[60,83,84] *Ex vivo* short-term screening has also been explored in pediatric precision oncology through programs including INFORM (Europe, international) and ZERO (Australia/New Zealand). These programs demonstrate the feasibility of integrating drug screening results into molecular tumor boards and identifying therapeutic vulnerabilities beyond molecular profiling alone.[61,64] Moreover, selected cases have shown concordance between *ex vivo* drug responses and

observed clinical responses.[61,64,85] Despite these promising results, several challenges continue to limit the broader implementation of *ex vivo* short-term screening in pediatric oncology. First, minimally invasive biopsies and heavily pre-treated resections often yield only limited amounts of viable tumor material for drug screening.[60,61,86] This reduces screening success rates and the number of compounds to be evaluated. Second, patient-derived samples contain substantial but varying proportions of non-malignant cells of the tumor microenvironment.[86,87] As conventional viability assays measure the combined response of all cells within a well, distinguishing tumor-specific drug responses from effects on non-malignant cells remains challenging.[26–29,88]

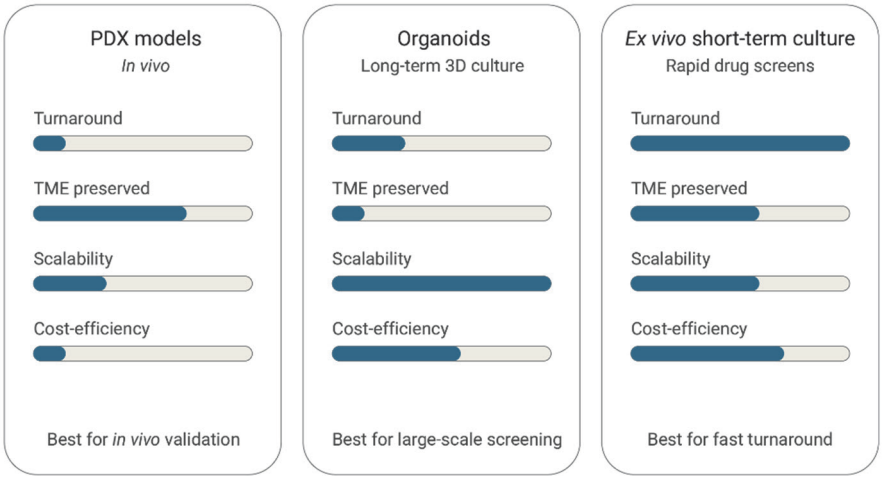


Figure 2: Comparison of patient-derived preclinical model systems for functional precision medicine. *Patient-derived xenograft (PDX) models, patient-derived organoids and ex vivo short-term drug screening differ in turnaround time, preservation of the tumor microenvironment (TME), scalability and cost-efficiency. Relative scores indicate the strengths and limitations of each model system.*[89]

Outline of this thesis

Functional precision medicine has emerged as a promising strategy to identify more effective therapies for children with high-risk cancer. However, its broader implementation is currently limited by slow model establishment, insufficient viable tumor material and the inability to distinguish tumor-specific drug

responses from effects on non-malignant cells. Therefore, this thesis explores strategies to improve the methodology, feasibility, scalability, and physiological relevance of preclinical drug testing across different preclinical model systems in neuroblastoma and other solid tumors.

To enable clinical implementation, drug responses must be generated within a timeframe compatible with clinical decision making. Therefore, in **Chapter 2**, we optimize protocols for and evaluate the feasibility and potential of *ex vivo* short-term drug screening across a broad range of pediatric extracranial solid tumors in a collaboration between the Princess Máxima Center and Curie Institute. Using PDX-derived models representing nine pediatric tumor types, we demonstrate that drug response can be obtained within two weeks and with cross-site reproducibility. Furthermore, *ex vivo* short-term drug screening identifies therapeutic opportunities beyond those identified by molecular profiling alone.

Different preclinical models offer distinct advantages and limitations for functional precision medicine. In **Chapter 3**, we compare the translational potential of different neuroblastoma model systems: PDX- and patient-derived *ex vivo* short-term cultures and organoids, as well as parallel *in vivo* drug testing in PDX models. We demonstrate concordance in drug responses across sample origin and drug screening method and assess the suitability of each model for different applications. While organoids remain essential for drug discovery and PDX models provide *in vivo* representation of the disease, *ex vivo* short-term drug screening provides a rapid alternative for functional precision oncology.

Limited amounts of viable tumor material restrict the success rate of *ex vivo* short-term drug screens. Therefore, in **Chapter 4** we investigate whether feasibility, scalability and standardization of neuroblastoma drug screening can be improved using MicroOrganoSphere™ technology. In this approach, tumor cells are encapsulated in small 3D Matrigel droplets, and the method has recently shown promise for drug testing of small needle biopsies. We demonstrate that MicroOrganoSphere™ technology supports tumor growth and reliable drug screening, using fewer cells than conventional drug screening cultures.

Patient-derived samples and co-cultures with immune cells better represent the tumor microenvironment but contain multiple cell populations. Conventional viability assays measure the combined response of all cells in the well and therefore cannot distinguish tumor-specific drug responses from effects on non-malignant cells. To address this challenge, in **Chapter 5**, we develop a scalable microscopy-based viability assay to quantify cell type-specific drug responses in neuroblastoma-immune co-cultures and *ex vivo* short-term cultured patient samples. This approach enables assessment of tumor-specific drug sensitivity while simultaneously evaluating effects on non-malignant cells.

Finally, **Chapter 6** discusses the findings presented in this thesis in the broader context of identifying more effective treatments for pediatric solid tumors. The opportunities and limitations of different preclinical model systems are considered, together with the applications for which they are best suited. The role and challenges of drug screening in precision medicine are discussed alongside the potential of these approaches for broader drug discovery studies.

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Ex vivo drug sensitivity profiling to complement molecular profiling in pediatric precision oncology

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Abstract

Pediatric patients with high-risk extra-cranial solid tumors face a 5-year survival rate below 50%. As molecular profiling alone is insufficient to guide treatment at relapse, complementary strategies like drug screening are urgently needed. We evaluated short-term drug screening as a rapid, reliable method to assess drug sensitivities in pediatric solid tumors using *ex vivo* cultures from previously established patient-derived xenograft (PDX) models. *Ex vivo* drug screening was performed within 14 days of receipt across two institutes, testing 77-224 compounds depending on cell availability. Drug responses were consistent across institutes (n = 6), and effective compounds were reproducibly identified in a replicate model. Tumor type-specific responses were observed. In neuroblastoma, ALK-mutation status did not correlate with ALK-inhibitor response, whereas correlations with transcriptomic changes were observed. Timepoint-specific drug sensitivities were observed in serial Ewing sarcoma models. Overall, drug hits were identified in 94% of screens (n = 63), broadening treatment options for 85% of cases without targetable alterations (n = 11). In case of a targetable event, drug screening refined compound choice. *Ex vivo* drug screening is a fast and feasible method, providing insights into compound efficacy and enabling quick identification of functional treatment suggestions. *Ex vivo* drug screening should be integrated into a future next-generation diagnostic platform for pediatric solid tumors, combined with genomics and transcriptomics.

Introduction

Although pediatric tumors are uncommon, they are the leading cause of disease-related death in children.[1] High-risk extracranial pediatric solid tumors have a poor prognosis despite intensive multimodal treatment approaches.[2] For survivors, treatment often results in severe long-term side-effects.[3] Heterogeneity in clinical response, even within cancer types, highlights the urgent need for more effective, patient-specific therapeutic approaches, based on response prediction. Currently, upon relapse or progression, pediatric precision oncology trials rely on identifying genetic biomarkers to guide clinical decisions.[4–8] However, molecular characterization alone has limitations. Actionable driver genes cannot be

identified in all patients, and even when they are, responses to matched therapies are often limited.[6,7,9–11] This is partly because actionable genetic alterations do not fully capture the biological complexity of pediatric tumors.[12] Therefore, additional strategies beyond genomics are needed to guide treatment decisions.

Functional precision medicine, such as drug screening on patient-derived material, offers a promising approach to improve response prediction.[12,13] *In vivo* patient-derived xenograft (PDX) and *in vitro* patient-derived organoid models have proven effective to identify potential treatments.[9,13–16] However, establishment of these models requires extended time frames that often exceed progression-free survival of patients,[6] a limitation we also observed in our study on high-throughput screening in neuroblastoma patient-derived organoids.[17] In addition, using *in vivo* PDX models limits the number of drugs that can be tested. Screening within a clinically relevant time frame would improve applicability. *Ex vivo* drug sensitivity screening, where fresh tumor material is directly used or only shortly expanded *in vitro* before compound screening, presents such a faster alternative. Recent studies, including those from INFORM and ZERO programs, have demonstrated the potential of *ex vivo* drug sensitivity profiling within shorter time frames, ranging from seven days to four months.[11,18,19]

We evaluated the feasibility of integrating rapid *ex vivo* drug sensitivity screening with molecular profiling, using *ex vivo* cultures from PDX-expanded pediatric tumors as tumor models. These models recapitulate molecular and phenotypic profiles of the patient tumor [15,20–22] and provide sufficient biomaterial for analysis. We demonstrate *ex vivo* drug screening as a feasible and reproducible method for a wide variety of pediatric tumors using *ex vivo* cultures from PDX models. Importantly, *ex vivo* drug sensitivity profiling of previously established PDX models can be completed within 14 days, aligning with the time frame of molecular profiling. Our findings confirmed known associations between genomic alterations and drug sensitivity, and we identified correlations between drug response and genomics, transcriptomics, and tumor growth rates. *Ex vivo* drug screening complements molecular profiling by broadening treatment options when no targetable genetic alterations are identified and by refining drug selection when one or several actionable targets are present. These results support the future integration of *ex vivo* drug sensitivity profiling alongside genomics and transcriptomics within a combined diagnostic platform, aiming to improve clinical decision-making for high-risk pediatric solid tumors.

Results

Ex vivo drug sensitivity profiling is fast, feasible and reproducible in determining drug sensitivity profiles for a broad range of solid tumors

To perform this feasibility study of *ex vivo* drug screening in a clinically relevant timeframe, Institut Curie (Curie) in France and the Princess Máxima Center (Máxima) in the Netherlands used previously established pediatric PDXs as the pediatric tumor models. Using a comparable sample processing and drug screening protocol at both institutes, we were able to determine drug sensitivity profiles within 14 days of PDX sample collection (Fig. 1A). We successfully performed 69 drug screens (37 in Máxima, 32 in Curie) on *ex vivo* cultures from PDX models. Screening was technically successful in 82% of cases (69/84 screens), with failures due to low viable cell yield or microbial contamination (Fig. 1B). The cohort of successful screens represent nine tumor types, including neuroblastoma (n = 26), Ewing sarcoma (n = 17), rhabdoid tumors (atypical teratoid/rhabdoid tumors and extra cerebral rhabdoid tumor samples; n = 11), rhabdomyosarcoma (n = 5), osteosarcoma (n = 4), hepatoblastoma (n = 2), malignant peripheral nerve sheath tumor (MPNST, n = 2) and synovial sarcoma (n = 2; Fig. 1C; Suppl. Fig. 1A showing center-specific cohorts). *Ex vivo* cultures were derived from PDX established from high-risk cancer patients with diverse clinical backgrounds, including untreated (at diagnosis), relapsed and progression under treatment samples. The PDX samples consisted mostly of human cells, as determined for PDX or *ex vivo* expanded samples (Suppl. Fig. 1B-C), except for a single screen (NB0537), which was therefore left out of analyses comparing drug response to genetic aberrations and gene expression.

At both institutes, drug screening was performed with libraries that include cytotoxic, cytostatic and targeted compounds (Suppl. Fig. 1D), that are either clinically approved or in (pre-)clinical evaluation to treat (pediatric) cancers (Suppl. Tab. 1-2). Depending on available PDX material, tumor cells were exposed to libraries consisting of 77 to 224 compounds, with at least 61 compounds shared across the two institutes (Fig. 1D). Drug concentrations ranged from 1e-6 to 1e3 μM, depending on the institute and the expected drug sensitivity (Suppl. Fig. 2A). Normalized area under the curves (AUC_{norm}) are used

to compare drug sensitivities within a single institute, including compounds that did not reach IC50s at the tested concentrations. However, when comparing the two different cohorts, IC50 is used as AUC_{norm} values are not directly comparable due to distinct dose ranges (Suppl. Fig. 2A-B).

To assess the reproducibility of our protocol, we screened two *ex vivo* cultures from PDX models derived from the same patient at the Máxima institute. The sensitivity profiles of the two PDX screens showed a strong correlation (r = 0.95; Fig. 1E). Then, to test robustness of our method, we screened six *ex vivo* cultures from PDX models at both institutes. We observed consistent trends in drug response (r = 0.50-0.85; Fig. 1F), despite differences in PDX passage, tissue expansion time, cell seeding density, and timing of compound addition. At the Máxima, samples are cultured for seven days before drug sensitivity screening, while at Curie compound screening was immediately performed upon receipt. Overall, these findings demonstrate that despite PDX passage and protocol differences, consistent drug sensitivity profiles can be reliably determined.

To correlate the drug sensitivities with the molecular profiles and to compare *ex vivo* drug sensitivity profiling to standard molecular profiling, the PDX models were molecularly profiled using whole-exome sequencing, methylation profiling, and RNA sequencing. Targetable genetic alterations were manually curated (Fig. 1G). Across all tumors, the most common genetic alterations were pathogenic homozygous TP53-inactivation (15/69 cases), MYCN amplifications (19/69), EWS translocations (15/69) and ALK mutations (7/69). PDX models faithfully retained key genetic aberrations of patient tumors, as determined for samples of which original patient tumor profiling was available (Suppl. Fig. 2C). This finding validates that our PDX models accurately reflect the genetic landscape of patient tumors. Overall, these results indicate that e*x vivo* drug sensitivity profiling is feasible and reproducible for pediatric solid tumors, obtaining drug sensitivities within a clinically relevant timeframe in *ex vivo* cultures from previously established PDXs.

Figure 1 (next pages): Ex vivo drug sensitivity profiling is feasible for a broad range of high-risk solid tumors. ***A****) Schematic representation of drug screening protocol.* ***B****) Success rate of ex vivo drug screens by tumor type, showing the proportion of samples with successful screens, samples with insufficient viability for plating, and samples with unsuccessful screens due to either infection or screening failure.* ***C****) Donut plot showing the cohort overview of ex vivo direct screens per tumor type.* ***D****) Number of compounds in drug libraries of the Princess Máxima Center (orange), Curie Institute (blue) and the number of overlapping compounds (brown).* ***E****) Scatterplot of drug sensitivity (AUC_{norm}) for ex vivo cultures from two distinct PDX samples of the same patient sample (NB0277) that were received two months apart and followed the same protocol in Máxima. Each dot represents the AUC_{norm} per compound, colored by drug target. Pearson correlation coefficient = 0.95.* ***F****) Comparison of drug sensitivities (AUC_{norm}) of overlapping patient samples screened in both institutes. For each patient sample, a scatterplot of IC50 values is shown (includes only those compounds for which IC50 was reached in both institutes). Each dot represents a compound, colored by drug target. Pearson correlation coefficients are indicated between brackets.* ***G****) Oncoplot depicting the molecular landscape of PDX models included in the cohort, showing key targetable genetic alterations. MPNST: malignant peripheral nerve sheath tumor; _M: screened in the Máxima institute; _C: screened in Institut Curie; _PP: PDX of primary tissue; _PT: PDX of progression under treatment tissue.*

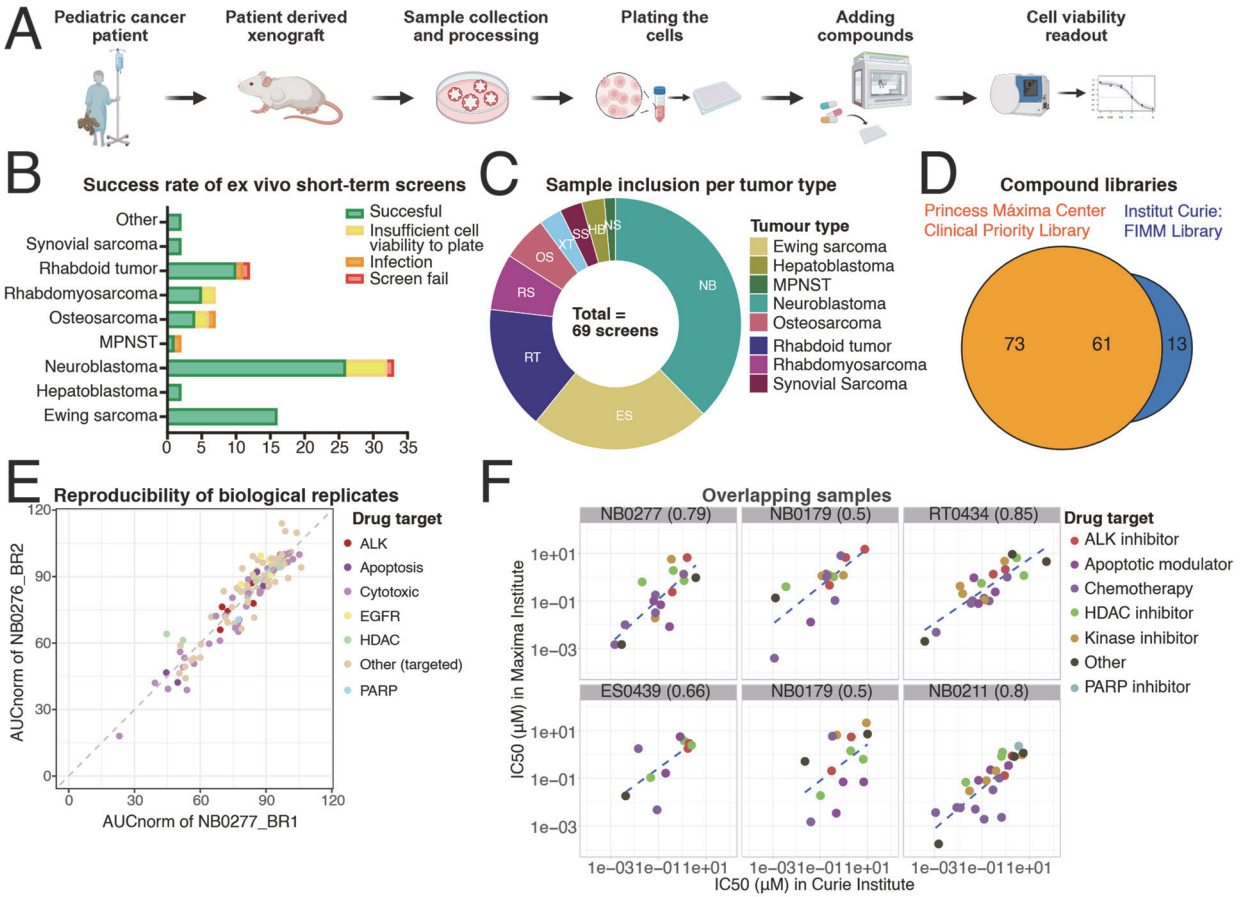

A
Pediatric cancer patient
Patient derived xenograft
Sample collection and processing
Plating the cells
Adding compounds
Cell viability readout
B
Success rate of ex vivo short-term screens
Other
Synovial sarcoma
Rhabdoid tumor
Rhabdomyosarcoma
Osteosarcoma
MPNST
Neuroblastoma
Hepatoblastoma
Ewing sarcoma
Succesful
Insufficient cell viability to plate
Infection
Screen fail
0 5 10 15 20 25 30 35
C
Sample inclusion per tumor type
Total = 69 screens
NB
ES
RT
RS
OS
KT
SS
HB
NS
Tumour type
Ewing sarcoma
Hepatoblastoma
MPNST
Neuroblastoma
Osteosarcoma
Rhabdoid tumor
Rhabdomyosarcoma
Synovial Sarcoma
D
Compound libraries
Princess Máxima Center Clinical Priority Library
Institut Curie: FIMM Library
73
61
13
E
Reproducibility of biological replicates
AUCnorm of NB0276_BR2
AUCnorm of NB0277_BR1
0 30 60 90 120
Drug target
ALK
Apoptosis
Cytotoxic
EGFR
HDAC
Other (targeted)
PARP
F
Overlapping samples
NB0277 (0.79)
NB0179 (0.5)
RT0434 (0.85)
ES0439 (0.66)
NB0179 (0.5)
NB0211 (0.8)
IC50 (µM) in Maxima Institute
1e+01
1e−01
1e−03
1e−031e−011e+01
IC50 (µM) in Curie Institute
Drug target
ALK inhibitor
Apoptotic modulator
Chemotherapy
HDAC inhibitor
Kinase inhibitor
Other
PARP inhibitor

Figure 1

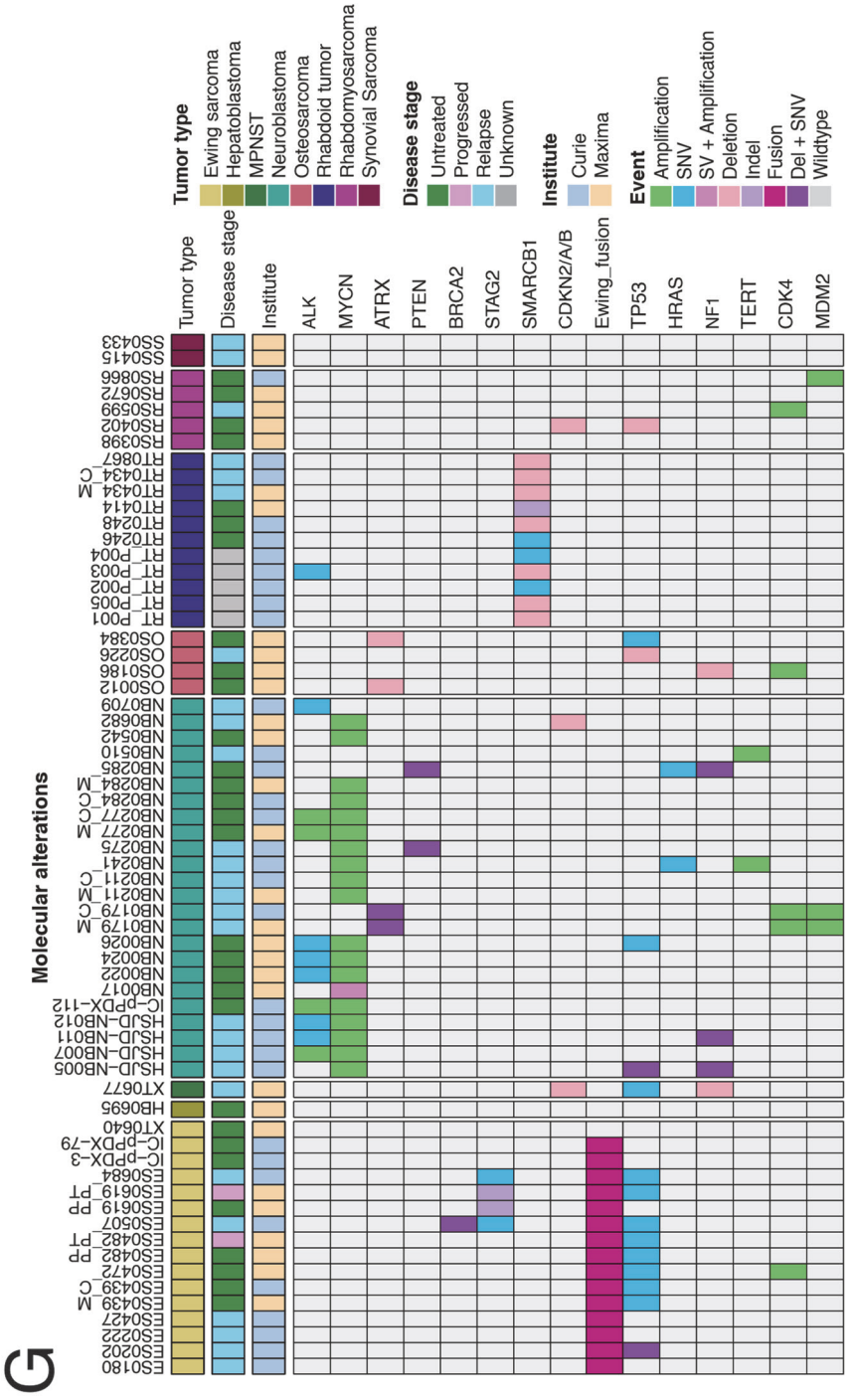


Figure 1 (continued)

Ex vivo short-term profiling to identify tumor-type specific drug sensitivities

We next examined drug response patterns across tumor types by looking at the top fifteen most effective compounds, defined as those with the lowest median AUC_{norm} across all tumor types at each institute. We included the compounds tested in both institutes and find the same fifteen most effective compounds across both cohorts, supporting the consistency of drug screening trends across cohorts. These top performing compounds included the proteasome inhibitor bortezomib, chemotherapeutic actinomycin D, and BCL-2 inhibitor navitoclax (Fig. 2A-B). To assess global patterns of drug sensitivity across tumor types, we performed principal component analysis (PCA) on the drug response profiles (AUC_{norm}; shown in Suppl. Fig. 3). PCA revealed partial clustering by tumor type (Fig. 2C), suggesting that while tumor type contributes to drug response, substantial inter-tumoral heterogeneity exists within types, likely reflecting underlying genetic diversity.

We then assessed which compounds showed tumor-type specific efficacy, only including tumor types where $n \geq 4$ samples were screened. Neuroblastoma samples were significantly more sensitive to the CDK4/6 inhibitor palbociclib and the MEK inhibitor cobimetinib in both cohorts ($p < 0.02$; Fig. 2D; Suppl. Fig. 4A) compared to other tumor types, although rhabdoid tumors showed a similar but non-significant trend. Ewing sarcomas were significantly more sensitive to PARP inhibitors in both cohorts ($p < 0.002$; Fig. 2E), PARP inhibitor talazoparib was particularly effective in Ewing sarcomas at Curie ($p = 5.08E-05$; Suppl. Fig. 4B). At Máxima, BRAF inhibitor vemurafenib was additionally found to be significantly effective ($p = 0.002$; Suppl. Fig. 4B). Rhabdoid tumors, tested at Curie, were significantly more sensitive to tyrosine kinase inhibitors ponatinib and axitinib ($p < 0.002$; Suppl. Fig. 5). Osteosarcoma, a tumor type that is difficult to model *ex vivo* and in urgent need of new treatment options,[23] was broadly resistant to most compounds compared to other tumor types (Suppl. Fig. 6), including for instance talazoparib (Suppl. Fig. 4B). For rhabdomyosarcoma, no compounds or drug classes showed significant efficacy compared to other tumor types (Suppl. Fig. 6-7). However, low sample sizes for osteosarcoma and rhabdomyosarcoma ($n = 4$) limit robust conclusions. Taken together, these results provide initial data on compounds with broad efficacy and tumor type-specific drug response trends.

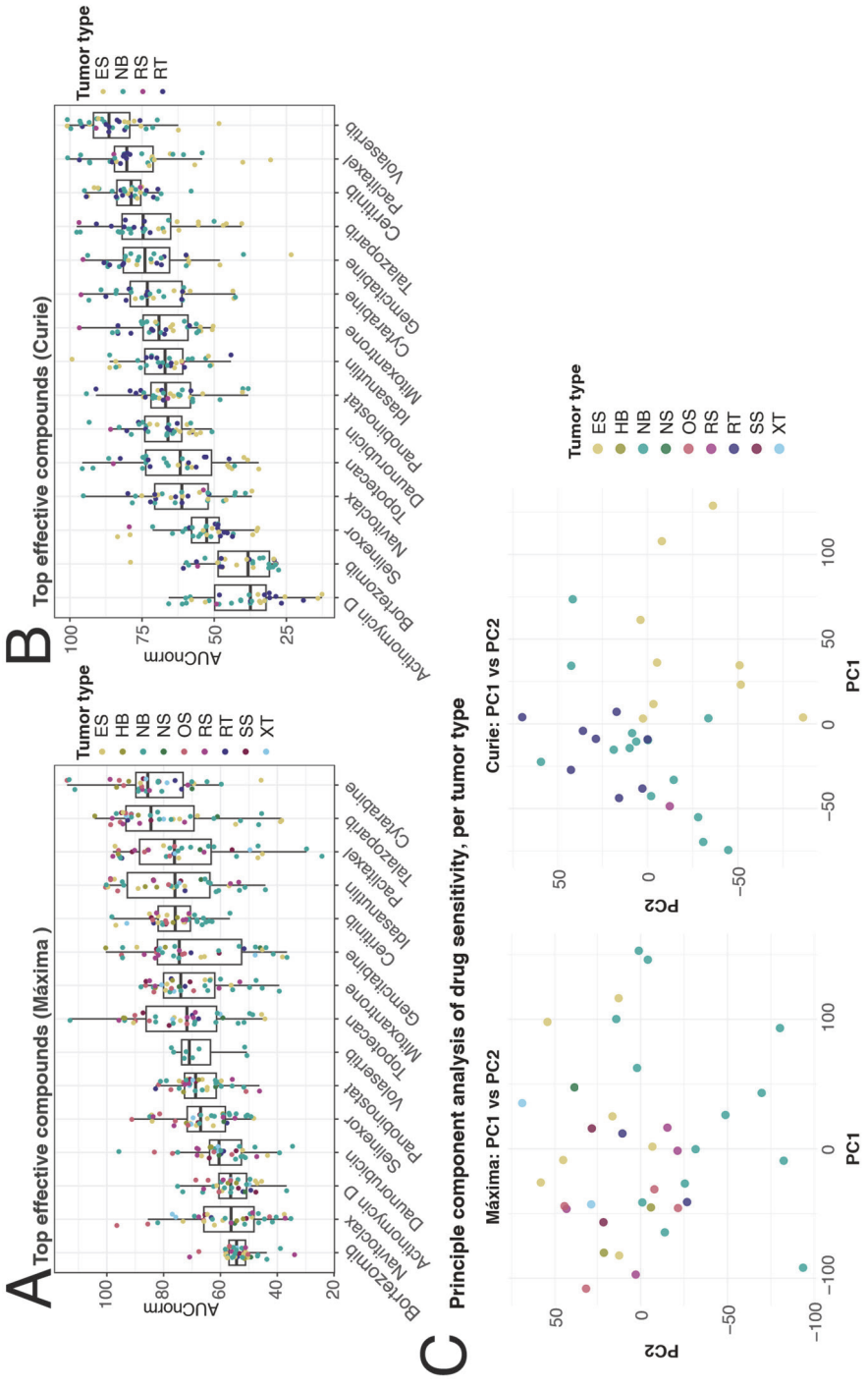

A
Top effective compounds (Máxima)
AUCnorm
Tumor type
ES HB NB NS OS RS RT SS XT
Bortezomib
Navitoclax
Actinomycin D
Daunorubicin
Selinexor
Panobinostat
Volasertib
Topotecan
Mitoxantrone
Gemcitabine
Cytarabine
Talazoparib
Paclitaxel
Idesanutlin
Ceritinib
B
Top effective compounds (Curie)
AUCnorm
Tumor type
ES NB RS RT
Actinomycin D
Bortezomib
Selinexor
Navitoclax
Topotecan
Daunorubicin
Panobinostat
Idesanutlin
Mitoxantrone
Cytarabine
Gemcitabine
Talazoparib
Ceritinib
Paclitaxel
Volasertib
C
Principle component analysis of drug sensitivity, per tumor type
Máxima: PC1 vs PC2
Curie: PC1 vs PC2
PC1
PC2
Tumor type
ES HB NB NS OS RS RT SS XT

Figure 2

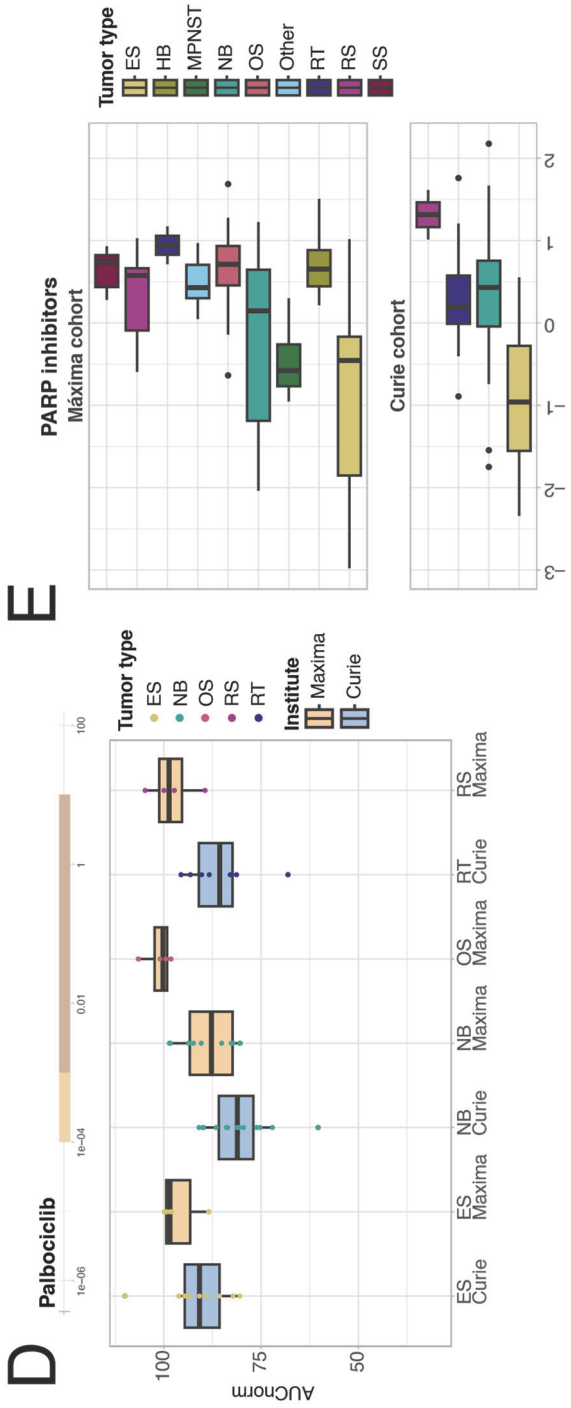


Figure 2 (continued)

***Figure 2 (previous pages): Ex vivo drug sensitivity profiling to reliably predict tumor-type specific response in two cohorts**. **A-B**) Boxplot representing the top 15 performing compounds based on AUC_{norm} in the Máxima (A) and Curie (B) cohorts. Only compounds tested in both institutes are included. Each dot represents an individual sample, colored according to the tumor type. For all box plots: the box represents the interquartile range (IQR), with the median (50th percentile) indicated by the horizontal line inside the box. The whiskers extend to 1.5 times the IQR, with data points outside this range considered as outliers and shown as individual points. **C**) Principal component analysis (PCA) showing PCA1 versus PCA2 on the drug response profiles, each dot represents a sample, colored by tumor type. **D**) Tumor-type specific sensitivity for palbociclib is shown in the two cohorts. Dose ranges (in μM) are indicated on top for both cohorts (Máxima in orange, overlapping in brown). Box plots show AUC_{norm} scores per tumor type, per cohort. Orange boxes represent the Máxima cohort and blue boxes represent Curie. Neuroblastoma samples demonstrate specific sensitivity for palbociclib. **E**) Tumor-type specific sensitivity for PARP inhibitors (olaparib and talazoparib) tested in two cohorts. Box plots show Z-scores (based on AUC_{norm}) per tumor type, for the Máxima (top) and Curie (bottom) cohorts. Ewing sarcomas demonstrate specific sensitivity for PARP inhibitors. A lower Z-score represents greater sensitivity compared to the other samples, the higher the Z-score, the less sensitive the sample is for this compound.*

Genetic aberrations alone poorly associate with *ex vivo* drug response

To assess the predictive value of molecular profiling, we first confirmed the known association of resistance to MDM2-inhibitor idasanutlin and XPO1-inhibitor selinexor upon homozygous TP53-inactivating mutations.[24–27] Samples with homozygous TP53-inactivation were significantly more resistant to the MDM2 inhibitor idasanutlin ($p < 0.002$; Fig. 3A), and selinexor ($p < 0.039$; Suppl. Fig. 8A) compared to TP53 wild type or heterozygous TP53-inactivated samples.

We then explored whether targetable genetic alterations predicted drug sensitivity across tumor types, focusing on compounds targeting these alterations (Suppl. Tab. 3). Beyond TP53, only 3 out of 22 (Máxima) and 1 out of 15 (Curie) showed any significant association with response to their matched compound (Fig. 3B; Suppl. Fig. 8B). None of these associations were shared across cohorts. In Máxima, significant associations involved MYCN amplifications and ALK alterations, which are neuroblastoma specific. None of the associations observed

across the pan-cancer cohort remained significant when this analysis was restricted to neuroblastoma, suggesting the correlations were confounded by tumor-type rather than the genetic alteration themselves (Suppl. Fig. 8C). All in all, these findings suggest that genetic aberrations alone are not sufficient to predict *ex vivo* drug response.

Ex vivo growth rate of tumor samples as a quality control metric

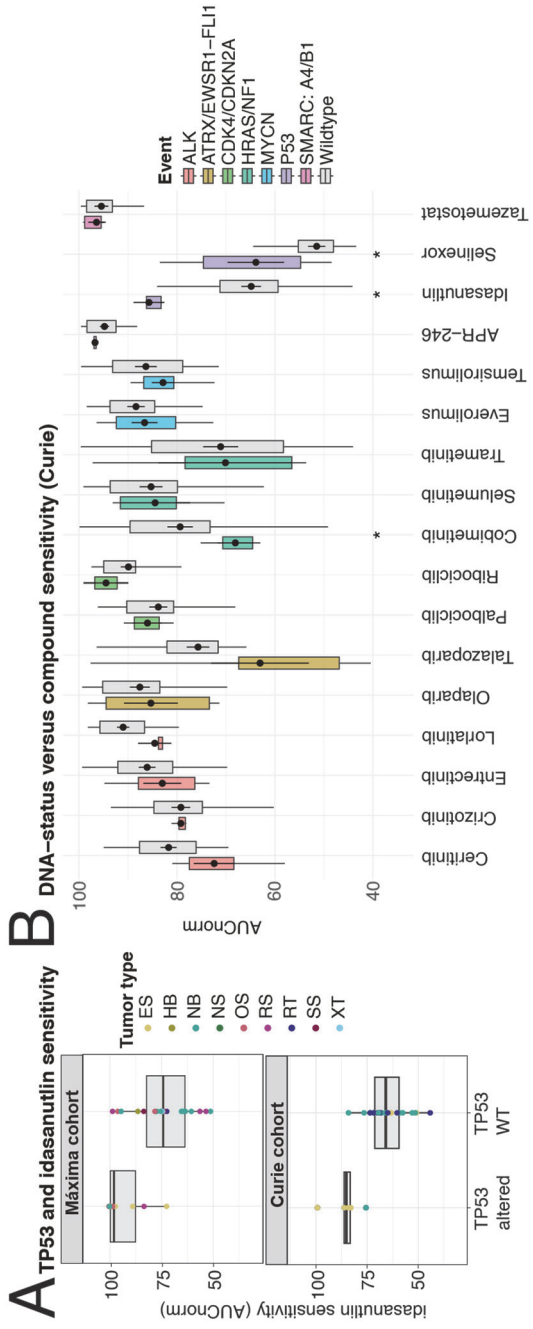
For the screens performed in the Máxima cohort, growth rate was determined by comparing cell viability at t = 0h and t = 72h in untreated controls, to assess screen quality and evaluate its impact on drug sensitivity. Growth rates of the included screens ranged from 60% to 479% (mean = 179%; Suppl. Fig. 8D), with values below 100% indicating spontaneous cell death and those above 100% representing proliferation. Screens with a growth rate <60% (n = 2) were excluded at an earlier stage and marked as failed (Fig. 1C). All four osteosarcoma screens, a tumor type that is generally hard to culture *ex vivo* [23], had growth rates above 100% (mean = 177%). For the full cohort, sensitivity to 24 compounds correlated significantly with growth rate (Suppl. Fig. 9A), particularly cytotoxic and cell cycle-targeting agents such as CHK1 inhibitor prexasertib, the tubulin inhibitor docetaxel (Fig. 3C), gemcitabine and topoisomerase inhibitor camptothecin. These associations confirm their known mechanism of action and validate growth rate as a relevant quality control parameter in *ex vivo* drug screens.

Serial *ex vivo* models illustrate tumor progression and shifting therapeutic windows

To explore whether drug sensitivity may change during tumor progression, and to explore the potential added value of screening drug sensitivities at multiple timepoints, we analyzed serial models from two Ewing sarcoma patients screened in the Máxima. While overall similarity was observed between primary tumor (treatment-naive) and progression under treatment samples, (ES0482: r = 0.78; ES0619: r = 0.72; Fig. 3D), distinct timepoint-specific sensitivities were found. For ES0482, the sample obtained at progression resembled the primary sample genetically (Fig. 1G), yet was generally slightly more sensitive. Higher sensitivity of topoisomerase and tubulin inhibitors in the sample from progression (Fig. 3D), could be explained by the higher growth rate (primary: 172%; progressed: 314%)

(Suppl. Fig. 9A). The progressed sample of ES0619 acquired a homozygous TP53 inactivating mutation and was resistant to idasanutlin (IC50 = 0.8 μM in primary versus not reached in progression; Fig. 3D, Suppl. Fig. 9C). Here, the primary sample generally showed slightly greater sensitivity. Although limited to two cases, these findings illustrate how tumor evolution – both genetic and phenotypic – could impact drug response, therefore highlighting the importance of drug sensitivity screening in newly acquired samples, rather than relying on archived models.

Figure 3 (next pages): Ex vivo drug sensitivity profiling to identify associations with drug response and ex vivo growth rate, and to identify timepoint specific therapeutic windows. ***A****) Box plots representing* AUC_{norm} *for idasanutlin of patients harboring wild type (WT) TP53 (right) and samples harboring homozygous TP53-inactivating events (left) in both cohorts. Samples are colored by tumor type.* ***B****) Boxplots representing sensitivity (*AUC_{norm}*) in Curie for targeted compounds, where colored boxes represent the ex vivo cultures from PDX models with genetic alterations colored by the alteration targeted by that compound (colored = altered, grey = wild type). Stars show a significant difference (*$p < 0.05$*) in drug response in the altered versus wild type samples.* ***C****) Scatter plot that shows correlation of growth rate and* AUC_{norm} *for prexasertib and docetaxel. Each dot represents a sample, colored per tumor type.* ***D****) Drug sensitivity for two Ewing sarcoma patients in two timepoints (primary and progression samples), left represents ES0482 and right represents ES0619. Each dot represents the* AUC_{norm} *of a compound, which is colored by its target subtype. Compounds that are highlighted by the red boxes: topoisomerase and tubulin inhibitors are influenced by the growth rate in ES0482 and idasanutlin, which is influenced by an acquired homozygous TP53-inactivation in ES0619.*


A
TP53 and idasanutlin sensitivity
Máxima cohort
Curie cohort
Tumor type
ES
HB
NB
NS
OS
RS
RT
SS
XT
idasanutlin sensitivity (AUCnorm)
100
75
50
TP53 altered
TP53 WT
B
DNA–status versus compound sensitivity (Curie)
AUCnorm
100
80
60
40
Event
ALK
ATRX/EWSR1–FLI1
CDK4/CDKN2A
HRAS/NF1
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P53
SMARC: A4/B1
Wildtype
Ceritinib
Crizotinib
Entrectinib
Lorlatinib
Olaparib
Talazoparib
Palbociclib
Ribociclib
Cobimetinib
Selumetinib
Trametinib
Everolimus
Temsirolimus
APR–246
Idasanutlin
Selinexor
Tazemetostat
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Figure 3

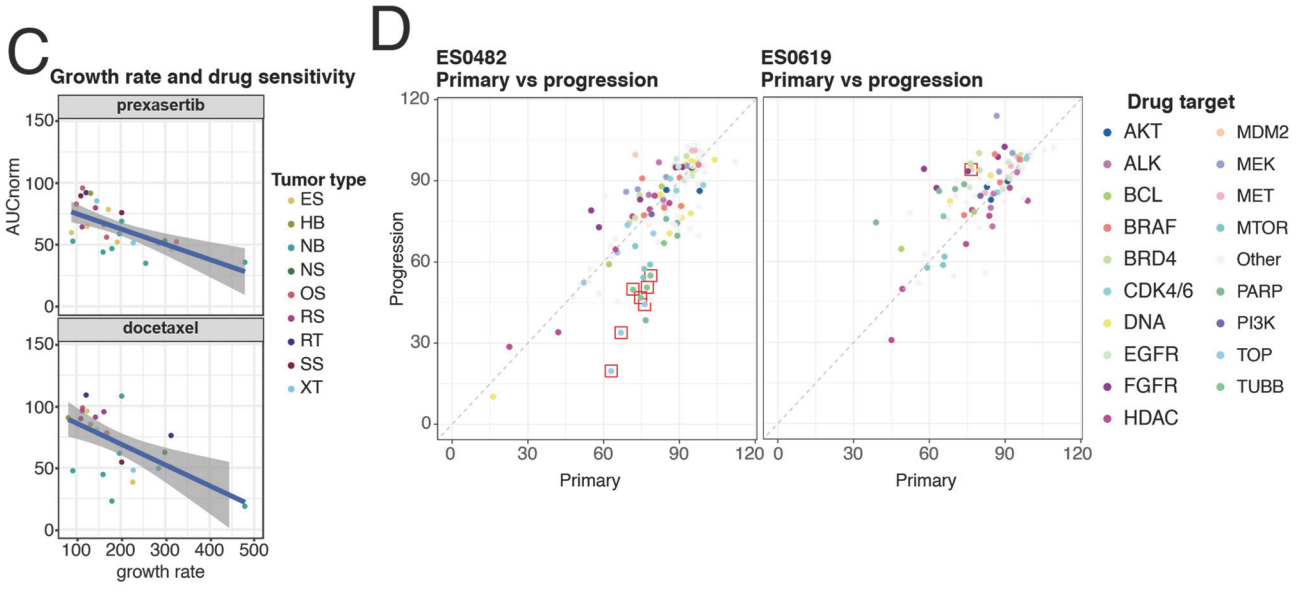


Figure 3 (continued)

RNA expression correlates with *ex vivo* drug sensitivity of targeted compounds

Since genetic alterations alone could not reliably predict drug response, we explored correlations between pre-treatment gene expression and drug response across tumor types. Higher *BCL-2* expression was associated with venetoclax sensitivity across both cohorts, consistent with previous reports ($p < 0.012$; Fig. 4A; Suppl. Fig. 10A).[28] In addition, higher ALK expression correlated with sensitivity to the ALK inhibitors ceritinib and alectinib in the Máxima cohort ($p = 0.002$; $p = 0.020$, Fig. 4B, Suppl. Fig. 10C), though not in Curie, where lower concentrations were tested (Suppl. Fig. 2A-B, 10B-D). We did not find a significant correlation with ALK-expression and lorlatinib sensitivity (Máxima $p=0.054$; Curie $p=0.39$).

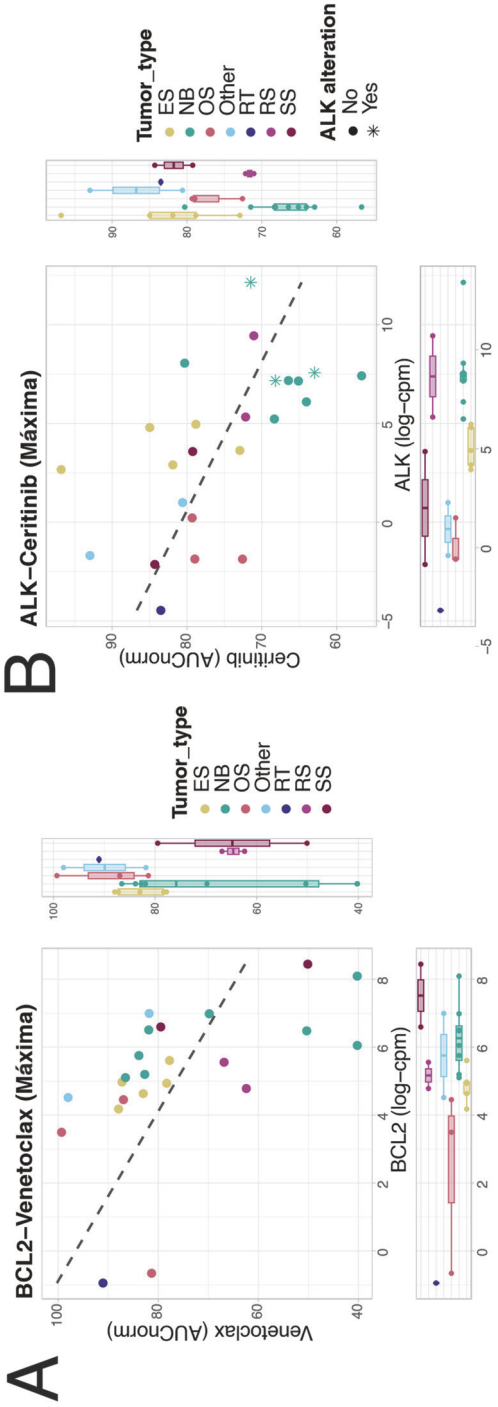
Genomic ALK alterations did not reliably predict sensitivity to ALK inhibition in neuroblastoma. Given the therapeutic relevance of ALK inhibitors, we explored whether baseline gene expression and master regulator (MR) activities were associated with response to alectinib, crizotinib, and lorlatinib across the two cohorts. These analyses focused on neuroblastoma samples. Single-gene analyses identified at least one overlapping gene associated with response to ALK inhibition in both cohorts: 11 for crizotinib, one for lorlatinib and two for alectinib (Suppl. Fig. 10E, 11A). However, we did not find a gene that was significantly associated to sensitivity to all ALK inhibitors in both cohorts. We excluded ceritinib from these analyses due to inconsistent results from the AUC_{norm}-based linear model.

Beyond individual genes, we examined pathway-level enrichments across both cohorts. Although both cohorts consist exclusively of neuroblastoma samples, the different culture conditions and experimental handling introduce substantial batch effects (Suppl. Fig. 11B). Pathway-level analyses help mitigate these technical differences by capturing coherent transcriptional programs that are more robust and reproducible than individual gene signals. Interferon *alpha* and *gamma*, and inflammatory responses were associated with response to crizotinib and lorlatinib in both cohorts (Fig. 4C; Suppl. Fig. 11C). Conversely, pathways such as IL-6/-JAK/-STAT3 signaling and TNF *alpha* signaling displayed drug- and cohort-specific enrichment patterns.

Next, given that MRs orchestrate coordinated gene expression programs, we used the VIPER method[29] to explore if MR activity was associated to ALK inhibitor response in neuroblastoma. In each cohort, we identified MRs whose

regulons were enriched within the AUC_{norm}-associated gene signatures for each compound. We found seven MRs (Fig. 4D) significantly associated with lorlatinib sensitivity in both the Máxima and Curie cohorts. Similarly, we identified 146 overlapping MRs for crizotinib (Suppl. Fig. 12A) and 19 for alectinib (Suppl. Fig. 12B) when comparing the two cohorts. Inferred activity of SMAD4 was associated with sensitivity to all three ALK inhibitors at Máxima, and a similar but non-significant pattern was observed in Curie (Suppl. Fig. 12C). These associations provide hypothesis-generating data on transcriptomic patterns that might be linked to ALK inhibitor response, which requires further validation.

Figure 4 (next pages): RNA expression and pathway analyses associate with ex vivo drug sensitivity of targeted compounds. ***A****) Scatterplot representing BCL-2-expression versus drug sensitivity for BCL-2-inhibitor venetoclax in the Máxima cohort (n = 23). Samples are colored by tumor type, boxplots represent BCL-2-expression across tumor types (bottom) or* AUC_{norm} *across tumor types (right).* ***B****) Scatterplot representing ALK-expression versus drug sensitivity for ALK-inhibitor ceritinib in the Máxima cohort (n = 24). Samples are colored by tumor type, shapes resemble ALK-genetic status (circle: wild type; star: altered), boxplots represent ALK expression across tumor types (bottom) or* AUC_{norm} *across tumor types (right).* ***C****) Dot plot showing pathway enrichment associated with sensitivity to alectinib, crizotinib, and lorlatinib in neuroblastoma ex vivo screens (Máxima: n = 8, Curie: n = 11). Pathways shown are those significantly enriched for at least one drug in either cohort; top 3 pathways with overlap across cohorts are highlighted, although the enrichment direction may differ between drugs or cohorts. Dot color indicates the Normalized Enrichment Score (NES) (red = positive enrichment, blue = negative); dot size reflects statistical significance (–log10 adjusted p-value), with larger dots indicating greater significance. MES: mesenchymal.* ***D****) Overlapping MRs associated with sensitivity to lorlatinib in neuroblastoma ex vivo screens. VIPER analysis results for the Máxima and Curie cohorts, highlighting TFs whose regulatory activity is significantly (padj. < 0.05) associated with drug sensitivity and shared between both cohorts for lorlatinib. Each row shows a TF inferred as an MR based on the coordinated differential expression of its target genes (regulon). The barcode plot (middle panel) shows the projection of each MR's activated (red) and repressed (blue) target genes along the gene expression signature (GES), ranked from the most downregulated (left) to the most upregulated (right) in drug-sensitive tumors. The heatmap (right panel) shows the MR Activity Score (Act) which indicates inferred regulatory activity (red = activation; blue = repression). Color intensity reflects the strength of activation or repression, with more saturated colors indicating greater inferred activity differences.*


A
BCL2–Venetoclax (Máxima)
Venetoclax (AUCnorm)
BCL2 (log–cpm)
Tumor_type
ES
NB
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SS
B
ALK–Ceritinib (Máxima)
Ceritinib (AUCnorm)
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Tumor_type
ES
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No
Yes

Figure 4

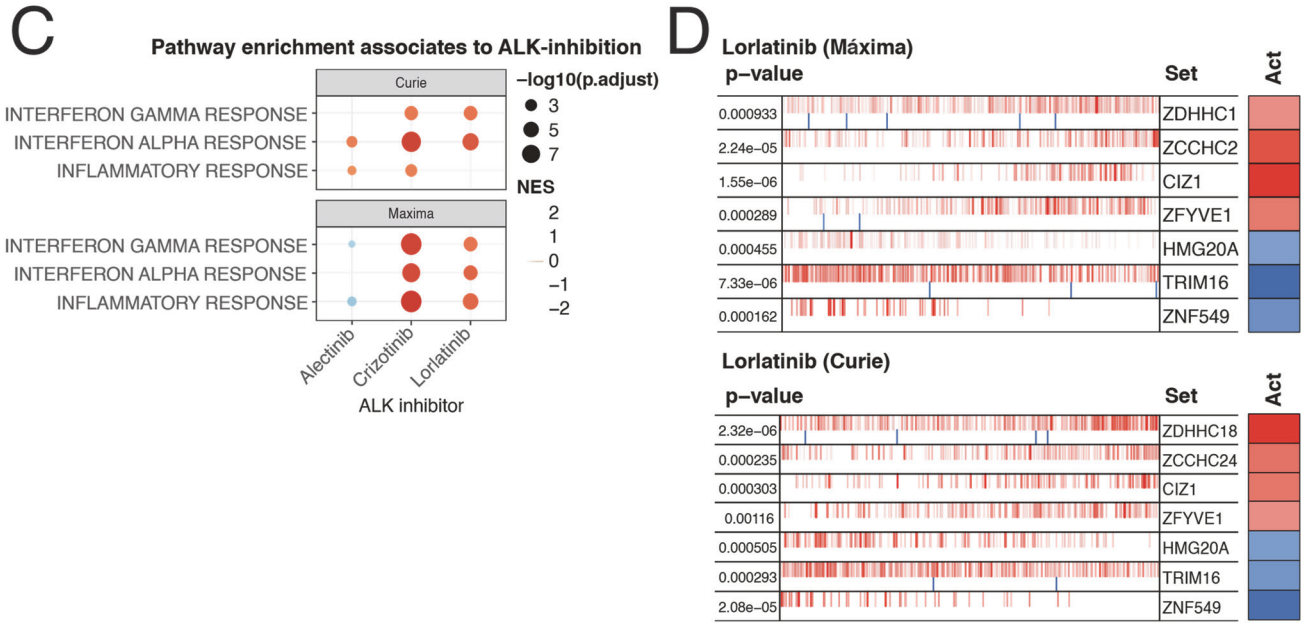


Figure 4 (continued)

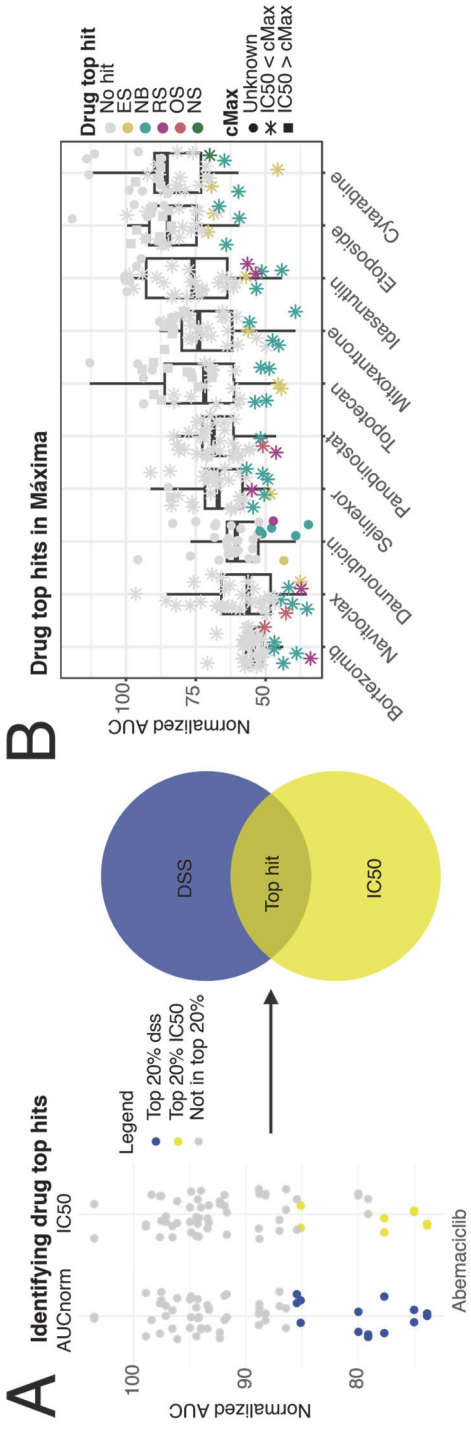
Ex vivo drug sensitivity profiling complements molecular profiling in identifying clinically relevant drug hits

As molecular profiling alone does not identify enough drug candidates for all patients and has limited predictive power for treatment response,[7,10–12] we investigated identifying patient-specific drug options using *ex vivo* drug sensitivity profiling. To increase therapeutic precision whilst minimizing toxicity of drugs, we defined top responders per compound as samples whose AUC_{norm} and the IC50 both fell within the top 20% most responsive quantile (Fig. 5A-B). In the Máxima cohort (n = 35, excluding replicates of NB0277), a maximum of seven top responders per compound could be identified, while in the Curie cohort (n = 32), this was six. Not all compounds produced top hits, either due to insufficient drug efficacy (i.e., the IC50 was not achieved) or discrepancies between AUC_{norm} and IC50. Across both cohorts, 94% of patients (63 out of 67) had at least one top hit and we found on average eleven functional top hits per patient (14.9 in Máxima, 6.8 in Curie) (Fig. 5C; Suppl. Fig. 13A-B). More importantly, we identified drug hits for 85% (11/13) of screens where no targetable alterations could be identified. For these samples, we identified eight PI3K /AKT and seven MAPK targeting compounds as top hits, which could be a potential pathway to further investigate. Furthermore, as a first-pass approximation of clinical achievability, we compared IC50 values with clinically achievable maximum doses (Cmax) for 134 compounds.[30] Among top hits with available Cmax values, 78% (391/501) had IC50s below Cmax, supporting the potential of functional top hits for clinical translation.

To demonstrate how *ex vivo* drug sensitivity profiling complements molecular profiling by expanding treatment options and refining drug selection, we compared functional top hits with molecular tumor board (MTB) reports including treatment recommendations, available for seven patients (Suppl. Tab. 4). We find overlapping compound targets and pathways in the MTB recommendations and functional top hits for two patients. For two others we find overlapping compound targets in the MTB recommendations and functional top hits. *Ex vivo* drug sensitivity profiling broadened treatment options for six patients by identifying compounds targeting alternative pathways beyond MTB recommendations. To illustrate its role in refining drug selection, we highlight patient NB0277, that was screened in both institutes. We compared *ex vivo* drug sensitivities for compounds recommended

by the MTB with those identified through *ex vivo* drug sensitivity profiling (Fig. 5D; Suppl. Fig. 12C-D; Suppl. Tab. 4). The MTB recommended ALK inhibitors based on ALK amplification, and mTOR inhibitors (everolimus, combined with topoisomerase inhibitor topotecan) due to MYCN amplification. Although *ex vivo* drug sensitivity profiling did not identify mTOR inhibitors as top hits, two ALK inhibitors (alectinib and lorlatinib) were identified as drug top hits (Fig. 5D). Next to this, IC50 values of ceritinib were below Cmax in both institutes. Together, these results strengthen ALK inhibition as a promising therapeutic avenue. *Ex vivo* drug sensitivity further broadened treatment options by identifying additional drug classes, including BCL-2 inhibitors venetoclax and navitoclax, MEK inhibitors trametinib and pimasertib, and EGFR inhibitors afatinib and erlotinib in the Máxima (Fig. 5D; Suppl. Fig. 13C), and topoisomerase inhibitor mitoxantrone and cisplatin in Curie (Suppl. Fig. 13D). The differences between the institutes reflect the distinct cohort compositions, where different tumor types are included. These functional top hits highlight expanded therapeutic opportunities compared to genomic profiling alone. In this illustrative case, ALK inhibition remains the most promising strategy as it is suggested both based on the molecular and drug sensitivity profiles. Taken together, these findings demonstrate that *ex vivo* drug sensitivity profiling could be used to complement and extend suggestions for molecularly guided treatment choices and identify additional therapeutic options.

Figure 5 (next pages): Ex vivo drug sensitivity profiling to complement molecular profiling in guiding clinical decisions. ***A****) Drug top hits are identified per compound, as patients whose* AUC_{norm} *and IC50 both fell within the 20% most responsive quantile. In this case, IC50 values for abemaciclib were not reached in all samples, therefore the top 20% consists of seven samples (versus fourteen in* AUC_{norm}*).* ***B****) Boxplot representing 10 most effective compounds in Máxima. For drug top hits, samples are colored by tumor type. The shape of each datapoint represents if the IC50 is lower than Cmax (star), higher than Cmax (square) or if Cmax is unknown/ IC50 is undetermined (circle).* ***C****) Number of top hits found for samples included in the Máxima cohort. Bold and green sample names represent samples for which molecular profiling failed to identify a targetable alteration.* ***D****) Ex vivo drug sensitivity of sample NB0277 (normalized AUC), in the Máxima cohort, exemplifying only BCL2- and ALK-inhibitors. Red datapoints resemble normalized AUCs of drug top hits for NB0277, black datapoints resemble normalized AUCs of compounds not found as drug top hit for NB0277, grey datapoints resemble the rest of the cohort.*


A
Identifying drug top hits
AUCnorm IC50
Legend
Top 20% dss
Top 20% IC50
Not in top 20%
Normalized AUC
100
90
80
Abemaciclib
DSS
Top hit
IC50
B
Drug top hits in Máxima
Drug top hit
No hit
ES
NB
RS
OS
NS
cMax
Unknown
IC50 < cMax
IC50 > cMax
Normalized AUC
100
75
50
Bortezomib
Navitoclax
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Selinexor
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Mitoxantrone
Idasanutlin
Etoposide
Cytarabine

Figure 5

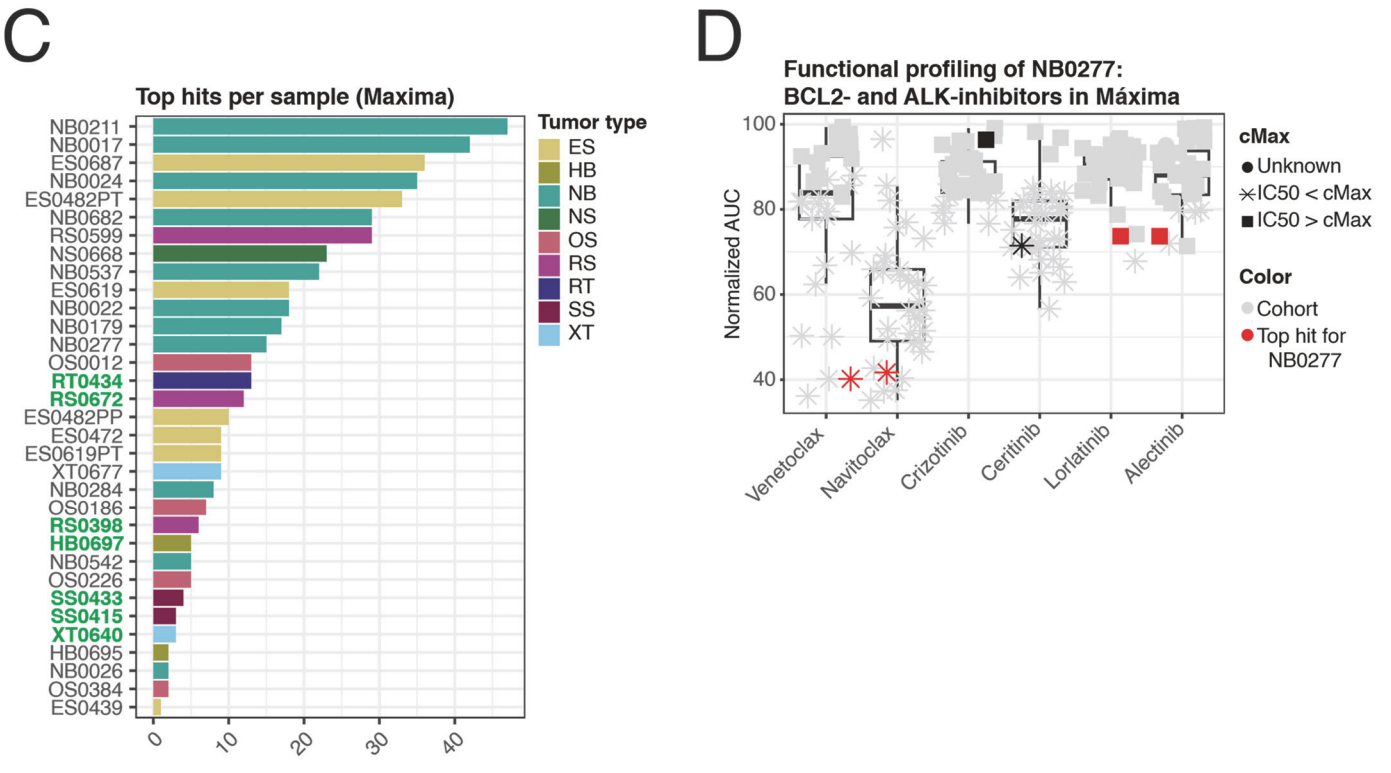

Figure 5 (continued)

Discussion

Molecular profiling alone often fails to predict effective therapies in high-risk pediatric tumors.[10,12] This has been confirmed in recent large-scale pediatric precision oncology programs including ITHER,[7] INFORM,[6] MAPPYACTS,[4] and ZERO.[8,9,11] These studies showed that only a small subset of patients with recurrent or high-risk tumors harbored high-evidence actionable mutations. Furthermore, a minority of patients with high-evidence targets received matched therapies (in part because of absence of clinical trials) and overall derived relatively low clinical benefit of the suggested treatments. The need for improved prediction of drug response is particularly pressing in pediatric oncology, where in addition to the underlying cancer complexity, the rarity and diversity of tumor types limits cohort sizes and hamper biomarker discovery. In this context, we conducted a proof-of-concept study to assess whether rapid *ex vivo* drug screening can complement genomic and transcriptomic profiling across pediatric solid tumors. We successfully tested drug sensitivity in 69 *ex vivo* cultures from previously established PDX samples and observed consistent trends in drug sensitivity across cohorts. *Ex vivo* responses were achieved within 14 days after receiving the PDX samples. Our results show that *ex vivo* drug sensitivity profiling can complement molecular profiling by providing drug hits for a majority of samples. These findings support the potential relevance of *ex vivo* drug sensitivity profiling alongside genomics and transcriptomics in an integrated diagnostic platform to facilitate clinical decision-making. Below, we evaluate the technical feasibility, success rate, clinical relevance of the screens, and the potential added value of *ex vivo* drug sensitivity profiling in a precision oncology setting.

We successfully performed 69 drug screens (82% success rate), achieving *ex vivo* drug sensitivity profiles within 14 days, and observed consistent trends in drug sensitivity across institutes. All results presented in this study were derived from *ex vivo* cultures of previously established PDXs. The dissociated PDX samples provide comparable material to that of patient resections,[31] but higher biomass than samples obtained directly from patient (fine needle) biopsies. This enabled us to test larger drug libraries and perform reproducibility screening. However, using PDX-expanded material comes with limitations. First, the high success rate is likely explained by a selection bias toward more aggressive tumors, as also reported in pediatric sarcomas.[32] For comparison, the INFORM study,

using fresh patient samples, reported sufficient viable tissue in 67% of cases, of which 78% passed internal quality control.[18] Second, the establishment and expansion of PDX models typically require several months,[32–34] often exceeding the time to progression described in INFORM registry.[6] However, once PDX-expanded tumors were received, *ex vivo* culture, drug screens and data processing were completed within 14 days, which is aligned with the timeline of molecular profiling, and comparable with the *ex vivo* drug screening timeframe reported in INFORM.[6,7,18] For clinical translation, we are conducting rapid screening of fresh patient samples, without an intervening PDX model, to achieve turn-around times compatible with clinical decision making. To improve success rates for these low-biomass samples, we recommend using round-bottom plates (requiring ~500 instead of 5,000-20,000 cells per well) and tumor-type guided compound prioritization.

Despite differences in drug screening protocols at Máxima and Curie, we observed high correlation and consistent top-effective compounds across overlapping models. This demonstrates the robustness of *ex vivo* drug screening across institutes. Nevertheless, differences in drug response may reflect protocol variations between Máxima and Curie: *ex vivo* expansion time (7 days versus direct seeding), cell densities (5,000 versus 20,000), dose ranges (Suppl. Fig. 2) and drug libraries (e.g., different compound batches). In addition, different passages of PDX models and concomitant clonal selection, could influence *ex vivo* drug response. Growth rate was assessed only in Máxima, serving as an indicator of sample quality, with growth rates below 100% reflecting spontaneous cell death in the absence of treatment. Growth rate correlated with response to cytotoxic and cell-cycle targeting agents, consistent with known mechanisms.[35] As proliferation rates of *ex vivo* cultures are typically higher than patient tumors, anti-tumor effects of mitotic inhibitors might be overestimated.[36] To improve comparability across models with variable growth dynamics, we recommend including growth rate as a standardized quality control metric.[35] Improving cross-institute comparison could result in large-scale cohorts, which are valuable for rare diseases, as exemplified by initiatives such as COMPASS and the Society for Functional Precision Medicine (ERAPERMED2018-121).[12] To further test cross-site variability we are performing cross-site benchmarking experiments within the COMPASS consortium. To enhance the potential for wider clinical

implementation, we therefore propose a harmonized protocol: 384-round bottom plates with standardized input (e.g. 500 cells per well), standardized quality control metrics (such as growth rate), agreed dose-ranges and compound libraries prioritized per tumor type, standardized timeframes of *ex vivo* expansion, and cross-site control samples.

Through collaboration between the two institutes, we increased sample sizes, enabling the exploration of tumor type specific sensitivities, including tumor types that are difficult to model *in vitro*, such as osteosarcoma.[23] However, the heterogeneity and small sample sizes for several tumor types does limit the identification of robust subtype-specific drug responses and our findings must therefore be considered as hypothesis-generating. Data correlating *ex vivo* sensitivity with patient response is rare in pediatric cancers yet is essential to validate drug response patterns. As an example, while Ewing sarcoma samples in our study showed sensitivity to PARP inhibitor *ex vivo*, consistent with other *in vitro* and *in vivo* studies,[37,38] clinical trials of PARP inhibition in Ewing have not shown durable benefit.[39–41] By contrast, INFORM reported parallels between *ex vivo* drug sensitivities and clinical courses in selected cases, such as complete resistance to the multidrug RIST regimen in relapsed Ewing sarcoma.[12,18] Validating clinical outcomes in large pediatric cohorts remains a critical challenge for functional precision medicine programs.

Beyond tumor-type-specific associations, we examined associations with drug response and molecular profiling across the full cohorts. Here, pan-cancer analyses can be confounded by tumor-type bias. While MYCN and ALK alterations appeared linked to drug sensitivity in the pan-cancer cohort, this link disappeared when restricted to neuroblastoma, indicating they were confounded by tumor-type sensitivity rather than the alteration. To address this, we explored ALK inhibitor response and transcriptomic changes within neuroblastoma alone. We found associations with ALK expression and response to selected ALK inhibitors, consistent with previous findings.[42] However, single genes and pathways did not result in consistent associations across all ALK inhibitors in both institutes. We therefore used master regulator analysis, which integrates the collective behavior of target gene networks which can reveal conserved regulatory programs that may be critical for modulating drug response.[29] Although only significant in Máxima, higher baseline SMAD4 activity was associated with sensitivity to all three ALK inhibitors in both

institutes. SMAD4 has tumor-suppressive roles in neuroblastoma and can be functionally inactivated by ALK-mediated phosphorylation.[43,44] Tumors with higher baseline SMAD4 activity could be more sensitive to ALK inhibition, as blocking ALK could restore SMAD4-dependent tumor suppression.[43,44] These results provide hypothesis-generating data and will require follow-up studies consisting of functional assays and *in vivo* PDX validations. While the potential of transcriptomics should be further validated in patients, our results suggest that relying solely on genomic status for trial eligibility may exclude patients who could benefit from ALK inhibition based on functional or transcriptomic profiles.

To assess potential predictive markers of drug sensitivity, we compared different approaches commonly used in a precision medicine context: genetic biomarkers, transcriptional biomarkers, and *ex vivo* drug screening. No targetable genetic alterations could be identified in 13 out of 69 screens, underscoring a common limitation in pediatric precision oncology.[4,6] Consistent with prior studies, when targetable alterations were detected, they did not consistently predict *ex vivo* drug response.[12] For example, CDK4 amplification did not associate with sensitivity to cell cycle inhibitors, in line with prior reports.[11,45] At the individual patient level, functional profiling may add clinical value by complementing molecular profiling. Using a selective outlier approach, we identified drug hits for 94% of samples, including 11 out of 13 cases where no targetable alteration was identified. These findings demonstrate that *ex vivo* drug sensitivity profiling can expand therapeutic suggestions, especially in patients whose tumors lack targetable alterations. When an alteration is found, *ex vivo* drug sensitivity profiling can be used to refine the choice of compound. We used Cmax to estimate clinical achievability of *ex vivo* determined IC50 values, a strategy previously applied in *in vitro* drug screening.[9] While useful as a first-pass filter, Cmax-IC50 comparisons do not capture *in vivo* pharmacokinetic and pharmacodynamic processes.[30,46] For example, plasma protein binding, drug lipophilicity and tissue distribution can strongly influence free drug exposure at the tumor site and should be integrated in future *in vitro* work.[47,48] Although this proof-of-concept study only included technical reproducibility as validation, biological validation will be pursued via matched *in vivo* testing. The ITCC-P4 consortium provides a platform for testing *in vitro*-identified drugs across a broad range of *in vivo* pediatric tumor models and is currently comparing the therapeutic efficacy of standard of care compounds with new investigational compounds or targeted

therapies in this panel. Sixty of the models used here have matched PDX counterparts in ITCC-P4, enabling future validation of *ex vivo* hits *in vivo*. As an example, standard of care compounds for neuroblastoma will include etoposide and topotecan and investigational drugs such as idasanutlin, cobimetinib, various ALK inhibitors, prexasertib and copanlisib.

Taken together, our study provides a proof-of-concept for the feasibility of rapid *ex vivo* drug screening in pediatric precision oncology. Currently, this method is being expanded to include fresh patient samples. Our data support the future integration of functional profiling into a next-generation diagnostic platform that combines *ex vivo* drug sensitivity profiles with genetic and transcriptomic data to predict drug response.

Methods

Tissue Processing and Drug Screening Protocols

Ex vivo short-term drug sensitivity screening was performed at two centers, Princess Máxima Center (Máxima) and Curie Institute (Curie), using fresh patient-derived xenograft (PDX) tumor material. This method aimed to assess *ex vivo* drug sensitivity profiling within a clinically relevant timeframe and to complement molecular profiling in pediatric precision oncology.

PDX establishment

PDX establishment and characterization was conducted in the ITCC-P4 consortium as previously reported.[22,49] 37 PDX models in the Máxima and 32 PDX models in Curie were selected and successfully screened *ex vivo*.

In brief, the PDX models used for the Máxima cohort were established at the participating academic centers within the ITCC-P4 consortium, where informed consent was obtained from patients and/or parents/legal caregivers prior to the collection of biological samples, in accordance with the approved local study protocols and ethical approvals. Following establishment, models were shipped to the CRO partners (EPO, Charles River, and Xentech) for expansion, and these CROs provided expanded material and study samples. For each patient, fresh tumor material (surgical resection or biopsy), a constitutional (germline) sample and clinical information was collected. Solid tumors were transferred in cell

culture medium and transplanted subcutaneously in NSG (NOD.Cg-Prkdcscid IL2rgtm1Wjl/SzJ), Swiss Nude (Crl:NU(Ico)-Foxn1nu) or CB17 SCID (CB17/Icr-Prkdcscid/IcrIcoCrl) immunodeficient mice.

At Institut Curie, PDXs were established from tumor biopsies obtained at diagnosis or at relapse, as described previously,[14] following informed consent of parents or guardians. Experimental procedures were approved by the ethics committee of the Curie Institute CEEA-IC #118 (Authorization APAFIS#11206-2017090816044613-v2) in compliance with international guidelines. The establishment of PDX models received approval by the Institut Curie institutional review board OBS170323 CPP ref 3272; n° 2015-A00464-45). In brief, tumor samples were engrafted subcutaneously in immunocompromised mice.

Upon successful establishment (> P2) and upon reaching ethical size, PDX tumors were cryopreserved for banking and further analyses. All PDXs screened at the Máxima, and 24 PDXs from Curie were integrated within the ITCC-P4 consortium. For these models, all human and corresponding PDX samples, along with associated clinical, and molecular data, were anonymized and labeled using a barcoding system. This system encoded information about the original source, tumor entity, tumor event, sample type, and data type. Barcodes were automatically generated upon sample and patient data registration in the R2 online platform (https://hgserver2.amc.nl). For models not included in the ITCC-P4 consortium, institutional anonymization was used (models IC-pPDX-### and models HSDJ_###).

PDX molecular profiling

Comprehensive molecular profiling was conducted on PDX tumor samples and whenever possible on patient tumor and germline control.[22] DNA and RNA were extracted from samples using standardized procedures.[4,7] For low coverage whole genome (lcWGS)/whole exome sequencing (WES) library preparation was performed using standardized procedures,[7] followed by capture using the Agilent SureSelect Clinical Research Exome V2 Kit, and sequenced using the Illumina R technologies. For DNA methylation array profiling, Methylation450K and MethylationEPIC BeadChip arrays (Illumina) were used. RNAseq Library construction was performed following the TruSeq Stranded mRNA LS protocol (Illumina, San Diego, CA). Sequencing was performed on Illumina sequencing machines: NextSeq 500 (150 nt paired-end) and NovaSeq (100 nt paired-end).

Calling of genetic events

NGS and methylation data was manually curated following visualization on the R2: Genomics Analysis and Visualization platform (https://hgserver2.amc.nl). Data was used for calling of CNVs, SNVs and SVs/fusions predicted to be actionable, as described previously.[4,7] For ALK, MYCN, TERT, and CDK4 high level genomic amplification were called as amplifications (copy number >log2). ALK alterations were also called in case of an activating mutation within the tyrosine kinase domain. Alterations in genes with suppressor gene function (SMARCA4, SMARCB1, NF1, ATM, PALB2, CHEK2, BRCA2, PTCH1 and PTEN) were considered in case of homozygous deletion, or loss of heterozygosity (LOH) in combination with an inactivating mutation.[50] Mutations in ATRX were called in case of homozygous inactivating events or deletion, or the combination of LOH, in-frame fusions, inactivating mutations, or hotspot mutations. Clinically relevant STAG2 was called in case of inactivating mutations.[51] CDKN2A/CDKN2B alterations were considered in case of homozygous deletions. TP53 mutations were called when a deletion occurred together with an inactivating (frameshift or SNV) mutation (and heterozygous was annotated but not used in analysis, unless stated otherwise). HRAS was called in case of activating mutations (hotspot regions).

Determining mice contamination in *ex vivo* short-term screen samples

Mouse contamination in screened samples was determined either based on DNA (27 cases, at Máxima) or based on RNA analyses (33, at Curie).

Prior to real-time quantitative PCR (RT-qPCR), genomic DNA extraction was carried out for all tested samples. Following DNA extraction and measurements, RT-qPCR was performed using the CFX384 Touch Real-Time PCR Detection System (BioRad). To measure the amount of either human or murine genomic DNA concentrations, two separate Go-Taq® qPCR-based (LOT: 0000602761) master mixes (MM) were prepared, one to probe human DNA and one to probe murine DNA. For each reaction, 4,6 µL H2O, 10 µL Go-Taq® qPCR MM, either 0,2 µL forward human primer (REF: 2378847) or 0,2 forward murine primer (REF: 2378849) and 0,2 µL reverse human-murine primer (REF: 2378848), totaling 15 µL of MM, was added to 5 µL of 1 ng/ µL of isolated DNA, together

with positive and negative controls. The positive human and murine controls were added to the plate in de-escalating human-murine ratios to more accurately determine human-murine ratios in the PDX samples. The ratios were set as follows: H100:0M, H99:1M, H95:5M, H80:20M, H20:80M, H0:100M. Cycling conditions included an initial denaturation of 5 minutes at 95 °C, followed by 40 cycles of 45 seconds at 95 °C and 30 seconds at 60 °C, ending on an extension of 90 seconds on 72 °C. Melting curve analysis was performed to assess specificity of amplification and Cq values. Human and murine control ratios were used to determine the true human-murine ratios in the PDX models.

RNAseq fastq files were processed by Institut Curie RNAseq pipeline version 4.1.0 with xengsort tool (https://gitlab.com/genomeinformatics/xengsort) to select reads coming from the human tumor, while filtering out mouse reads.

Tissue dissociation, cell culture and cell seeding

At both centers, tumor tissues were mechanically and enzymatically dissociated to obtain single-cell suspension for *ex vivo* drug screening in flat bottom 384-well plates. While core principles were shared, protocol details regarding enzyme combinations, media composition, and expansion time before seeding differed between the Máxima and Institut Curie.

At the Máxima, tumor tissues were first mechanically dissociated using sterile scalpels, followed by a two-step enzymatic digestion. Initially, tissues were transferred to Liberase DL (Sigma Aldrich, cat no. 5466202001) at a 1:43 dilution in tumor stem media (TSM) base, as reported before.[18,52] Tumor pieces are digested at 37°C for 30 minutes, with intermittent swirling. Supernatant was collected and filtered through a pre-washed 70 µm strainer. On the remaining tumor pieces, a second digestion step was performed in a digestion buffer of Liberase TM (Sigma Aldrich, cat no. 5401119001) at a 1:43 dilution in TSM base for another 30 minutes at 37°C, with intermittent swirling. Following each harvest, cells were centrifuged at 300 g for 5 minutes. Red blood cells were lysed using a red blood cell lysis buffer, with a maximum of 4 minutes incubation at room temperature. Post-lysis, cells were suspended in TSM complete [18,52] and counted using trypan blue. Tumor cells were cultured for 7 days before drug screening in TSM complete, at 37°C and 5% CO_2. 40 µL of cells, aiming for 5,000

cells/well, are plated in a flat bottom 384-well tissue culture treated microplate (#3764, Corning, NY) using a multidrop combi reagent dispenser (Thermo Scientific). Total DNA was isolated and processed at the time of plating.

At Institut Curie, freshly collected PDX tumors were first dissociated as described previously [16]. After mechanical dissociation using sterile scalpels, enzymatic dissociation with trypsin (10 mg/ml) and type II collagenase (275 U/mg, Worthington Biochemical) was performed. The tube was then placed in a warm 37°C water bath for 60 minutes. Dissociation was stopped by adding Soybean Trypsin Inhibitor (10 mg/ml, Sigma). Deoxyribonuclease I (2 mg/ml, Sigma) and magnesium chloride (1 M) were added in equal amounts. The tumor suspension was filtered through a 40 μm cell strainer and then centrifuged at 500g for 5 minutes. The supernatant was discarded, and cell pellet was resuspended in PBS-minus/10% FBS for cell counting. The suspension was then re-centrifuged and resuspended in serum-free stem-cell (SC) medium, which contained Dulbecco's Modified Eagle Medium/Nutrient Mixture F-12 (DMEM/F12, Gibco) supplemented with 40 ng/ml basic fibroblast growth factor (bFGF), 20 ng/ml epidermal growth factor (EGF), 1× B27 supplement (Gibco) and 500 U/ml of penicillin/streptomycin. Dissociated cells were directly seeded in flat bottom 384-well plates for drug screening without any pre-culture amplification phase. 25 μl of cells (20,000 cells/well) prepared in serum-free stem cell (SC) medium was dispensed into each well using a MultiDrop Combi (Thermo Fisher).

High-throughput compound screening

At both centers, high-throughput compound screening was performed to determine drug sensitivity profiles in *ex vivo* cultured PDX-derived tumor cells. Common principles included the use of flat bottom 384-well plates, incubation with compound libraries across multiple concentrations, and luminescence-based viability assays after 72h. Clinically relevant compounds were added using acoustic dispensing systems and screening was conducted in technical duplicates. Differences between the two sites involved compound library composition, dosing schemes, cell numbers, and assay reagents.

In Máxima, screening was performed in two versions of its internal Princess Máxima Center library - version L13 and L14 -, which contains 224 and 225 drugs

respectively. These libraries include compounds that are FDA approved, or in (pre-)clinical evaluation (Suppl. Tab. 1). For samples with limited biomass, a 134-compound clinical priority subset was applied. Most of the compounds from the libraries are dissolved in dimethylsulfoxide (DMSO) and stored at room temperature under nitrogen atmosphere. Five (metformin, perifosine, carboplatin oxaliplatin and copanlisib) drugs are dissolved in MQ and one (cisplatin) is dissolved in a saline solution, which are stored at -20 °C. Before the high-throughput screening, the 384-well plates (Well Low Dead Volume Source Microplates, Labcyte) containing the dissolved drugs are shaken (30 minutes at rt) and centrifuged (2.30 minutes at 1500 rpm). Subsequently, the plates are surveyed with the Echo 550 dispenser to determine whether the amount of solution in the wells is sufficient to start the screen (minimal 2.5 µl) and the DMSO percentage is >80%.

In Institut Curie, the Compass03 oncology collection, covering a total of 77 oncology drugs (mainly FDA/EMA approved) and control drugs designed for COMPASS study, was used (Suppl. Tab. 2) [18,53]. All compounds were obtained from commercial compound vendors. The compounds were dissolved in 100% DMSO or MQ.

In Máxima, high-throughput drug screens were conducted in duplicate in collaboration with the high-throughput screening facility of the Princess Máxima Center. Cells are cultured for 24h under standard culturing conditions (5% CO_2, 37 °C). Next, the Cell Titer Glo 3D (CTG3D, Promega, USA, G9683) cell viability assay is performed for the t = 0 microplate using the supplier's protocol by measuring luminescence. Units of luminescence signal generated by a thermo-stable luciferase are proportional to the amount of ATP presented in viable cells. Furthermore, the drugs are added to tissue culture microplates and the library using the high-throughput screening facility (https://www.prinsesmaximacentrum.nl/en/professionals/core-facilities-shared-resources/high-throughput-screening). Using the Echo 550 dispenser, 100 nL of the drugs (in DMSO or MQ, at different concentrations) are added to the wells containing the cells, to yield final concentrations of 0,1 nM, 1 nM, 10 nM, 100 nM, 1 µM and 10 µM (0.25% DMSO or MQ). Cells treated with only DMSO are used as negative controls, whereas cells treated with staurosporine (final concentration of 10 µM) are used as positive controls. The cells are

incubated with the compounds for 72h at standard culturing conditions. Next, after 72h of incubation, the CTG3D assay is performed for the microplates. Growth rates of the DMSO-treated cells during the screen are determined by dividing the CTG3D signal of the DMSO controls at day 3 by the signal at t0 (cell growth). In addition, to assess culture quality, microscopic inspection and imaging are performed at both timepoints.

At Institut Curie, 384-well plates with cells were also incubated for 24h at 37°C. Next, the compounds were dispensed using an acoustic liquid handling device, Echo 550 (Beckman Coulter) at FIMM High Throughput Biomedicine unit (University of Helsinki, Finland; https://www.helsinki.fi/en/infrastructures/drug-discovery-chemical-biology-and-screening/infrastructures/fimm-high-throughput-biomedicine-unit) in three 384-well microplates (Corning - 3764). All drugs were plated in duplicates in five different concentrations in 10-fold dilutions covering a 10,000-fold concentration range (e.g. 1- 10,000 nM) mostly encompassing the achievable plasma concentrations in pediatric patients [18,53]. The cells are incubated with the compounds for 72h at standard culturing conditions. Drug effects on cell viability were measured using the CellTiter-Glo 2.0 Luminescent Cell Viability Assay (Promega). The reagent was equilibrated to RT for 30 minutes prior to use. A volume of 25 µl per well of CTG reagent was added in each well, mixed for 2 minutes at 300 rpm on an orbital shaker (Titramax 100). Plates were further incubated for 10 minutes at RT to stabilize luminescence signals. Luminescence was recorded using a CLARIOStar (BMG Labtech) (gain = 3600) using top measurement.

Data processing

In all cases, following high-throughput drug screening, the data is normalized to the 0.25% DMSO-treated cells (defined as 100% viability) and the empty controls (0% viability). Dose-response curves were estimated by a 5 parameters log-logistic model using the R package *drc* (version 3.0.1 with R 4.3.1; [54] From these fitted models several values were estimated: the concentrations of the drug needed to achieve a 50% reduction in cell viability (IC50), the definite integral of the curve (area under the curve, AUC). AUCs were normalized by the range of concentration for each compound as AUC_{norm} score to compare the two cohorts. Quality of the

screens was approved after assessment of the cell growth, the negative, positive, and empty controls, and the amount of variability between the duplicates. After processing, in the Máxima, a standard quality check is performed by high-throughput screening facility as well as a manual check per compound.

AUC_{norm} are used when analyzing within one cohort to define and compare drug sensitivities, as IC50s are not reached for ineffective compounds. When comparing two different cohorts with different dose ranges, the IC50 is used to compare drug sensitivities. In rare cases, normalized viability exceeded 100% at non-effective concentrations, resulting in an AUC_{norm} slightly above 100. This was retained to avoid artificial truncation of the curve and reflects occasional biological or technical variability. NB0277 was received three times due to a sample mislabeling. Only the first received sample was included in all analyses, except for the reproducibility testing (Fig. 1E) and growth-rate correlations (Fig. 3C), where the other two samples are also included.

Tumor type compound specificity

To assess the specificity of compound responses across tumor types, an analysis of variance (ANOVA) model was employed, between AUC_{norm} and tumor type as covariate. Compounds yielding a p-value ≤ 0.05 were considered to exhibit statistically significant differential responses among tumor types. To further investigate tumor-type-specific efficacy, t-tests were conducted comparing each tumor type against the pooled set of all other types. This analysis aimed to identify compounds with significantly higher or lower AUC_{norm} values in specific tumor types relative to others.

Identifying transcriptomic signatures linked to drug sensitivity

To identify genes whose expression correlates with AUC_{norm} differential gene expression analysis was performed using DESeq2 (v 1.44.0)[55] in R version 4.4.1. Raw count data were filtered to remove lowly expressed genes, retaining only those with a minimum expression threshold in multiple samples. Counts were normalized using the median-of-ratios method to account for differences in sequencing depth. Both drug response (AUC_{norm}) and gene expression were modeled as a continuous variable (no dichotomization), using a generalized linear model (GLM) of the form:

$$Expression \sim AUC_{norm}$$

where AUC_{norm} represents the drug sensitivity score for each sample. Wald tests were used to assess statistical significance, and Benjamini–Hochberg (BH) correction was applied to control for false discovery rate (FDR). Genes with adjusted p-values (padj) < 0.05 were considered significant. We excluded ceritinib from the analysis due to inconsistent results from the AUC_{norm}-based linear model.

Pathway enrichment analysis was performed using Gene Set Enrichment Analysis (GSEA) with the fgsea (v 1.30.0)[56] package in R. Genes were ranked based on Wald test statistics from DESeq2, preserving the direction of association with AUC_{norm}. Hallmark gene sets were obtained from the Molecular Signatures Database (MSigDB) [57] using the msigdbr (v 7.5.1) [58] package in R, for enrichment analysis. The fgsea algorithm was run with 1,000 permutations, using a minimum gene set size of 15 and a maximum of 500 to exclude nonspecific pathways. Pathways with adjusted pvalue < 0.05 were considered significantly enriched.

Master regulator (MR) analysis was performed using VIPER (v 1.38.0; [59]) to infer the activity of transcription factors (TFs) associated with drug sensitivity in neuroblastoma. Two complementary approaches were employed: msVIPER, which identifies MRs at the cohort level by integrating gene-level differential expression data, and VIPER, which infers MR activity at the individual sample level. To identify TFs driving AUC_{norm}-associated transcriptional changes, the *msviper()* function was applied to the gene signature defined as described in the "Identifying transcriptomic signatures linked to drug sensitivity" section above, using a neuroblastoma-specific regulatory network constructed from ARACNe-inferred[60,61] TF-target interactions derived from the SEQC dataset of 498 bulk RNA-seq neuroblastoma samples. [62] Enrichment significance was assessed using adaptive-weighted Kolmogorov–Smirnov statistics with 1000 permutations. All analyses were conducted in R version 4.4.1 using VIPER with a neuroblastoma-specific regulon derived from publicly available expression datasets, applying default parameters unless otherwise specified.

Data access statement

The data presented in this study are available via the R2 Genomics Analysis and Visualization Platform (https://hgserver2.amc.nl).

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Author contributions

MCS, PG, GS and JJM were responsible for conceptualization of this manuscript. Bioinformatic analyses were performed by **MCS**, PG, ArB, MvL, EC and ESA. Laboratory investigation was performed by **MCS**, LV, EDN, AnB, SS and EL.

Methodology was developed by **MCS**, VAA, EDN, AnB, VMP, LLT, JSS and SE. Resources were provided by VAA (Máxima drug library), VMP, LLT and JSS (COMPASS drug library), JS, EI, DG, KS, AE, RB, FB, SZ, DS, AMC and BG (PDX models), and JK (R2 platform). Project administration was coordinated by **MCS**, LV, YI, ASC, AG and EMR. Visualization was performed by **MCS**, PG, ArB, MvL and EC. The manuscript was written by **MCS**, PG and EC, and review and editing was conducted by EL, FC, HNC, LS, GV, SP, SE, SvH, MvdB, GS and JJM. Supervision was provided by FC, HNC, LS, GV, SP, SE, SvH, MvdB, GS and JJM. MK and AF: methodology and resources (PDX molecular characterization).

Conflict of interest statement

Gudrun Schleiermacher receives research funding from Roche, BMS, Pfizer and MSD. The other authors declare no competing interests related to this study.

Key words

Precision oncology, high-throughput drug screening, pediatric oncology, precision medicine, high-risk solid tumors, neuroblastoma, transcriptomic biomarkers, drug repurposing

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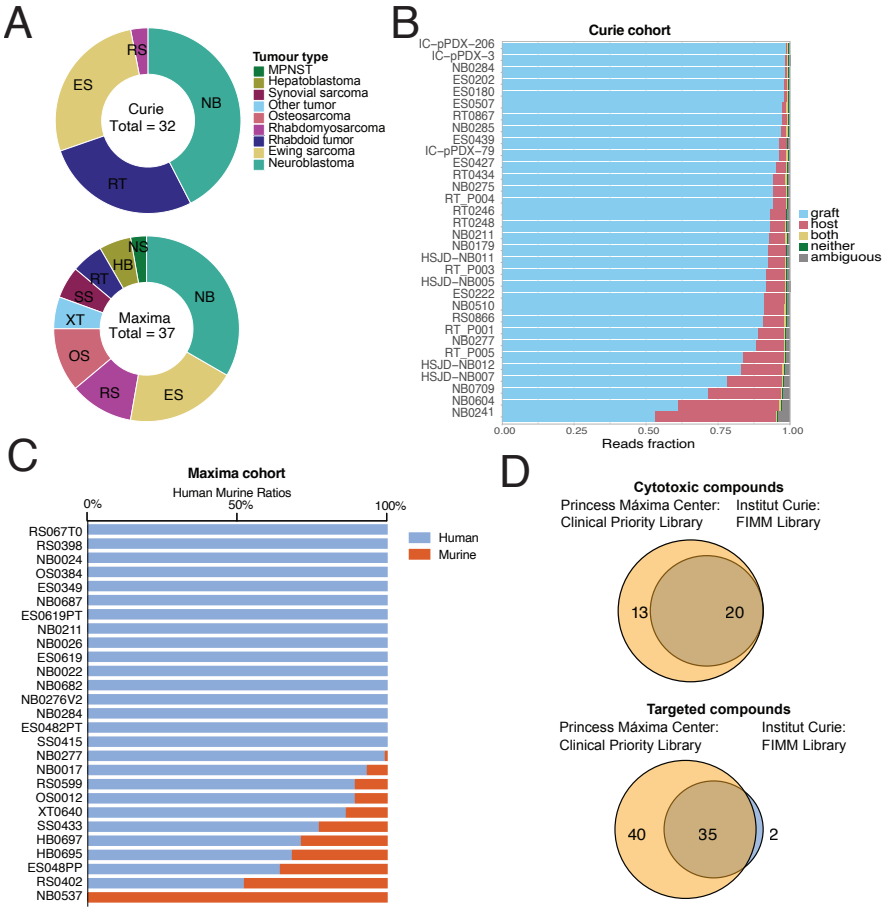
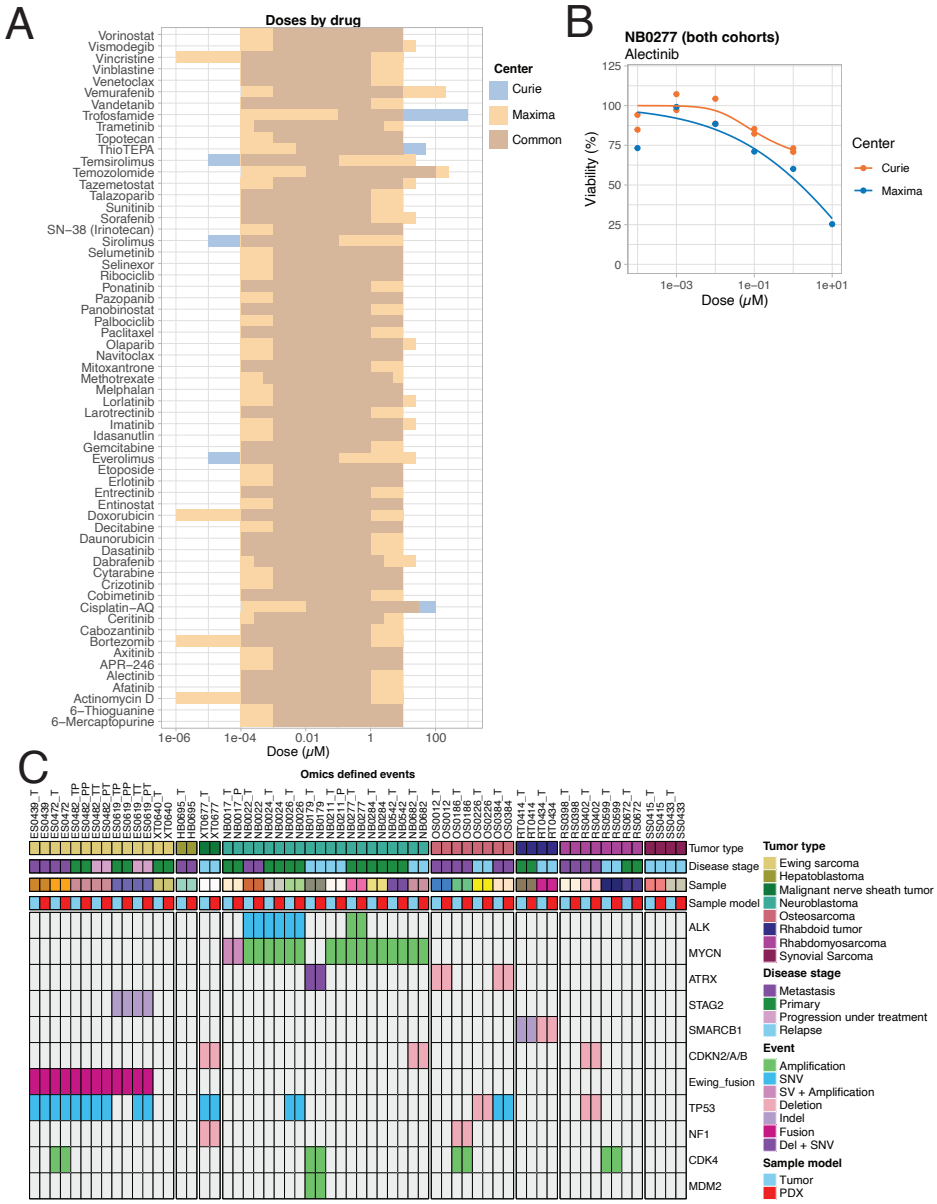
Supplementary material

The published version of this chapter, including supplementary tables, can be accessed via the QR code below or at: https://www.nature.com/articles/s41698-025-01266-0.

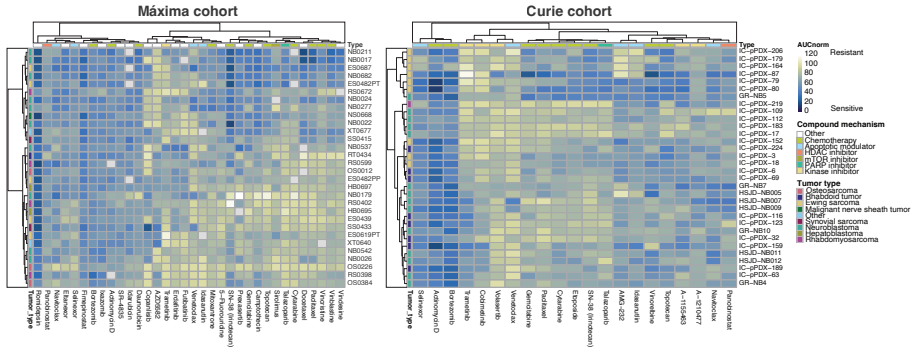
The supplementary figures are included on the following pages.



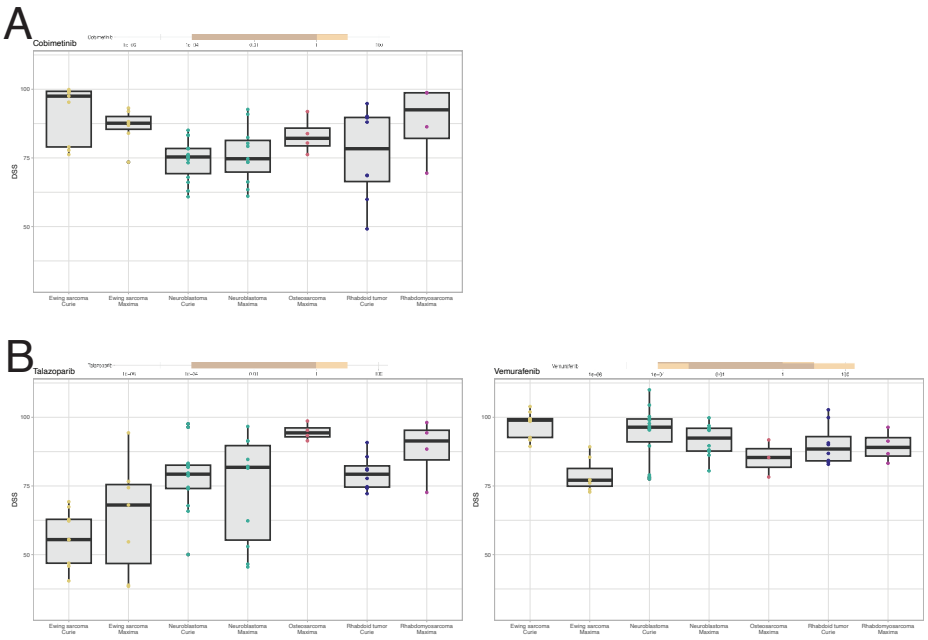
Supplementary figure 1: *Ex vivo* drug sensitivity profiling of high-risk solid tumors in the Máxima and Curie institutes. A) Donut plot showing cohort overview of *ex vivo* direct screens showing tumor type. Top: Maxima institute; bottom: Curie institute. B) Bar graph showing tumor RNA content of PDX models included in cohort of Curie institute, indicating percentage of graft/human RNA reads in blue, host/mice reads in red, both in yellow, neither in green and ambiguous in grey. C) qPCR of *ex vivo* samples screened in the Princess Máxima Center, harvested at day of plating for drug screen. D) Composition of cytotoxic and targeted compounds in the compound libraries of the Princess Máxima Center (orange), Institute Curie (blue), indicating the number of overlapping compounds.



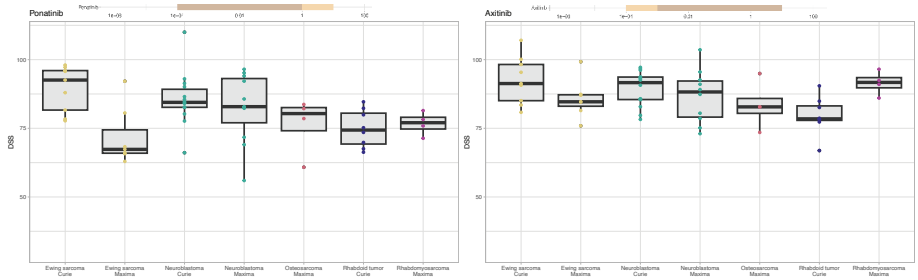
Supplementary figure 2: *Ex vivo* drug sensitivity profiling: concentrations of drug libraries and genetic events in PDX versus patient samples. A) Dose ranges of compounds in Curie (blue), the Princess Máxima Center (orange) and overlapping (brown) doses. B) Dose-response curve for afatinib of NB0179 shown for both cohorts (Curie: orange; Maxima: blue). The difference in concentration is indicated, here Maxima tests up to 10 μM, whereas Curie tests afatinib in concentrations up to 1 μM. C) Oncoplot representing the molecular landscape of patient samples as well as PDX models included in the Maxima cohort, showing key targetable genetic alterations that are retained in the PDX models. Samples are annotated and colored by tumor type, disease stage, sample (each color represents a patient ID) and sample model. _T = patienttumor; second _xP = primary (TP= patient tumor, PP = PDX, primary); _xT = progression under treatment (TT: patient tumor, PT: PDX).



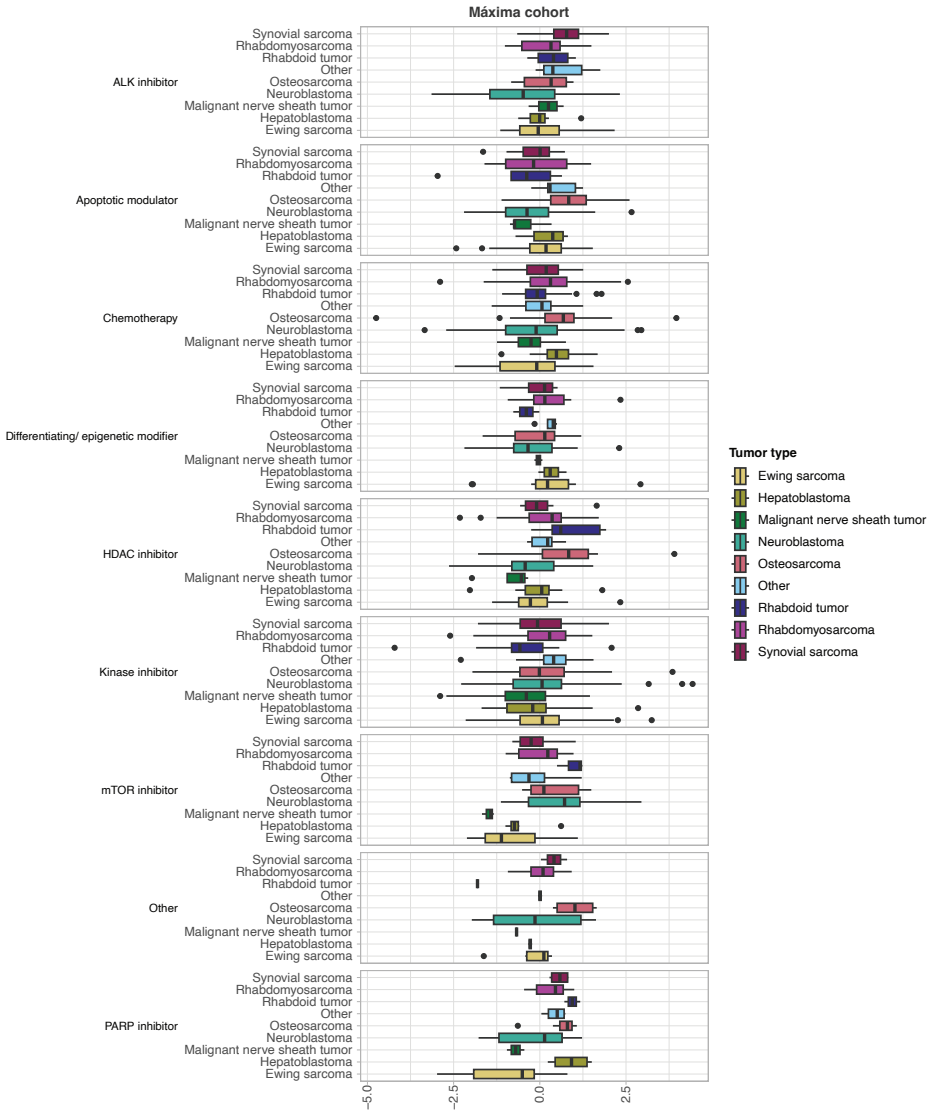
Suppl. Figure 3: Comparing drug sensitivities across samples in the Maxima and Curie cohorts. A heatmap representing AUCnorm of compounds that have a response of AUCnorm >50 for at least one patient. Samples are colored based on tumor type and clustered based on drug sensitivity. Compounds are colored per category and clustered based on drug sensitivity. Left heatmap represent samples in the Maxima cohort, right heatmap represents Curie cohort. In the Maxima cohort, samples for which one or more AUCnorm scores were undetermined (due to technical failures) are excluded to allow proper correlation analyses between samples.



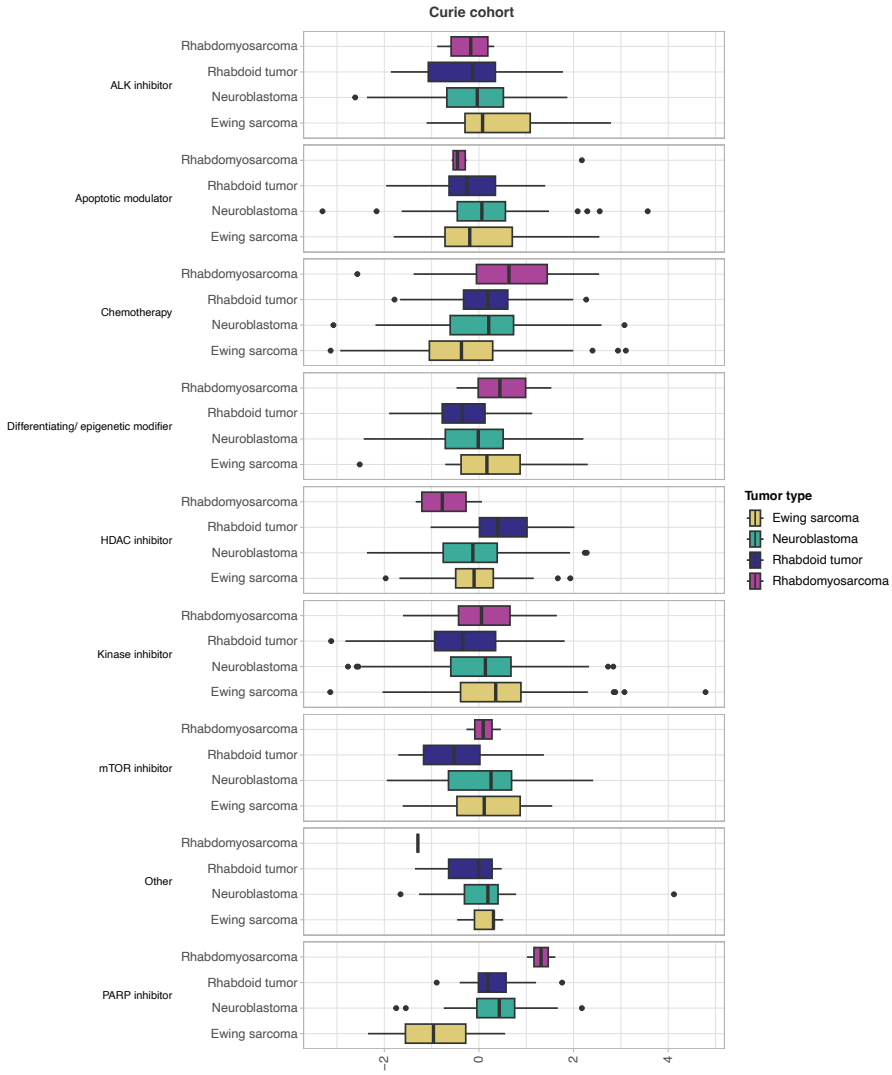
Suppl. Figure 4: Tumor type specific drug efficacies in two cohorts. For both cohorts separately, tumor type-specific compound efficacy is investigated by testing for significant differences in drug sensitivity among tumor types represented by n ≥ 4 samples. Box plots show AUCnorm per tumor type, per cohort, for compounds of interest for Neuroblastoma (A) and Ewing Sarcomas (B). Dose ranges (in µM) are indicated in orange for tested doses in Maxima, blue for Curie, brown for overlapping doses. For all box plots: the box represents the interquartile range (IQR), with the median (50th percentile) indicated by the horizontal line inside the box. The whiskers extend to 1.5 times the IQR, with data points outside this range considered as outliers and shown as individual points. In panel B, vemurafenib appeared significantly more effective in Maxima (p = 0.002), which was not observed in in Curie, where lower doses were tested.



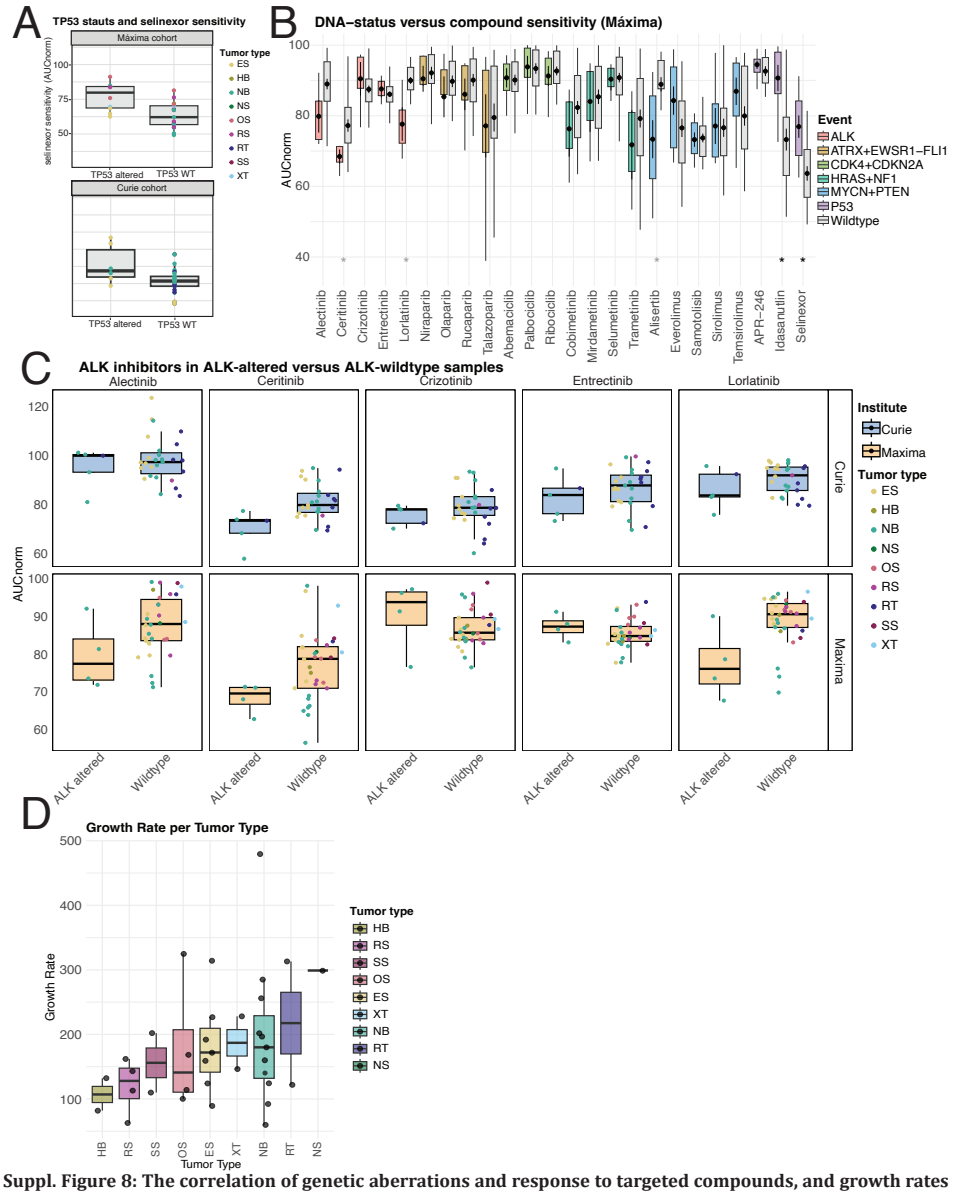
Suppl. Figure 5: Tumor type specific drug efficacies for Rabdoid tumors tested at Curie. For the Maxima cohort, tumor type-specific compound efficacy is investigated by testing for significant differences in drug sensitivity among tumor types represented by n>2 samples. Box plots show AUCnorm per tumor type, per cohort, for compounds of interest for Rhabdomyosarcomas. Dose ranges (in µM) are indicated in orange for tested doses in Maxima, blue for Curie, brown for overlapping doses. For all box plots: the box represents the interquartile range (IQR), with the median (50th percentile) indicated by the horizontal line inside the box. The whiskers extend to 1.5 times the IQR, with data points outside this range considered as outliers and shown as individual points. The left panel shows sensitivity for BCR-ABL/multi-tyrosine kinase inhibitor ponatinib and the right panel for the pan-VEGFR tyrosine kinase inhibitor axitinib.



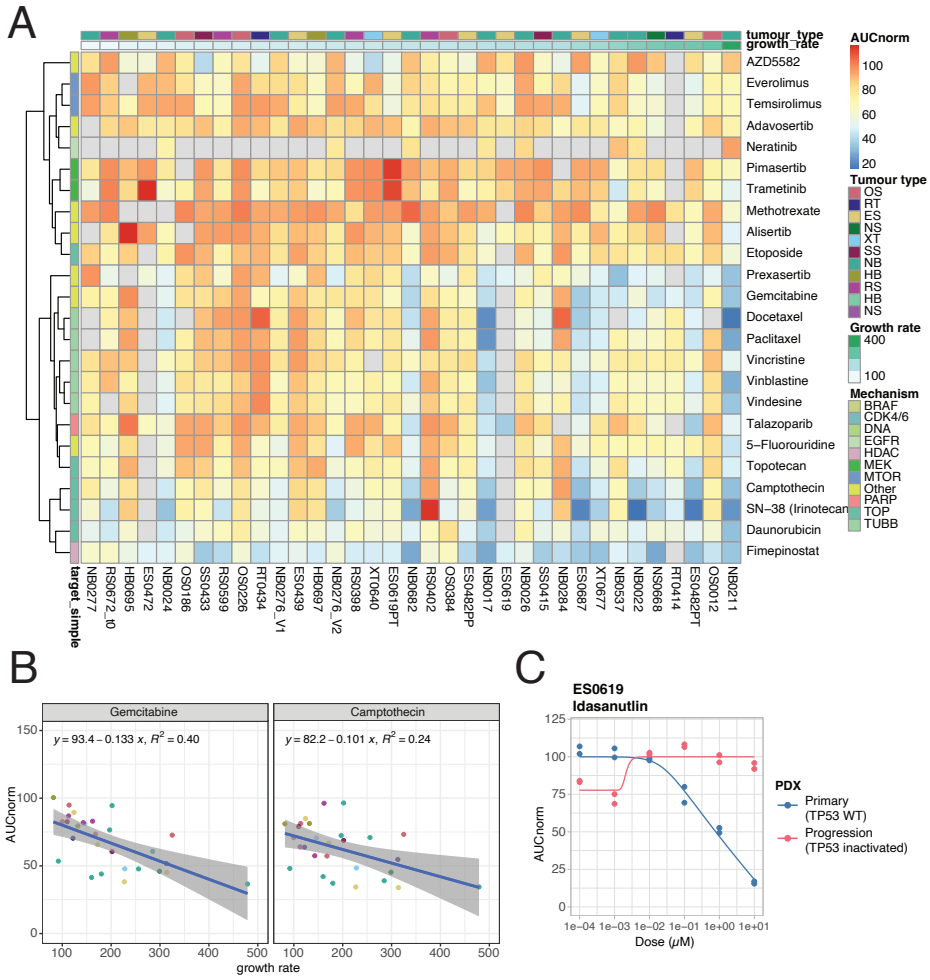
Suppl. Figure 6: Tumor type-specific sensitivity for compound mechanisms in the Maxima cohort. For the Maxima cohort, tumor type-specific efficacy for compound classes were investigated by testing for significant differences in drug sensitivity among tumor types. Mean AUCnorm of each tumor type against the mean of all tumor types is tested per compound mechanism, AUCnorm are adjusted for differences in compounds by using compound as cofactor (Z-scores). Box plots show Z-scores (based on AUCnorm) scores per tumor type. The box represents the interquartile range (IQR), with the median (50th percentile) indicated by the vertical line inside the box. The whiskers extend to 1.5 times the IQR, with data points outside this range considered as outliers and shown as individual points. A lower Z-score represents greater sensitivity compared to the other samples, the higher the Z-score, the less sensitive the sample is for this compound.



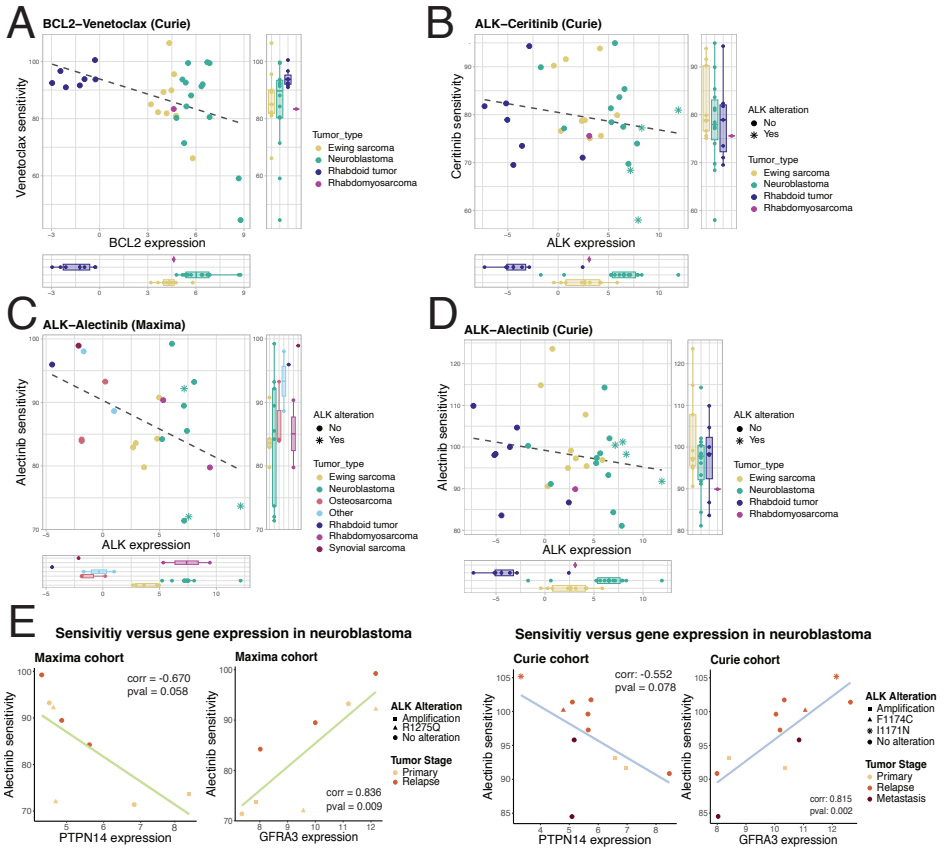
Suppl. Figure 7: Tumor type-specific sensitivity for compound mechanisms in the Curie cohort. For the Curie cohort, tumor type-specific efficacy for compound classes were investigated by testing for significant differences in drug sensitivity among tumor types. Mean AUCnorm of each tumor type against the mean of all tumor types is tested per compound mechanism, AUCnorm are adjusted for differences in compounds by using compound as cofactor (Z-scores). Box plots show Z-scores (based on AUCnorm) scores per tumor type. The box represents the interquartile range (IQR), with the median (50th percentile) indicated by the vertical line inside the box. The whiskers extend to 1.5 times the IQR, with data points outside this range considered as outliers and shown as individual points. A lower Z-score represents greater sensitivity compared to the other samples, the higher the Z-score, the less sensitive the sample is for this compound.



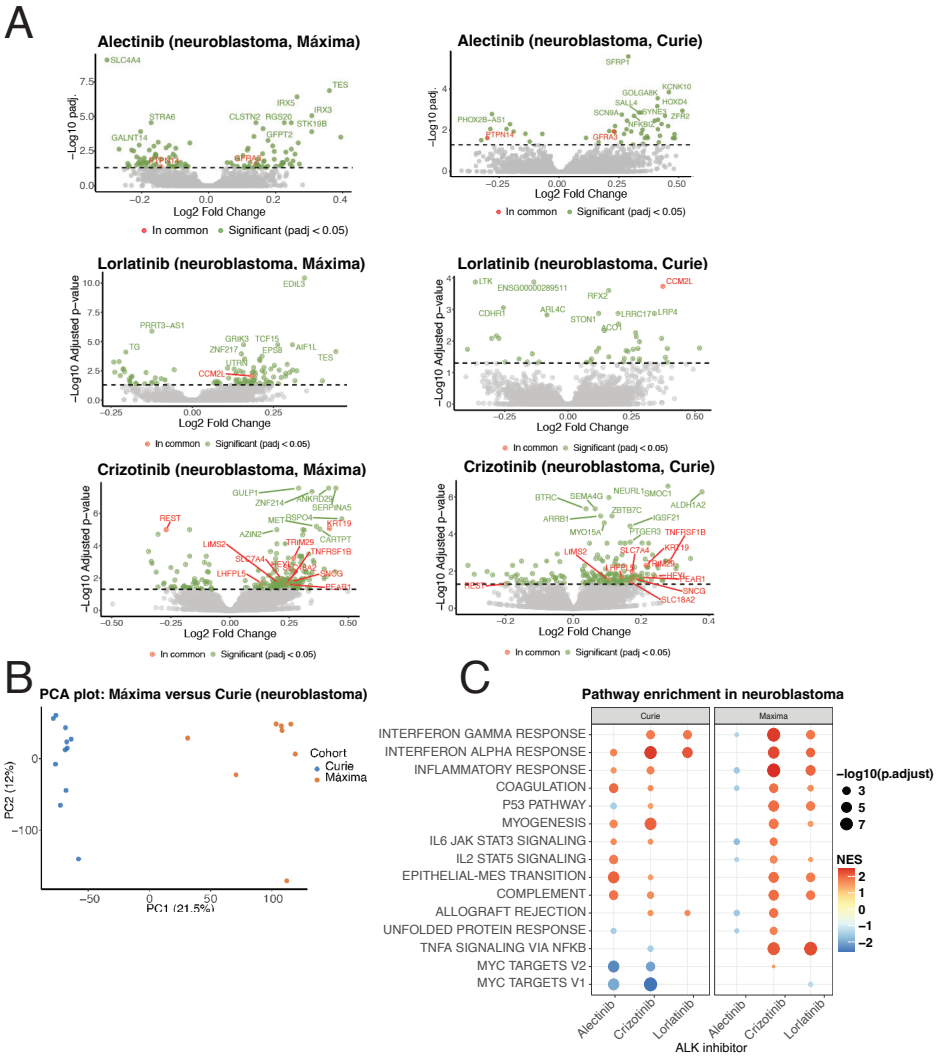
Suppl. Figure 8: The correlation of genetic aberrations and response to targeted compounds, and growth rates per tumor type A) AUCnorm for selinexor are shown for both cohorts, of patients harboring wild type (WT)-TP53 and patients harboring inactivating TP53 homozygous events. For all box plots: the box represents the interquartile range (IQR), with the median (50th percentile) indicated by the horizontal line inside the box. The whiskers extend to 1.5 times the IQR. B) Boxplots of sensitivity (AUCnorm) in Máxima for targeted compounds, where colored boxes represent the *ex vivo* cultures from PDX models with genetic alterations colored by the alteration targeted by that compound (colored = altered, grey = wild type). Stars show a significant difference ($p < 0.05$) in drug response in the altered versus wild type samples (grey stars: these alterations are only present in neuroblastoma, and significance did not hold in neuroblastoma only analyses). C)AUCnorm for ALK inhibitors are shown for both cohorts, of patients harboring ALK alterations and patients with wildtype ALK. Samples are colored per tumor type. D) Boxplots showing *ex vivo* growth rates, as determined by measuring cell viability at t= 0h and t= 72h in untreated controls, per tumortype.



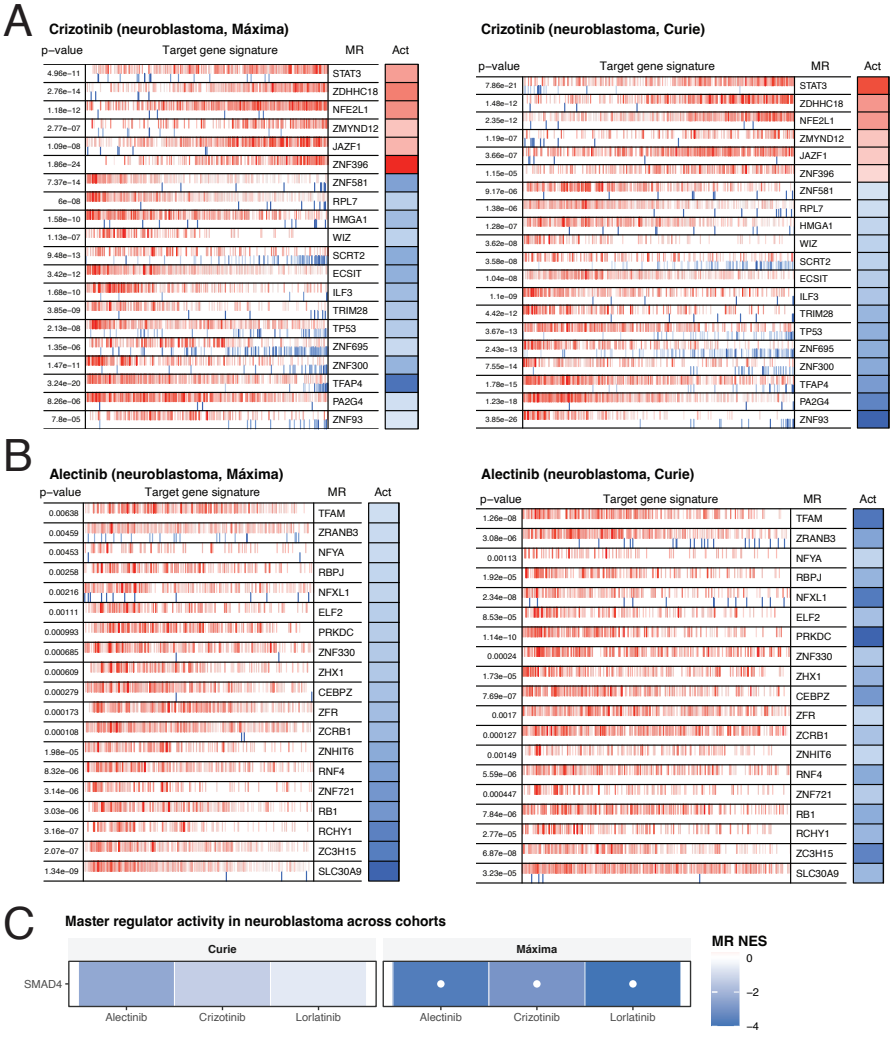
Suppl. Figure 9: Correlation drug sensitivity and growth rate, and shifting therapeutic windows in ES0619 A) Heatmap of drug sensitivity for samples of which growth rate is determined. Samples are colored per tumor type and ordered on growth rate (left = lowest, right = highest growth rate). Compounds are colored by compound mechanism. Only compounds are included that demonstrated a significant correlation with growth rate, and where the slope of the graph was >0.05 or <-0.05. Grey represents a non-determined AUCnorm. PP = PDX of primary tissue; PT = PDX of progression under treatment tissue. B) Scatter plot that shows correlation of AUCnorm with growth rate for gemcitabine and camptothecin. C) Dose-response curve for idasanutlin, of sample ES0619: primary versus the progressed sample.



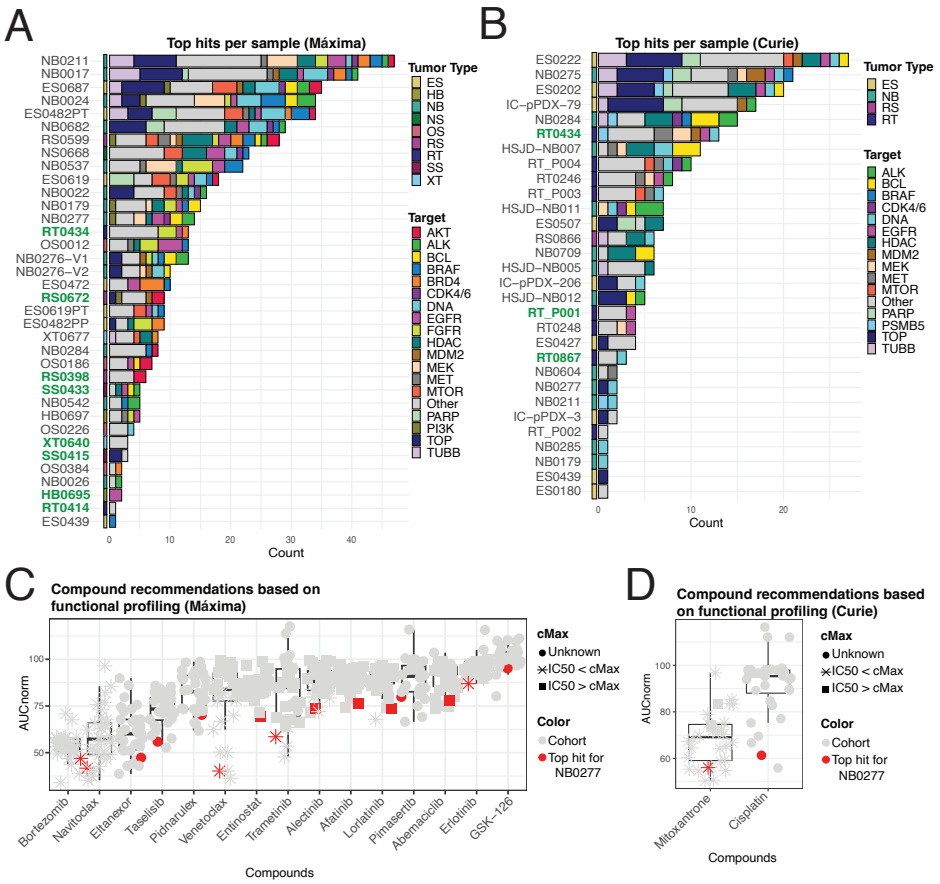
Suppl. Figure 10: RNA expression and its correlation to drug response. Here, we compare baseline (untreated) gene expression with drug response. As an example, a negative correlation (downward trend) indicates that higher baseline gene expresssion is associated with increased sensitivity (lower AUCnorm). A-D) Scatterplots of drug response (AUCnorm) according to gene expression (log-cpm) with samples colored by tumor type. Boxplots show the distribution by tumor type of gene expression (bottom) and drug response (right); dashed line shows the linear trend between gene expression and drug response. A) Represents BCL2-expression versus drug sensitivity for BCL2-inhibitor venetoclax in the Curie cohort. B-D) Represents ALK-expression versus drug sensitivity for ALK-inhibitors: ceritinib in Curie (B), alectinib in Maxima (C) and Curie (D). The shapes resemble ALK-genetic status (circle: wildtype; star: altered). E) Correlation between gene expression and ALK inhibitor sensitivity (AUCnorm) in neuroblastoma *ex vivo* screens. Scatter plots illustrating the Pearson correlation between gene expression levels (VST normalized expression) and AUCnorm for alectinib across the Maxima (left) and Curie (right) cohorts. Each point represents a tumor sample, with colors indicating tumor stage (primary, relapse, or metastasis) and shapes denoting ALK alteration status. The regression line in each plot represents the direction and magnitude of association, while the correlation coefficient (corr) and p-value (pval) indicate the statistical significance of the relationship.



Suppl. Figure 11: RNA expression and its correlation to response to ALK inhibitors in neuroblastoma. We compared baseline (untreated) gene expression with drug response in two institutes: Máxima (n = 8) and Curie (n = 11). A) Volcano plots showing differential gene expression associated with ALK inhibitor response in neuroblastoma models from Máxima (n=8) and Curie (n=11). Each dot represents a gene; the x-axis shows the log2 fold change, the y-axis the –log10 adjusted p-value. A positive log fold change indicates transcriptional up-regulation in resistant models, a negative value transcriptional up-regulation in sensitive models. Green dots indicate genes significantly correlated with drug sensitivity ($p < 0.05$); red dots mark significant genes shared between the two cohorts for the given drug. B) Principal component analysis (PCA) plot of gene expression profiles from neuroblastoma models originating from Máxima (n=8; orange) and Curie (n=11; blue), colored by institute. C) Dot plot showing pathway enrichment associated with response to ALK inhibitors in neuroblastoma *ex vivo* screens (Máxima: n = 8, Curie: n = 11). Pathways shown are significantly enriched for at least one drug in either cohort; top 15 pathways are shown. Dot color indicates the Normalized Enrichment. Score (NES) (red = enrichment in resistant models, blue = enrichment in sensitive models); dot size reflects statistical significance (–log¬ adjusted p-value), with larger dots indicating greater significance. ROS = reactive oxygen species.



Suppl. Figure 12: Master regulators and its correlation to drug response A-B) This figure shows the VIPER analysis results for master regulators whose regulatory activity is significantly (padj. < 0.05) associated with sensitivity to crizotinib (A) or alectinib (B) across both the Máxima (n = 8) and Curie (n = 11) cohorts. The top 20 overlapping master regulators are shown, and each row represents a master regulator shared between cohorts. The left column indicates the p-value. The barcode plot in the second column represents the activation or repression pattern of the master regulator's target genes, ranked from genes downregulated (left) to upregulated (right) in samples that are sensitive to the compound. Red indicates activated target genes; blue indicates repressed target genes. The third column represents the name of the master regulator, indicated by MR. The heatmap (right panel) shows the Activity Score (Act) of the master regulator: which indicates inferred regulatory activity (red = activation; blue = repression). The intensity of the color reflects the strength of activation or repression, with more saturated colors indicating greater inferred activity differences. If the Act is red, this indicates that samples with higher baseline (inferred) activity are more sensitive to the compound. If the Act is blue, this indicates that samples with lower baseline (inferred) activity are more sensitive to the compound. C) Heatmap representing master regulator inferred activity in neuroblastoma across two cohorts. Master regulator SMAD4 is shown, the heatmap is colored by NES (red = higher inferred activity in more sensitive samples; blue = higher activity in less sensitive samples). White dots mark a significant association (FDR < 0.10) between inferred activity and sensitivity for ALK inhibitors (alectinib, crizotinib, lorlatinib) across Curie and Máxima cohorts.



Suppl. Fig. 13: *Ex vivo* drug sensitivity profiling to complement molecular profiling in guiding clinical decisions: Curie cohort A-B) Frequency of top hits found for samples included in Máxima(A) and Curie (B). Bold and green are samples for which molecular profiling failed to identify a targetable alteration. Bar colors represent drug target. C-D) *Ex vivo* drug sensitivity in the Máxima(C) and Curie (D) cohorts for compounds as advised by functional profiling, i.e. drug top hits. Red datapoints resemble top hits for NB0277 in the full cohort (colored in grey). For box plots: the box represents the interquartile range (IQR), with the median (50th percentile) indicated by the horizontal line inside the box. The whiskers extend to 1.5 times the IQR.

Sensitivity profiling reveals consistent drug responses across preclinical neuroblastoma models

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Abstract

Despite intensive treatment, overall survival for high-risk and relapse neuroblastoma patients remains below 50%. Although comprehensive molecular profiling enables treatment stratification, druggable alterations have been identified for only a subset of patients. *In vitro* drug screening offers a complementary approach. Here, we compare the translational potential of three preclinical drug screening methods: *ex vivo* short-term, organoid and patient-derived xenograft (PDX) drug testing. 55 drug screens were performed from 38 neuroblastoma samples and five pediatric non-malignant samples, testing 77-224 drugs per screen. *Ex vivo* short-term drug screens achieved higher success rates than organoid screens for both patient-derived (31% versus 13%) and PDX-derived models (78% versus 27%), while reducing turnaround times from 3-12 months to 14 days. Matched samples showed consistent drug sensitivities across sample origin (patient versus PDX-derived; mean r = 0.84) and method (*ex vivo* short-term versus organoids; mean r = 0.87). In parallel, ten compounds were tested *in vivo* in eight matched PDX models. For seven of ten clinically available compounds, *ex vivo* drug responses were comparable with *in vivo* responses. These results demonstrate that, while organoids and PDX models remain essential for drug discovery, *ex vivo* short-term drug screening provides a rapid alternative for functional precision oncology in neuroblastoma.

Introduction

Neuroblastoma arises from the developing sympathetic nervous system and remains one of the most challenging pediatric solid tumors to treat. Despite advances in treatment, effective therapy options remain limited for high-risk and relapsed neuroblastoma, with overall survival rates at ~50%.[1–3] Pediatric precision oncology programs have been established with the aim of identifying druggable tumor driving alterations through molecular profiling.[4–6] Although promising, for many patients no targetable alterations could be identified, and the clinical translation was limited.[4–6] This highlights the need for additional strategies in personalized medicine to guide treatment selection.

As a complementary approach to molecular profiling, drug screening can be used to guide treatment.[7] For this purpose, *in vivo* Patient-Derived Xenografts

(PDXs) and *in vitro* organoid models have been used in neuroblastoma.[8–10] However, these methods are time-consuming, and expensive to develop and maintain.[9,11] Moreover, the clinical application of these models is further impeded by low establishment rates and long establishment times, which often exceed the median time to progression of four months, as we reported previously.[12] *Ex vivo* short-term drug screening (in short: short-term screens), where freshly obtained tumor tissue is briefly cultured before drug screening, offers a rapid alternative, delivering drug response profiles within a clinically relevant timeframe of 7-24 days after sample collection.[13–15]

In this study, we evaluated integrated drug sensitivity and molecular profiling in different neuroblastoma models to assess their consistency in drug sensitivity and their potential to guide personalized treatment selection. We performed short-term and organoid screens and combined this with our previous work on patient-derived organoids,[9] *ex vivo* short-term screens derived from patients[14] or PDX models.[15] In parallel, *in vivo* compound testing on matched PDX models was performed within the ITCC-P4 initiative, a European public-private partnership evaluating preclinical drug efficacy in pediatric cancer PDX models.[16,17] Furthermore, we included pediatric non-malignant samples to assess non-selective drug toxicity. Our results demonstrate consistent drug sensitivities across different neuroblastoma models, assay durations and tissue sources, and highlight the translational potential of *ex vivo* short-term drug screening.

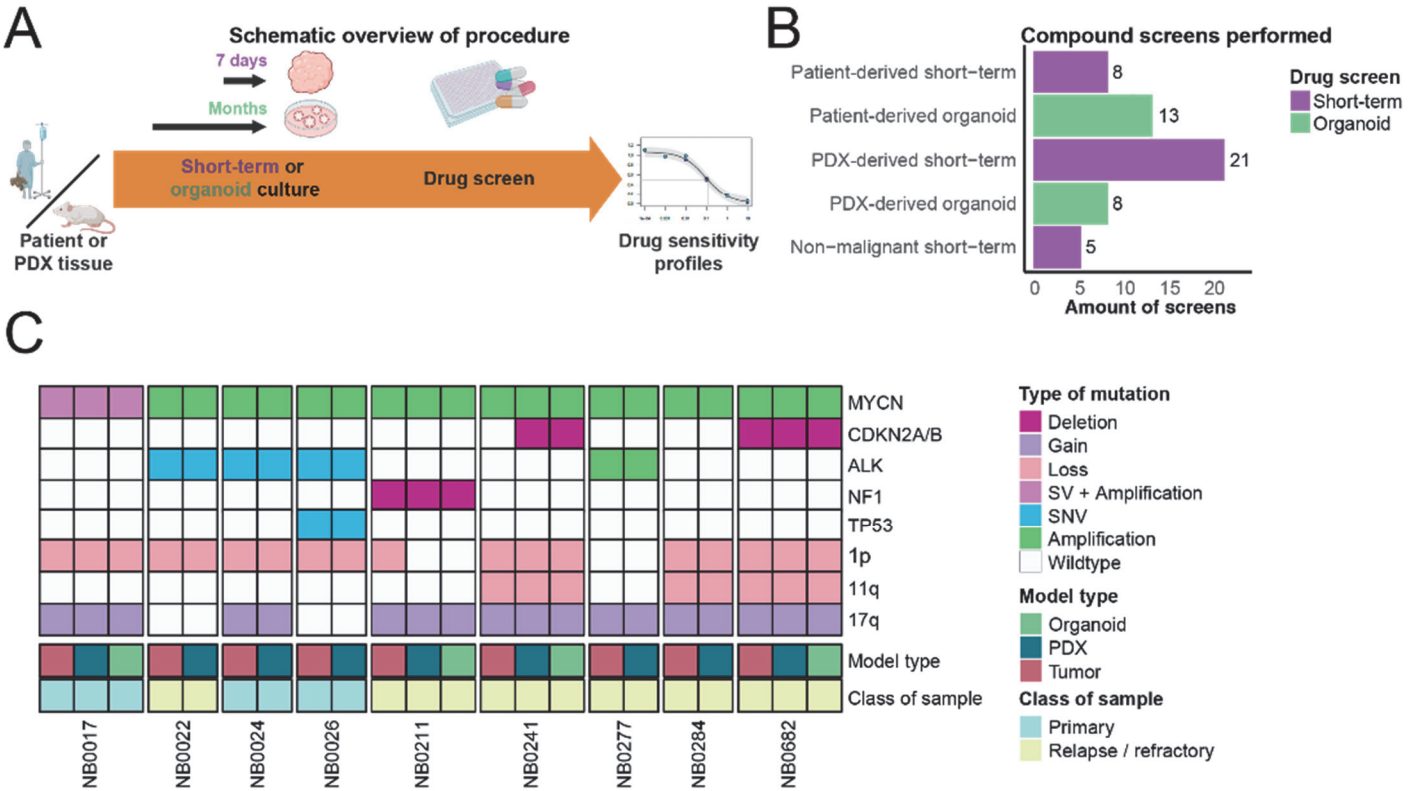
Results

Establishment and characterization of short-term, organoid, and PDX neuroblastoma models

To compare drug sensitivities of pre-clinical neuroblastoma models, 55 drug screens, from samples originating from 38 patients, were performed at the Princess Máxima Center (Máxima) or Hopp Children's Cancer Center (KiTZ) (Fig. 1A). Four types of drug screens were performed on tumor cells: patient-derived short-term (n = 8) and organoid (n = 13) screens, and PDX-derived short-term (n = 17) and organoid (n = 8) screens (Fig. 1B; Supplementary Figure 1A). In addition, five short-term screens were performed on pediatric patient-derived non-malignant tissue, including fibroblasts (n = 2), astrocytes (n = 2), and

PBMCs (n = 1). The feasibility of the different approaches was assessed by comparing establishment timelines and success rates (defined as successful culture establishment and high-quality drug screen). Short-term screens of both patient and PDX material provided drug sensitivity profiles within 14 days of sample receipt, whereas organoid models required 3-12 months. In the KiTZ cohort, patient-derived short-term screens were successfully established and performed in 31% of cases, whereas patient-derived organoid establishment succeeded in 13% of attempts. The short-term screens from patient-derived non-malignant tissue were successful in 86% of attempts (one failed quality control). Across both cohorts, PDX-derived short-term screens were successfully established in 78% of attempts, compared with 27% for PDX-derived organoids. Most short-term screen failures were due to insufficient viable cell yield prior to plating and all organoid failures were caused by growth arrest at long-term culture. Of note, four screens were excluded from further analyses due to murine content (n = 2) or because they represented duplicate screens from the same patient (n = 2).[15] These results show that both the turnaround times and success rates of short-term screens are superior to organoid screens, indicating that short-term screens could be more suitable for real-time clinical decision making.

To interpret drug sensitivity profiles in a molecular context, we analyzed whole exome sequencing, RNA sequencing and methylation array on patient tumors, PDX tumors, and the organoid models for which drug screens were performed in the Máxima. The most frequent molecular events were *MYCN* amplification, *ALK* mutation, loss of chromosome 1p, loss of chromosome 11q, and gain of chromosome 17q (Supplementary Figure 1B). To determine if PDX and organoid models retain the genetic alterations of the originating patient tumor, we examined driver events in models where tumor data was available. In matched samples, all driver events were maintained with the exception of a chromosome 1p loss observed in the NB0211 patient tumor but absent in the PDX and organoid models (Fig. 1C). We also assessed whether gene expression patterns were consistent for organoids in the Máxima cohort. We indeed show that models generally cluster close to the originating tumor (Supplementary Figure 2). These results demonstrate that neuroblastoma short-term, organoid, and PDX models can be successfully established while maintaining the key genomic and phenotypic features of the original patient tumor.


A
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Short-term or
organoid culture
Drug screen
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Drug sensitivity
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Model type
Class of sample
NB0017
NB0022
NB0024
NB0026
NB0211
NB0241
NB0277
NB0284
NB0682
Type of mutation
Deletion
Gain
Loss
SV + Amplification
SNV
Amplification
Wildtype
Model type
Organoid
PDX
Tumor
Class of sample
Primary
Relapse / refractory

Figure 1

Figure 1 (previous page): Establishment and genetic characterization of models for drug screening. A) *A schematic overview of model establishment and drug screening. Samples were obtained from patients or PDX models, digested, and cultured either for seven days (short-term) or until stable growth was observed (months; organoids) before drug screening. Drug response was determined using the area under the curve. This figure was adapted from Biorender.* ***B)*** *Bar plot representing the total number of drug screens per model type (purple: short-term; green: organoid). Part of the short-term and organoid screens performed at KiTZ were previously described in Peterziel et al.[14], Máxima patient material organoids were previously described in Langenberg et al.[9], and short-term screens performed at Máxima in Schoonbeek et al.[15].* ***C)*** *Oncoplot of genomic landscape of matched neuroblastoma samples from Máxima, showing key gene alterations and chromosomal aberrations colored by alteration type. ALK mutations include: F1174L (NB0022) and R1275Q (NB0024 and NB0026).*

PDX = Patient-Derived Xenograft; SNV = single-nucleotide variant; SV = structural variant.

Consistent drug sensitivities across patient- and PDX-derived models, and across short-term and organoid models

To evaluate drug sensitivities across preclinical neuroblastoma models, screening was performed with drug libraries of 77-224 pediatric cancer-approved and clinical trial compounds, of which 60 were shared between Máxima and KiTZ (Fig. 2A; Supplementary Table 1). The drug libraries included cytotoxic and targeted therapies, tested at concentrations ranging from 1 x 10^{-6} µM to 250 µM (Supplementary Figure 3). Drug response was quantified using the area under the curve (AUC), ranging from 0 (sensitive) to 100 (resistant) (Supplementary Figure 4-5).

Figure 2 (next pages): Comparison of drug response across neuroblastoma model systems and sample origin*. **A)** Overlapping drugs of the Máxima and KiTZ libraries.* ***B)*** *Screens performed on matched samples, colored by their model.* ***C)*** *Correlation plots of drug response (AUC) on organoid or short-term screens from matched samples derived from different tissue origins (patient or PDX material). Each dot represents a drug, colored by category: cytotoxic or targeted. A lower AUC defines higher sensitivity.* ***D)*** *Correlation plots of the drug response (AUC) of organoids and short-term cultures established from matched samples. Each dot represents a drug, colored by category: cytotoxic or targeted. A lower AUC defines higher sensitivity.*

AUC = area under the curve; R = Pearson correlation; P = p-value.

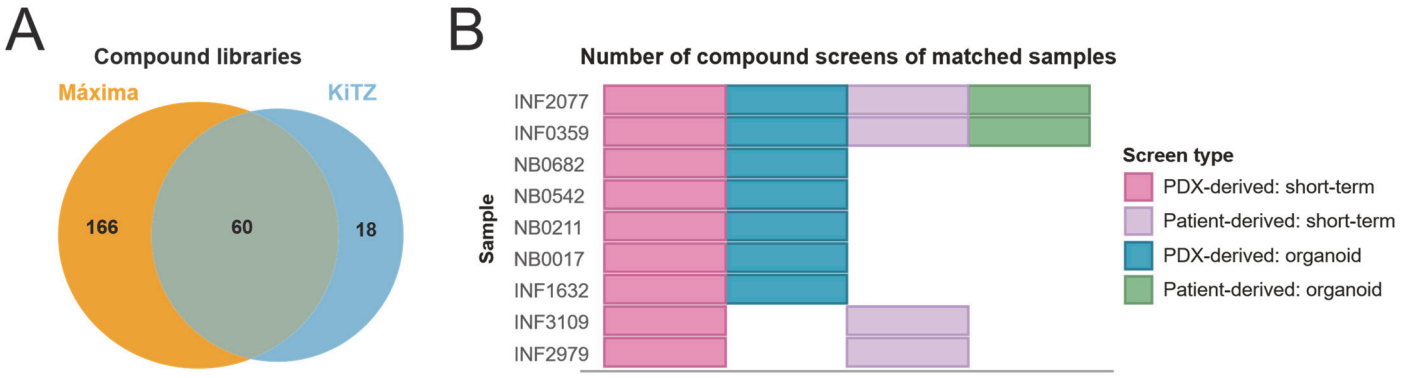
A
Compound libraries
Máxima
KiTZ
166
60
18
B
Number of compound screens of matched samples
Sample
INF2077
INF0359
NB0682
NB0542
NB0211
NB0017
INF1632
INF3109
INF2979
Screen type
PDX-derived: short-term
Patient-derived: short-term
PDX-derived: organoid
Patient-derived: organoid

Figure 2

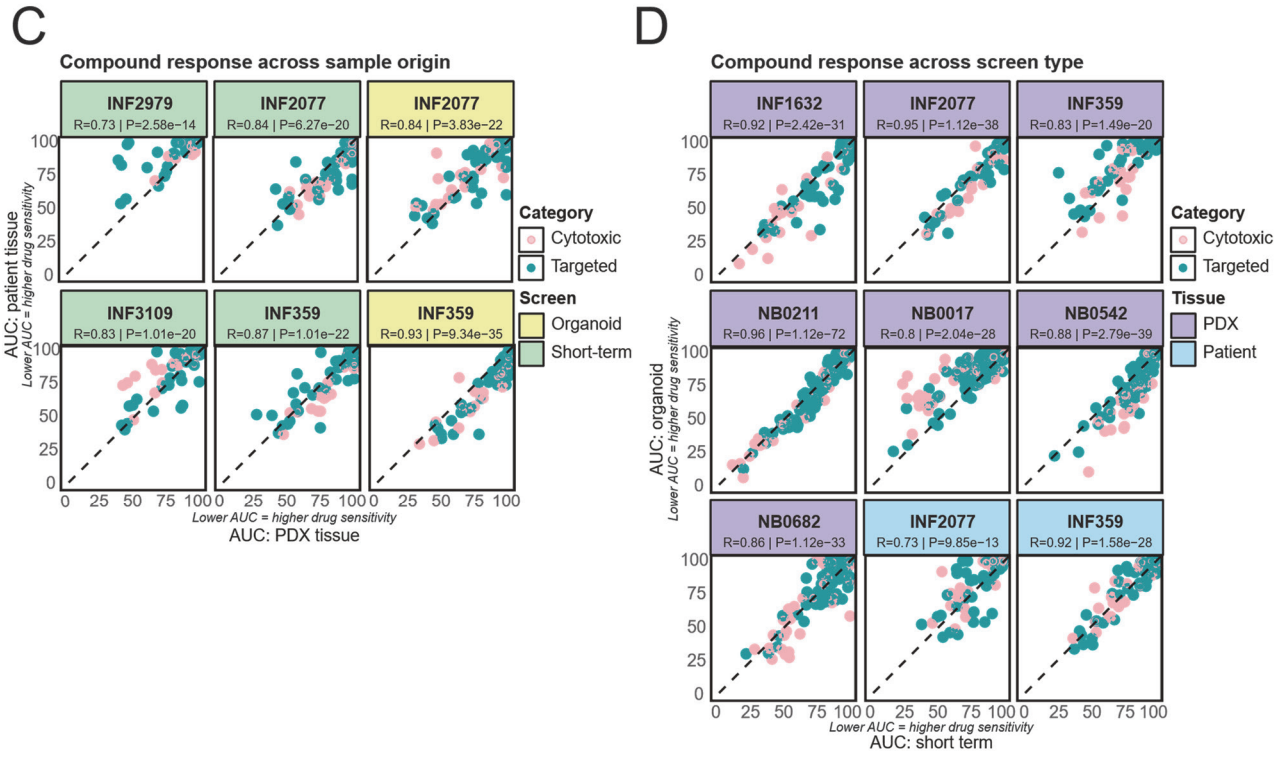

Figure 2 (continued)

To assess whether drug sensitivity is influenced by *in vivo* expansion, we compared drug response of six matched samples, including short-term (n = 4) and organoid (n = 2) screens from either patient or PDX material (Fig. 2B). Matched drug response profiles are highly correlated despite differences in sample origin (mean r = 0.84; Fig. 2C; Supplementary Figure 6; Supplementary Figure 7A). No drug was significantly associated with origin of the model indicating that this factor did not have a major influence on drug sensitivity (Supplementary Figure 7B). These results demonstrate that PDX-derived models recapitulate drug sensitivity observed in patient-derived material.

To evaluate the influence of the duration of *in vitro* culturing (max. 7 days versus 3-12 months), we compared drug sensitivities of matched short-term and organoid screens (PDX n = 7, patient n = 2). In the Máxima cohort, additional methodological differences between short-term and organoid screens existed: the culture medium and the compound incubation time (72 hours versus 120 hours in the organoids). Despite these differences, short-term and organoid screens show similar trends in drug response (average Pearson correlation = 0.87; Fig. 2D; Supplementary Figure 7C), and no drugs significantly differed in their efficacy across the two model systems (Supplementary Figure 7D-E). Next to consistent drug sensitivities across patient- and PDX-derived tissues, these results indicate that short-term screens recapitulate drug responses of the organoids, the current standard for functional profiling,[9,18,19] while shortening the turnaround from months to days.

Genomic and transcriptomic features in relation to *in vitro* drug response

To evaluate whether genomic or transcriptomic features of untreated neuroblastoma samples are associated with drug sensitivity in short-term and organoid models, we associated drug response with corresponding molecular data. When multiple screens of one sample were available, one representative screen (the short-term PDX-derived screen, available for all these samples) was included in this analysis. To evaluate whether targetable genetic alterations are associated with *ex vivo* drug response, we compared sensitivities in altered versus wild-type samples. Across the two cohorts, three targetable alterations were significantly associated with drug response (Fig. 3A; Supplementary Figure 8A).

Increased sensitivity to MDM2 inhibition was observed in TP53 wild-type samples compared to homozygous TP53-altered samples, with a significant association for idasanutlin and XPO1 inhibitor selinexor in the Máxima cohort, and trends in the same direction in the KiTZ cohort. These findings validate previous reports of resistance of tumors with a TP53 inactivation to MDM2 and XPO1 inhibitors.[20,21] However, subgroup sizes for these comparisons were small and these exploratory associations may be driven by a limited number of cases. Next to this, ALK alterations were significantly associated with sensitivity to ALK inhibitor lorlatinib in the Máxima cohort, with the same trend in KiTZ, consistent with the use of ALK status as a biomarker for treatment with lorlatinib.[22] However, ALK alterations were not associated with sensitivity to other ALK inhibitors, which might be due to the limited ALK-altered samples in our dataset. Other alterations, such as SMARCA4 and MYCN, were not significantly associated with drug sensitivity in either cohort. Although this should be validated in larger cohorts, these results indicate that genetic alterations alone may not be sufficient to predict *ex vivo* drug sensitivity in neuroblastoma.

To explore whether expression of target genes or pathways reflects drug sensitivity, drug responses were correlated with RNA expression of each compound's known target gene or pathway, where transcriptomic data were available (3B-C; Supplementary Figure 8B). The strongest correlations included an association between sensitivity to futibatinib and higher FGFR4 expression and sensitivity to ribociclib with elevated expression of cell cycle genes (Fig. 3B). ALK mutational status was associated with lorlatinib sensitivity but not for the other four ALK inhibitors tested. Next, we explored if transcriptomic features might provide additional context, by examining ALK expression and activity of the PI3K-AKT pathway. ALK expression did not correlate significantly with ALK inhibitor sensitivity, except for entrectinib (Supplementary Figure 8B), driven by one ALK-amplified sample with elevated ALK expression. Interestingly, reduced activity of the PI3K-AKT pathway did correlate with increased sensitivity to ceritinib, alectinib, and lorlatinib (Fig. 3C). The PI3K-AKT pathway functions downstream of ALK signaling and has been previously reported as a resistance mechanism.[23,24] These exploratory findings suggest that transcriptomic context may complement mutational status in the investigation of biomarkers of response.

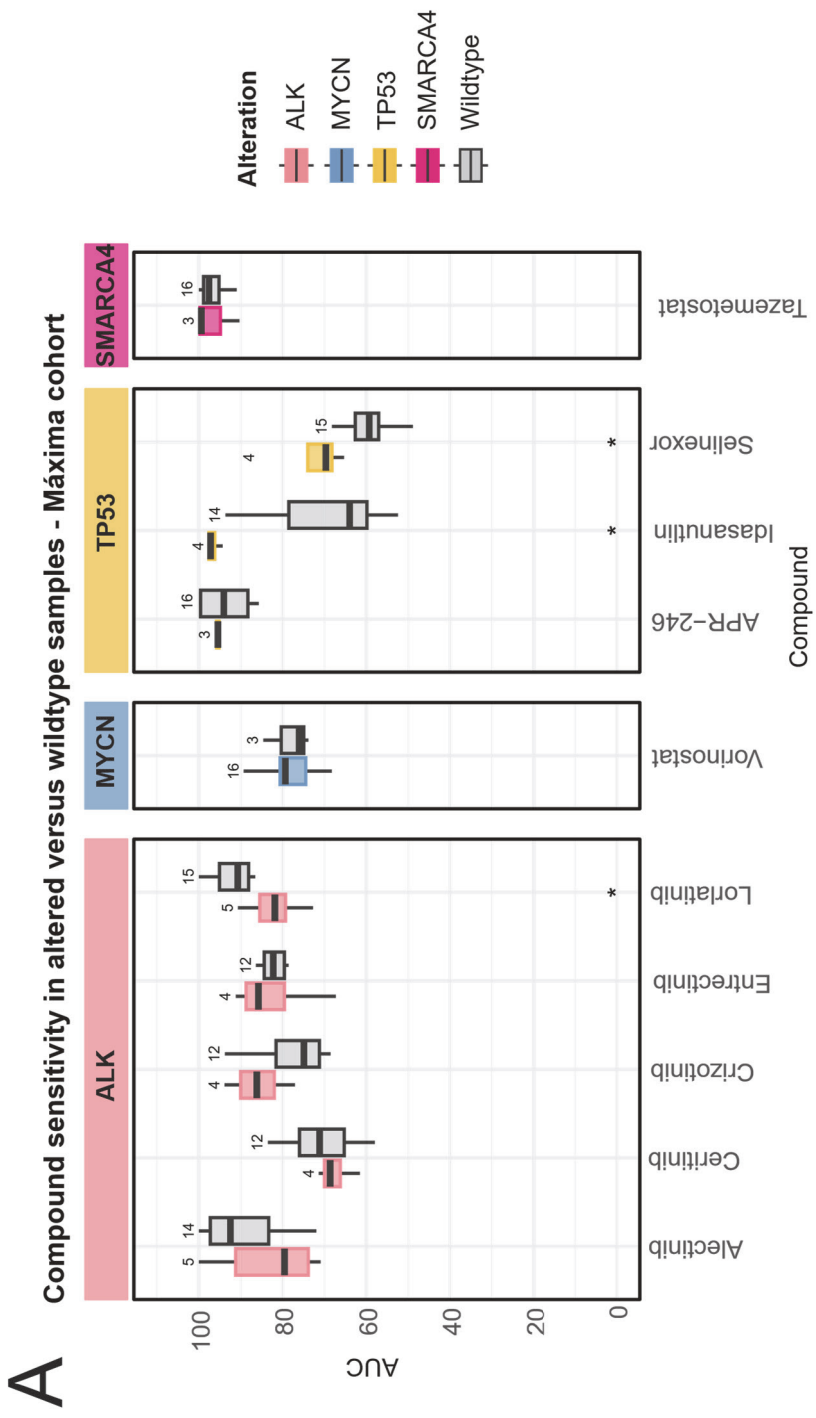
A
Compound sensitivity in altered versus wildtype samples - Máxima cohort
ALK
MYCN
TP53
SMARCA4
AUC
Alectinib
Ceritinib
Crizotinib
Entrectinib
Lorlatinib
Vorinostat
APR−246
Idasanutlin
Selinexor
Tazemetostat
Compound
Alteration
ALK
MYCN
TP53
SMARCA4
Wildtype

Figure 3

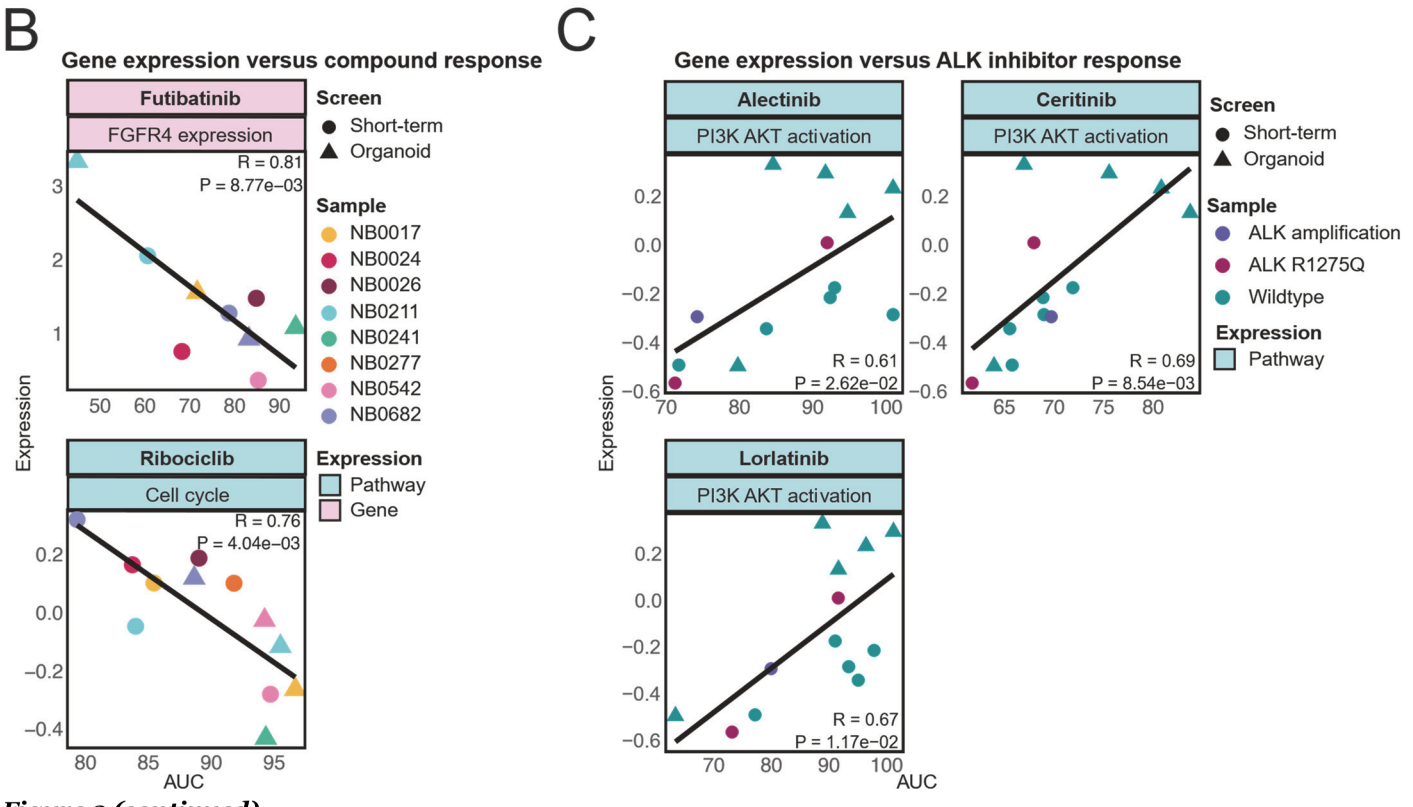

Figure 3 (continued)

Figure 3 (previous pages): Genomic and transcriptomic features associations with in vitro drug sensitivity. ***A****) AUC values of samples with targetable gene alterations, in boxes colored by alteration, compared to wildtype samples, in grey boxes, of the Máxima cohort. Statistical testing was only possible when there were at least three samples in a group, and stars indicate a significant difference between the altered and wildtype samples (adjusted p-value < 0.05).* ***B****) Correlation of AUC with FGFR4 RNA expression (log2 TMM-normalized expression; upper panel) and with cell cycle pathway activity (GSVA enrichment score; lower panel). Each dot represents a sample, colored by name and shaped by the type of screen (short-term or organoid).* ***C****) Correlation of AUC of the ALK inhibitors alectinib, ceritinib, and lorlatinib with PI3K–AKT pathway activity (GSVA enrichment score). Each dot represents one sample, colored by ALK alteration.*

AUC = area under the curve; TMM = trimmed mean of M-values; GSVA = Gene Set Variation Analysis; R = Pearson correlation coefficient.

Figure 4 (next pages): Drug screening of neuroblastoma and pediatric patient-derived non-malignant samples in the KiTZ cohort. ***A****) Box plots representing drug sensitivity (AUC) of tumor and non-malignant samples for compounds currently used in the clinic. Each dot represents the drug response of one sample, colored by tissue type. Stars indicate that the IC50 was reached within the tested concentrations. The line indicates the mean AUC. Cisplatin AQ was used.* ***B****) Bar plot representing the top 15 tumor selective drugs. The top 15 drugs with the highest mean AUC difference between tumor and non-malignant samples (mean AUC non-malignant – mean AUC tumor) are shown. Bars are colored based on the target of the drug. Asterisks indicate that the drug is significantly more effective in tumors.* ***C****) Plots representing drug sensitivity (AUC) of tumor and non-malignant samples for drugs that were significantly more effective in neuroblastoma. Each dot represents the drug response of a sample, colored by tissue type. Stars indicate that the IC50 was reached within the tested concentrations. The horizontal line indicates the mean AUC of the samples.*

AUC = area under the curve.

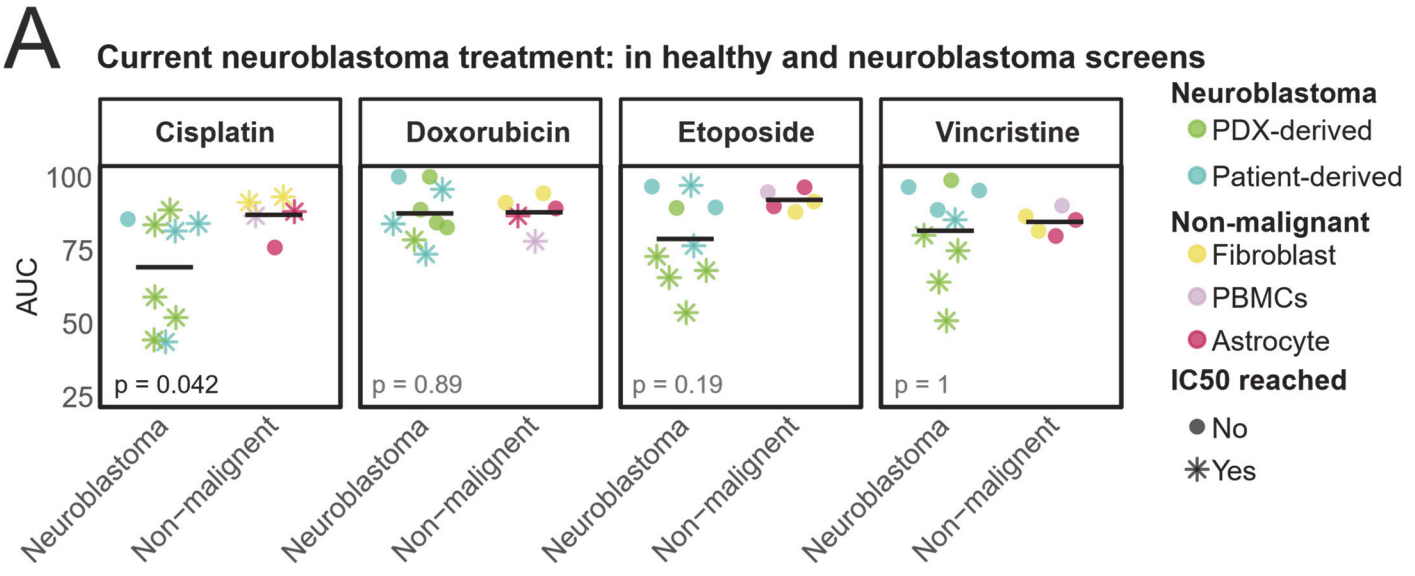
A
Current neuroblastoma treatment: in healthy and neuroblastoma screens
Cisplatin
Doxorubicin
Etoposide
Vincristine
AUC
100
75
50
25
p = 0.042
p = 0.89
p = 0.19
p = 1
Neuroblastoma
Non−malignent
Neuroblastoma
Non−malignent
Neuroblastoma
Non−malignent
Neuroblastoma
Non−malignent
Neuroblastoma
PDX-derived
Patient-derived
Non-malignant
Fibroblast
PBMCs
Astrocyte
IC50 reached
No
Yes

Figure 4

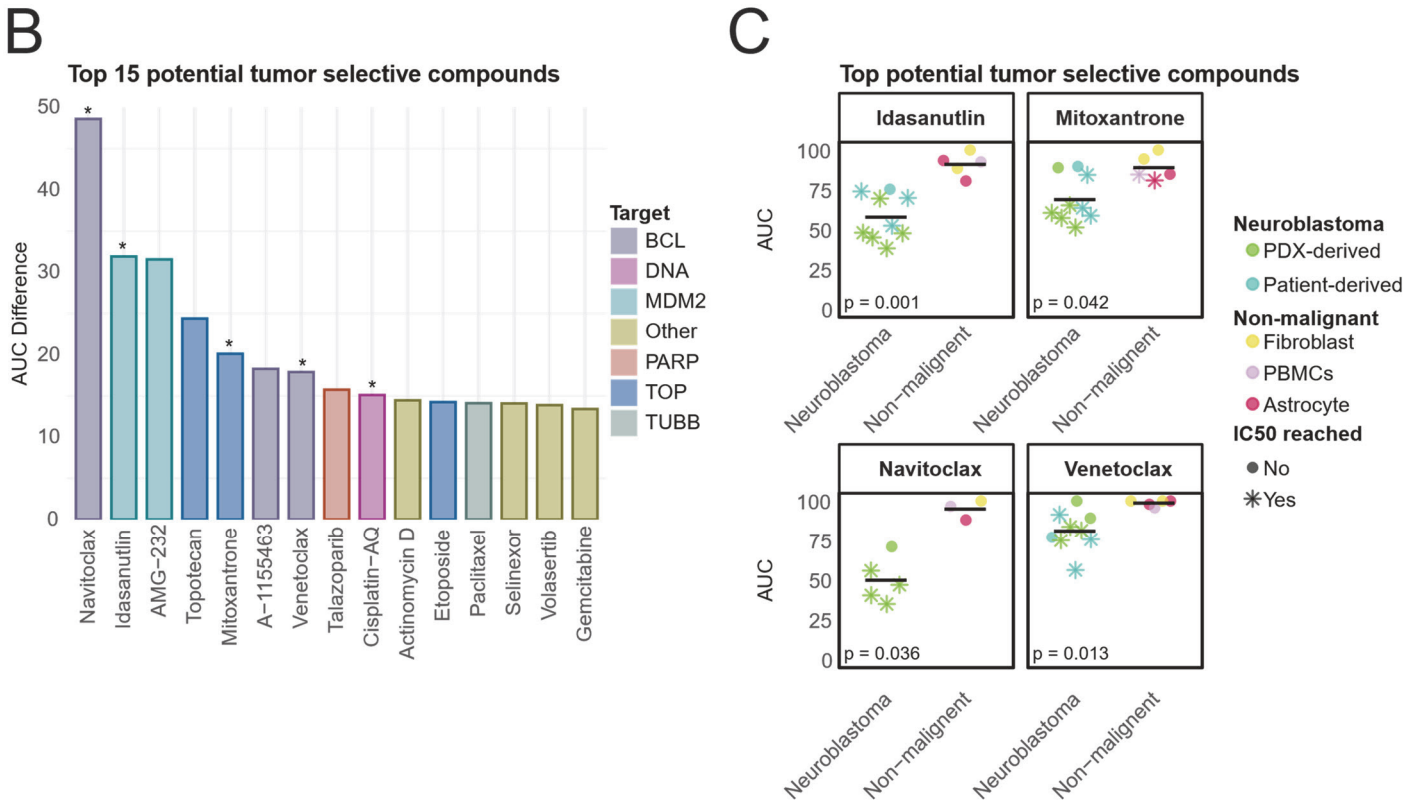

Figure 4 (continued)

Drug sensitivity profiling to assess tumor selectivity in neuroblastoma

To identify if drugs are more effective in tumor or non-malignant cells, we compared the drug sensitivities of neuroblastoma to patient-derived non-malignant pediatric samples (fibroblasts (n = 2), astrocytes (n = 2), and PBMCs (n = 1)) screened at KiTZ. First, we analyzed the compounds comprising the clinical backbone of high-risk neuroblastoma treatment as used in the SIOPEN HR-NBL2 trial[25] (Fig. 4A). Cisplatin ($p = 0.04$) and etoposide ($p = 0.19$) were more effective in tumor than non-malignant samples. In contrast, doxorubicin and vincristine were similarly effective.

To identify tumor-specific drugs, we compared drug sensitivity in tumor and non-malignant models. Figure 4B shows the 15 most tumor selective drugs. Besides cisplatin (Fig. 4A), four compounds were significantly more effective in tumor samples: BCL2 inhibitors navitoclax and venetoclax, MDM2 inhibitor idasanutlin, and topoisomerase inhibitor mitoxantrone (Fig. 4C). Conversely, we identified drugs for which non-malignant cells are more sensitive (Supplementary Figure 9A): MEK inhibitors cobimetinib and trametinib, tyrosine kinase inhibitor dasatinib, mTOR inhibitors everolimus and temsirolimus, and HDAC inhibitor panobinostat (Supplementary Figure 9B). These findings highlight the value of including pediatric non-malignant tissues as a reference in drug screens to identify compounds with favorable tumor selectivity and flag those with potential toxicity.

Concordance between *ex vivo* and *in vivo* drug responses in preclinical neuroblastoma models

In vivo compound studies were performed within the ITCC-P4 consortium.[16,17] For every compound a number of different PDX models were tested in a single mouse trial setup to optimize the number of tested compounds and models while limiting animal numbers. Ten compounds were tested in eight neuroblastoma PDX models (Fig. 5A; Supplementary Figure 10A; Supplementary Table 2). *In vivo* response was determined as end-of-treatment relative tumor volume, normalized to the starting tumor volume.

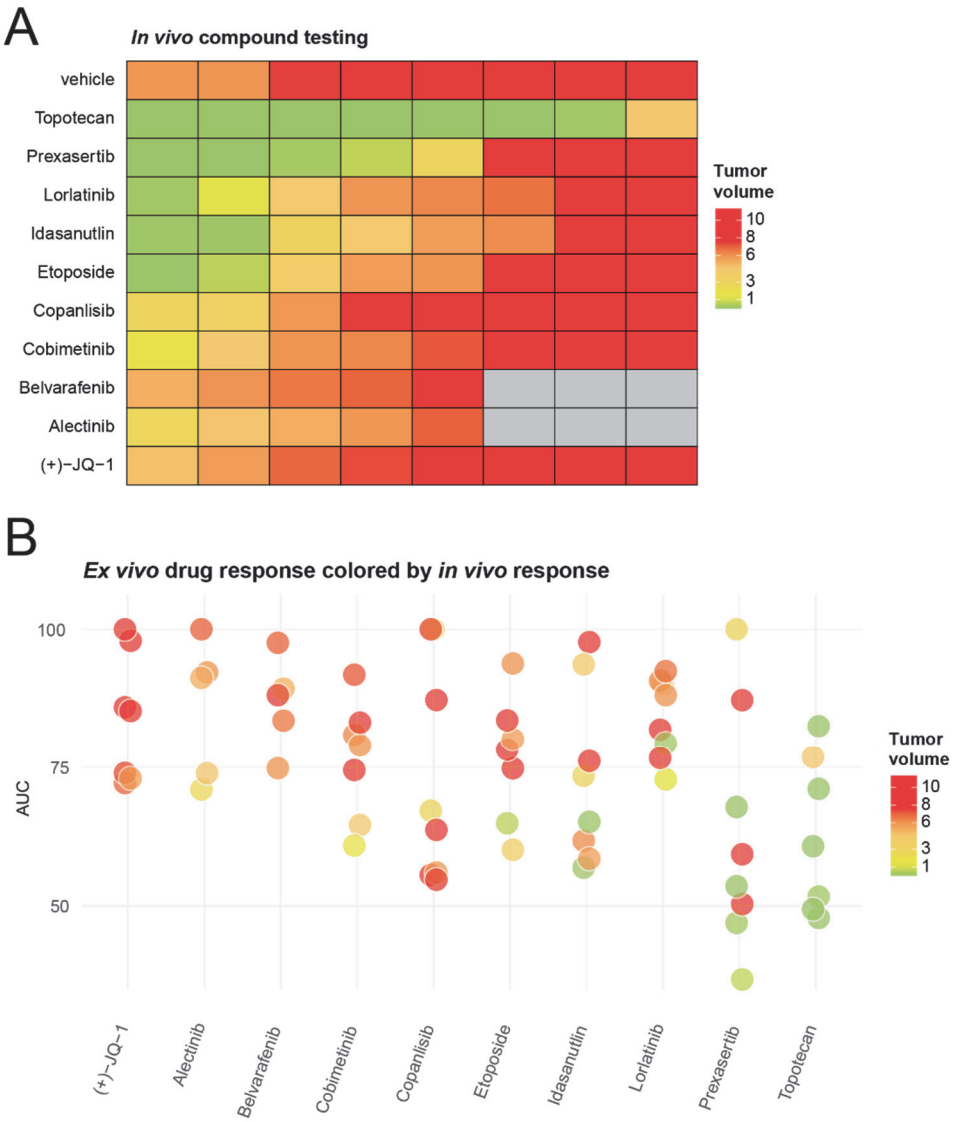

Figure 5: PDX treatment response reflects ex vivo response. ***A****) Heatmap representing the in vivo treatment responses of PDX models. Each row represents a compound (ordered by response), colors indicate end-of-treatment response (tumor volume, normalized to starting volume; grey shading indicates missing data).* ***B****) Dot plot that represents ex vivo response (area under the dose-response curve, AUC), colored by in vivo end-of-treatment response (Tumor volume, normalized to starting volume). Each dot represents a matched sample that is successfully tested both in vivo and in vitro. Higher sensitivity to the compound is defined by a lower AUC or a lower Tumor volume.*

To evaluate concordance between drug sensitivity in *ex vivo* short-term screens and therapeutic responses in corresponding PDX models, we compared the *ex vivo* AUCs with *in vivo* tumor volume. Although the assays differ substantially in design (constant 72-hour exposure across six concentrations *ex vivo* versus intermittent dosing of a single concentration over 28 days *in vivo*), we observed concordant response patterns for several compounds. Topotecan was highly effective in both model systems, reducing *in vivo* tumor volume and *ex vivo* viability in most samples (Supplementary Figure 10A-B). In contrast, belvarafenib was generally not effective in both settings, with high tumor volumes *in vivo* (tumor volumes >5) and low *ex vivo* sensitivity (AUC > 75). As compounds that are effective for only part of the patients are most relevant for precision oncology, we assessed sample-level concordance (Fig. 5B; Supplementary Figure 10C). Three drugs showed strong rank-based concordance (spearman $\rho > 0.6$): the ALK inhibitor alectinib ($\rho = 1$), the MEK inhibitor cobimetinib ($\rho = 0.71$) and the chromatin-remodeling BET inhibitor JQ-1 ($\rho = 0.61$). Moderate concordance was observed for the topoisomerase inhibitor etoposide ($\rho = 0.54$) and the MDM2 inhibitor idasanutlin ($\rho = 0.45$). For idasanutlin, most *ex vivo* responders also responded *in vivo*. However, an exception was NB0542, which was resistant *in vivo* (tumor volume = 17), despite moderate *ex vivo* sensitivity (AUC = 76, slightly above the mean of 72) – consistent with its TP53 wild-type status. Weak concordance ($\rho < 0.4$) is found for three compounds: copanlisib, prexasertib and lorlatinib. Overall, these findings demonstrate encouraging *in vivo* concordance for seven out of ten drugs tested.

Discussion

Despite intensive treatment, survival of high-risk neuroblastoma patients remains around 50%. Molecular profiling alone is insufficient to stratify treatment for all patients.[12,26–28] Functional profiling using organoids and patient-derived xenograft (PDX) models has shown promise but is limited by establishment timelines and success rates.[9,11,12] Here, we evaluated the suitability for precision medicine of three types of patient-derived neuroblastoma models: the short-term cultures, organoids, and PDX models. We performed 55 drug screens on tissues derived from 38 neuroblastoma patients and 5 non-malignant

samples. In parallel ten compounds were tested *in vivo*, in eight neuroblastoma PDX models.[17] Short-term drug screens achieved higher success rates and shortened establishment times compared to organoids while retaining the drug response profiles. Short-term screens also showed encouraging concordance with *in vivo* PDX model treatment response for seven out of ten compounds tested.

PDX models remain widely used in neuroblastoma to enable *in vivo* assessment of drug efficacy and tolerability.[8,9] In our study, PDX models retained the major genetic and transcriptomic characteristics of their originating tumors. However, their suitability for personalized precision medicine is constrained by low engraftment success, high costs and long establishment times.[29,30] Moreover, treatment responses in PDXs do not always fully mirror patient activity or toxicity, as murine pharmacokinetics and pharmacodynamics differ from those in children.[29,30] In addition, the use of animals raises ethical considerations, and regulatory agencies such as the FDA increasingly encourage alternatives to animal-based preclinical testing.[31] These limitations motivate the search for alternative patient-derived model systems that are eligible for precision medicine.

We compared four different *ex vivo* drug screening approaches and observed concordance in the responses of short-term and organoid screens, as well as between drug screens derived from patient material and those derived from PDX. Although this cohort was enriched for MYCN-amplified tumors due to the availability of matched PDX models with corresponding *in vivo* drug response data, it enabled direct comparison of *ex vivo* and *in vivo* drug responses. To the best of our knowledge, such cross-model comparisons have rarely been systematically explored. Previous work from the ZERO program also demonstrated strong concordance between short term screens of matched patient- and PDX-derived tissue for a single rhabdomyosarcoma case.[13] Our study extends on these observations and provides a unique dataset which allows direct comparison of drug responses before and after *in vitro* and/or *in vivo* expansion.

Organoid models can be derived from both patient samples and from PDX models. Organoids are attractive for large-scale drug discovery and mechanistic

studies. In our study, patient-derived organoids were successfully established in only 13% of attempts, while PDX-derived organoids achieved a success rate of 27%. In contrast, patient-derived short-term cultures succeeded in 31% of attempts and PDX-derived short-term cultures in 78%. The lower success rates for patient samples are likely reflecting the limited amount and variable quality of material obtained directly from patient samples. The turnaround time for organoid establishment and screening was 3-12 months, and similar timelines have been reported by others for neuroblastoma organoid models.[9,10,32] These timelines often exceed the median progression-free survival of relapsed or high-risk patients (4 – 7 months), which restricts their use for real-time clinical decision making.[11] A second potential downside is clonal selection during prolonged culture. Karlsson *et al.*[32] demonstrated that within 42 days (11 passages), the major subclone can expand to fixation while others are lost. Such evolution over months of culture may influence drug responses by not fully reflecting the polyclonal composition of the tumor. Short-term cultures are likely better preserving the cellular and clonal composition of the original tumor, although quantification of compositional drift was not performed in this study.[14] Overall, these factors limit the suitability of organoids for real-time clinical decision making.

Ex vivo short-term cultures offer a faster approach for preclinical drug testing. In our study, patient-derived short-term screens were successfully established in 31% of cases, whereas PDX-derived short-term screens achieved a success rate of 78%. *Ex vivo* short-term screens can support precision medicine, by identifying patient-specific drug sensitivities, although prospective clinical studies are required to establish their predictive value for patient response.[15] Although treatment decisions were not based on the drug screening results generated in this study, previous work from ZERO and INFORM demonstrated the feasibility of integrating *ex vivo* short-term screens cultures in precision medicine programs as the rapid turnaround time allows integration into multidisciplinary tumor boards.[13,14] The integration of molecular and functional data may provide a higher level of clinical confidence in treatment recommendations. In ZERO, 82% of screens were completed while patients were still alive. In addition, in INFORM, we observed preservation of the tumor microenvironment (with a trend for tumor

cell enrichment) during preculture and drug screening. As short-term cultures can retain residual microenvironmental cells, they may provide a more representative tumor model than long-term models. To account for residual microenvironment cells, future studies should develop methods to distinguish between tumor and non-tumor cells to determine the tumor-specific drug response. The inclusion of the tumor microenvironment may enable testing of clinically relevant immunotherapies such as anti-GD2 antibodies. In contrast, the evaluation of anti-angiogenic agents is not feasible in this system. In addition, further optimizing success rates and reducing the required cell numbers per well will be essential for implementation of drug screening when using small needle biopsies.

To assess whether *ex vivo* drug responses correspond with *in vivo* activity, we compared *ex vivo* drug sensitivity from short-term screens with matched PDX model responses, performed within the ITCC-P4 consortium.[16,17] We observed encouraging concordance with *in vivo* PDX models for seven agents, including kinase inhibitors targeting ALK (alectinib), MEK (cobimetinib) and MDM2 inhibitor idasanutlin. For others (e.g. copanlisib), concordance was limited, which could be due to several factors. Direct comparisons are constrained by major differences between the models, such as discordant drug concentrations, treatment durations, and pharmacokinetic and pharmacodynamic parameters.[33] For rapidly metabolized compounds, *ex vivo* concentrations may be higher than concentrations achievable in plasma or tumor tissue, which may explain the lack of *in vivo* efficacy even when *ex vivo* sensitivity is observed. Another potential difference is the discrepancy between *in vitro* and *in vivo* growth rates, which especially impacts sensitivity to cytotoxic agents[34,35]. Besides these differences, *in vivo* models are known for their biological variability in treatment response.[36] Therefore, using an n-of-1 setup limits our direct comparison between *in vivo* and *ex vivo* sensitivity. Lastly, the PDXs tested *in vivo* were not from the same passage from which the *ex vivo* models were derived, introducing the possibility of clonal shifts between the passages. Ultimately though, direct correlation with in-patient, not PDX, response holds clinical value. Future studies with larger, clinically annotated cohorts should investigate whether prior treatment exposure results in resistance to the same agents in *ex vivo* drug screens. Previous work by ZERO[13]

and INFORM[14] demonstrated that *ex vivo* sensitivities were consistent with responses in matched PDXs and, critically, in patients.

Beyond predicting tumor response, functional assays may also provide insights into potential tumor selectivity and toxicity. Two of the clinical backbone agents used in high-risk neuroblastoma treatment, doxorubicin and vincristine, showed comparable effectiveness in neuroblastoma and patient-derived non-malignant samples. This is consistent with previous reports of doxorubicin-related cardiotoxicity and vincristine-induced peripheral neuropathy.[37,38] In our study, topotecan, navitoclax, venetoclax and idasanutlin were significantly more effective in tumor than non-malignant samples. Consistent with these findings, clinical studies report tolerability with promising clinical outcomes for topotecan, navitoclax and venetoclax in neuroblastoma or other pediatric solid tumors.[39–43] Interestingly, topotecan also demonstrated high efficacy in both *ex vivo* and *in vivo* models. Although idasanutlin showed more efficacy in tumor than non-malignant samples, a phase I trial of idasanutlin in patients with leukemias or solid tumors was terminated due to toxicity and lack of efficacy.[44] In addition, some compounds were more effective in non-malignant cells than in tumor cells. This may reflect the dependence of non-malignant proliferating cells and immune cells on the targeted pathways. For dasatinib, PBMCs specifically showed the highest sensitivity, consistent with the known dependence of lymphocytes on Src-family kinase signaling for T-cell activation and proliferation.[45,46] For MEK inhibitors cobimetinib and trametinib, neuroblastoma cells in our cohort could be less dependent on RAS-MEK-ERK signaling than non-malignant proliferating cells, and/or may have acquired bypass mechanisms that further reduce single-agent sensitivity. For mTOR inhibitors , a clinical trial combining temsirolimus with re-induction chemotherapy reported excessive toxicity in children with relapsed acute lymphoblastic leukemia.[47] However, interpretation is limited by the small sample size for non-malignant samples. In addition, this analysis considered all tumor samples collectively, irrespective of genetic or transcriptomic features, and all non-malignant samples irrespective of tissue origin. The non-malignant samples used in this study were selected based on the feasibility of obtaining pediatric samples through minimally invasive procedures. They provide an initial reference for evaluating tumor selectivity, but do not represent the major organs involved

in drug metabolism or dose-limiting toxicity. Future studies should focus on establishing larger reference panels including pediatric liver, kidney, and intestine tissues.[48–51] Including non-malignant pediatric tissues in *ex vivo* drug screening may help identify drugs that could cause unacceptable toxicity. Another possible advantage of collecting a large non-malignant panel is that it allows the investigation of how inherited genetic variation influences drug toxicity. This is particularly relevant in neuroblastoma, where young survivors face long-term treatment-related risks of (cardiovascular) complications and secondary malignancies.[52]

In conclusion, our findings demonstrate that drug sensitivities were generally consistent between short-term and organoid screens, but these models differ markedly in feasibility and turnaround times. Prospective clinical studies will be essential to determine whether *ex vivo* drug responses predict patient response. Additionally, future studies should extend single-agent screens to include validated drug combinations. While organoids and PDX models remain essential for large-scale drug discovery and biomarkers research, *ex vivo* short-term screening offers higher success rates and rapid establishment times, making it well suited to complement molecular profiling for precision oncology in neuroblastoma and possibly other pediatric and adult tumors.

Methods

PDX establishment at the ITCC-P4 consortium and KiTZ

Informed consent to participate in research and for the collection and use of biological samples was obtained from all patients, or their parents or legal guardians, in accordance with the Declaration of Helsinki.

The establishment and characterization of mouse Patient-Derived Xenografts (PDXs) used for drug screening in the Princess Máxima Center were performed by the ITCC-P4 consortium, as described before,[15–17] and samples were expanded by EPO[53,54] in Berlin, Germany, or Charles River in Freiburg, Germany. Before sample collection, studies were approved by the Ethics Board of the Medical Faculty of the University of Heidelberg (IRB: S-610/2016) or by the respective local authorities. Informed consent included permission for anonymized data

sharing in academic and nonprofit settings, according to the Helsinki guidelines. Fresh tumor material (obtained through surgical resection or biopsy), a constitutional (germline) sample, and clinical information were collected for each patient. PDX models were originally established at participating centers according to local protocols: tumor tissue was subcutaneously transplanted to either NSG (NOD.Cg-Prkdc^{scid} IL2rg^{tm1Wjl}/SzJ; Charles River Laboratories, Germany), Swiss Nude (Crl:NU(Ico)-Foxn1nu), CB17 SCID (CB17/Icr-Prkdc^{scid}/IcrIcoCrl), or NOG (NOD.Cg-*Prkdc^{scid} Il2rg^{tm1Sug}*/JicTac; Taconic Denmark) immunodeficient mice. Thereafter, all PDX passaging, tissue collection for *ex vivo* drug screening, and *in vivo* single-mouse drug efficacy studies reported in this manuscript were performed in NSG mice according to the standardized ITCC-P4 protocol.[17] All sample and clinical information, as well as analyzed molecular data of patient tumors, PDXs, and organoids were barcoded and subsequently operated via R2 genomics analysis and visualization platform (r2-itccp4.amc.nl).

At KiTZ, patient material was collected and dissociated as described before, in agreement with local institutional ethical regulations, approved by the institutional review board, and with patient consent under the guidelines of the MAST protocol (the INFORM study was registered with the German Clinical Trial Register, number DRKS00007623).[14] Up to 1 million dissociated cells in PBS were mixed with Matrigel (1:1) and subcutaneously injected into the right flank of NSG mice under isoflurane anesthesia. Tumors were passaged three times in mice, and drug screening as well as PDX long term cultures were established from human-to-mouse passage 0 and/or mouse-to-mouse passage 3. All animal experiments were conducted in accordance with the German Animal Welfare Act and approved by local authorities.

Table 1. *The treatment schedule of the compounds used in vivo, with the compound name, the dose given to the mouse, the schedule, the route of administration, and the formulation of the given compound. On represents days on treatment, off represents days off treatment.*

Drug	Dose mouse (mg/kg)	Schedule	Route	Formulation
Etoposide	5	Daily 5-on/2-off for 4 cycles	Intravenous	0.9% NaCl
Topotecan	0.5	Daily 5-on/2-off for 4 cycles	Intravenous	5% Glucose
Idasanutlin	100	Twice a day, 5-on/2-off for 4 cycles	Orally	GTX011795 (water 97.8%, 2% hydroxypropyl cellulose, 0.1% polysorbate, 0.09% methyl paraben, 0.01% propyl paraben, 0.0072% sodium acetate, 0.0568% acetic acid
JQ-1	50	Daily, 5-on/2-off for 4 cycles	Intraperitoneally	Vehicle: 1:10 DMSO: 10% hydroxypropyl-β-cyclodextrin
Prexasertib	10	Twice a day, 3-on/4-off for 3 cycles	Subcutaneously	20% captisol in purified water
Cobimetinib	7.5	Daily, for 28 days	Orally	0.5% methylcellulose/ 0.2% Tween 80 (MCT)
Lorlatinib	3	Twice a day, for 28 days	Orally	MilliQ DI water, 1:1.2 lorlatinib:HCI molor ratio
Copanlisib	10	Daily, 2-on/5-off 4 cycles	Intravenously	0.9% NaCl
Alectinib	20	Daily, for 28 days	Orally	5% DMSO, 5% cremophor EL solution, 90% NaCl (0.9%)
Belvarafenib	15	Daily, for 28 days	Orally	10% DMSO, 10% cremophor, 15% PEG400, 15% HPCD, 0.02N HCl in sterile water

ITCC-P4 drug screen in PDX for response characterization

ITCC-P4 PDX models were characterized for their response to single compound or combination therapies in preclinical single-mouse trials.[16,17] Based on individual formulation and specific schedules, single compounds and combinations were administered either orally, intravenously, subcutaneously, or intraperitoneally (Table 1). When the PDX tumor volume achieves a size of 100-250 mm^3, a blinded randomization procedure was used to distribute mice to test groups with the subsequent start of treatment. The actual amount of compound was calculated daily depending on the individual body weight of animals throughout a four-cycle 28-day treatment phase. All animals were terminated at individual time points when they reached the maximum ethical tumor volume threshold (TV > 1.5cm^3), body weight loss (>20%) limits, or when animals suffered from adverse effects before the end of four cycles. Control animals receive the most demanding schedule and therefore the lorlatinib vehicle formulation. From the first treatment day onwards, tumor size is measured twice a week until the end of the treatment.

Tissue dissociation

PDX-derived tumor tissue (both cohorts) and patient-derived tumor tissue (KiTZ) were processed using a site-specific enzymatic digestion protocol. Tissue was incubated in Liberase DL (Sigma Aldrich, cat. no. 5466202001; 1:43 dilution) in tumor stem media (TSM)[14] at 37°C for 30 minutes with occasional swirling (Máxima) or in trypsin (10 mg/ml) at 37°C for ten minutes (KiTZ). TSM consists of 47.5% Neurobasal-A medium, 47.5% D-MEM/F-12, 1% HEPES buffer solution (1 M), 1% sodium pyruvate MEM (100 mM, CE), 1% MEM non-essential amino acids solution (10 mM), L-glutamine solution BIOXTRA (2 mM), and 1% penicillin-streptomycin-glutamine (all from Life Technologies, Thermo Fisher Scientific Inc.). This was supplemented with 2% B27 Supplement Minus Vitamin A (50×; Life Technologies, Thermo Fisher Scientific Inc., Waltham, MA, USA), 0.02% (w/v) recombinant human EGF (100 µg/mL), 0.02% (w/v) recombinant human FGF-basic (100 µg/mL), 0.05% H-PDGF-AA (20 µg/mL) (all from PeproTech Inc., Rocky Hill, NJ, USA), and 0.1% heparin solution (2 mg/mL; Sigma-Aldrich, Munich, Germany). Enzymatic activity was stopped by harvesting the supernatant, adding medium (Máxima), or by adding 12 µl trypsin inhibitor,

followed by repeated additions of a 1:1 mixture of 1 M $MgCl_2$ and DNase (2 mg/mL; 60 µL per addition) until the tumor fragment settles (KiTZ). At the Máxima, the residual tissue underwent a second digestion in Liberase TM (Sigma Aldrich, cat. no. 5401119001; 1:43 dilution in TSM). After another 30 minutes, cells were harvested as described above. Cells were filtered through a pre-washed 70 µm cell strainer (Máxima) or a 70- or 100 µm strainer (KiTZ) and centrifuged at 300 g for 5 minutes (Máxima) or at 400-500 g for 10 minutes (KiTZ). Red blood cells were lysed using lysis buffer with a maximum incubation time of 2-5 minutes at room temperature. Cells were resuspended in TSM complete[14] and counted using trypan blue.

Short- and long-term organoid culture

After tumor tissue dissociation, cells were cultured for short-term or until organoids were established. In all cases, tumor cells were cultured at 37°C, 5% CO_2. Short-term cultures and screens at the Máxima and KiTZ were done in TSM. At the Princess Máxima Center, DNA was harvested before drug screening. Freshly dissociated cells were cultured for 5-8 days (at Máxima) or for 2-7 days (depending on growth and morphology, at KiTZ), before screening. Two short-term screens were excluded due to mice contamination.

Long-term organoid cultures at the Máxima were cultured in Neuroblastoma Organoid Medium (NBOM), consisting of DMEM with GlutaMAX™ (Life Technologies, cat. no. 21885108), 25% Ham's F-12 Nutrient Mix, B-27™ Supplement (minus vitamin A), N-2 Supplement, 100 U/mL penicillin, 100 µg/mL streptomycin, 20 ng/mL recombinant human EGF, 40 ng/mL recombinant human FGF-basic, 200 ng/mL recombinant human IGF-I, 10 ng/mL recombinant human PDGF-AA, and 10 ng/mL recombinant human PDGF-BB.[55] At KiTZ, long-term cultures were cultured in TSM. In both centers, cells were cultured biweekly, with medium refreshment or passaging based on microscopic evaluation. Medium was refreshed by harvesting medium, centrifuging at 250 g for 5 minutes, and resuspending the pellet in fresh medium. Oversized cultures with necrotic cores were dissociated by pipetting in a 50 mL tube before splitting (Máxima) or using Trypl Express (Life Technologies, Thermo Fisher Scientific Inc., Waltham, MA, USA; 5 min, 37 °C; KiTZ). Mycoplasma testing and STR profiling (PowerPlex system and GeneMapper

software, Promega) were performed routinely. Drug screening of long-term organoid cultures was initiated only after stable growth of human tumor cells was confirmed. Tumor content was confirmed by comparing patient-specific breakpoints of the culture (using SNP array) to those of the original tumor. On the day of organoid screening, cryovials, RNA, and DNA were harvested at the Máxima. For all organoid models established at KiTZ, unsupervised clustering of genome-wide DNA methylation data (Illumina 850K array) was performed, using t-distributed stochastic neighbor embedding (t-SNE) to confirm clustering of the long-term culture with the original fresh frozen sample of the patient and with the appropriate reference methylation classes. Concordance of key molecular alterations is additionally assessed by direct comparison of CNA profiles derived from the same methylation array datasets between patient-derived fresh frozen samples and matched organoid models.

Part of the KiTZ short-term screens were previously reported in Peterziel *et al.*[14], Máxima organoids from patient material were previously reported in Langenberg *et al.*[9], and Máxima short-term screens on *ex vivo* cultures from PDX models were previously reported.[15,56]

Molecular profiling

For models used for drug screening in the Máxima, comprehensive molecular profiling was performed as part of the ITCC-P4 consortium[16,17] in Paris (Curie) and Heidelberg (KiTZ), on PDX and organoid samples, and if available, on primary patient tumors and matched germline controls. For models used for drug screening at KiTZ, genome sequencing was exclusively performed on the primary patient tumor. For low-coverage whole-genome sequencing (lcWGS) and whole-exome sequencing (WES), libraries were prepared using standardized methods[27] and captured with the Agilent SureSelect Clinical Research Exome V2 Kit, followed by sequencing using Illumina platforms at both centers. For PDX models, organoids and, if available, primary patient tumors, used for drug screening at Máxima, RNA-seq was also performed by the ITCC-P4 consortium[16,17] in Paris (Curie) and Heidelberg (KiTZ). RNA-seq was performed on the NextSeq 500 (150 nt paired-end) or NovaSeq (100 nt paired-end).

Drug libraries

The Máxima drug libraries L13 and L14 consist of 224 and 226 drugs, respectively, including FDA-approved agents and those in (pre-)clinical development (Supplementary Table 1). All organoids are screened with the full library. For short-term screens, the full library is used when sufficient biomass is available; otherwise, the clinical subset of 134 drugs is used. The KiTZ drug library consists of 78 clinically relevant anticancer drugs, including targeted as well as chemotherapy (Supplementary Table 1). Both libraries consist of clinically relevant anticancer drugs, including FDA-approved agents and those in (pre-) clinical development. Most drugs are dissolved in DMSO and stored under nitrogen at room temperature (Máxima) or in an oxygen- and moisture-free environment at room temperature (KiTZ). At the Máxima, exceptions include five drugs (metformin, perifosine, carboplatin, oxaliplatin, and copanlisib) dissolved in MQ, and cisplatin in saline, which are stored at –20 °C. Before screening, 384-well working plates (Labcyte 384LDV) are shaken (30 min at room temperature) and centrifuged (2.5 min at 1500 rpm). The Echo 550 dispenser is then used to verify well volume (>2.5 µL) and DMSO percentage (>80%).

Drug screening

Drug screens were performed in duplicate. Cells (5,000/well in 40 µL at the Máxima; 1000/well in 25 µL at KiTZ) were plated into 384-well flat-bottom tissue culture plates (#3764, Corning) using a multidrop combi dispenser (Thermo Scientific) and cultured for 24 hours (Máxima) or directly screened (KiTZ, as described in Peterziel *et al.*[14]). At the Máxima, cell viability was assessed at t = 0 using the Cell Titer Glo 3D (CTG3D) assay (Promega, G9683) according to the manufacturer's protocol.

Using the Echo 550, 100 nL of each drug (in DMSO or MQ) was added to achieve final concentrations ranging from 0.1 nM to 100 µM (0.25% DMSO/MQ). DMSO-only treated cells served as negative controls; 10 µM staurosporine-treated cells were used as positive controls. Cells were incubated with the drugs for 72 hours at 37°C and 5% CO_2, except for organoid screens at the Máxima (120 hours).

Data were normalized against DMSO-treated (100% viability) and empty well controls (0% viability). Dose-response curves were modeled using a five-

parameter log-logistic model with the R package drc (v3.0.1; R 4.3.1[57]). From these models, the IC50 (concentration for 50% viability reduction), area under the curve (AUC), and relative cell growth (Máxima, CTG3D signal day 5/t0) were calculated. Screen quality was assessed based on growth dynamics, control performance, and consistency across replicates. At the Máxima, a screen was considered successful if: (1) clear separation was observed between staurosporine (positive control) and DMSO (vehicle control) responses; (2) relative cell growth exceeded 60% over the assay duration; and (3) replicate consistency and individual dose-response curves passed independent visual inspection by two researchers. For PDX-derived short-term screens at the Máxima, tumor cell content was additionally verified at the time of screening using qPCR quantification of human versus murine cell proportions.[15] Cell viabilities prior to screening were 80–97% across both patient- and PDX-derived short-term and organoid screens. At KiTZ, screen quality was assessed using a Robust Z-score approach based on DMSO and benzylamine nucleoside controls, yielding high- and medium-quality screens; for medium-quality screens, replicate correlation and raw signal strength were additionally evaluated, as previously described.[14] Establishment success was defined as the successful generation of a culture of which a drug screen that passed the predefined quality-control criteria could be performed. AUC was the primary metric used for defining and comparing drug sensitivities, as IC50 could not be reliably determined for inactive drugs. We normalized the area under the curve (AUC) between 0-100, accounting for differences in the number of concentrations across cohorts.

Calling of genomic alterations

Whole-exome sequencing was used to identify genomic alterations as described above. High-level amplifications of *MYCN*, *ALK*, *CDK4*, *KRAS*, and *MDM2* were called when copy number (CN) > 8. ALK alterations were also called in case of an activating mutation within the tyrosine kinase domain. Alterations in genes *TERT*, *NF1*, *PTPN11*, *PIK3CA*, CDKN2A/B and *AKT* were considered based on homozygous deletions, or loss of heterozygosity (LOH) in combination with an inactivating mutation. ATRX was called in case of homozygous inactivation or deletion, or the combination of LOH, in-frame fusions, inactivating mutations, or hotspot mutations. TP53 mutations were called in case of a deletion combined

with an inactivating (frameshift or SNV) mutation. Chromosomal gains were called when CN > 2.5 and losses when CN < 1.5.

RNA-sequencing normalization and gene set enrichment

RNA-sequencing data were aligned to the human reference genome (GRCh37, GENCODE 19) using STAR[58] and the raw counts were normalized to CPM values using the Trimmed Mean of M-values (TMM) method implemented in the edgeR[59] R package (version 4.6.2) and log2 transformed. Gene set scores were calculated using GSVA[60] (version 2.2.0) and gene sets were obtained from the Reactome database.[61]

Statistical testing

Statistical analysis was performed using the Wilcoxon rank-sum test.[62] Correlation was calculated using Pearson correlation[63] (package stats, version 3.6.2) to assess linear relationships. For *in vivo-ex vivo* comparison, Spearman concordance was used. P-values were used to evaluate the statistical significance of the observed correlations. Differential response analysis between tissue sources and models was performed using Limma[64] (version 3.21). To correct for multiple hypothesis testing, p-values were adjusted using the Benjamini-Hochberg[65] method. Significance was considered when the p-value was below 0.05.

Data availability statement

The datasets generated and/or analysed during the current study are not publicly available because they contain patient-derived genomic and drug screening data that could compromise participant privacy, but are available from the corresponding author on reasonable request.

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Author contributions

Marlinde C. Schoonbeek: conceptualization, data curation, analyses, investigation, methodology, writing- original draft, visualization, project administration. Marloes van Luik: data curation, analyses, writing- original draft, visualization. Heike Peterziel and Dennis Gürgen: investigation, conceptualization, writing – review & editing. Vicky Amo-Addae: methodology, resources (Máxima library). Lindy Vernooij: project administration. Eleonora J. Looze and Sabina Valová: investigation. Sina Kreth, Angelika Eggert, Julia Schueler, Frank Westermann: methodology, resources (PDX models). Jan Koster: resources (R2 platform, molecular profiling). Aniello Federico, Apurva Gopisetty, Benjamin Schwalm, Marcel Kool: methodology, data curation, analyses, resources (molecular profiling). Alexandra Saint-Charles: project administration. Gudrun Schleiermacher: methodology, resources (molecular profiling and PDX models). Gilles Vassal, Hubert N. Caron, Lou Stancato, Stefan M. Pfister: conceptualization, supervision (ITCC-P4 leadership). Jens Hoffman: investigation, supervision. Arjan Boltjes: visualization, supervision. Ina Oehme: conceptualization, supervision, writing – review & editing. Marlinde L. van den Boogaard, Selma Eising, Jan J. Molenaar, Sander R. van Hooff: conceptualization, supervision, writing – review & editing. All authors reviewed the manuscript.

Competing interest statement

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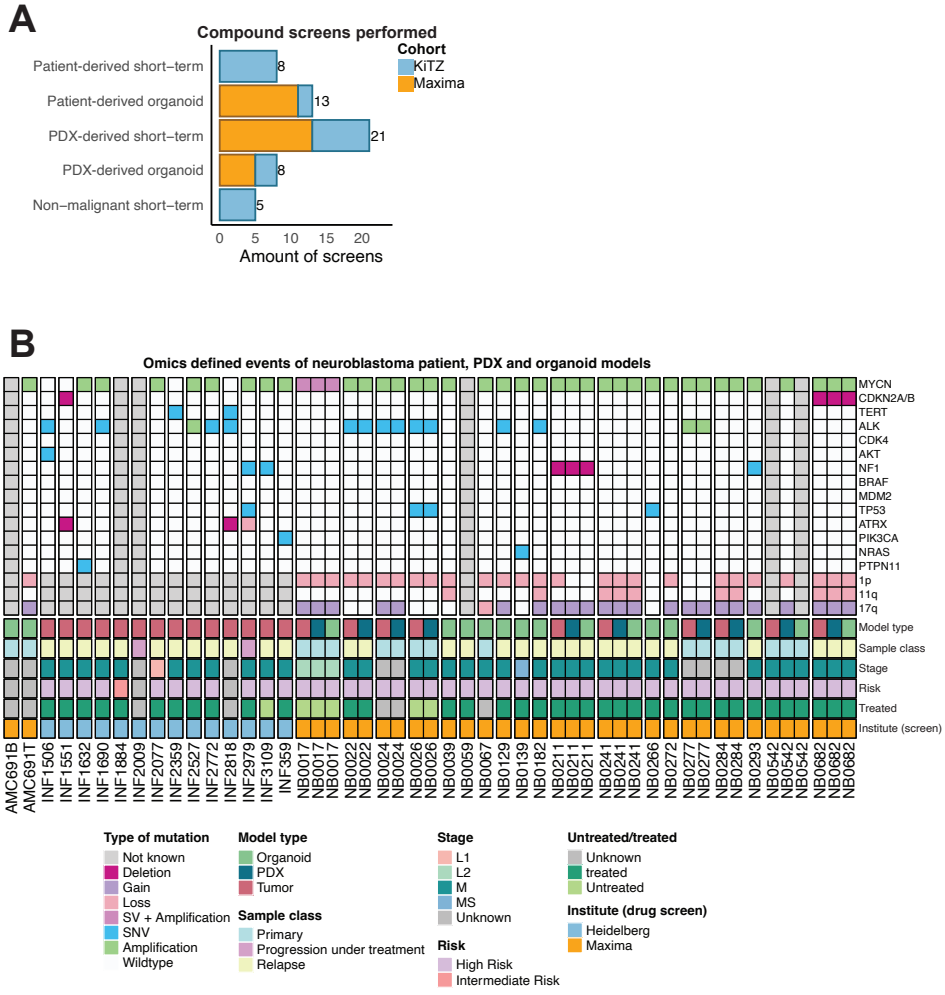
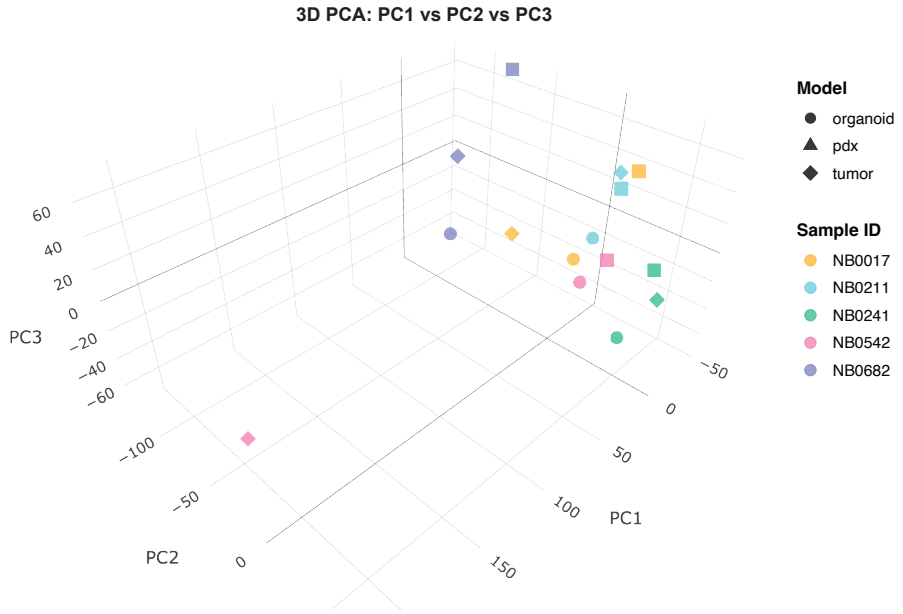
Supplementary material

The online version of this chapter, including supplementary tables, can be accessed via the QR code below or at:
https://www.biorxiv.org/content/10.64898/2026.02.24.707695v1.

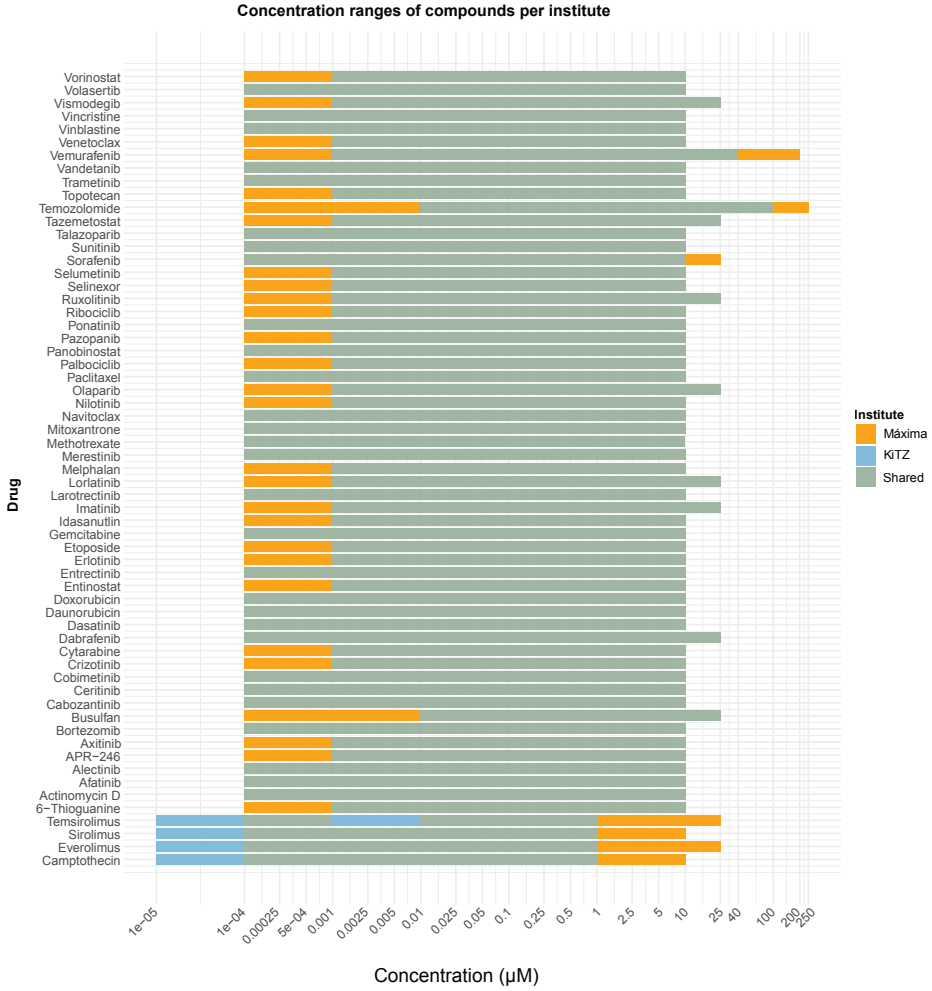
The supplementary figures are included on the following pages.



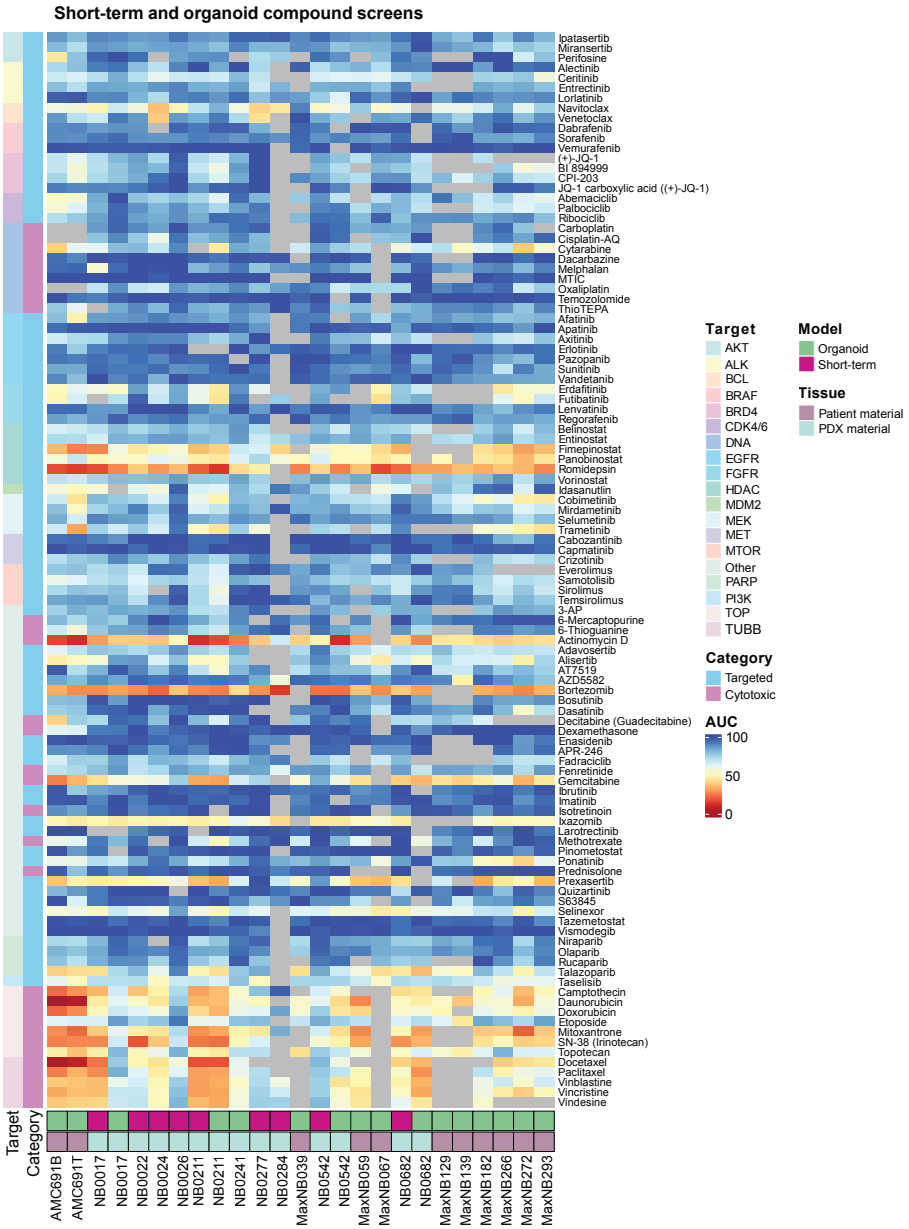
Supplementary Figure 1. Establishment and genetic characterization of models for compound screening. A) Five model types were generated across centers: patient-derived short-term and organoid models, PDX-derived short-term and organoid models, and short-term screens of non-malignant samples. B) Bar plot representing the total number of compound screens per model type of the Máxima Center and KiTZ. Part of the short-term and organoid screens performed at KiTZ were previously described in Peterziel *et al.*, Máxima patient material organoids were previously described in Langenberg *et al.*, and short-term screens performed at Máxima in Schoonbeek *et al.* C) Oncoplot of genomic landscape of neuroblastoma samples, showing key gene alterations and chromosomal aberrations colored by alteration type. Institute (drug screen) refers to the institute where the drug screen is performed. PDX = Patient-Derived Xenograft, SNV = single-nucleotide variant, SV = structural variant.



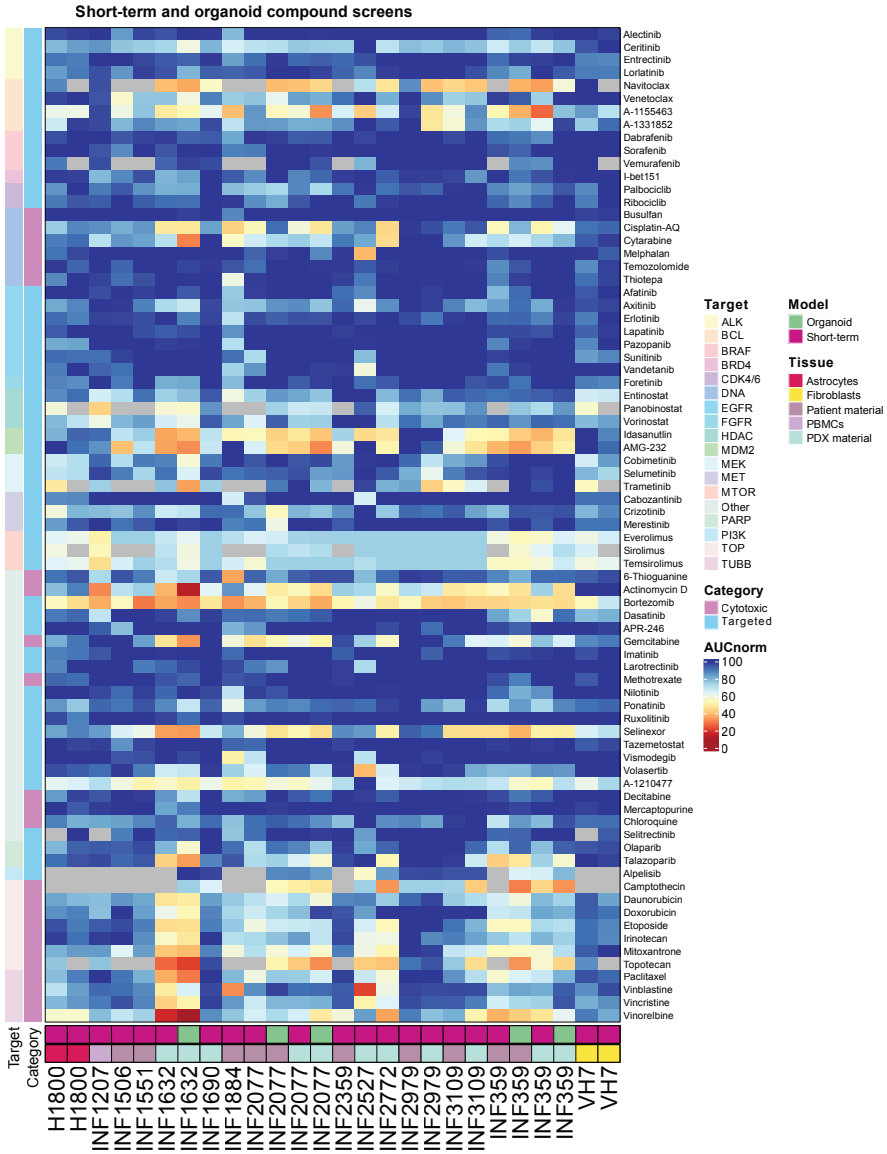
Supplementary Figure 2. Transcriptomic profiling across matched organoid, PDX and patient samples. Three-dimentional principal component analysis (PCA) plot of gene expression profiles from matched neuroblastoma patient tumors, patient derived xenografts (PDX) and organoid models. Principal components 1, 2 and 3 (PC1-3) are shown. Each point represents an individual sample; shapes indicate the model type (circle = organoid; square = PDX; triangle = patient tumor) and the colors indicate matched sample IDs.



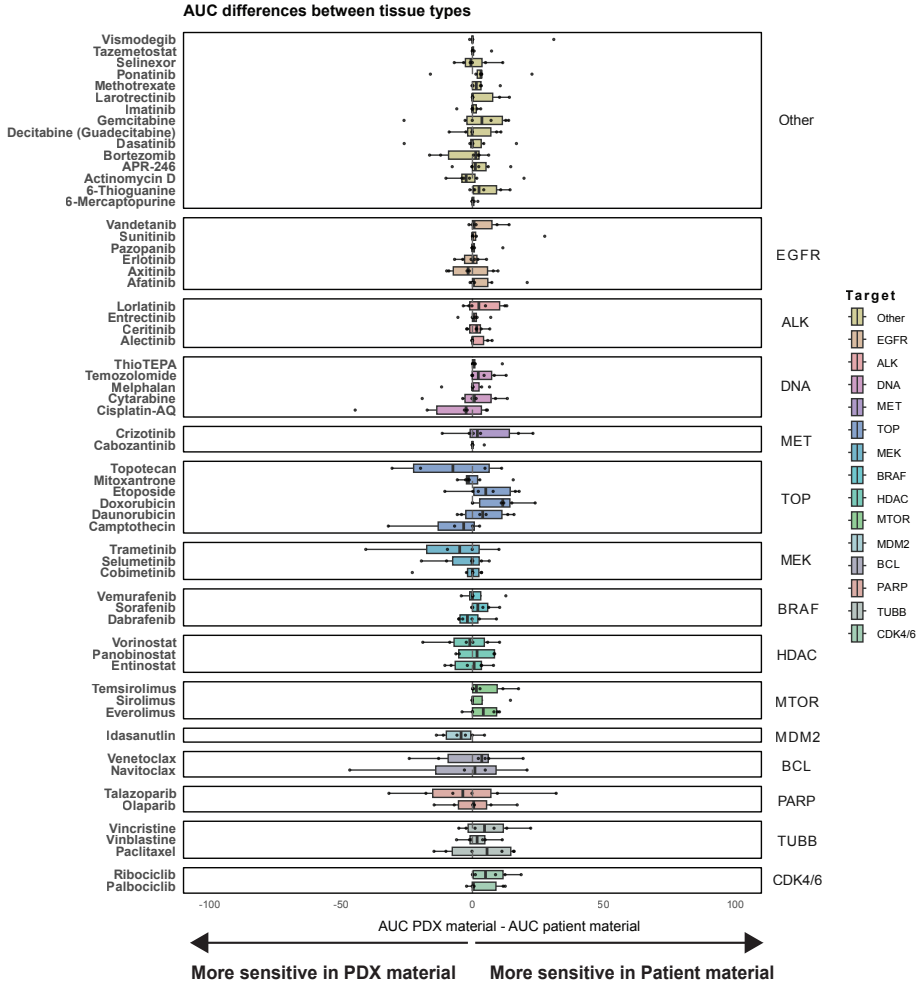
Supplementary Figure 3. Concentration ranges of the overlapping drugs per institute. Concentrations of drugs in Máxima (orange), KiTZ (blue), and overlapping (green). Concentrations are between 1 x 10-6 μM to 250 μM.



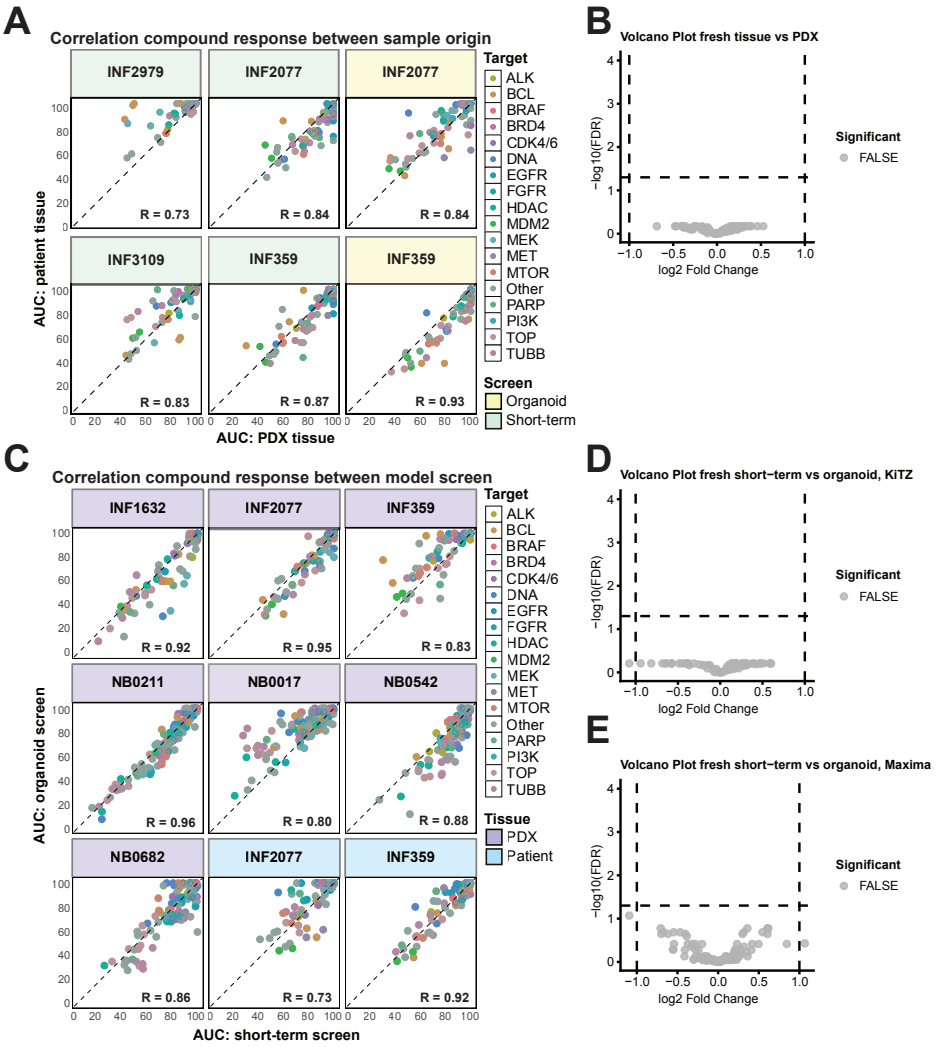
Supplementary Figure 4. Compound screening on Máxima neuroblastoma samples. The normalized area under the curve (AUC) of the short-term and organoids screens of the Máxima cohort, lower AUC values indicates more sensitivity to the compound. Samples are colored on screen type and tissue origin of the sample. Compounds are colored based on target and category of the compound (cytotoxic or targeted). Compounds of the clinical priority library are shown as they are screened in most samples.



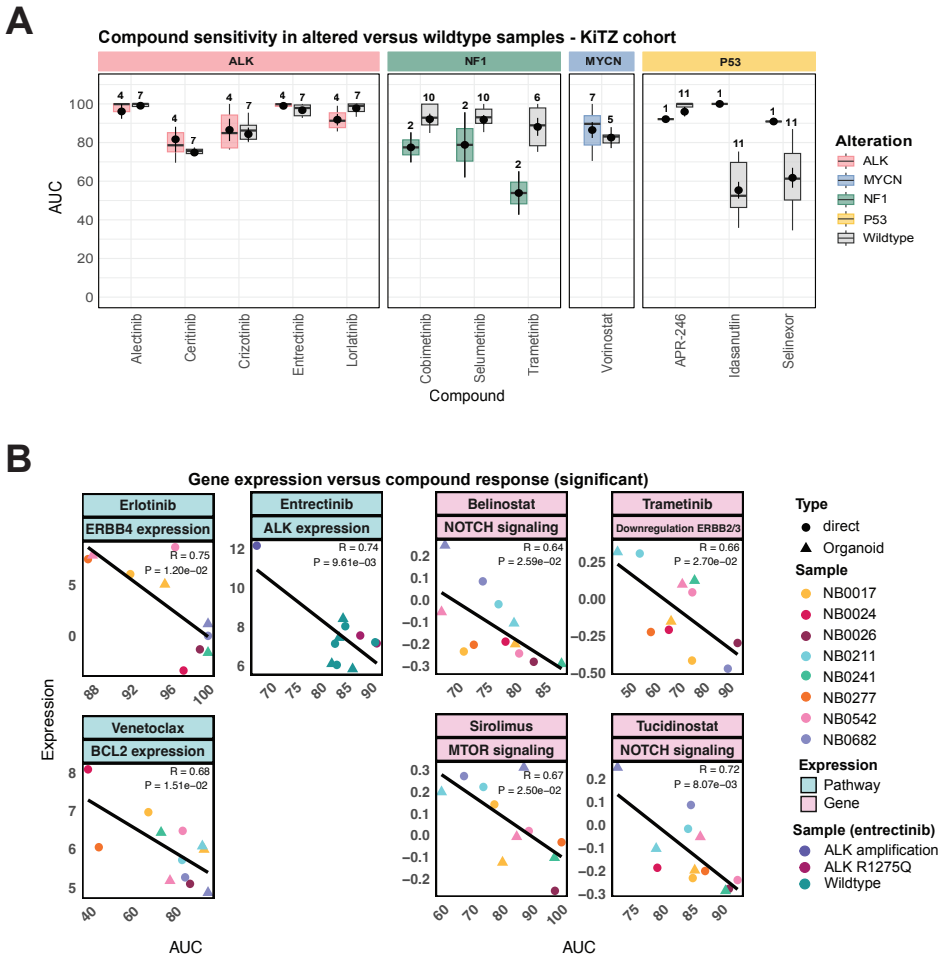
Supplementary Figure 5. Compound screening on KiTZ neuroblastoma and non-malignant samples. The normalized area under the curve (AUC) of the short-term and organoids screens of the KiTZ cohort. Samples are colored on screen type and tissue origin of the sample. Compounds are colored based on target and category of the compound (cytotoxic or targeted).



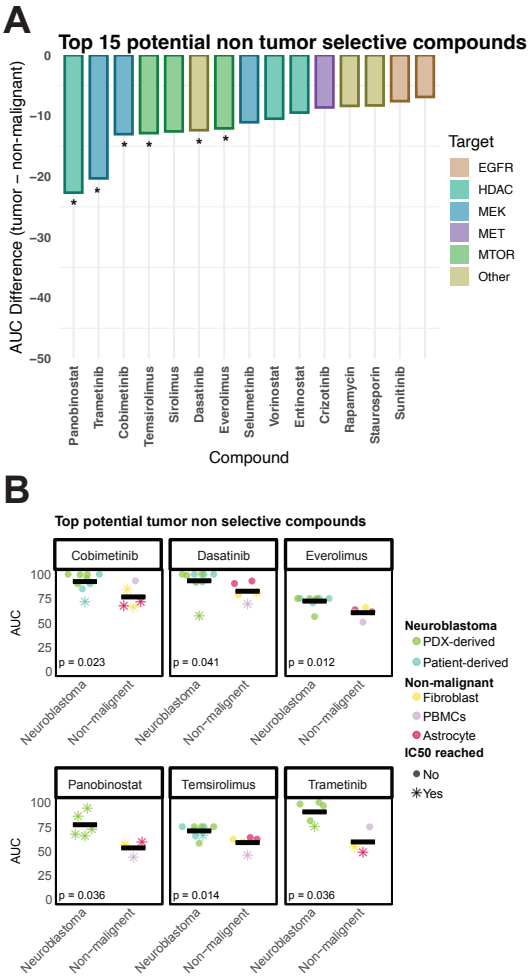
Supplementary Figure 6. Differences in drug sensitivity between screens from patient material and PDX material of matched samples. For each sample pair, the AUC of the PDX material screen is substracted by the AUC of the patient material screen. Positive numbers indicate higher sensitivity in the patient material screen, negative numbers indicate higher sensitivity in the PDX material screen. Boxes represents the distribution over the differences per compound, colored per target. Each dot represents the difference between a sample pair.



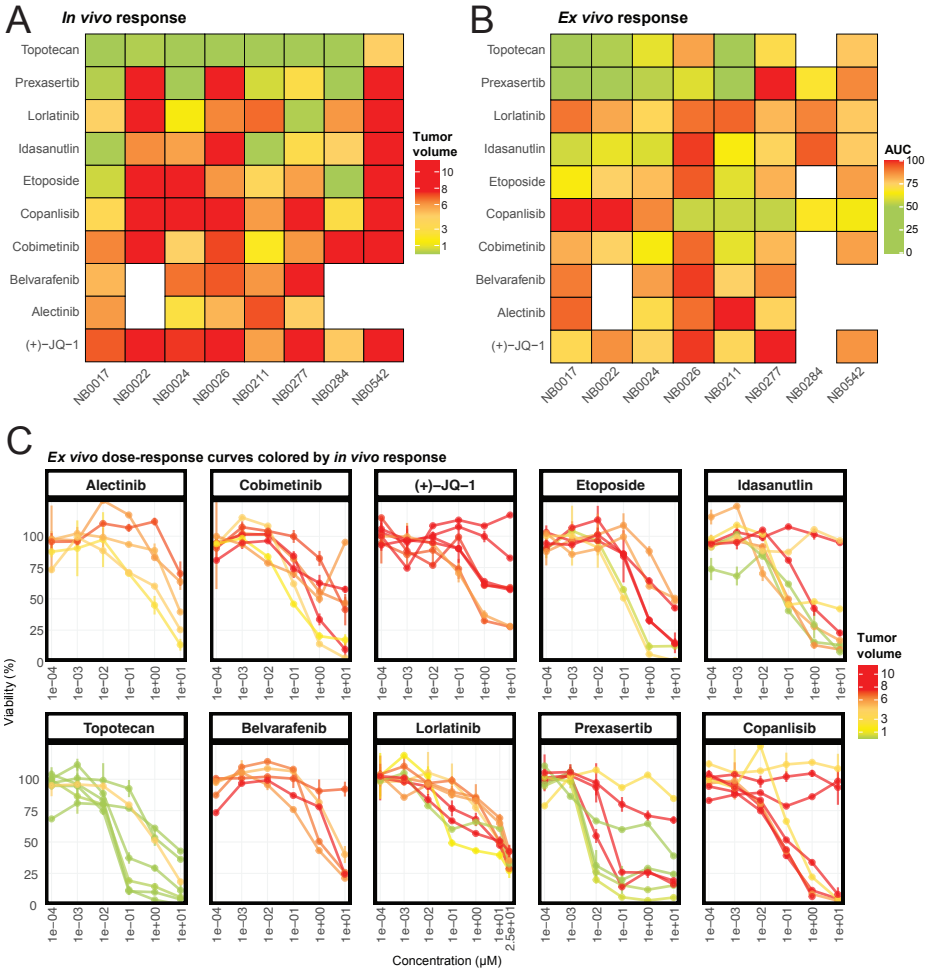
Supplementary Figure 7. Comparison of drug response across neuroblastoma model systems and sample origin. A) Correlation plots of compound response (AUC) on organoid or short-term screens from matched samples derived from different tissue origins (patient or PDX material). B, D-E) Volcano plot showing the influence of tissue origin (B: patient versus PDX) or screen type (D-E) on compound efficacy. A significant (adjusted $p < 0.05$; indicated by the dashed line) compound would mean that the AUC of that compound is influenced by tissue or screen type. In all cases, no compounds are found to be significant. Differential response analysis between tissue sources and models was performed using Limma (version 3.21). C) Correlation plots of the compound response (AUC) of organoids and short-term screens established from matched samples. In the correlation plots, each dot represents a compound, colored by its target. A lower AUC defines higher sensitivity. R represents Pearson correlation.



Supplementary Figure 8. The association of genomics and transcriptomics with compound response. A) Area under the curve (AUC) values of samples with targetable gene alterations, in boxes colored by alteration, compared to wildtype samples, in grey boxes, of the KiTZ cohort. The NF1/RAS altered samples consist of two NF1 altered samples (KiTZ). A lower AUC defines higher sensitivity to the compound. The dot indicates the mean, and the vertical line indicates the standard error around the mean. Stars indicate a significant difference between the altered and wildtype samples (adjusted p-value < 0.05). B) Correlation between AUC and expression of target/targeted pathway of compounds. Each dot represents a sample, colored by its name and shaped by model type. An exception is entrectinib, hwere samples are colored based on ALK-status. R represents the Pearson correlation. All correlations are significant (adjusted p-value < 0.05).



Supplementary Figure 9. Potential (non) tumor selective compounds. A) Bar plot representing the top 15 tumor non-selective compounds. The top 15 compounds with the highest mean AUC difference between non-malignant and tumor samples (mean AUC non-malignant – mean AUC tumor) are shown. Bars are colored based on the target of the compound. Asterisks indicate that the compound is significantly more effective in non-malignant samples. B) Box plots representing drug sensitivity (AUC) for tumor and non-malignant samples for compounds that were more effective in non-malignant cells than in neuroblastoma. Displayed are compounds for which tumor samples had a significantly higher AUC than non-malignant samples and that ranked among the top 15 compounds with the largest mean differences between the two groups. Each dot represents the response of one sample, colored by tissue type. The horizontal line indicates the mean AUC of the samples.



Supplementary Figure 10. Comparing *ex vivo* and *in vivo* drug responses. A) Heatmap representing the *in vivo* treatment responses in PDX models. Each block represents a model (ordered by PDX model), colored by end-of-treatment response (Tumor volume, normalized to starting volume). B) Heatmap representing the *ex vivo* treatment responses of short-term cultured PDX models. Each block represents a model (ordered by sample), colored by AUC. C) *Ex vivo* dose response curves. Each line represents the *ex vivo* dose response curve of a sample and is colored by *in vivo* end-of-treatment-response (tumor volume) in the matched PDX model. Each dot represents the viability of the cells at each concentration, with an error bar showing the variability of the two replicates (if available). Compounds are ordered by spearman correlation (AUC versus tumor volume; left top: highest spearman rho = 1; bottom right: lowest spearman rho = 0.05).

MicroOrganoSphere™ technology to advance drug screening in neuroblastoma precision medicine

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Abstract

Background. Neuroblastoma is among the most fatal pediatric solid tumors and high-risk patients face poor survival despite intensive treatment. Genomic profiling alone has not translated into effective treatment stratification for all neuroblastoma patients. Drug screening offers a complementary strategy, but conventional preclinical models are constrained by limited establishment success from low-input material and by timelines often incompatible with diagnostic decision-making. Here, we evaluated MicroOrganoSphere™ (MOS™) technology, in which tumor cells are encapsulated in Matrigel droplets generated by microfluidic droplet formation, by a head-to-head comparison with the conventional suspension cultures to improve growth and drug testing in neuroblastoma.

Methods. MOS™ were established from patient-derived tumoroids, freshly dissociated patient-derived xenograft (PDX) tumors, PDX-derived tumoroids, and a freshly dissociated patient tumor to compare growth and drug responses with suspension.

Results. Tumor cell growth in MOS™ was comparable or superior to suspension cultures. MOS™ drug screens were performed at a 2.5-fold lower seeding density compared to suspension, whilst maintaining reliable results with consistent drug sensitivity profiles and screen quality to suspension. Promisingly, drug testing in MOS™ was feasible within 9 days of receiving fresh PDX tumor tissue.

Conclusion. These results demonstrate in a proof-of-concept setting how MOS™ technology could support the establishment and drug screening of neuroblastoma cultures from low-input samples within clinically relevant timelines. MOS™ technology provides a promising functional platform to accelerate preclinical drug testing in neuroblastoma precision medicine.

Keywords: precision medicine, 3D cancer models, pediatric oncology, neuroblastoma.

Introduction

Neuroblastoma, the most prevalent pediatric extra-cranial solid tumor, poses a therapeutic challenge [1]. Current treatment for high-risk patients includes induction chemotherapy, surgery, high-dose chemotherapy with autologous stem cell transplantation and radiation therapy followed by consolidation therapy with anti-GD2 antibody treatment and retinoic acid [2]. Despite their rigorous treatment regimen, high risk patients face a 5-year overall survival of 50% [3]. The heterogeneity in clinical response highlights the urgent need for a more effective and targeted approach. Median progression-free survival across high-risk pediatric cancers, including neuroblastoma, has been reported as 116 days [4] with ranges from 1-65 months reported for neuroblastoma patients [5]. This emphasizes the limited clinical window for identification of effective therapeutic strategies.

Currently, pediatric precision oncology trials use genome sequencing to identify patient-specific targetable driver mutations to guide clinical decisions [4,6]. Such actionable driver genes are detected in 37-94% of patients, and the clinical benefit from matched targeted therapies remains limited for many patients. This can partly be explained as genetic characterization does not fully capture the disease's complexity, and epigenetic and microenvironmental factors should also be considered [7,8]. As an alternative screening approach that could overcome these limitations, functional drug testing is increasingly considered to guide treatment [8–14]. *In vivo* patient derived xenograft (PDX) models are representative and have been used extensively to study tumor response [15]. However, they are low-throughput, time-consuming and expensive to develop and maintain, as engraftment and stable expansion require approximately 1-11 months, followed by several weeks to assess drug response [9,16–18]. While 2D cell lines are easy to culture, they fail to mimic the tumor's complex architecture [16]. 3D patient-derived tumoroids, that have been established for neuroblastoma [10,18,19], offer a more promising middle ground with better scalability and lower costs compared to PDX. They more accurately mimic the architecture and physiological responses of solid tumors compared to 2D models [20]. However, tumoroid establishment has several limitations that hinder its clinical implementation for precision medicine: the required large amount of input material, limited success rate, loss

of the tumor microenvironment and extended establishment time, typically requiring 3-12 months, after which drug testing typically lasts 5 days [4,9,10,12–14,16,18,20,21]. This often exceeds patients' progression-free survival [4]. Patients are treated during these extended establishment periods of PDX and tumoroid models, which may limit the representation of the current state of the patient's tumor. Additionally, in precision oncology programs, treatment strategies are typically discussed in a Molecular Tumor Board, where molecular profiling results become available after a mean turnaround time of 25.4 days. To allow integrated discussion alongside molecular profiling, preclinical models used to guide treatment decisions should ideally provide drug response predictions within a similar timeframe. To shorten establishment timelines, *ex vivo* short-term drug screening can be performed on freshly dissociated tumor tissue cultured in suspension. This method has been successfully explored in novel pediatric precision medicine studies [8,9,11,22], but is limited by the low quantity of available tumor material from minimally invasive procedures used in children, variable viability after dissociation, and the time required to expand sufficient biomass prior to screening. Further optimization is therefore required to improve success rates and reduce the time to screen.

MicroOrganoSphere™ (MOS™) technology can be used to overcome the previously mentioned limitations of *ex vivo* screens. MOS™ are droplets generated by microfluidic droplet formation that encapsulate patient-derived tumor cells in Matrigel. MOS™ are automatically and rapidly generated from minimal patient tissue (i.e. low input samples with limited cell numbers) in a high-throughput fashion [23]. Compared to conventional tumoroids, MOS™ require lower input tissue and can be rapidly used for drug screening for personalized medicine. In colorectal cancer, MOS™ responses to chemotherapeutics correlated with clinical outcome [24,25] and Ding *et al.* demonstrated that drug sensitivities could be obtained within ~10 days from small, 18-gauge, biopsies. In addition, MOS™ accurately recapitulate patient samples and their microenvironment [24], recapitulate the heterogeneity within tumors [25] and facilitate the testing of immuno-oncology therapies [26]. The feasibility of MOS™ in pediatric solid tumors, with a different developmental origin and tumor biology than adult colorectal cancer, and its performance

relative to suspension cultures have not yet been evaluated. If using MOS™ technology in neuroblastoma could reduce the number of cells required per well, it would allow testing of a broader array of therapeutic agents, increasing the throughput of drug screening processes with the same biomass input.

In this study, we explore the potential of the MOS™ platform as an alternative to previously optimized suspension cultures for neuroblastoma patient-derived models: patient-derived tumoroids, PDX-derived fresh tumor and tumoroid models, and a fresh patient tumor. We directly compare MOS™ and suspension cultures to assess viability, proliferation and drug sensitivity testing of tumoroids and freshly dissociated samples. Our findings demonstrate how MOS™ technology can support short-term culture and functional drug testing, providing a framework for integrating MOS™ into preclinical testing pipelines for neuroblastoma to overcome some of the key limitations of currently available technologies.

Methods

Detailed experimental procedures, including model generation, culture conditions, imaging settings and data processing pipelines, are provided in the Supplementary Methods.

Model generation and culture

All cultures were maintained at 37°C and 5% CO_2. Tumoroid cultures were passaged twice per week, routinely tested for Mycoplasma and authenticated by short tandem repeat profiling.

Patient derived tumoroid cultures

In brief, three patient-derived tumoroids with distinct molecular backgrounds were selected for outgrowth experiments (Suppl. Table 1) and cultured in neuroblastoma organoid medium (NBOM) [10,19].

PDX-derived fresh and tumoroid cultures

Neuroblastoma PDX models were previously established and genetically profiled in the ITCC-P4 consortium [27]. Previously reported comprehensive molecular profiling was used in this study: low coverage whole genome sequencing, whole exome sequencing, DNA methylation profiling and RNA

sequencing [22,27]. The dissociation method for the PDX samples and the established PDX tumoroids are previously described [18,22]. In brief, fresh PDX tumors were mechanically dissociated, followed by a two-step enzymatic digestion. Fresh PDX cultures were resuspended and cultured in NBOM, with the exception of NB0026, that was cultured in suspension in TSM for 8 days prior to drug screening. For tumoroid establishment, freshly dissociated cells were cultured until stable growth was observed and human tumor cultures were confirmed by SNP analysis and, subsequently, tumoroids were molecularly profiled [18].

Patient-derived fresh primary culture

A fresh neuroblastoma tumor sample was obtained during surgical resection of a refractory patient through an institutionally approved biobank study at the Princess Máxima Center (PMCLAB2025.660). All procedures were performed in accordance with the Declaration of Helsinki, with age-appropriate written informed consent from patients and/or parents. Preparation of the tumor pieces was started within 2 hours after surgery. The primary tumor sample was dissociated via the Liberase protocol, as described in the Supplementary Methods, and immediately MOS™ were generated, and MOS™ and suspension cultures were plated in NBOM.

Generation and plating of MicroOrganoSpheres™

MOS™ were generated as previously described [24]. Briefly, dissociated tumor cells from patient-derived tumoroids, PDX and human primary tissue digest were resuspended in ice-cold, growth factor–reduced Matrigel (Corning) at a defined density corresponding to 50 viable tumor cells per MOS™. The cell–Matrigel mixture was kept on ice to prevent premature polymerization. MOS™ were generated using a proprietary droplet-based microfluidic generator, which produces highly uniform Matrigel droplets encapsulating tumor cells. Briefly, the cell–Matrigel suspension was loaded into a temperature-controlled microfluidic system in which pressure-driven flow generates monodisperse droplets in an immiscible oil phase. Generation was performed under cooled conditions to maintain Matrigel in liquid state during droplet formation. Droplet size is defined by the microfluidic geometry and pressure settings,

resulting in reproducible MOS™ diameters of 230-250 µm. Following droplet generation, MOS™ were collected and incubated at 37°C for 30 minutes to allow Matrigel polymerization. After gelation, solidified MOS™ were demulsified by transfer onto a hydrophobic PVDF membrane to separate the oil phase from the aqueous droplets. After 2–3 minutes, polymerized MOS™ were recovered using pre-warmed NBOM. Recovered MOS™ were counted, centrifuged at 150xg for 5min and subsequently resuspended at the appropriate density for plating for downstream applications.

Plating of MOS™ and suspension cultures

For MOS™ cultures, MOS™ were dispensed in a 384-well flat bottom plate (6007460, Perkin Elmer) at a density of 40 MOS™ per well, final cell density: 2,000 cells per well in 40 µL. For suspension cultures, 40 µL of single cell suspensions were plated in a flat bottom 384-well tissue culture treated microplate (#3764, Corning). For cell growth experiments (Fig. 1), 2,000 cells per well were plated, matching MOS™ conditions. For drug screening (Fig. 2), the suspension cultures were seeded at 5,000 cells per well, based on previously optimized conditions for neuroblastoma drug screening [10,18].

Microscopy-based cell viability readout and analyses

In brief, cell viability was assessed by microscopy using Calcein AM viability dye (Calcein; 0.625 µM; #65-0853-78, ThermoFisher) and Ethidium-Homodimer-2 (EthD; 2.5 µM; #E3599, ThermoFisher). The readout was performed with either ImageXpress® Micro Confocal High-Content Imaging system (IXM, Molecular Devices) or the Thunder Widefield Fluorescence microscope (Thunder, Leica™; Suppl. Table 1). Wells without any detected viability (<4%) were excluded from analysis. Image analysis was performed using threshold-based segmentation of live (Calcein-positive) and dead (EthD-positive) objects, after which integrated fluorescence intensities were used for downstream analysis.

Drug screening

In brief, drug screening was performed using libraries of 45–90 compounds. Compounds were tested in duplicate across six concentrations (0.1 nM to 10 µM).

DMSO-treated wells served as negative controls and staurosporine-treated wells as positive controls. Cells were incubated with compounds for 72 hours in fresh PDX screens and 120 hours in PDX-derived tumoroid screens.

For MOS™ cultures, viability was determined using the microscopy-based live/dead readout. A t = 0 plate was included to assess growth during the screen. Viability was calculated from the relative live and dead fluorescence signals. For suspension cultures, and for comparison in MOS™ cultures after microscopy, CellTiter-Glo 3D (CTG) was used as a conventional ATP-based viability readout [18].

Results

Tumor cell growth in MicroOrganoSpheres™ is comparable or superior to growth in conventional suspension culture

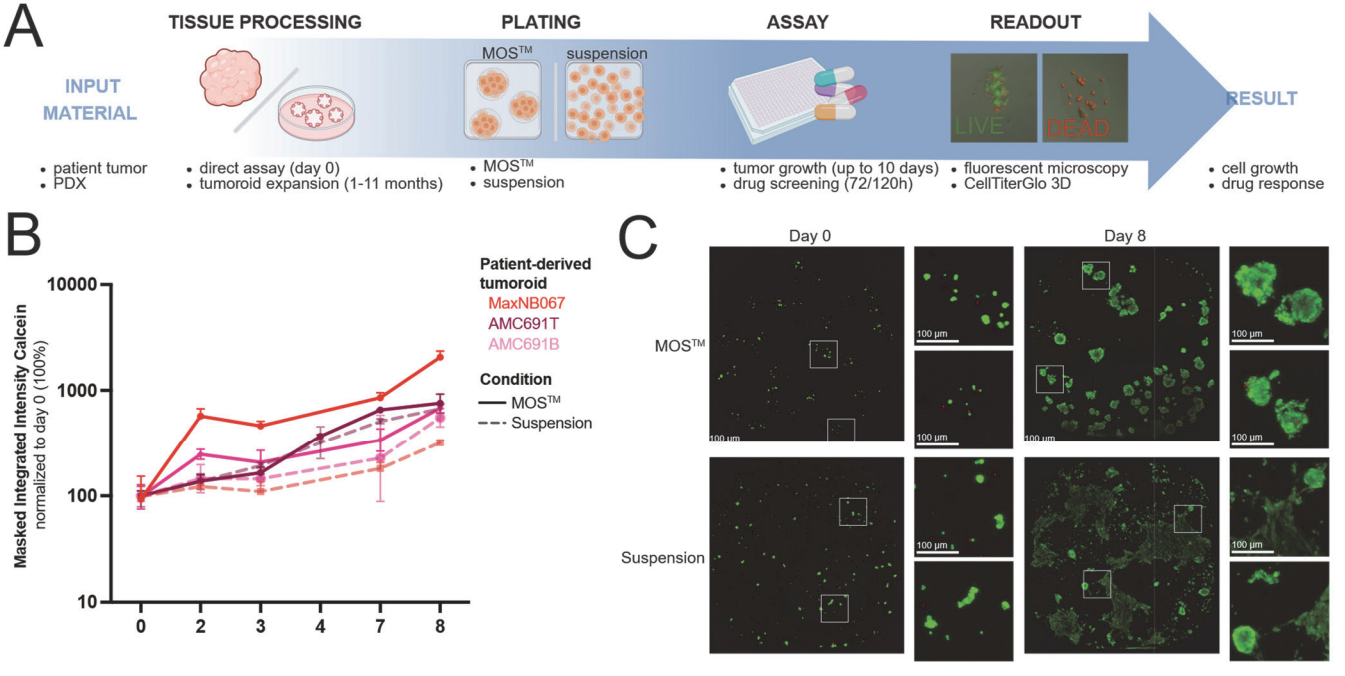
To assess MOS™ technology for neuroblastoma cells, growth of tumor cells derived from various model systems (patient-derived tumoroid, PDX, patient) was monitored when grown in MOS™ and compared to the conventional growth method: in suspension cultures (Fig. 1A). Tumor tissue was dissociated into single cells and encapsulated in Matrigel droplets using microfluidic droplet generation. After polymerization and demulsification, MOS™ were cultured under standard conditions in direct comparison with suspension cultures. The optimal cell density for cell growth, determined by microscopic inspection, was identified as 50 cells per MOS™ (Suppl. Fig. 1A). Microfluidic droplet generation produced uniform MOS™ with a diameter of 230–250 µm. Spheroids in MOS™ did not exceed this size, whereas suspension cultures could form larger spheroids, although reliable diameter quantification in suspension was limited due to partial attachment. Growth was observed using fluorescent-based microscopy, where adapted-threshold based signal intensity of Calcein AM and Ethidium Homodimer, were used to identify and quantify the amount of live and dead objects respectively (Suppl. Fig. 1B-C).

Viable biomass of patient-derived tumoroids cultured in MOS™ versus suspension was quantified over eight days. MOS™ cultures could successfully be established from all tested patient-derived tumoroids (n = 3), where a biomass increase was observed over time in all models (Fig. 1B-C; Suppl. Fig. 2A).

MaxNB067 demonstrated superior biomass expansion in MOS™ cultures compared to suspension, while AMC691B and AMC691T showed comparable growth in both conditions. Interestingly, AMC691B cells formed more defined 3D structures in MOS™ compared to the flatter 2D structures when cultured in suspension (Fig. 1C). These results show comparable to improved cell growth in MOS™ versus suspension cultures derived from patient-derived tumoroids.

Next, MOS™ cultures were established from freshly dissociated PDX-derived tissues. PDX-expanded tumors provide sufficient biomass for larger experiments compared to fresh patient samples, while recapitulating molecular and phenotypic profiles of the patient tumor [28–30]. Parallel MOS™ and suspension cultures were established from four samples and all models showed comparable viability (Fig. 1D-E; Suppl. Fig. 2B). One model (NB0276) failed to expand in both conditions, consistent with previous reports describing unsuccessful short-term neuroblastoma cultures due to failed growth after dissociation [18,22]. To determine whether possible benefit was due to the presence of Matrigel in the MOS™ conditions, and not the MOS™ themselves, a control experiment with media enriched with Matrigel was performed. Here, biomass decreased in fresh PDX cultures enriched with 0.5% and 5% Matrigel compared to NBOM (Suppl. Fig. 2C). These results indicate that MOS™ supports cell growth to a similar extent as suspension in freshly dissociated PDX tissues.

Lastly, for one fresh patient tumor that was available, outgrowth was compared in MOS™ versus suspension cultures when established directly from the primary patient material. Cell growth was monitored over 10 days (Fig. 1F-G). Biomass rapidly increased in MOS™, peaking at day 3, while biomass rapidly decreased in suspension cultures. These results indicate that MOS™ technology enables short-term expansion of fresh patient-derived tumor cells *ex vivo*. Taken together, these findings demonstrate that MOS™ supports viable growth across diverse neuroblastoma models, including fresh and tumoroid cultures derived from patients and PDXs.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• PDX
• direct assay (day 0)
• tumoroid expansion (1-11 months)
• MOS™
• suspension
• tumor growth (up to 10 days)
• drug screening (72/120h)
• fluorescent microscopy
• CellTiterGlo 3D
• cell growth
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Figure 1

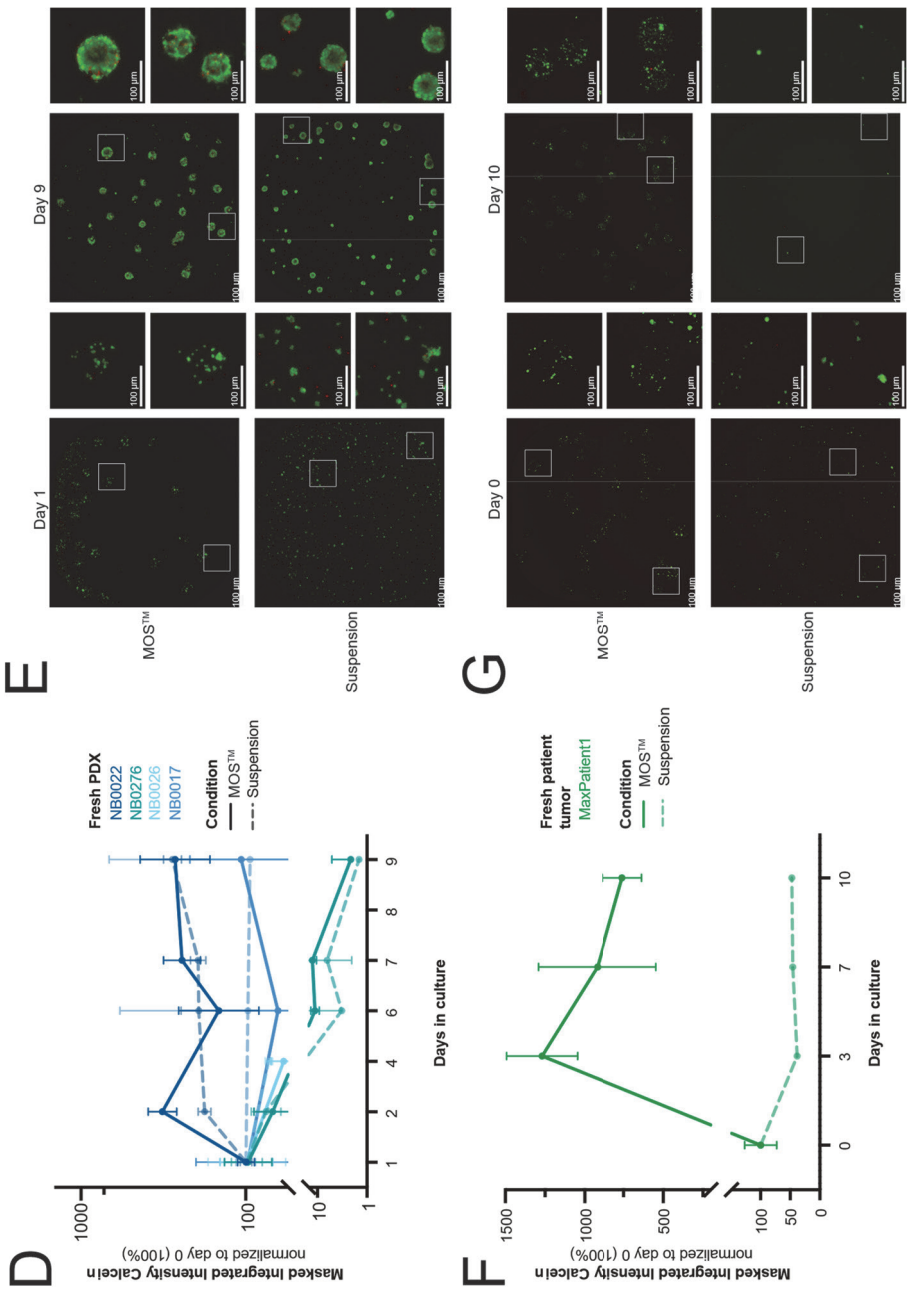


Figure 1 (continued)

Figure 1 (previous pages): MicroOrganoSphere™ establishment from patient-derived tumoroid and PDX cultures shows comparable proliferation compared to suspension cultures. ***A****) Schematic of experimental set up comparing MOS™ versus suspension cultures (created with Biorender). Drug screening assays are 72 hours in case of direct assays and 120 hours in case of tumoroid screens. Fluorescent microscopy-based readouts were performed for MOS™ screens, CellTiter-Glo 3D for suspension and MOS™ screens.* ***B, D, F****) Quantified cell viability of patient-derived tumoroids (B; n = 3), PDX-derived fresh cultured (D; n = 4), and a freshly dissociated sample (F) in MOS™ versus suspension. Cell viability was measured as Calcein integrated intensity, within the viable (Calcein-positive) mask, normalized by the first day of imaging. Each dot represents the geometric mean; error bars indicate geometric standard deviation. Solid lines indicate MOS™ cultures; dotted lines (50% opacity) indicate suspension cultures. Colors represent distinct patient-derived tumoroids. All conditions used 2,000 cells per well (MOS™: 50 cells per MOS™, 40 MOS™ per well). Three biological replicates were used for B and D, and five biological replicates for F.* ***C, E, G****) Representative fluorescent microscopy images of patient-derived tumoroid AMC691B (C), fresh PDX NB0022 cultures (E) and the primary sample (G) cultured in MOS™ and suspension showing live cells (green, Calcein) and dead cells (Orange, EthD) on the first and final day of the experiment. The scale bar in the lower left corner of each image indicates 100 µm.*

Abbreviations: MOS™= MicroOrganoSphere™; PDX= patient derived xenograft; EthD= ethidium homodimer.

MicroOrganoSphere™ technology enables reliable drug screening, with *in vitro* drug sensitivities consistent with those obtained using suspension cultures albeit requiring less biomass

To determine whether MOS™ cultures can be reliably used for drug sensitivity testing, automated drug screening in both MOS™ and suspension culture was performed on three PDX-derived tumoroid models and one freshly dissociated PDX sample (Fig. 1A). Drug sensitivity of suspension cultures was assessed using CTG-based viability, the conventional method for neuroblastoma tumoroid screens [10]. For MOS™ cultures, drug sensitivity was evaluated using the microscopy-based live/dead ratio quantification (Calcein for viable objects, EthD for dead objects; described in Methods). For each compound tested, six concentrations were screened in duplicate. Between 45 and 90 compounds were

tested per sample, based on the available biomass, and vehicle (0.25% DMSO) and staurosporine (10 μM) were included as live and dead control, respectively (Fig. 2A). Abemaciclib, a CDK4/6 inhibitor inducing approximately 50% cell death in this model (1 μM), was selected as a representative compound to visualize intermediate drug response between vehicle and positive control.

To compare drug screen quality and reproducibility, the coefficient of variation (CV) in vehicle-treated control wells and the inter-replicate correlation were used. Drug screens in MOS™ cultures displayed identical average CVs of 9% in vehicle-treated control wells compared to suspension cultures (Fig. 2B), which is within the preferred acceptance threshold of 10% [31]. Average inter-replicate correlation (same drug, same concentration, different well in the other plate) was comparable but lower in MOS™ conditions (r = 0.83) versus in suspension (r = 0.92; Suppl. Fig. 3A). Although MOS™ drug screens were seeded at 2,000 cells per well compared to 5,000 cells per well in suspension cultures, screen quality metrics were comparable between both conditions. We next explored whether MOS™ drug screens reflected expected genotype-associated drug response patterns. NB0026, the only sample with homozygous TP53 inactivation, showed limited response to the MDM2 inhibitor idasanutlin, whereas the TP53-wild-type samples were sensitive (Fig. 2C). This is consistent with prior reports, but this descriptive observation could not be statistically evaluated because of the limited cohort size (n = 4) and only one TP53-altered sample was included [32–34]. Similarly, NB0026 harbored an ALK mutation, but due to the limited cohort size, no association between ALK status and ALK inhibitor sensitivity could be assessed (Suppl. Fig. 3B). Drug response was consistent between MOS™ and suspension cultures for three PDX-derived tumoroids and one freshly dissociated PDX (r = 0.88-0.93; Fig. 2D-E; Suppl. Fig. 3C). To assess whether MOS™ conditions affect response of individual compounds, we performed a comparative analysis between MOS™ and suspension screens using Limma [35]; adjusted p-value > 0.05). No individual compound showed a significantly different response between MOS™ and suspension cultures. Inclusion of the freshly dissociated PDX sample NB0026 indicated the ability to rapidly screen low biomass samples. This sample was briefly cultured in suspension prior to drug screening, which was performed 9 days after tissue receipt in MOS™ and suspension, with a 2.5-fold

lower seeding density in MOS™. These results indicate that, under the conditions tested here, MOS™ enables drug screening with comparable drug responses and screening quality compared to suspension cultures at a 2.5-fold lower seeding density.

Next to microscopy-based readouts, we used CellTiter-Glo (CTG) luminescent cell viability assay to determine cell viability and subsequent drug response as this is the conventional readout for suspension cultures [18,22]. CTG was performed as a second readout for MOS™ conditions, on the same wells as the microscopy-based readout, to assess drug screen quality using CTG as a readout of MOS™ cultures. Due to low cell numbers in MOS™, CTG readouts were less reliable compared to a microscopy-based readout. The variability in vehicle-treated control wells was too high in the CTG-based readout (CV = 0.35, compared to CV = 0.09 in the microscopy-based readout) and inter-replicate correlation was poor (r = 0.54, compared to r = 0.83 in the microscopy-based readout; Suppl. Fig. 3D). These results show that for drug screening in MOS™, the microscopy-based readout is a more reproducible and reliable approach compared to CTG. The microscopy-based workflow was fully automated and required comparable time to CTG-based workflows, enabling high-throughput screening. Taken together, these results show that MOS™ technology, combined with a microscopy-based readout, enables reliable drug screening with drug sensitivities and screen quality consistent with conventional suspension cultures, while using a 2.5-fold lower seeding density.

Figure 2 (next pages): MicroOrganoSphere™ technology enables reliable drug screening with sensitivities consistent with suspension cultures A) *Representative fluorescent microscopy images of PDX-derived tumoroid NB0542 screened in MOS™ showing live cells (green, Calcein) and dead cells (orange, EthD). Visualized: vehicle (0.25% DMSO, negative control), cells treated with CDK4/6 inhibitor abemaciclib (1 μM), and cells treated with staurosporine (10 μM, positive control). The scale bar in the lower left corner of each image indicates 100 μm.*

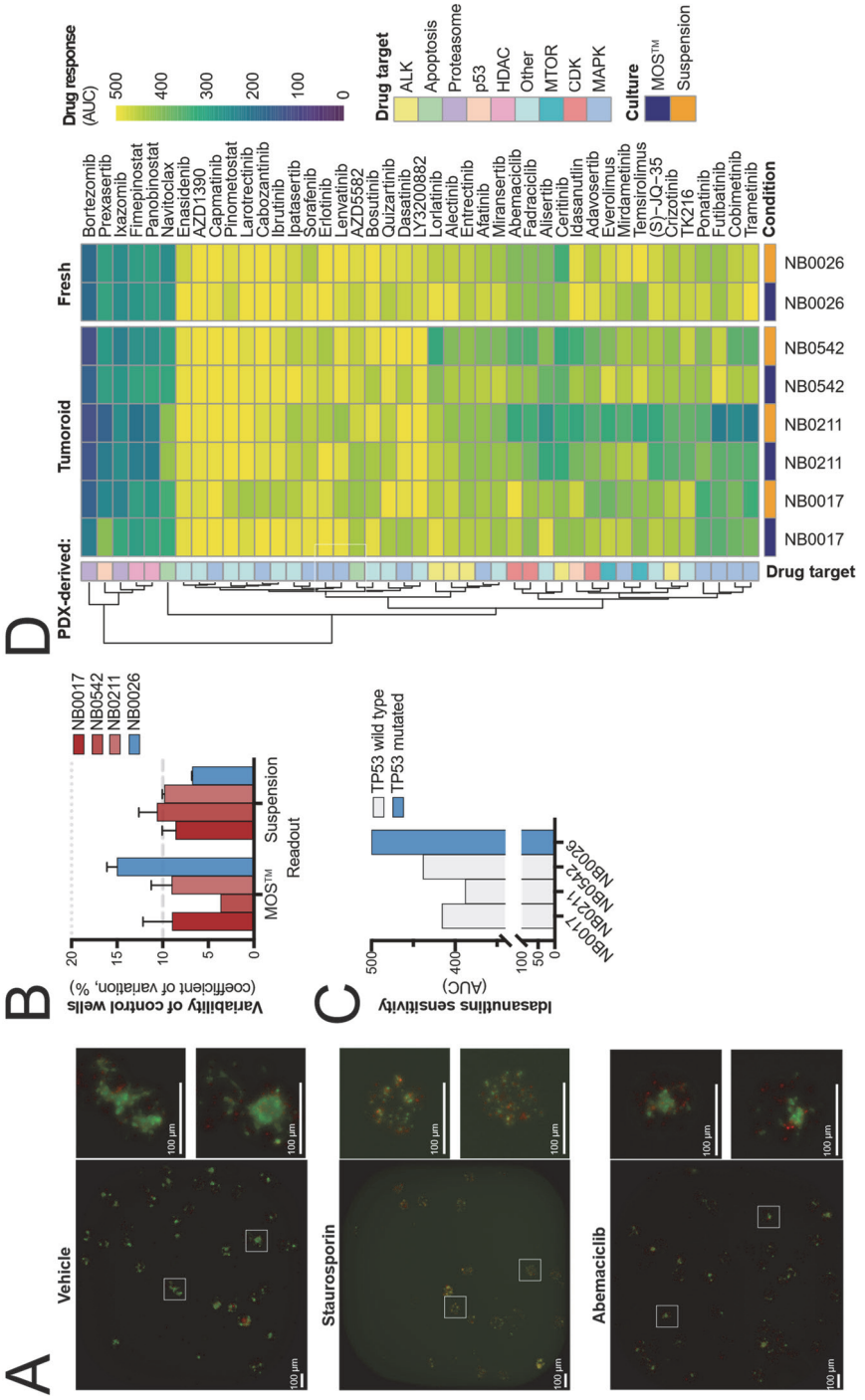
B) *Bar plots showing variability in 32 vehicle-treated control wells (0.25% DMSO) in MOS™ versus suspension, shown as CV (%). Bars are colored per sample: red = PDX-derived tumoroids, blue = PDX-derived fresh tissue. The drug screen of NB0542 in MOS™ was performed in a single replicate due to a technical error during compound addition, thus no error bar is shown. The grey dotted line represents the acceptance threshold of 20%, the dashed line represents the preferred acceptance threshold of 10% [31,37].*

C) *Bar plot of idasanutlin response (AUC) in TP53-wildtype and TP53-mutated (homozygous inactivation) samples. Bars represent individual samples (MOS™ only): blue = TP53-mutated (NB0026, specific TP53 mutation: C176F; VAF 0.93), grey = TP53-wildtype (NB0017, NB0211, NB0542).*

D) *Heatmap showing drug response (AUC) in MOS™ versus suspension cultures. Left: PDX-derived tumoroids. Right: PDX-derived fresh cultures. Only overlapping compounds (n = 44) with successfully determined AUCs across all screens are shown.*

E) *Correlation of drug response (AUC) in MOS™ versus suspension cultures of PDX-derived tumoroid cultures (red labels) and PDX-derived fresh cultures (blue label). Each dot represents one compound; "n = x" indicates the number of compounds with valid AUCs in both conditions. Pearson correlation coefficients (r) and p-values (p) are indicated.*

Abbreviations: MOS™ = MicroOrganoSphere™; PDX = patient-derived xenograft; AUC = area under the dose–response curve; CV = coefficient of variation; DMSO = dimethyl sulfoxide.


A
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Staurosporin
Abemaciclib
100 µm
B
Variability of control wells
(coefficient of variation, %)
NB0017
NB0542
NB0211
NB0026
MOS™ Suspension
Readout
C
Idasanutlins sensitivity
(AUC)
TP53 wild type
TP53 mutated
NB0017
NB0211
NB0542
NB0026
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PDX-derived:
Tumoroid
Fresh
Drug response
(AUC)
Drug target
ALK
Apoptosis
Proteasome
p53
HDAC
Other
MTOR
CDK
MAPK
Culture
MOS™
Suspension
Bortezomib
Prexasertib
Ixazomib
Fimepinostat
Panobinostat
Navitoclax
Enasidenib
AZD1390
Capmatinib
Pinometostat
Larotrectinib
Cabozantinib
Ibrutinib
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Alisertib
Ceritinib
Idasanutlin
Adavosertib
Everolimus
Mirdametinib
Temsirolimus
(S)-JQ-35
Crizotinib
TK216
Ponatinib
Futibatinib
Cobimetinib
Trametinib
Condition
NB0026
NB0542
NB0211
NB0017
Drug target

Figure 2

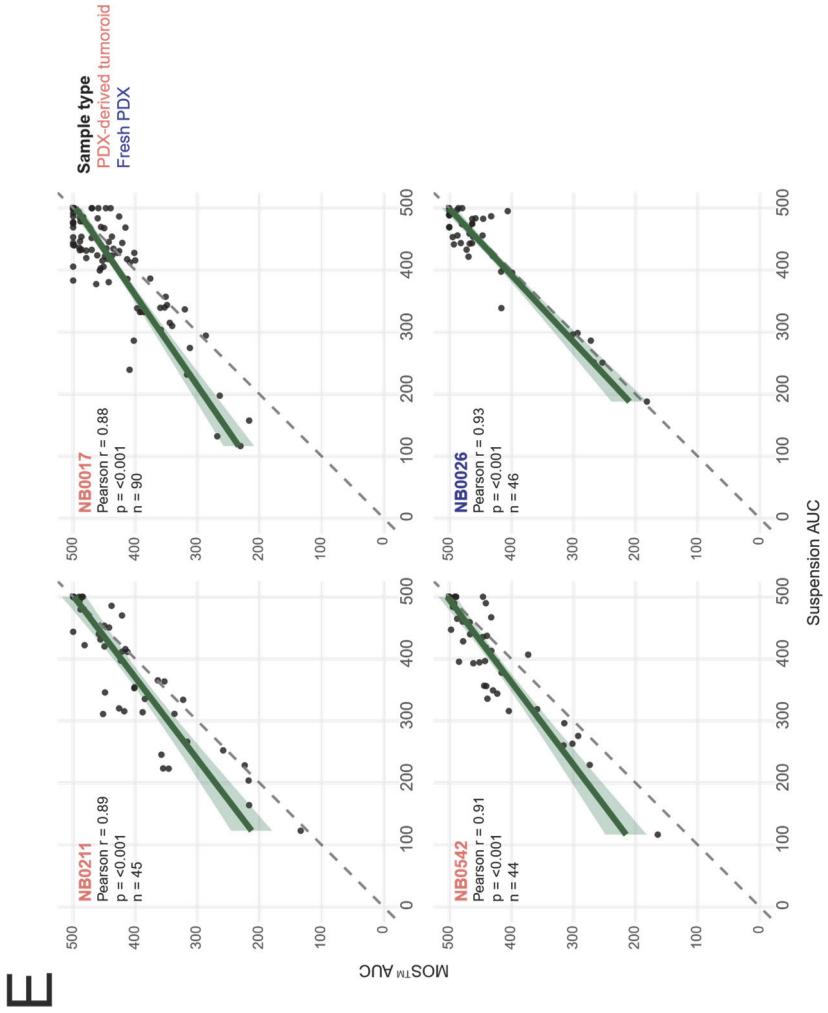

Figure 2 (continued)

Discussion

High-risk neuroblastoma remains associated with poor survival despite intense multimodal treatment, and robust biomarkers for treatment stratification are lacking in a substantial subset of patients [1,3]. Functional approaches such as *in vitro* drug testing have shown promise in guiding treatment in pediatric cancers [8,9,12,14], but broader implementation is limited by the large amount of required input material, long turnaround times and low establishment success [10,12–14,16,21,36]. MicroOrganoSphere™ (MOS™) technology allows for rapid automated *ex vivo* testing, with responses to chemotherapy reflecting clinical outcome [24]. Here, we evaluated the potential of MOS™ as an alternative for conventional suspension cultures in neuroblastoma to enable drug screening from limited cell numbers. By this novel direct comparison, we demonstrate that MOS™ technology supports short-term tumor cell growth at least as well as suspension cultures and that drug screening can be performed using fewer cells while maintaining reproducibility and drug sensitivity profiles compared to suspension. Furthermore, we demonstrate that the MOS™ technology is suitable for standardized drug screening in the rare and understudied pediatric cancer neuroblastoma.

Across multiple patient-derived sources, including patient-derived tumoroids, PDX-derived fresh and tumoroid cultures and one freshly dissociated patient tumor, MOS™ supported short-term outgrowth that was comparable or superior to suspension culture. This data demonstrates that MOS™ can maintain neuroblastoma tumor viability even from low-input samples, consistent with previous work showing successful establishment of MOS™ cultures from small 18-gauge biopsies [24]. Although only a single primary patient sample could be included here, this proof-of-concept study supports feasibility for future expansion to larger patient cohorts. As expected, variability in growth of the models was observed, especially for the PDX-derived and freshly dissociated patient tumor. Such variability in growth rates has been previously described and has been associated with partial failure of model development [18,22].

Drug screening in MOS™ was performed at 2,000 cells per well compared with 5,000 cells per well in suspension cultures, while maintaining comparable reproducibility and consistent drug response profiles. The suspension condition

reflected the established neuroblastoma drug screening conditions using 5,000 cells per well and a CTG-based viability assessment [18], whereas MOS™ screening was optimized for lower-input screening combined with a microscopy-based viability readout. Potentially a microscopy-based readout could also improve the quality of suspension drug screens at lower densities. A lower seeding density allows the same number of compounds to be tested with less material, which is particularly relevant for fresh patient samples where material is often limited. While reproducibility of MOS™ drug screens was comparable to suspension and below the preferred acceptance threshold of 10% [31,37], well-to-well variability could be further reduced by quantifying cell viability at the level of individual MOS™ rather than at the whole-well level, as applied in this study. Recent work has demonstrated that single-MOS™ analyses are feasible and result in high reproducibility (CV ≤ 2.5; [25]). MOS™ cultures showed a tendency toward slightly reduced drug sensitivity compared to suspension, which might be due to reduced drug penetration into the tumoroids because of the different 3D structure or the presence of Matrigel. Only for the freshly dissociated PDX sample NB0026 the effect could be due to presence of tumor micro-environment (TME), although the trend in this sample is less obvious. Interestingly, the freshly dissociated patient tumor showed rapid growth in MOS™ compared to suspension, where it even decreased in biomass. By testing more samples in the future, it needs to be investigated whether this large advantage of the MOS™ technology is consistent across neuroblastoma subtypes.

In a freshly dissociated PDX sample, MOS™ drug screening could be successfully completed within nine days of tissue receipt. Likewise, a fresh patient tumor expanded biomass in the first three days after MOS™ generation, comparable with patient-derived tumoroids and fresh PDX samples that increased biomass. These proof-of-concept experiments in one fresh patient tumor and one fresh PDX drug screen provide initial evidence that MOS™ can support *ex vivo* testing of fresh material within a short time frame. This is important, as the long turnaround times that often exceed the median time to progression [4], have limited the clinical use of patient-derived tumoroids [9,10,12]. *Ex vivo* drug screening of fresh tumor tissue shortens turnaround times [11,22], and the reduced cell input required by MOS™ could further improve

feasibility of screening low-input samples. We could not compare drug response in MOS™ with patient response, as this data was not available for these samples. However, previous studies have shown that MOS™ responses correlated with clinical outcome [24,25], and that drug sensitivities could be obtained within ~10 days of sample receipt in adult cancers. Although future studies should expand this work, as larger pediatric cohorts will be essential to validate clinical applicability and to correlate drug response with patient outcomes, these findings suggest that MOS™ is suited for drug screening within a clinically relevant time frame, especially for low-input material.

Next to MOS™, many complementary 3D *ex vivo* modelling approaches are being developed to improve the physiological relevance of *in vitro* drug testing [38]. Whole-tissue platforms such as ALTEN preserve intact tumor tissue architecture and multicellular niches by encapsulating primary tissue fragments in alginate hydrogels [39]. These approaches are well suited for studying the native tissue niche and interactions with the TME but are generally less scalable and allow testing of fewer compounds. In contrast to whole-tissue platforms, patient-derived tumoroids provide high scalability and long-term expansion, making them suitable for compound discovery and biomarker research. However, patient-derived tumoroids require longer establishment times. MOS™, in contrast, is optimized for the rapid generation and short-term expansion of tumor cells from freshly dissociated low-input samples, enabling reproducible and scalable drug sensitivity testing within clinically relevant timelines which is especially useful for guiding clinical decision-making. Successful MOS™ generation depends on the quality and viability of the starting material after mechanical and enzymatic dissociation, which may vary between samples. In addition, MOS™ cultures rely on Matrigel, which introduces an extracellular matrix environment that must be tolerated by tumor cells. In our study, using MOS™ allowed parallel testing up to 90 compounds while maintaining 3D growth conditions. As such, MOS™ is particularly suited for precision medicine applications, where speed is crucial.

This study did not investigate tumor TME retention, although previous work has shown that MOS™ can preserve stromal components [24]. Exploring if neuroblastoma MOS™ retain elements of the TME will be an important next step

for assessing compound activity in a more patient representative setting. In patient samples where other cell types are present, tumor-specific markers, such as NCAM, B7H3 and L1CAM, could be used to distinguish tumor from non-tumor responses. In addition, MOS™ have been shown to support efficient CAR-T cell infiltration and killing, indicating their potential for immunotherapy testing [26]. In addition, culture conditions such as hypoxia have been proposed to better recapitulate aspects of the neuroblastoma tumor niche [40,41]. While hypoxia was not investigated in the present study, MOS™ cultures can be maintained under controlled oxygen conditions after generation. Finally, it should be noted that *ex vivo* drug screening does not capture *in vivo* pharmacokinetic and pharmacodynamic processes. Factors such as plasma protein binding, drug lipophilicity and tissue distribution can influence effective drug exposure at the tumor site and should be considered when translating *ex vivo* drug sensitivity results to clinical applications [42–44]. Future studies of neuroblastoma MOS™ screening could include evaluation of TME preservation, testing of immunotherapies, hypoxia culture, and better representing pharmacokinetic and pharmacodynamic processes.

Conclusion

This proof-of-concept demonstrates that MOS™ technology supports the *ex vivo* growth and drug screening of neuroblastoma cells, including low-input samples, within clinically relevant time frames. Future studies should focus on expanding pediatric patient cohorts, assessing clinical applicability and comparing drug response to clinical outcome, to validate MOS™ as a functional precision medicine platform for preclinical drug testing.

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Data availability statement

Supporting imaging and drug screening data is available upon request.

Competing interest statement

M.K., C.L., E.D. and S.E. are employees of Xilis BV, and equity options granted (<5% of total company shares). All remaining authors have declared no conflicts of interest.

Declaration of generative AI use

During the preparation of this work the author(s) used ChatGPT in order to receive feedback on phrasing of sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Authorship contributions

Jan J. Molenaar: Writing – review & editing, Supervision, Conceptualization. Eleonora J. Looze: Investigation. Else Driehuis: Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization. Karin Langenberg: Resources, Conceptualization. Sander van Hooff: Writing – review & editing, Supervision, Conceptualization. Marlinde L. van den Boogaard: Writing – review & editing, Supervision, Conceptualization. Sarah Swaak: Formal analysis, Data curation. Angelika Eggert: Resources. Gudrun Schleiermacher: Resources. Selma Eising: Writing – review & editing, Supervision, Methodology, Conceptualization. J. Christoph Lampert: Resources. Birgit Geoerger: Resources. Gilles Vassal: Resources. Vicky Amo-Addae: Resources, Methodology. Mandy Koomen: Methodology, Investigation. Julia Schueler: Resources. Dennis Gürgen: Resources. Marlinde C. Schoonbeek: Writing – original draft, Visualization, Project administration, Investigation, Formal analysis, Data curation.

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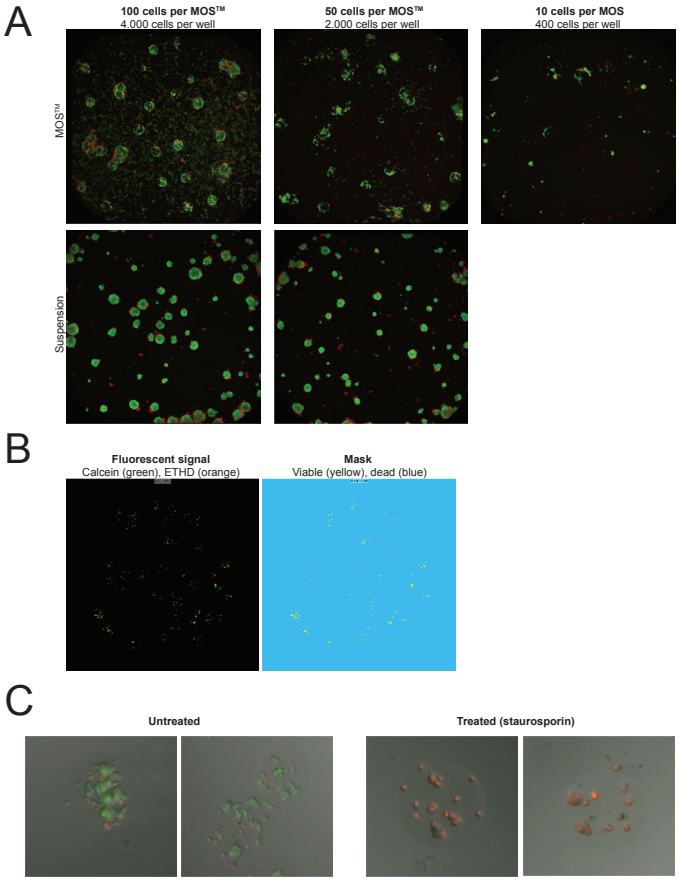
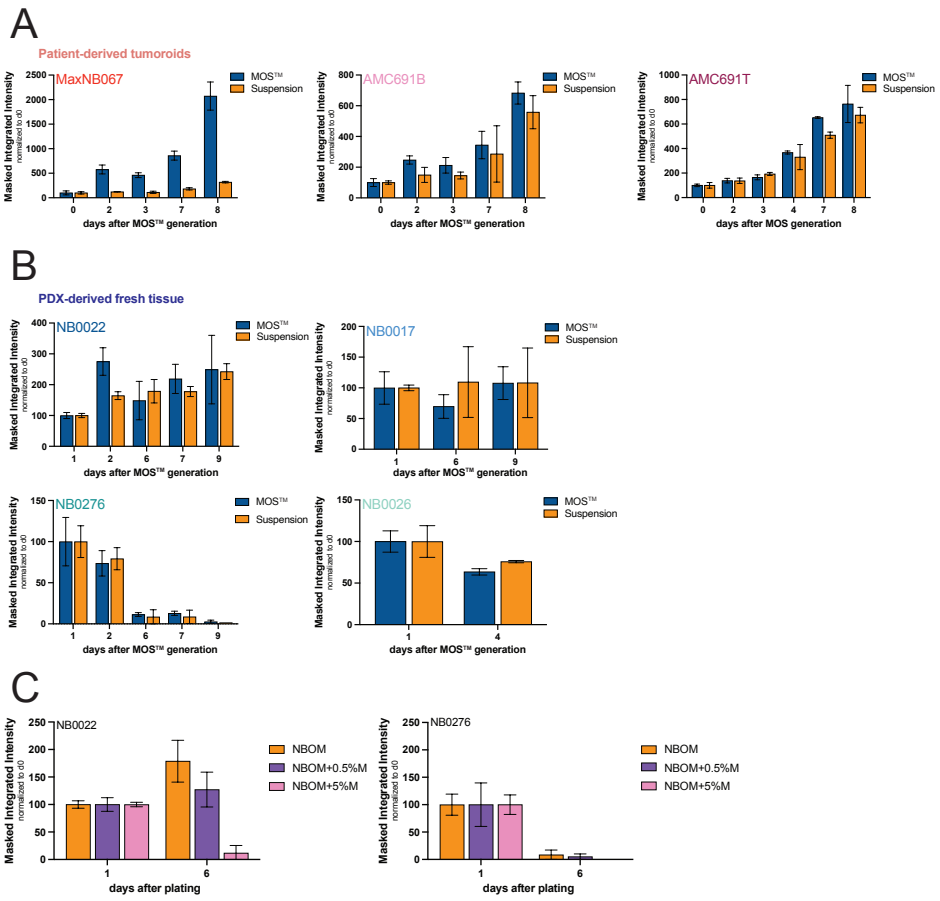
Supplementary Material

The published version of this chapter, including supplementary methods and supplementary tables, can be accessed via the QR code below or at: https://www.ejcped.com/article/S2772-610X%2826%2900053-X/fulltext

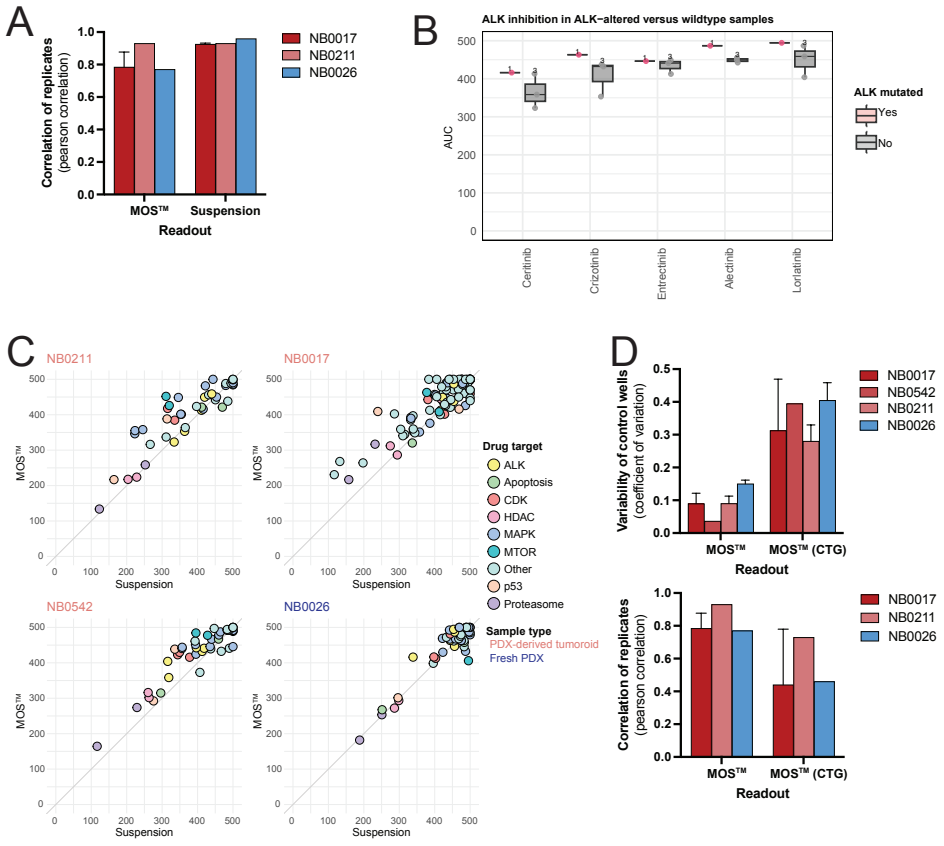
The supplementary figures are included on the following pages.



Supplementary Figure 1: MOS™ establishment and optimization from patient-derived tumoroids using a microscopy based readout. A) Fluorescent microscopy images representing different cell seeding densities of NB0682. MOS™ cultures comparing 100 cells per MOS™ (4.000 cells per well), 50 cells per MOS™ (2.000 cells per well) with suspension cultures in the same cell density, as well as 10 cells per MOS™ (400 cells per well). B) Fluorescent microscopy image and subsequent masking strategy of AMC691T patient-derived tumoroid at day 0 in MOS™ culture. Left: composite of Calcein (green) and EthD (orange) fluorescence. Right: viability masks showing live cells (yellow; Calcein-positive) and dead cells (blue; EthD-positive). C) Fluorescent microscopy images of treated versus non-treated cells. Calcein/EthD-based viability readout in NB0682 MOS™ cultures (50 cells/MOS™), untreated or treated with staurosporin (10 μM). Abbreviations: MOS™= MicroOrganoSphere™; PDX= patient derived xenograft; EthD= ethidium homodimer.



Suppl. Fig. 2: Biomass quantification in MOS™ and suspension cultures derived of patient-derived tumoroids and PDXs. A-B) Bar plots of quantified viable biomass (Calcein-based) of A) patient-derived tumoroids (n=3) and B) PDXs (n=4), cultured in MOS™ versus suspension. Bars represent mean Calcein integrated intensity signal within the viable (Calcein-positive) mask, normalized to the first day of imaging. Blue bars are representing MOS™ cultures, orange bars suspension. Cells were seeded in 2.000 cells per well in both culture conditions (MOS™: 50 cells per MOS™, 40 MOS™ per well). C) Bar plots of viability in freshly dissociated PDX samples NB0022 (left) and NB0276 (right), cultured under three suspension conditions. Orange bars = culture in conventional medium, NBOM; purple bars = conventional medium + 0.5% Matrigel; pink bars = conventional medium + 5% Matrigel. Abbreviations: MOS™= MicroOrganoSphere™; PDX= patient derived xenograft; EthD= ethidium homodimer; NBOM= neuroblastoma organoid medium.



Supplementary Figure 3: Drug screening in MOS™. A) Bar plots showing inter-replicate reproducibility in MOS™ versus suspension, shown as Pearson correlation coefficients, based on 270 or 540 (NB0017) replicates. Bars are colored per sample: red = PDX-derived tumoroids, blue = PDX-derived fresh tissue. The drug screen of NB0542 was performed in a single replicate due to a technical error during compound addition. B) Box plots showing sensitivity (AUC) for ALK inhibition in one ALK mutated (R1275Q in NB0026: in pink) versus three ALK-wildtype samples (in grey). C) Scatter plot of drug response (AUC) in MOS™ versus suspension. Blue = PDX-derived organoids. Red = PDX-derived fresh culture. Each dot represents one compound, colored by target subtype. D) Bar plots showing the quality and reproducibility metrics of MOS™ drug screens using microscopy versus CTG. Top: variability in vehicle treated control wells (0,25% DMSO), shown as CV. Bottom: inter-replicate correlation, shown as Pearson correlation coefficient. Bars are coloured per sample: red = PDX-derived organoids, blue = PDX-derived fresh culture. Abbreviations: MOS™ = MicroOrganoSphere™; PDX = patient-derived xenograft; AUC = area under the dose–response curve; CV = coefficient of variation; DMSO = dimethyl sulfoxide; CTG = CellTiter-Glo 3D.

A microscopy-based readout to assess tumour-specific viability in neuroblastoma co-cultures and short-term cultured patient samples

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Abstract

High-risk neuroblastoma patients face poor survival despite intensive treatment. Drug testing using patient-derived models can support therapy prioritization for precision medicine and drug development. Models incorporating tumour microenvironmental components, such as co-cultures and short-term cultured patient samples containing substantial non-malignant cell fractions, could better recapitulate microenvironment-dependent drug responses. However, conventional viability assays measure the combined signal from all viable cells in a well and therefore cannot determine tumour-specific drug responses. Here, we establish a microscopy-based readout to quantify cell-type-specific viability in two complementary settings: neuroblastoma-PBMC co-cultures and freshly dissociated patient tumour samples. In the co-cultures, PBMCs were pre-labelled with a cell-tracking dye, and Calcein staining was used to independently quantify the viability of tumour cells and PBMCs in the same well. The Calcein-based viability readout correlated strongly with conventional CellTiter-Glo measurements and was compatible with automated high-throughput drug screening. The imaging workflow enabled identification of compounds with differential efficacy in co-culture versus monoculture and distinguished tumour-specific effects from PBMC toxicity. The microscopy-based viability readout was further adapted to short-term cultured patient samples. Neuroblastoma tumour cells were distinguished from the non-malignant cells using a combination of tumour-specific surface markers NCAM, L1CAM and B7H3. This enabled determination of tumour fractions and measurement of tumour-specific drug responses. Tumour fractions varied substantially between patient samples, highlighting the importance of tumour-specific viability measurements. Together, the microscopy-based viability readout for co-cultures and patient samples enables scalable assessment of tumour-specific drug responses.

Introduction

Neuroblastoma arises from neural crest progenitor cells of the developing sympathetic nervous system and is the most common extra-cranial solid tumour in children.[1,2] Despite intensive multimodal treatment, high-risk neuroblastoma remains associated with poor outcome, with 5-year overall survival rates of approximately 50%.[2,3] In addition, survivors often experience severe long-term treatment-related toxicity.[4,5] This highlights the need for more effective and better tailored treatment strategies.

Patient-derived organoids are widely used for drug testing in neuroblastoma.[6–8] Organoid establishment usually takes several months, but once established, they are scalable for high-throughput assays which makes them well suited for drug development.[7–9] Conventional organoid monocultures, however, lack components of the tumour microenvironment, which can influence drug response and are essential for evaluating immune-mediated therapies.[10–12] Co-culture models incorporating immune cells increase physiological relevance while retaining scalability. However, these assays require readouts that distinguish tumour viability from immune-cell viability in the same well.[13–16] Furthermore, drug testing on *ex vivo* short-term cultured patient material, containing unknown tumour fractions, is increasingly being explored as a complementary approach to molecular profiling for precision oncology.[8,17–20]

Most high-throughput viability assays, including CellTiter-Glo, measure the combined metabolic activity from all cells present in the well and therefore cannot distinguish tumour from non-malignant cells.[8,21,22] Genetic labelling strategies, such as luciferase expression, can enable cell-type-specific readouts but require prior modification of the tumour cells which is time- and labour-intensive and not feasible for primary patient samples.[16] Therefore, scalable, non-genetic, cell-type-specific viability assays are needed to assess drug responses in physiologically relevant *in vitro* models. Immunofluorescent microscopy-based approaches are increasingly compatible with high-throughput workflows and provide a solution for tumour-specific viability measurements in co-cultures and patient-derived samples.[21,23–25] Tumour-specific markers, already established in diagnostics and therapeutic targeting, enable tumour-specific labelling in mixed cell populations. However, expression of these individual markers is heterogeneous across

neuroblastoma samples, which limits the reliability of single-marker approaches.[26–29]

By combining viability dyes with either pre-labelling of PBMCs or a cocktail of tumour-specific surface markers, we establish a microscopy-based viability readout that enables separate quantification of tumour and non-tumour viability in neuroblastoma-PBMC co-cultures and patient samples.

Methods

Detailed experimental procedures, including model establishment, culture conditions, CellTracker Orange (CTO) retention after T-cell activation, drug screening methods, imaging settings and data processing pipelines, are provided in the Supplementary Methods.

Shared methodology

In short, two organoid lines from distinct tumour sites of the same patient[6] were used throughout the study: AMC691T, (primary tumour; mesenchymal phenotype; HLA^+) and AMC691B (bone marrow metastasis; adrenergic phenotype; HLA^-).

A compound library of 134 compounds was screened in 6 concentrations in a 384-well format. Viability was normalized to vehicle-treated controls and the area under the curve (AUC) was determined from the dose-response curves. Quality control included assessment of growth rates, positive and negative controls, replicate consistency, and visual inspection of curve fits of individual curves and the overall screen.

Neuroblastoma-immune co-cultures

In short, peripheral blood mononuclear cells (PBMCs) were obtained from healthy donors (Mini Donor Bank; institutional review board approval at UMCU: TC18-774), isolated using Ficoll density gradient centrifugation[30] and cryopreserved at -180 °C until use.

Organoids were seeded in 384-well plates and co-cultured with PBMCs, which were pre-labelled with CTO to enable distinction between tumour and immune cells during image analysis. Co-cultures were established at an effector-to-target ratio of 1:5 and exposed to compounds for 120 hours.

At readout, viable cells were stained using Calcein Violet (CalceinV) and imaged using the high-content confocal microscope Opera Phenix (5x air objective; confocal mode). One fixed Z-plane was selected based on comparison with a five-plane z-stack and its maximal intensity projection. CellTiter-Glo 3D (CTG) assays were performed after imaging for multiple plates, to benchmark the microscopy-based viability readout against conventional whole-well viability measurements.[7,8,22]

Image analysis was performed in Harmony software (Revvity). PBMCs were identified as CTO^+ regions. All viable cells were defined as $CalceinV^+$ regions. Viable PBMCs were then defined as overlapping $CTO^+/CalceinV^+$ regions and viable tumour cells as $CalceinV^+$ regions subtracting the viable PBMC mask. The viability (CalceinV) was quantified using the sum pixel intensity of images. Compounds showing intrinsic autofluorescence that interfered with signal detection (Suppl. Table 1) were excluded from downstream analyses in the affected channel.

Patient-derived samples

In short, patient-derived tumour samples were obtained through institutionally approved biobank protocols of the Erasmus Medical Center Rotterdam (MEC-2016-793) and the Princess Máxima Center (PMCLAB2022.370). Freshly dissociated samples were short-term cultured prior to plating in round-bottom 384-well plates. To distinguish tumour from non-malignant cells, a neuroblastoma-specific antibody cocktail (NB-cocktail) targeting NCAM, L1CAM and B7H3 conjugated to a far-red fluorophore was combined with the viability dye CalceinAM. Neuroblastoma samples (MaxPatient1-3) were cultured for 4–7 days prior to screening, whereas one melanotic neuroectodermal tumour of infancy sample (MaxPatient4) was cultured for one month until screening was performed.

Images were acquired using optimized confocal microscopy settings (10x objective; single-field acquisition; 6 z-planes) and analysed using mask-based segmentation workflows. Viable tumour cells were defined as $NB\text{-}cocktail^+/CalceinAM^+$ and total viable cells as $CalceinAM^+$. The sum of the integrated pixel intensities of the CalceinAM signal was determined in both populations. Tumour fractions were calculated as the ratio of viable tumour cells to

total viable cells. For proof-of-principle drug testing, patient-derived samples were exposed to selected compounds and tumour-specific viability was quantified within tumour-marker$^+$ populations.

Results

Optimization of a microscopy-based viability readout

To enable microscopy-based measurement of viability in neuroblastoma-immune co-cultures, we first optimized and validated technical aspects of the readout method. We utilized two organoid lines derived from distinct tumour regions of the same patient: AMC691T and AMC691B (Fig. 1A).[6] AMC691T represents the mesenchymal phenotype and expresses MHC-I (HLA$^+$), enabling interaction with immune cells, such as cytotoxic T cells and natural killer cells. In contrast, AMC691B represents the more typical adrenergic neuroblastoma phenotype and lacks MHC-I expression (HLA$^-$), thereby limiting immune recognition. This allowed us to model phenotypic heterogeneity regarding tumour-immune interactions while keeping other properties of the models as similar as possible.

To validate CalceinV as a reliable measure of cell viability, we benchmarked this imaging-based readout against the luminescence-based assay CTG, the current standard for drug screens in monocultures of organoids.[8] Calcein Violet (CalceinV) reports on esterase activity in viable cells, whereas CTG quantifies cellular ATP content (Suppl. Fig. 1A). Microscopy-based CalceinV staining clearly distinguished viable (vehicle treated) from non-viable (staurosporine treated) cells (Suppl. Fig. 1B). The coefficients of variation in untreated wells were comparable between CalceinV and CTG signals, indicating similar assay reproducibility for both readouts (Fig. 1B; Suppl. Fig. 1C).

To compare drug sensitivities in tumour monocultures, both organoid lines were screened using a comprehensive drug library (n = 134 compounds). Comparing area under the dose response curve values (AUC), CalceinV-derived drug sensitivity correlated strongly with that of CTG (r = 0.88-0.96; Fig. 1C). Despite their phenotypic differences, overall drug sensitivity was largely conserved between AMC691T and AMC691B, with differential responses limited to a small subset of compounds (Suppl. Fig. 1D). Eight compounds, including prexasertib, showed higher efficacy (ΔAUC $\geq$ 100) in AMC691B (Suppl. Fig. 1E).

In contrast, five compounds, including cobimetinib, were more effective in AMC691T (Suppl. Fig. 1E). Together, these results demonstrate that the microscopy-based readout provides a quantitative and robust measure of cell viability and drug response.

Distinguishing cell type-specific viability in tumour-immune co-cultures

To assess cell-type-specific viability in co-culture conditions, peripheral blood mononuclear cells (PBMCs) from healthy donors were pre-stained with the proliferation dye CellTracker Orange (CTO) and distinguished from tumour cells using an image-based masking strategy (Fig. 1D). Downstream analysis pipelines for co-cultures segmented viable organoids as CalceinV$^+$/CTO$^-$ regions and viable PBMCs as CalceinV$^+$/CTO$^+$ regions. Cell-type specific CalceinV sum intensities were extracted from both regions. We used this approach to distinguish between viable and non-viable fractions for both cell types present in the co-culture well.

We first assessed the baseline killing of the AMC691T and AMC691B by PBMCs. Immune mediated differences were expected, since, in contrast to AMC691T, AMC691B is HLA$^-$, limiting T-cell recognition. Indeed, AMC691T showed reduced viability in a PBMC co-culture compared to monoculture (Suppl. Fig. 2A), consistent with PBMC-mediated killing. To achieve ~20% baseline killing, we tested multiple effector-to-target ratios and found 1:5 consistently optimal across three donors (Suppl. Fig. 2B). In contrast, the growth of AMC691B was comparable with and without PBMCs (Suppl. Fig. 2C).

We assessed the stability of CTO labelling over the 120-hour assay duration, including conditions of immune activation that may induce proliferation and subsequent dye dilution. PBMC monocultures were stimulated via CD3 signalling to induce TCR-mediated activation. Under these conditions, CTO loss was minimal: below 3% of viable cells misclassified as CTO$^-$ (Fig. 1E). PBMCs were selected for pre-labelling rather than organoids because organoids are cultured for an additional 48 hours and generally display higher proliferative capacity, potentially resulting in greater dye dilution. These results demonstrate that CTO pre-staining enables stable and reliable identification of immune cells throughout the screening assay.

Finally, to enable high-throughput screening, we automated the readout and optimized imaging speed by selecting a single Z plane, as described in the methods

section. This reduced imaging time approximately two-fold, resulting in ~8 minutes acquisition time per 384-well plate, without compromising signal intensity (Suppl. Fig. 2D-E). Collectively, these validation experiments established a robust, scalable cell-type specific microscopy-based readout suitable for high-throughput drug screening in neuroblastoma-immune co-cultures.

Figure 1 (next pages): Set up and technical validation of a microscopy-based readout to reliably distinguish cell types for high-throughput drug screening in co-culture models A*) Representative microscopy images of organoid lines AMC691B and AMC691T, which are derived from different tumour locations of the same patient and differ in morphology and immune interaction.* ***B****) Bar plots representing the intensity of CalceinV (sum) and the CTG luminescence in untreated AMC691T and AMC691B organoids. The error bars represent the standard deviation. The intra-assay coefficient of variance for CalceinV (indicated above the bars) were: 13.84% (AMC691B) and 12.3% (AMC691T), and for CTG: 13.16% (AMC691B) and 12.73% (AMC691T).* ***C****) Scatter plot representing the correlation of drug response (AUC) measured by CalceinV intensity versus CTG for AMC691T and AMC691B. Each dot represents the mean AUC of two replicates normalized to the vehicle-only control of a compound (n = 45). Compounds are coloured by drug target, as indicated in the legend. Compound treatment lasted 120 hours. The Pearson correlation coefficient (r) and p-value are mentioned above the scatter plot.* ***D****) Schematic representation of the image analyses control conditions and workflow. Left part: representative microscopic images of control conditions (AMC691T): vehicle-only control (DMSO, 0.25%), positive control for cytotoxicity (staurosporine, 10 μM) and maximal immune-engagement control ($CD3^+$). Images are shown for CalceinV, CTO and merged channels. Right part: analysis workflow. Images were segmented to define populations as follows: live PBMCs = $CTO^+/CalceinV^+$, dead PBMCs = $CTO^+/CalceinV^-$, and live NB = $CTO^-/CalceinV^+$. CalceinV intensity was measured within the live-NB and live-PBMC masks. Cell-type-specific dose–response curves for each compound were generated by normalizing signals to the DMSO control.* ***E****) Bar graphs representing CTO retention over time in CD3-stimulated PBMCs. PBMCs were seeded at 2,000 cells/well. The $CTO^-/CalceinV^+$ fraction indicates CalceinV signal misclassified as non-PBMCs. This fraction (indicated above the bars) was 1.04% at t = 0 hours, 2.94% at t = 72 hours and 2.89% at t = 120 hours. Error bars represent standard deviation.*

AUC = area under the dose-response curve; CTG = CellTiter-Glo 3D; CTO = CellTracker Orange; CalceinV = Calcein Violet; DMSO = Dimethyl sulfoxide; NB = neuroblastoma; PBMC = Peripheral Blood Mononuclear Cells; Stauro = Staurosporine.

A
AMC691T
-Mesenchymal
-HLA+
-TCR-mediated
immune interaction
AMC691B
-Adrenergic
-HLA-
-No TCR-mediated
immune interaction
200 µm
200 µm
B
CalceinV Intensity (sum)
6×10^8
4×10^8
2×10^8
0
13.8%
12.3%
AMC691B
AMC691T
CTG Luminescence
4×10^6
3×10^6
2×10^6
1×10^6
0
13.2%
12.7%
AMC691B
AMC691T
C
AMC691T
r = 0.957, p = <0.001
CTG AUC
Microscopy AUC
0
100
200
300
400
500
AMC691B
r = 0.884, p = <0.001
Target
AKT
ALK
BCL
BRAF
BRD4
CDK4/6
EGFR
FGFR
HDAC
MDM2
MEK
MET
MTOR
Other

Figure 1

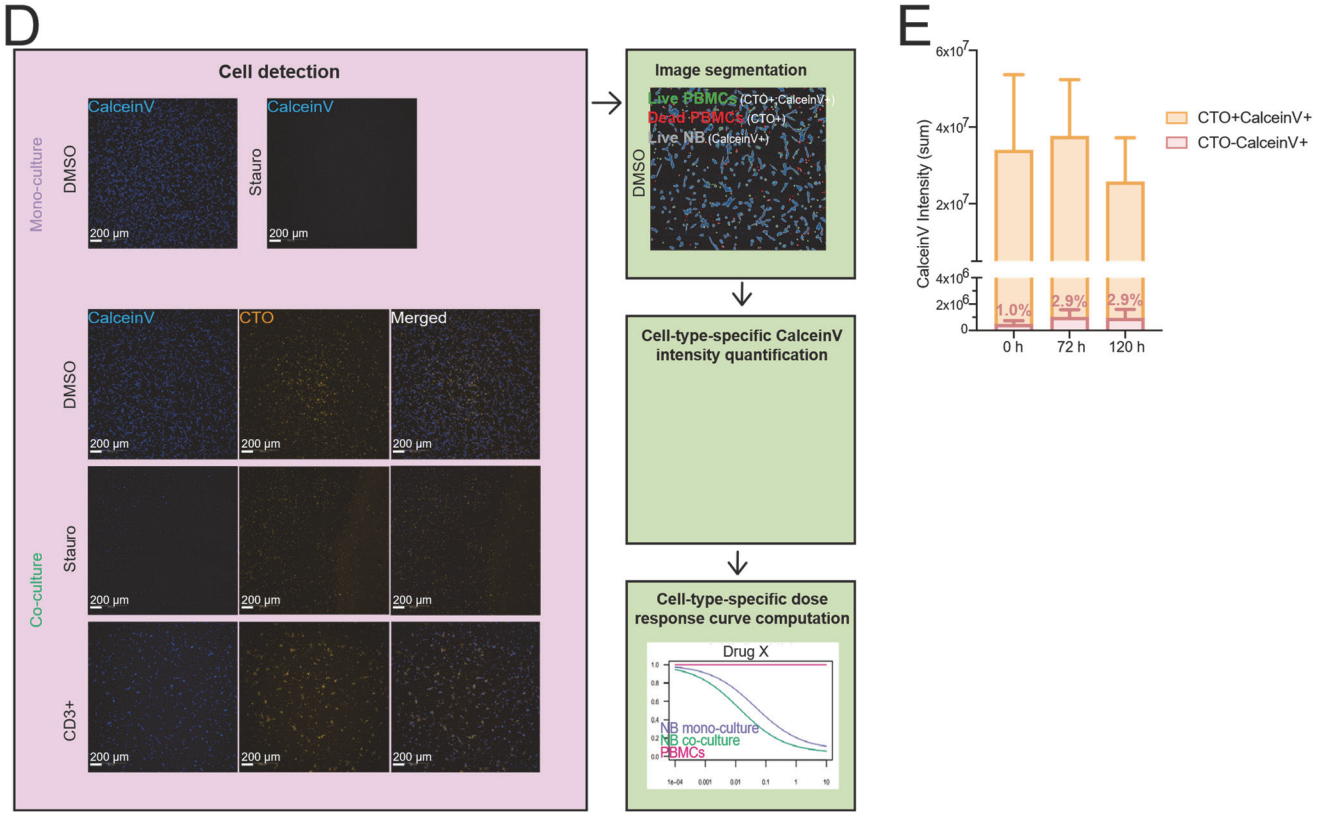


Figure 1 (continued)

Identifying compound sensitivities in co-cultures using the microscopy-based readout

Having established and technically validated this microscopy-based readout for cell-type-specific drug responses, we next applied it to high-throughput drug testing in organoid-PBMC co-cultures (Fig. 2A). In short, organoids were seeded in 384-well plates. Two days later (t = 0 h), pre-labelled PBMCs were added to the organoids and compounds were added for 120 hours. At readout, the viability dye CalceinV was added and automated image acquisition was performed. The drug library was screened across both conditions in the two organoid lines, AMC691T and AMC691B.

To identify compounds with altered efficacy in the presence of PBMCs, we compared drug responses between mono- and co-culture conditions (Fig. 2B). A clear distinction was observed between the two organoid models, in accordance with their immune status. Specifically, there were several compounds with altered efficacy in co- versus monoculture for AMC691T, whereas in AMC691B almost no differences were observed (Fig. 2B). In AMC691T, five compounds showed increased efficacy in the co-culture (ΔAUC ≥ 50), and 22 compounds showed reduced efficacy, in contrast to AMC691B where only 3 and 2 compounds showed increased and reduced efficacy, respectively. No overlapping compounds with altered efficacy were identified between the two organoid lines. In AMC691T, AZD5582 demonstrated greater efficacy in co-culture compared to monoculture (ΔAUC = 197; Fig. 2C-D; Suppl. Fig. 3), suggesting a PBMC-mediated effect. In line with this, AZD5582 did not exert toxic effects on PBMCs at concentrations below 10 μM (Fig. 2C). Similarly, vinblastine showed increased efficacy in co-culture (ΔAUC = 50), with PBMCs remaining viable at high concentrations (Fig. 2C-D; Suppl. Fig. 3). Together, these results demonstrate the applicability of this microscopy-based approach to identify compounds with altered efficacy in co-culture conditions, potentially informing the selection of compounds for combination with immunotherapy.

To identify compounds with tumour-selective efficacy, we directly compared tumour cell and PBMC responses from the same wells, as limiting PBMC toxicity may preserve immune-mediated anti-tumour activity *in vivo*. We identified 6 compounds that were consistently more effective in tumour cells in both organoid lines (ΔAUC ≥ 100 in organoids vs PBMCs in both lines), comprising 5

microtubule-targeting agents (docetaxel, paclitaxel, vindesine, vinblastine and vincristine) and antimetabolite gemcitabine (Fig. 2E). For example, docetaxel and paclitaxel induced tumour killing at low concentrations (0.001 μM and 0.01 μM, respectively), while PBMC viability was unaffected up to 10 μM (Fig. 2F-G). Additional tumour-selective compounds were identified in only one of the two organoid lines, with two compounds specific for AMC691T and 30 for AMC691B (Suppl. Fig. 4A). The AMC691B-specific compounds included prexasertib, decitabine, alisertib and carboplatin, which were already more effective in AMC691B monocultures and did not affect PBMC viability. Together, these results demonstrate that the microscopy-based readout enables identification of tumour-selective compounds that may be relevant for combination strategies with immunotherapies.

We next assessed compounds with comparable effects on tumour and immune cells, as the toxicity of the compounds on the immune cells may be a reason to not apply them *in vivo*. Compounds were selected based on efficacy in both cell types (AUC < 400) and limited differences in viability reduction between PBMCs and organoids (|ΔAUC| < 100). We identified six compounds, the proteasome inhibitors bortezomib and ixazomib, the HDAC inhibitors romidepsin and fimepinostat, β-catenin antagonist tegavivint and CDK12/13 inhibitor SR-4835, with similar activity in tumour cells and PBMCs across both organoid lines (Suppl. Fig. 4B-C). No compounds were consistently more toxic to PBMCs versus organoids across the full dose-response range (ΔAUC ≥ 100 in PBMCs versus organoid in both lines). Overall, this data demonstrates the applicability of this readout for high-throughput drug screening in tumour-immune co-cultures, enabling assessment of compounds with enhanced or decreased efficacy in co-cultures, tumour-specific effects, and potential PBMC toxicity.

Figure 2 (next pages): Identification of compounds with altered efficacy in mono- versus co-culture using a microscopy-based readout.

A*) Protocol workflow of plating, compound treatment and image acquisition. In brief, organoids are seeded (t = -48 hours), CTO pre-stained PBMCs were seeded and then compounds were added (t = 0 hours). At the day of readout (t = 120 hours), viability is assessed using CalceinV and images are acquired.*

B*) Scatterplots comparing drug efficacy (AUC) of the tumour cells in mono- versus co-culture for AMC691T and AMC691B. AUC values were calculated from two technical replicates per compound and normalized to the vehicle control (0.25% DMSO) of that condition (i.e. viabilities of treated wells from the co-culture condition were normalized to the vehicle-treated co-culture condition). Each point represents one compound. Compounds with |ΔAUC| ≥ 50 were considered to show altered efficacy; red and green dots indicate decreased and increased efficacy in co-culture, respectively.*

C*) Dose-response curves of the tumour cells of AMC691T in mono- (light blue) and co-culture (blue), and of PBMCs (red) treated with apoptosis protein antagonist AZD5582 (left) and tubulin inhibitor vinblastine (right). Symbols indicate the mean and error bars the standard deviation. AUC values are determined from this dose-response curve and represent the mean of two technical replicates (AZD5582: AUC_co = 216 versus AUC_mono = 412; Vinblastine: AUC_co = 282 versus AUC_mono = 332).*

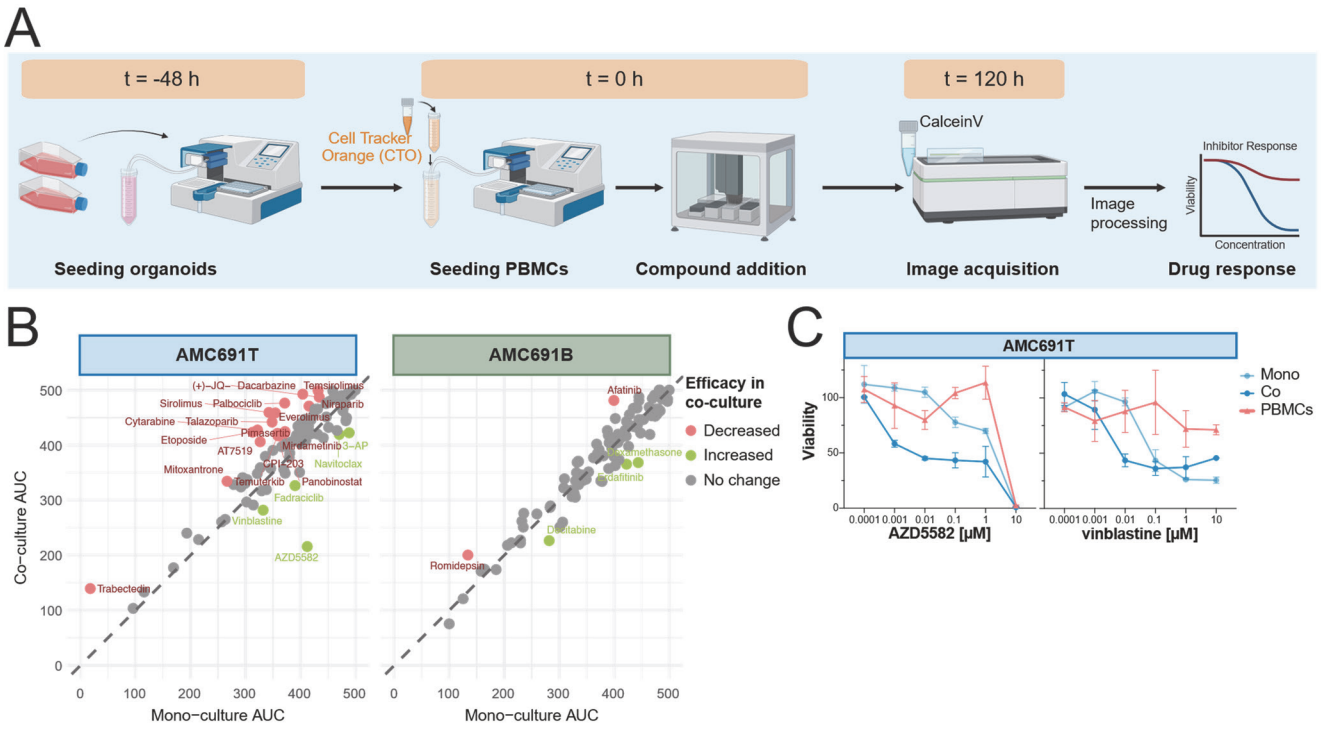
D*) Representative microscopy images of AMC691T in mono- and co-culture stained with CalceinV. Conditions shown: vehicle control (0.25% DMSO), apoptosis protein antagonist AZD5582 (0.001 μM); tubulin inhibitor vinblastine (0.01 μM).*

E*) Heatmap showing drug response (AUC) for compounds with tumour-selective effects (ΔAUC ≥ 100 in organoids versus PBMCs, per line) for both organoid lines.*

F*) Dose-response curves of organoids and PBMCs in co-culture. Colours represent AMC691T (blue), AMC691B (green) and PBMCs (red), treated with the microtubule-targeting agents docetaxel (top) and paclitaxel (bottom). Symbols indicate the mean and error bars the standard deviation. Data points exceeding the plotted y-axis range are not displayed.*

G*) Representative microscopy images of AMC691T in co-culture treated with microtubule-targeting compounds docetaxel (0.01 μM, top) and paclitaxel (0.1 μM, bottom).*

AUC = Area Under the dose-response Curve; CTO = CellTracker Orange; CalceinV = Calcein Violet; DMSO = Dimethyl sulfoxide; PBMC = Peripheral Blood Mononuclear Cells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Cell Tracker Orange (CTO)
CalceinV
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Inhibitor Response
Viability
Concentration
Seeding organoids
Seeding PBMCs
Compound addition
Image acquisition
Drug response
B
AMC691T
AMC691B
Co−culture AUC
Mono−culture AUC
(+)−JQ−
Dacarbazine
Temsirolimus
Sirolimus
Palbociclib
Niraparib
Cytarabine
Talazoparib
Everolimus
Etoposide
Pimasertib
AT7519
Mirdametinib
3−AP
CPI−203
Navitoclax
Mitoxantrone
Temuterkib
Panobinostat
Fadraciclib
Vinblastine
AZD5582
Trabectedin
Afatinib
Dexamethasone
Ibrutinib
Decitabine
Romidepsin
Efficacy in co-culture
Decreased
Increased
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C
AMC691T
Viability
AZD5582 [µM]
vinblastine [µM]
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Figure 2

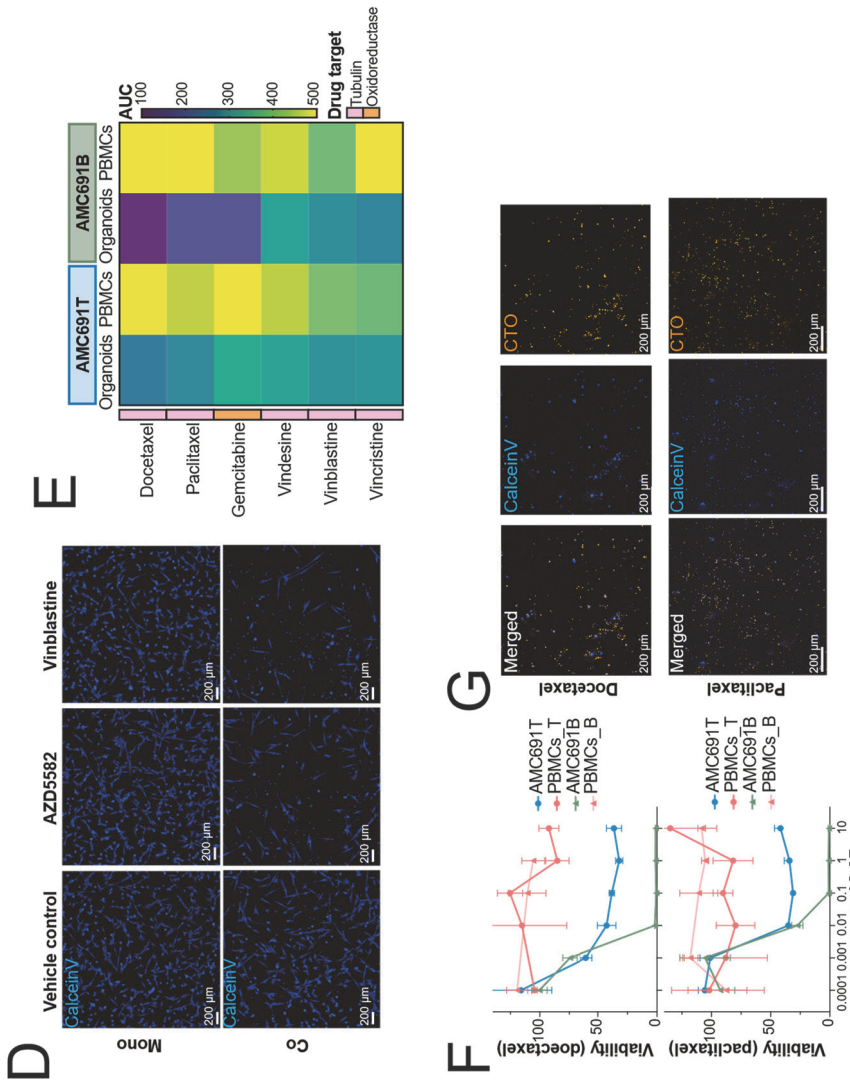

Figure 2 (continued)

Development and validation of a microscopy-based readout to determine tumour-specific viability in patient-derived samples

In short-term cultured patient material, which still contains the tumour microenvironment, it is essential to assess tumour-specific viability, similar to the co-cultures described above. To distinguish tumour from their microenvironmental cells, we developed a microscopy-based readout for use in primary patient material.

To enable drug testing in low-input patient samples, we optimized the assay for reduced cell numbers per well. Standard flat-bottom assays typically require ~5000 cells per well, which is not feasible for testing multiple compounds on minimally invasive biopsies.[8,22] Therefore, we adapted the assay to round-bottom 384-well plates to improve cell viability and promote spheroid growth from a limited cell density (500 cells/well). Round-bottom plates pose challenges for automated microscopy due to potential focus errors, as consecutive autofocus failures lead to termination of automated image acquisition. To address this, we optimized imaging settings using the two organoid models by systematically comparing objective magnification and numbers of field-of-view and selecting for the lowest amount of focus errors. Imaging with a 10× objective and a single central field resulted in zero focus errors, whereas imaging with multiple fields or the 5x magnification increased error rates substantially (n = 28 and 5, respectively; Suppl. Fig. 5A). Additionally, using only one field of view imaged 42 wells in 4 minutes versus 13 minutes for the 4 fields of view condition. Therefore, we selected the 10x objective with one central field of view, imaging the centre of the well while excluding the well edges. These optimized imaging settings enabled robust and fast image acquisition in round-bottom plates with a reduced cell input of 500 cells per well.

Patient material is more complex than neuroblastoma-PBMC co-culture systems, because tumour cells must be distinguished from multiple non-malignant cell types. As these cells cannot be selectively pre-labelled prior to plating, tumour cells were identified using a far-red-conjugated neuroblastoma-specific antibody cocktail targeting NCAM, L1CAM and B7H3 (NB-cocktail). These markers are widely used in neuroblastoma diagnostics and are being explored as therapeutic targets due to their high expression on tumour cells and limited expression in surrounding stromal and immune cells.[26–29] By combining

these markers with the viability dye CalceinAM Green (CalceinAM), it is possible to identify viable neuroblastoma cells as NB-cocktail^{+}/CalceinAM^{+} and non-neuroblastoma viable cells as NB-cocktail^{-}/CalceinAM^{+} (Fig. 3A).

We first evaluated the tumour markers both individually and combined on two organoid models by quantifying CalceinAM intensity within each marker-defined population. The adrenergic organoid line AMC691B showed strong positivity for all three markers, whereas its mesenchymal counterpart AMC691T was positive for B7H3 but showed minimal expression of NCAM and L1CAM (Fig. 3B-C; Suppl. Fig. 5B). Combining all three markers into a single neuroblastoma-specific antibody cocktail resulted in robust detection of tumour cells in both organoids while PBMC controls showed minimal signal (Fig. 3B). We then tested the sensitivity and specificity of the NB-cocktail setup in a co-culture of AMC691B organoids with pre-labelled PBMCs and observed that the NB-cocktail distinguished tumour from immune cells (Suppl. Fig. 5C-E), with minimal overlap between the NB-cocktail and PBMC-specific eFluor450 signals. The combination of three tumour markers is particularly advantageous for patient-derived samples, where (individual) marker expression is not determined prior to the assay.

Tumour-specific viability and drug responses in short-term cultured patient samples

To apply the established microscopy-based readout to *ex vivo* tumour material, we obtained three neuroblastoma patient samples (two biopsies and one resection) and one melanocytic neuroectodermal tumour of infancy sample (MNTI; resection; Suppl. Table 2). This rare paediatric tumour was included to demonstrate broader applicability of the workflow, since it was confirmed B7H3 positive by pathology.

We analysed the viability of the tumour and non-tumour fractions at baseline and during short-term culture. The MNTI sample (MaxPatient4) consisted almost entirely of tumour cells at t = 0 h (99%), whereas tumour fractions varied substantially across the neuroblastoma samples, ranging from 22% (MaxPatient3) to 66% (MaxPatient2), with MaxPatient1 at 55% (Fig. 3D-E; Suppl. Fig. 6A-B). These findings underscore the importance of tumour-specific viability measurements, as drug responses of tumour cells may be masked by large non-malignant cell fractions.

Marker expression also varied between patients. MaxPatient2 showed higher expression of B7H3 and NCAM compared to L1CAM, whereas MaxPatient3 was positive for L1CAM and NCAM, but showed lower B7H3 expression (Suppl. Fig. 6C). Together, these results support the use of the NB-cocktail in patient samples with different marker expression profiles and demonstrate the applicability of the readout to determine tumour fractions and tumour-specific viability.

We next assessed whether this readout can be applied to determine tumour-specific drug responses in short-term cultured patient samples. Because of the limited amount of tumour material, we selected a small number of compounds to test at a single concentration as a proof-of-concept. Staurosporine was included as a cytotoxic compound. Vincristine was selected based on the use in standard neuroblastoma treatment regimens and, together with cytarabine, showed limited PBMC toxicity in our co-culture experiments. Due to the limited availability of viable tumour cells in MaxPatient1, only a single compound was tested. Viability was quantified within tumour-specific masks and compared with total cell viability (Suppl. Fig. 7A). The low tumour fraction at t = 0 h for MaxPatient3 (Fig. 3D; Suppl. Fig. 6A) was reflected by a clear difference between total and tumour-specific viability at the drug screen readout (t = 72 h; Suppl. Fig. 7A). In contrast, the MNTI sample MaxPatient4 showed more similar total and tumour-specific viability, consistent with the high tumour fraction. In addition, the neuroblastoma samples showed limited changes in viability during the assay (75-115% viability from 0 h to 72 h), whereas the MNTI sample did proliferate (146%).

Cytotoxicity could clearly be detected using this microscopy-based readout, as staurosporine (10 μM) induced cell death compared to the vehicle control (0.25% DMSO; Fig. 3F-G; Suppl. Fig. 7A-B). Tubulin inhibitor vincristine (1 μM) and DNA polymerase inhibitor cytarabine (10 μM) showed limited efficacy in the neuroblastoma samples, likely reflecting the limited cell proliferation (Suppl. Fig. 7A). In contrast, vincristine induced cell killing in the MNTI sample, while cytarabine showed minimal activity. Finally, the MNTI sample provided sufficient material for a larger drug screen of 44 compounds across the full dose-response range (Fig. 3H; Suppl. Fig. 7C). Mitoxantrone and daunorubicin were the most effective compounds. The high tumour fraction allowed for direct comparison of CTG versus CalceinAM based drug response quantified within the tumour population, which was highly concordant (r = 0.88; Suppl. Fig. 7D).

Together, this data supports the feasibility and specificity of this microscopy-based readout for the use of short-term cultured patient samples to determine the tumour fraction over time and, if sufficient material is available, to test tumour-specific drug response.

Figure 3 (previous pages): A microscopy-based readout to determine tumour fractions and tumour-specific drug responses in freshly dissociated patient samples ***A)*** *Schematic overview of the microscopy-based set up. Viable neuroblastoma (NB) cells are identified as NB-cocktail$^+$/CalceinAM$^+$ and viable tumour microenvironment (TME) cells are defined as NB-cocktail$^-$/CalceinAM$^+$. The NB-cocktail consists of antibodies against the neuroblastoma-specific markers B7H3, L1CAM and NCAM.* ***B)*** *Bar graphs of the quantified CalceinAM sum intensity within different masks, including the NB-cocktail (1:1000 final concentration) and individual markers (B7H3, L1CAM, NCAM) for NB organoids and immune cells. Values are normalized to the total CalceinAM intensity per well. Error bars indicate standard deviation.* ***C)*** *Representative microscopy images of AMC691T showing L1CAM staining (top) and NB-cocktail staining (bottom; B7H3, L1CAM, NCAM).* ***D)*** *Graph representing tumour-specific viability, measured as CalceinAM sum intensity within the tumour-specific mask. Neuroblastoma samples were dissociated directly after tissue collection and plated within 4-7 days, the melanocytic neuroectodermal tumour of infancy sample was cultured for a month. t = 0 h is 24 hours after plating.* ***E)*** *Representative microscopy images of MaxPatient1 and MaxPatient3 representing CalceinAM (green) and NB-cocktail (red). Merged channels are shown, single channels are in Suppl. Fig. 6B. Samples were dissociated directly after tissue collection and plated within 4-7 days. t = 0 h, 24 hours after plating, is shown.* ***F)*** *Bar graphs representing tumour-specific viabilities normalized to vehicle-treated controls. Cells were treated with vehicle control (0.25% DMSO; blue), cytarabine (10 μM; purple) and MaxPatient2, MaxPatient3 and MaxPatient4 were additionally treated with staurosporine (10 μM; red) and vincristine (1 μM; green). Viability is quantified as CalceinAM sum intensity in CalceinAM$^+$/tumour-marker$^+$ mask and raw intensities are in Suppl. Fig. 7A.* ***G)*** *Representative microscopy images of neuroectodermal sample MaxPatient4 at the day of readout (t = 72 h), cells were treated with vehicle control (0.25% DMSO), cytarabine (10 μM), vincristine (1 μM) or staurosporine (10 μM). Merged channels (green: CalceinAM; red: B7H3) are represented, single channels are in Suppl. Fig. 7B.* ***H)*** *Heatmap representing tumour-specific drug response (AUC, determined by CalceinAM sum intensity in CalceinAM$^+$/B7H3$^+$ mask) of MaxPatient4 for the top 10 most effective compounds of the 44 tested compounds (all compounds in Suppl. Fig. 7C).*

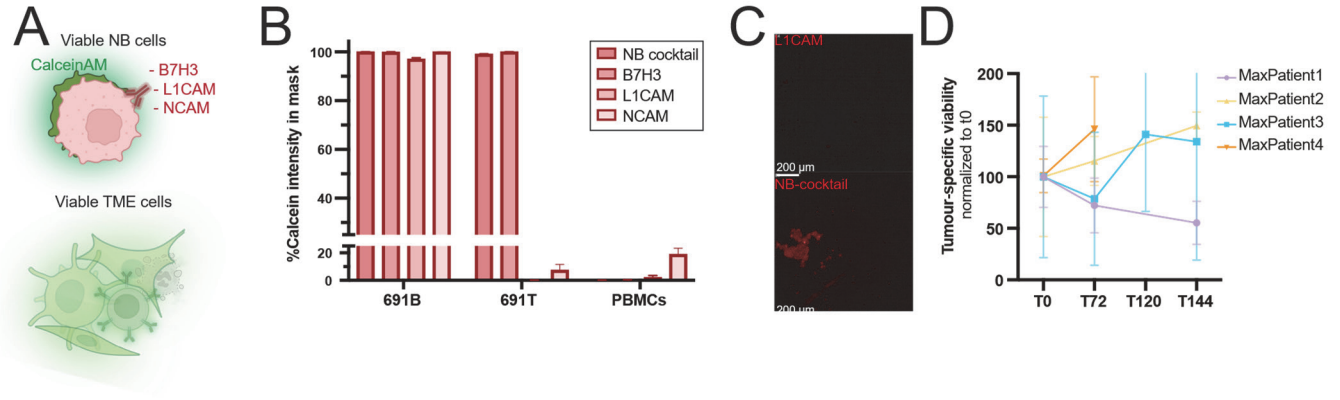
A
Viable NB cells
CalceinAM
- B7H3
- L1CAM
- NCAM
Viable TME cells
B
%Calcein intensity in mask
100
80
60
40
20
0
691B
691T
PBMCs
NB cocktail
B7H3
L1CAM
NCAM
C
L1CAM
200 µm
NB-cocktail
200 µm
D
Tumour-specific viability
normalized to t0
200
150
100
50
0
T0
T72
T120
T144
MaxPatient1
MaxPatient2
MaxPatient3
MaxPatient4

Figure 3

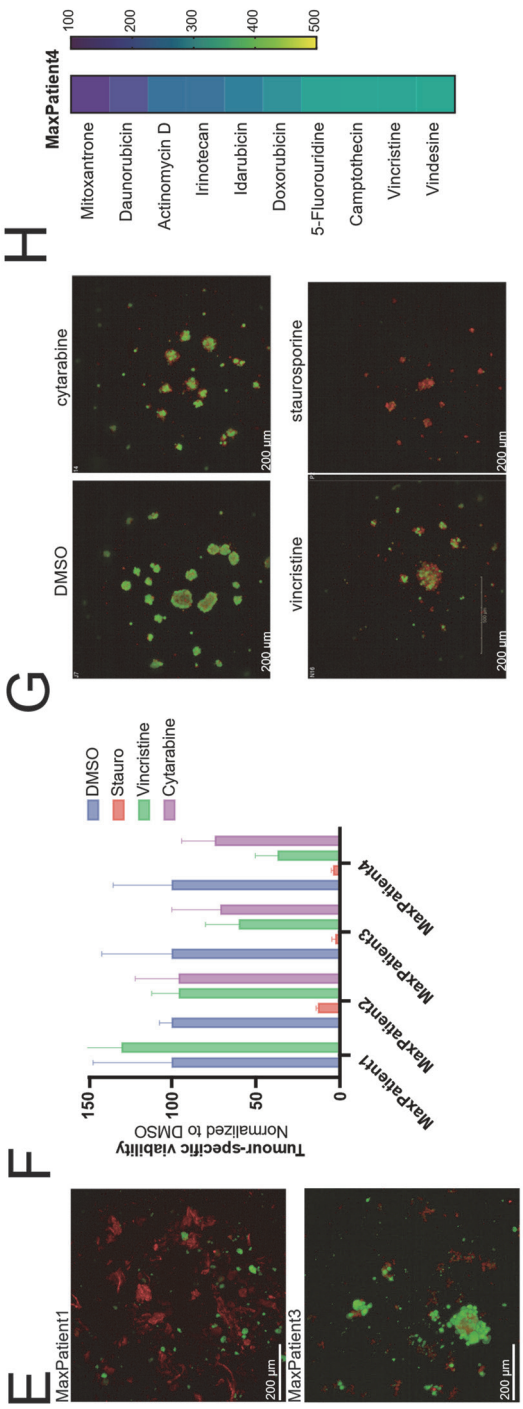

Figure 3 (continued)

Discussion

To advance drug testing in neuroblastoma, a difficult-to-treat paediatric cancer, we developed a microscopy-based viability readout that enables assessment of drug responses in the presence of the tumour microenvironment (TME), in both neuroblastoma-immune co-cultures and short-term cultured patient samples. This is increasingly relevant as the TME influences drug response and because immunomodulatory therapies are emerging for neuroblastoma.[10–12] The readout strongly correlated with conventional viability measurements while providing additional value through cell type-specific viability quantification.[7,8]

The microscopy-readout for organoid-PBMC co-cultures enables direct comparison of drug effects on tumour cells and PBMCs within the same well, in contrast to approaches that assess PBMC toxicity in separate monoculture assays independently from tumour cell viability.[8,31,32] Compounds effective against tumour cells either showed comparable PBMC toxicity or largely spared PBMC viability. The latter may preserve anti-tumour immune activity *in vivo* and thereby support combination strategies with immunotherapies. Microtubule-targeting agents were especially effective against tumour cells and had little effect on PBMC viability. These agents disrupt microtubule dynamics, leading to mitotic arrest and apoptosis, processes to which highly proliferative cells are particularly vulnerable.[33,34] Both organoid lines proliferated substantially more than PBMCs *in vitro*, which may partly explain their dependency on microtubule dynamics and the resulting cytotoxic selectivity. Although PBMC viability represents only a limited measure of non-tumour toxicity, this high-throughput workflow supports identification of tumour-selective compounds to be further validated in future studies.

Several compounds showed increased efficacy in co-culture compared to monoculture, including the apoptosis protein antagonist AZD5582 and microtubule targeting agent vinblastine. Previous studies have indicated immunomodulatory effects for AZD5582, including enhanced tumour infiltration by CD8+ T-cells, reduced regulatory T-cell abundance, dendritic cell activation and improved T-cell priming.[35–37] Similarly, vinblastine has been shown to stimulate anti-tumour immune responses through activation of dendritic cells,

although dendritic cells are present at low frequencies in PBMCs.[38] These findings illustrate the value of the co-culture screen in scalable detection of compounds whose efficacy is enhanced by the presence of immune cells. Prioritized candidates can subsequently be characterized using immune profiling approaches, including evaluation of their effects on different immune cell subsets, cytokine production and activation and exhaustion markers.[39]

The co-culture readout used allogeneic PBMCs from healthy adult donors to allow standardization, reproducibility and scalability, but these may not fully recapitulate the immune composition and dysfunction observed in neuroblastoma patients.[12,40] Future studies could incorporate autologous patient-derived immune cells, although limited material from minimally invasive paediatric procedures remains a challenge. PBMCs represent only part of the TME and incorporating additional cell types such as stromal cells would increase physiological relevance. The workflow is also suited for studying emerging immune-mediated therapies, such as CAR-T cells, which are increasingly relevant in neuroblastoma and could be incorporated into co-culture screens.[12,41] This co-culture workflow provides a scalable assay for tumour-immune drug screening, with potential to inform both immunotherapy development and tumour-selective compound prioritization in neuroblastoma.

We extended the microscopy-based viability readout to short-term cultured patient material to quantify tumour fraction and tumour-specific drug response in mixed cell populations. This addresses an important limitation of current *ex vivo* drug screening approaches, which rely on whole-well viability measurements and cannot distinguish tumour from non-malignant cells.[20,19]

Combining neuroblastoma-associated markers (L1CAM, NCAM and B7H3) into a cocktail enabled distinction of tumour cells from surrounding cells, compensating for intra- and interpatient variability in surface marker expression. Although these markers are widely used neuroblastoma markers, NCAM can also be expressed by immune cells (particularly NK cells) and B7H3 can be expressed by myeloid and endothelial cells in the tumour-microenvironment.[26–29,42–44] Using a threshold-based separation limits false tumour-positive signals, as neuroblastoma cells have higher surface expression than immune cells. This microscopy-based viability readout for patient samples can be readily adapted to other tumour types by substituting tumour-specific markers.

Limited viable tumour material remains a challenge for *ex vivo* short-term patient screens.[8,19,20,22] Therefore, we reduced the cell density per well. Neuroblastoma patient samples retained non-malignant cells during short-term culture, showing variable tumour fractions (22-66%), but tumour cells showed little to no proliferation during the assay, which likely contributed to the limited compound responses observed. Longer pre-culture durations may increase viable tumour cell numbers and proliferation rates but simultaneously change TME representation, as illustrated in the MNTI sample. After one month of culture, the MNTI sample consisted almost entirely of tumour cells with an increased proliferation rate, providing more viable tumour material for screening that enabled testing of a larger compound panel. The balance between preserving TME components and obtaining sufficient tumour material requires optimization based on the research question and available material.

Current precision oncology initiatives such as INFORM and ZERO rely on whole-well viability measurements for *ex vivo* drug testing.[19,20] For cultures containing both tumour and non-malignant cells, tumour-specific viability assessment provides additional value by distinguishing tumour-specific drug responses from effects on surrounding non-malignant cells. The predictive value of *ex vivo* drug screening for treatment response in patients remains to be established in large paediatric cohorts. Microscopy-based tumour-specific viability readouts could contribute to this effort by enabling more precise drug response measurements, thereby ultimately supporting treatment stratification in neuroblastoma precision medicine.

Conclusion

We developed a scalable microscopy-based readout that enables cell-type-specific drug response profiling in neuroblastoma-immune co-cultures and short-term cultured patient samples that could support future immunotherapy development and precision medicine.

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Author contribution statement

Jan J. Molenaar: Writing - review & editing, Supervision, Conceptualization. Selma Eising: Writing - review & editing, Supervision, Conceptualization. Marlinde L. van den Boogaard: Writing - review & editing, Supervision, Conceptualization. Sander van Hooff: Writing - review & editing, Supervision, Conceptualization. Judith Wienke: Writing - review & editing, Supervision. Karin Langenberg: Resources. Eoghan O'Duibhir: Resources, Investigation. Jeroen F. van Velzen: Resources. Barrera Román: Resources. Luca M. van den Brink: Resources. Merle M.J. Watzeels: Investigation. Eleonora J. Looze: Investigation. Sarah Swaak: Investigation, Visualisation. Sabina Valová: Writing - original draft, Visualization, Investigation, Formal analysis. Marlinde C. Schoonbeek: Writing - original draft, Visualization, Project administration, Investigation, Methodology, Formal analysis, Data curation.

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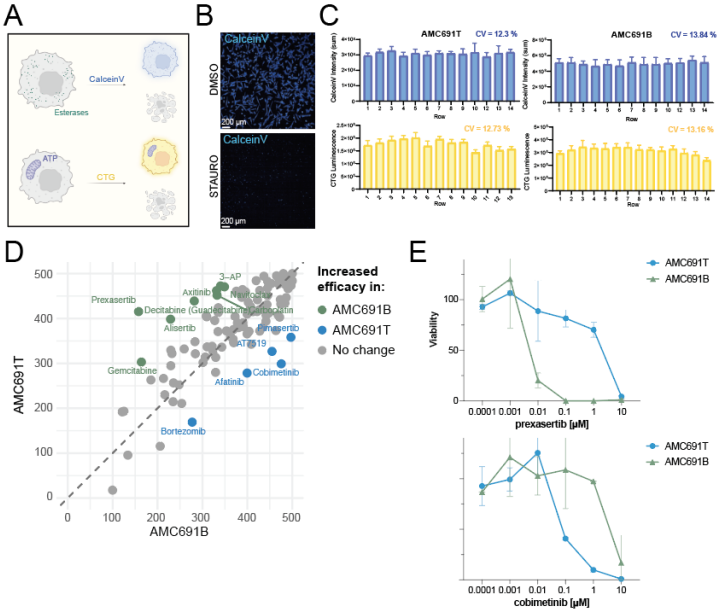
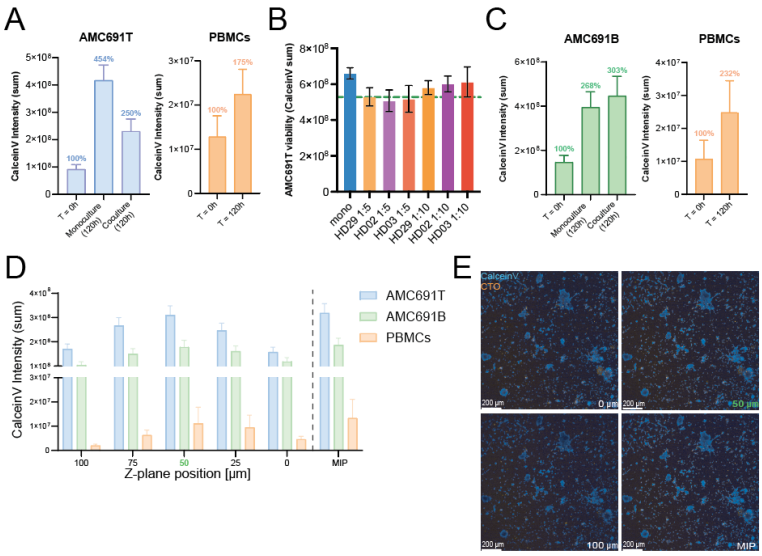
Supplementary material

The online version of this chapter, including supplementary tables, can be accessed via the QR code below or at:
https://www.biorxiv.org/content/10.64898/2026.06.24.734168v1.

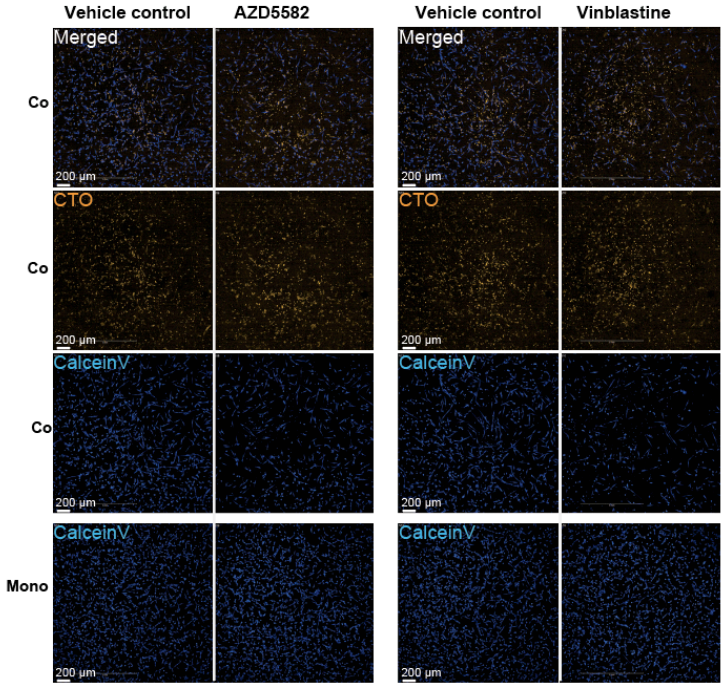
The supplementary figures are included on the following pages.



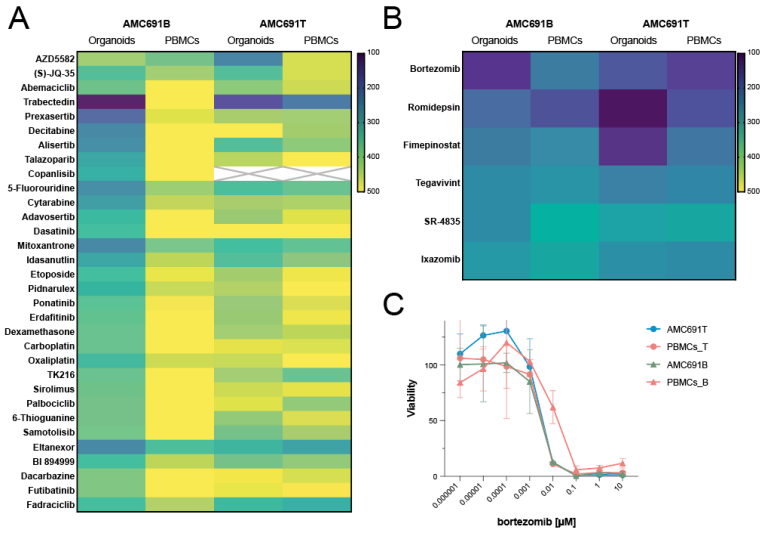
Supplementary Figure 1: Set up and technical validation of a microscopy-based readout to reliably determine drug response in high-throughput drug screens. A) Schematic comparison of CalceinV and CellTiter-Glo (CTG) mechanisms. CalceinV, quantified using the sum pixel intensity of images, reports on esterase activity in viable cells, whereas the luminescence-based CTG quantifies cellular ATP content B) Representative microscopic images of vehicle-only control (DMSO, 0.25%) and positive control for cytotoxicity (staurosporine, 10 µM). C) Bar plots representing the intensity per row of CalceinV (sum) and the CTG luminescence in untreated AMC691T and AMC691B organoids. The error bars represent the standard deviation and intra-assay CV (%) are shown for each condition above the error bars. D) Scatterplot comparing drug efficacy (AUC) in monoculture for AMC691T versus AMC691B. AUC values were calculated from two technical replicates per drug and normalized to the vehicle control. Each point represents one compound. Compounds with |ΔAUC| ≥ 100 were highlighted: green compounds indicate those with higher efficacy in AMC691B and blue those which were more effective in AMC691T. E) Dose response curves of AMC691T (blue) and AMC691B (green) in monoculture, treated with CHEK1 inhibitor prexasertib (top) or MEK inhibitor cobimetinib (bottom). Symbols indicate the mean and error bars the standard deviation.



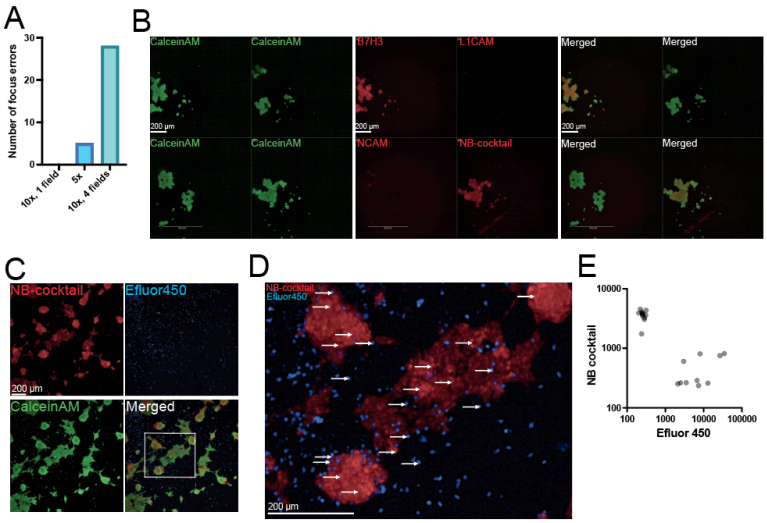
Supplementary Figure 2: Set up and technical validation of a microscopy-based readout to reliably distinguish cell types for high-throughput drug screening in co-culture models. A&C) Bar plots showing CalceinV intensity of AMC691T (A) and AMC691B (C) across timepoints and culture conditions. CalceinV intensity of organoids in mono- and co-culture, and of PBMCs are shown at t = 0h versus t = 120h. Errors bars represent standard deviation and growth rates relative to t = 0h are indicated above the bar. B) Bar graphs representing effector to target ratios (indicated below the bars) for three different donors. Error bars represent standard deviation. The green dashed line indicates 80% cell viability, with 20% baseline killing by PBMCs. D) Bar graph representing CalceinV intensity (sum) across individual Z-stacks versus maximal intensity projection for AMC691T/B and PBMCs. Error bars represent standard deviation. The selected Z-stack is indicated in green. E) Representative images of AMC691B co-cultured with HD-PBMCs showing CalceinV (live cells, 450 nm) and CellTracker Orange (CTO; pre-stained PBMCs, 590 nm) across individual Zstacks versus MIP (maximal intensity projection). The selected Z-stack is indicated in green.



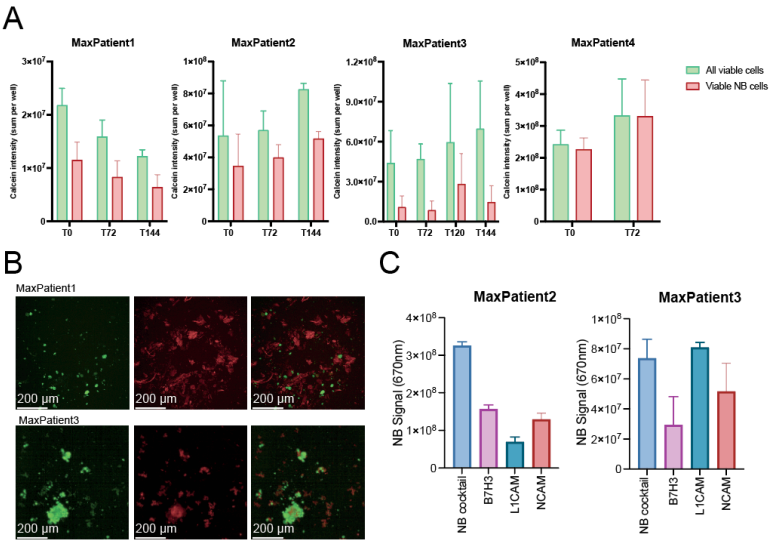
Supplementary Figure 3: Identification of compounds with altered efficacy in mono- versus co-culture using a microscopy-based readout. Representative microscopy images of AMC691T in mono- and co-culture showing CalceinV, CTO and Merged channels. Conditions shown: vehicle control (0.25% DMSO), apoptosis protein antagonist AZD5582 (0.001 µM); tubulin inhibitor vinblastine (0.01 µM).



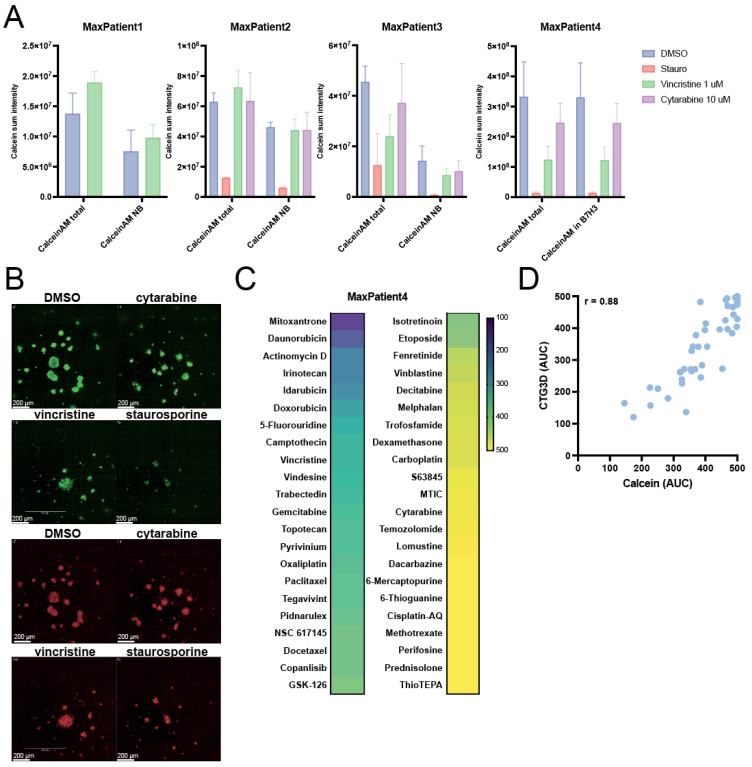
Supplementary Figure 4: Identifying compounds with tumour- or immune- selective efficacy. A) Heatmap of compounds representing drug response (AUC) of compounds with tumour-selective effects. Compounds are shown for which |ΔAUC| ≥ 100 between organoids and PBMCs in one of the two organoid models. Top two compounds are based on AMC691T (AZD5582 and (S)-JQ-35), remainder on AMC691B (30 compounds). **B**) Heatmap showing compounds with comparable activity in tumour cells and PBMCs for both organoid models. Compounds were selected based on efficacy towards both cell types (AUC < 400) and on a difference in AUC between organoids and PBMCs of less than 100 in both models (|ΔAUC| < 100). Values represent AUC per condition for organoid–PBMC co-cultures and PBMC-only controls. **C**) Dose response curves of organoids and PBMCs in co-culture. Colours represent AMC691T (blue), AMC691B (green) and PBMCs (red), and cells were treated with the proteasome inhibitor bortezomib. Bortezomib is tested across 8 concentrations. To allow for efficacy comparison for all compounds, AUCs are determined across the 6 concentrations that are tested across all compounds (0.0001-10 µM). Symbols indicate the mean and error bars the standard deviation.



Supplementary Figure 5: Optimization and validation of the microscopy-based readout for patient samples
A) Bar graph representing imaging performance under different acquisition settings, measured as the number of autofocus errors per plate. The same 42 wells were imaged in different conditions. Conditions that are compared: 10x objective with 1 field of view versus the 5x objective versus the 10x objective with 4 fields of view. **B**) Representative microscopy images of AMC691T stained with individual markers and the combined neuroblastoma-cocktail (NB-cocktail). Upper left: B7H3; upper right: L1CAM; lower left: NCAM; lower right: NB-cocktail (B7H3, L1CAM, NCAM). Scale bars are identical across conditions. **C**) Representative microscopy image of AMC691B cells (NB-cocktail$^+$, red) co-cultured with pre-labelled PBMCs (eFluor450$^+$, blue). CalceinAM was used as a viability dye. Cells were cultured in flat-bottom plates. **D**) Zoom-in of the image shown in (C), illustrating AMC691B cells (NB-cocktail$^+$, red) and PBMCs (eFluor450$^+$, blue). Arrows indicate regions for which fluorescence intensities were quantified, as shown in (E). E) Scatter plot showing quantified fluorescence intensities for the regions indicated in (D). Each dot represents one region (n = 22).

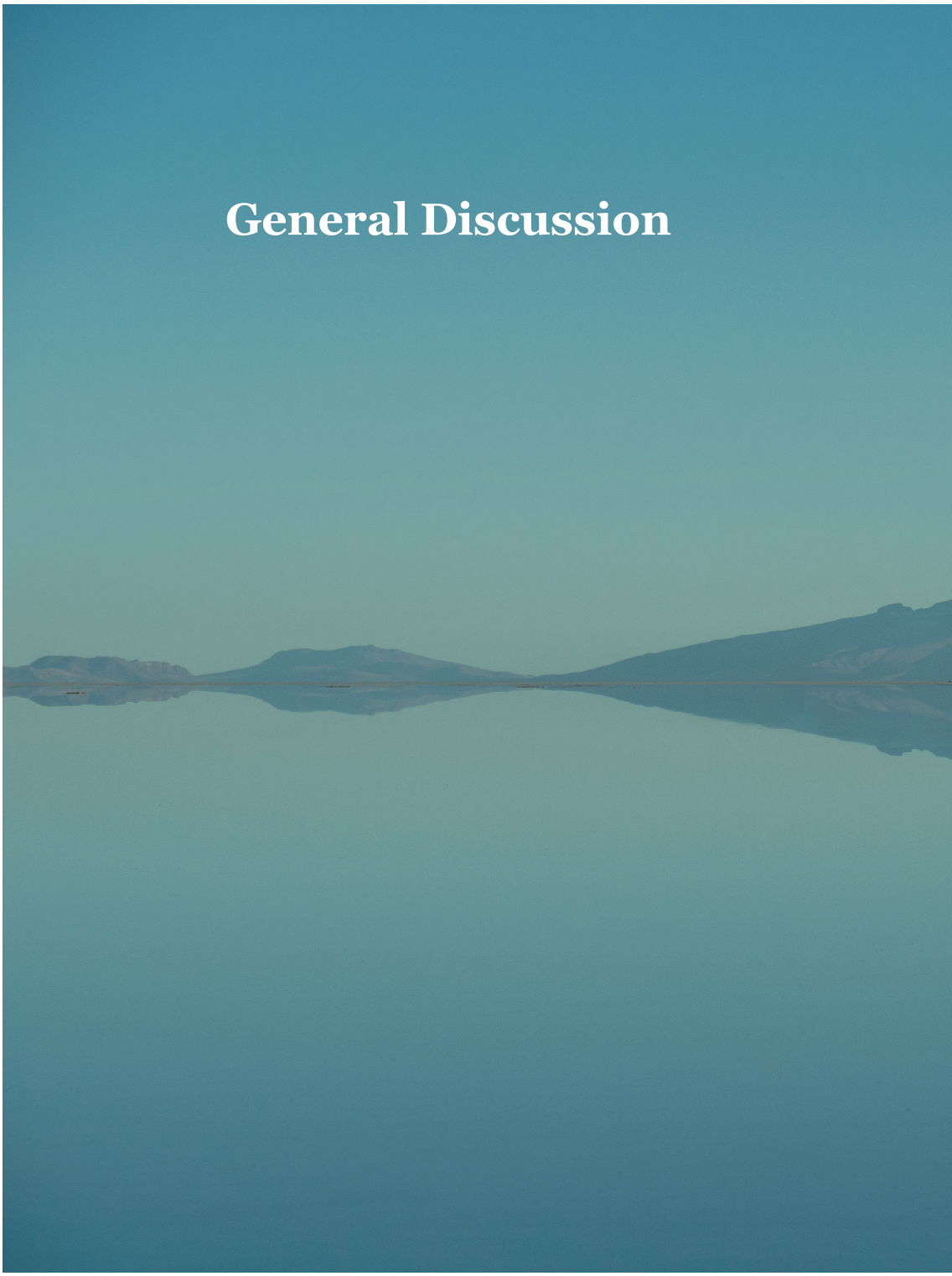


Supplementary Figure 6: Tumour fractions of patient-derived samples in different culture conditions or over time. A) Bar graphs comparing the total Calcein sum intensity (green) with the Calcein sum intensity within the tumour-specific mask (red) over time for four patients. All samples were dissociated directly after tissue collection, neuroblastoma samples were plated within 4-7 days, and the melanocytic neuroectodermal tumour of infancy sample was cultured for a month. t = 0h is 24 hours after plating. **B**) Representative microscopy images of MaxPatient1 and MaxPatient3 representing CalceinAM, NB-cocktail. Samples were dissociated directly after tissue collection and plated within 4-7 days. t = 0h, 24 hours after plating, is shown. **C**) Bar graphs comparing the sum intensity of the markers in the NB+ mask of each individual marker and the combined NB-cocktail for MaxPatient2 (t = 144h) and MaxPatient3 (t = 72h). Each of the markers and the cocktail are added in a final concentration of 1:1000. In the NBcocktail condition, each individual marker contributed one-third of the total antibody concentration



Supplementary Figure 7: Drug testing in short-term cultured patient samples. A) Bar graphs representing total and tumour-specific viability (CalceinAM sum intensity in CalceinAM+ mask, and CalceinAM+/tumour-marker+ mask) of all four patients at the day of readout (t = 72h). Cells were treated for 72 hours with vehicle control (0.25% DMSO; blue), staurosporine (10 µM; red), vincristine (1 µM; green) or cytarabine (10 µM; purple). **B**) Representative microscopy images of neuroectodermal sample MaxPatient4 cells are treated with vehicle control (0.25% DMSO), cytarabine (10 µM), vincristine (1 µM) or staurosporine (10 µM). CalceinAM (green, top) and the NB-cocktail (red, below) are shown. **C**) Heatmap representing tumour-specific drug response (AUC, determined by CalceinAM sum intensity in CalceinAM+/B7H3+ mask) of MaxPatient4 after 72 hours of treatment with 44 compounds, ordered by efficacy. **D**) Scatterplot representing the drug response (AUC) determined by CTG or CalceinAM sum intensity in CalceinAM+/B7H3+ mask after 72 hours of treatment. Pearson correlation coefficient (r) is included in the top left corner.

General Discussion



6

Expanding the precision medicine landscape of pediatric solid tumors with *ex vivo* drug screening

Despite scientific advances and intensive multimodal treatment, cancer remains a leading cause of disease-related death in children.[1] Treatment strategies in pediatric oncology are shifting away from historical "one-size-fits-all" chemotherapies by increasingly applying precision medicine to guide treatment. One approach is molecular profiling (the sequencing of the genome and transcriptome), which has become standard practice in the Princess Máxima Center. Although this has improved treatment for selected patients, molecular profiling alone often fails to identify effective therapies in high-risk pediatric tumors.[2,3] The poor survival rates for high-risk disease emphasize the urgent need for more effective treatment strategies and have stimulated growing interest in functional precision medicine which captures biological responses in preclinical models.

The studies in this thesis therefore investigated functional precision medicine in the form of drug screening in preclinical models, aiming to better predict patient drug response. The following questions form the foundation of this thesis: which model system is most appropriate for clinical treatment stratification versus drug and biomarker discovery? How can we further improve our drug screening methodology? And, importantly, what is the position of drug screening in guiding treatment decisions? This discussion first evaluates the strengths and limitations of the preclinical model systems investigated throughout this thesis. Subsequently, the role and challenges of drug screening in rapidly guiding treatment decisions are discussed. Finally, opportunities for drug and biomarker discovery aimed at improving treatment for future patients are considered.

Preclinical models to test drug response

Different preclinical model systems have been developed to evaluate therapeutic vulnerabilities in pediatric solid tumors, including neuroblastoma, the primary tumor type throughout this thesis. Each model balances physiological relevance, scalability, cost and turnaround time differently.

Patient-derived xenograft (PDX) models retain the major genetic and transcriptomic characteristics of their originating tumors, as demonstrated in this thesis (Chapter 2-3) and previous studies.[4–6] However, their suitability for personalized precision medicine is constrained by low engraftment success, low scalability, high costs and long establishment times.[4–9] To reduce animal testing, encouraged by regulatory agencies, patient-derived non-animal models are increasingly being explored as alternatives.[10]

In this thesis, previously established PDX models of the ITCC-P4 consortium were used for *ex vivo* short-term drug screens, organoid generation and *in vivo* compound testing (Chapter 2-4). Dissociated PDX samples resemble primary patient material while offering easier access and sufficient quantity of viable tumor cells, enabling protocol optimization and testing of larger compound libraries.[11] We established five new PDX-derived neuroblastoma organoids and used previously established patient-derived neuroblastoma organoids for drug screening (Chapter 3). Once established, organoids provided a scalable platform for drug screens, but, in line with literature, we found that their application for rapid treatment stratification is limited due to relatively low establishment rates (23%) and prolonged establishment timelines (3-12 months).[3,7–9,12–17] These timelines often exceed the median progression-free survival of relapsed or high-risk patients (4-7 months).[3] Furthermore, patients continue to receive treatment during organoid establishment, which may drive tumor evolution and reduce the organoid's representation of the current disease state. This was illustrated in Chapter 2, where two longitudinally obtained Ewing sarcoma samples displayed different drug responses following disease progression.

Ex vivo short-term screens, where freshly dissociated tumor material is briefly cultured prior to screening, offer a faster approach for preclinical drug testing. In this thesis, *ex vivo* short-term drug screens were completed within 14 days after tissue receipt (Chapter 2 and 3). We demonstrated the feasibility of this approach in a pan-cancer cohort, achieving an 82% success rate using PDX-derived tumor material (Chapter 2). Similarly, ZERO and INFORM demonstrated the feasibility of integrating *ex vivo* short-term screens in precision medicine programs as the rapid turnaround time allows integration into multidisciplinary tumor

boards.[18,19] Zooming in on neuroblastoma, *ex vivo* short-term screens achieved higher success rates (65% versus 23%) and shorter turnaround times compared to organoids while retaining drug sensitivity patterns (Chapter 3). In addition, *ex vivo* drug responses were comparable with *in vivo* responses in matched PDX models for seven out of ten clinically available compounds.

The studies presented in this thesis (Chapter 2-5) indicate that, despite model-specific differences, major drug sensitivity trends were generally preserved across different readouts and neuroblastoma model systems. Consequently, selecting the best model for a particular application can be guided by practical considerations such as scalability, feasibility and turnaround time. Different model systems may be best suited for rapid treatment stratification for the patient of today, or for discovering therapies for the patient of tomorrow.

Guiding treatment decisions in precision oncology for today's patient

Rapid drug screening within a clinically relevant timeframe

To identify the most effective treatment for the patient of today, preclinical models to predict drug sensitivity must be rapid, reproducible and feasible for low-input samples. Turnaround time is particularly important, as treatment decisions generally need to be made in a timeframe of days rather than months.[17] The findings presented in this thesis indicate that *ex vivo* short-term drug screening is currently the most feasible approach for rapid treatment stratification as results can be obtained within 14 days (Chapter 2). Furthermore, emerging international precision medicine initiatives incorporating drug screening have reported clinical feasibility. The ZERO program demonstrated that 82% of screens could be completed while patients were still under clinical care and that *ex vivo* drug responses more accurately reflected *in vivo* responses than actionable molecular markers.[19] The INFORM program identified drug vulnerabilities in 80% of cases lacking actionable high-evidence molecular alterations, with a median preculture time of four days.[18] These studies support the potential clinical utility of drug screening as a complementary strategy alongside molecular profiling in pediatric precision oncology.

Towards clinical implementation: challenges associated with limited patient material

Several challenges remain before *ex vivo* short-term drug screening can be broadly implemented in clinical practice. One of the major barriers is the limited quantity and quality of viable tumor material obtained from pediatric patient samples.[18] Small biopsies obtained through minimally invasive procedures and the reduced viability of tumor tissue in chemotherapy-treated surgical resections limit the amount of material available for drug screening. Furthermore, current pathology workflows, optimized for tissue fixation and genomics rather than viable tissue preservation, might limit the amount of material available for drug screening even in cases with sufficient tissue. Close collaboration with pathologists is important to optimize tissue allocation.

This thesis included both freshly dissociated PDX tumor samples, which provided relatively large quantities of viable tumor cells, and freshly dissociated patient samples, where material is often limited (Chapter 2-5). We found that drug sensitivity patterns were largely conserved between matched drug screens from patient and PDX samples, indicating that PDX-derived material is a suitable surrogate for primary patient samples (Chapter 3). However, the high success rates using PDX samples (82% for *ex vivo* short-term drug screens in pan-cancer samples; Chapter 2) likely overestimate achievable success rates of drug screens from patient samples. Besides the larger amounts of viable tumor compared to clinical biopsies, PDX samples may be biased toward more aggressive, and thereby better growing, tumor subclones.[20] Indeed, experiments using patient samples for *ex vivo* short-term cultures and drug screens demonstrated that both drug screen success and the number of compounds tested were strongly limited in cases of low quantity and quality of viable tumor cells present in the initial sample (Chapter 4-5). In particular, cryopreserved biopsy samples failed to establish successful short-term cultures after thawing (data not included in this thesis). This illustrates that fresh samples of sufficient quality are needed for successful drug screening, emphasizing the importance of experimental prioritization and careful allocation of available tumor material to ensure sufficient cell numbers.

Extending the short pre-culture period prior to drug screening may improve recovery from tissue dissociation, increase viable tumor cell numbers and thereby improve drug screening success. Future studies could therefore optimize the pre-culture duration and culture conditions to maximize viable tumor material while still obtaining drug responses within a clinically relevant timeframe. Importantly, prolonged culture periods may simultaneously change representation of microenvironmental cells that influence drug response, thereby potentially decreasing physiological relevance.

Optimizing drug screening methods for limited patient material

To maximize the utility of limited patient material, screening approaches may be optimized by either reducing the number of cells required per condition or limiting the number of conditions to test. In this thesis, round-bottom plates (Chapter 5) and MicroOrganoSpheres™ (MOS™; Chapter 4) reduced the number of cells required per condition, increasing the number of compounds that can be screened using the same amount of input material. Previous work in our group demonstrated that, compared to flat-bottom plates, round-bottom plates supported sufficient tumor growth at substantially lower cell densities, enabling drug screening with fewer cells per condition (data not included in this thesis). Next to this, the use of round-bottom plates supports 3D spheroid formation which may better represent the architecture of solid tumors.[13] MOS™, 3D droplets generated by microfluidic droplet formation that encapsulate patient-derived tumor cells in Matrigel, have previously demonstrated rapid drug screening results that correlate with in-patient response, while requiring low quantities of tumor tissue.[21–24] We demonstrated that MOS™ technology supports short-term tumor cell growth at least as well as suspension cultures. Furthermore, drug screening in MOS™ could be performed using fewer cells while maintaining reproducibility and comparable drug responses compared to suspension. Other miniaturized 3D tumor avatars are increasingly being explored, such as engineered 3D tumor models and whole-tissue platforms.[25–31]

Additionally, the number of conditions could be reduced by creating tumor type-specific drug libraries consisting of clinically relevant compounds, including

standard-of-care agents, targeted therapies and drugs previously identified as active in that tumor type. Ultimately, patient-specific libraries restricted to clinically accessible treatment options for an individual patient may further reduce the number of conditions. Although conceptually attractive, generating patient-specific libraries would require knowing the clinically available treatment options upfront and is therefore operationally challenging in routine practice. A more feasible application may be the use of *ex vivo* short-term drug screening as a complementary tool within multi-compound clinical trials, to help prioritize which trial arm is most likely to benefit an individual patient.

In this thesis, we performed drug screening of only single agents to reduce the number of conditions to test. However, single agents are rarely efficacious due to genetic intratumor heterogeneity or acquired resistance and therefore combination therapies better reflect clinical treatment strategies.[32] Testing combinations *in vitro* remains challenging due to the limited input material and practical complexities such as dose selection and scheduling. While combination and synergy screens are increasingly performed in established model systems,[33–35] comprehensive testing of clinically relevant multi-agent regimens remains difficult in *ex vivo* short-term samples as the number of experimental conditions rapidly increases with each additional compound. Clinical implementation therefore requires strategic prioritization of experimental conditions that can be tested from a single patient sample.

Interpreting drug responses in heterogeneous patient samples

Ex vivo short-term cultures can contain substantial proportions of residual microenvironmental cells, potentially providing a more representative tumor model than long-term cultures. However, the presence of these non-malignant cells may also complicate the interpretation of drug screening results, as these cells influence drug response and may confound interpretation of tumor-specific drug sensitivity.[36,37] Currently, tumor purity and microenvironmental contributions are generally not incorporated into the interpretation of *ex vivo* short-term drug screens. This is because the conventional viability readout for high-throughput drug screening, CellTiter-Glo, measures the combined viability of all cells present in a well. To address this limitation, we developed a

microscopy-based readout to determine tumor-specific viability in *ex vivo* short-term cultured neuroblastoma samples, which can be readily adapted to other tumor types (Chapter 5). Neuroblastoma tumor cells were distinguished from the non-malignant cells using a combination of tumor-specific surface markers NCAM, L1CAM and B7H3, which enabled determination of tumor fractions and measurement of tumor-specific drug responses. Because tumor purity is often unknown prior to screening, microscopy-based tumor-specific viability readouts may provide a more reliable assessment of tumor-specific drug response.

Interpreting drug screening results to define drug hits

Beyond generating drug response data, clinical implementation requires robust strategies for data interpretation, hit prioritization and outcome validation. Currently, there is no universally accepted definition of a clinically meaningful drug hit in functional precision oncology, limiting cross-site comparison. Different institutes use different response metrics, including area under the curve (AUC), half-maximal inhibitory concentration (IC50), half-maximal lethal concentration (LC50) and drug sensitivity scores (DSS), and each has its advantages and limitations. Across this thesis (Chapter 2-5), AUC-based analyses were most frequently applied, as AUC values can be determined for both responders and non-responders, whereas IC50 values cannot be determined when 50% inhibition is not reached. However, IC50 values are valuable for comparing drug response when different dose ranges are used, e.g. across institutes/drug libraries (Chapter 2-3). Importantly, harmonization of dose ranges and response metrics will be essential for reliable cross-institute comparisons.

To properly interpret drug sensitivity profiles, both we and others applied outlier approaches to identify patient-specific vulnerabilities relative to reference cohorts. Depending on the question, comparisons may be performed within a tumor type to identify subtype-specific hits, across pan-cancer cohorts to identify patient-specific vulnerabilities (Chapter 2), or relative to non-malignant control samples to estimate potential therapeutic selectivity and toxicity (Chapter 3 and 5). Large pan-cancer reference cohorts are currently being established in the Princess Máxima Center, using drug screening data from *ex vivo* short-term and

organoid cultures from PDX and patient samples, to further improve interpretation of patient screens. The relevance and predictive value of non-malignant reference cohorts remain to be established. If so, pediatric reference cohorts could be expanded to include samples derived from tissues such as liver and colon, which are commonly affected by treatment-related toxicities.

Drug screening in the lab is limited by the incomplete representation of *in vivo* pharmacokinetics.[38–41] In this thesis (Chapter 2), Cmax values, typically derived from plasma measurements, were used as a first approximation to assess whether *ex vivo* determined IC50 values could be clinically achievable. However, comparisons between Cmax and IC50 do not account for factors such as plasma protein binding, drug clearance, tissue distribution and penetration and active metabolite formation, all of which influence free drug exposure at the tumor site.[42,43] Future studies should therefore incorporate pharmacokinetic parameters more comprehensively, for example by testing concentrations that are clinically achievable in the tumor, informed by physiologically based pharmacokinetic modelling.[39–41] In addition, *in vitro* drug screens typically expose cells to static drug concentrations for fixed periods of time, whereas patients experience dynamic drug concentrations due to dosing schedules, drug clearance and treatment duration.[38] Approaches that better mimic clinical drug exposure, including compound washout experiments or microfluidic systems, may therefore further improve the translational value of *ex vivo* drug screening.[38,44]

The greatest challenge for functional precision oncology, and at the same time the most important limitation of this thesis, remains the correlation of *ex vivo* short-term drug responses with clinical outcome. Clinical response data should ultimately guide selection of both response metrics and hit definition. We demonstrated concordant *ex vivo* and *in vivo* drug responses for seven out of ten compounds (Chapter 2), but clinical correlations could not yet be assessed within this thesis. Fortunately, ongoing efforts address this in our center. Encouragingly, other pediatric precision medicine initiatives demonstrated concordance between *ex vivo* short-term drug responses and observed in-patient responses for selected cases.[18,19,45] Furthermore, promising results are increasingly emerging

from adult cancer studies, where the utility of functional precision medicine approaches is also currently being tested in prospective clinical trials.[3,46] Given the small patient numbers and biological heterogeneity that characterize pediatric cancers, international collaboration (such as in Chapter 2 and 3) and harmonization of drug screening workflows will be essential to generate sufficiently powered outcome correlations.

Integrating functional testing into clinical decision-making for today's patient

To guide treatment decisions alongside molecular profiling, *ex vivo* drug responses should ideally be generated within a similar timeframe, allowing parallel discussion at Molecular Tumor Boards. The integration of both datasets may provide a higher level of clinical confidence in treatment recommendations. We demonstrated that *ex vivo* short-term drug screening can both expand treatment options for patients lacking targetable molecular alterations and refine compound selection in the presence of targetable alterations (Chapter 2), which is consistent with previous reports from precision medicine initiatives INFORM and ZERO.[18,19] Within the Princess Máxima Center, drug screening results are discussed in the "Innovation4Care" initiative together with international experts and the patient's treating physician. As an example, treatment selection for the melanocytic neuroectodermal tumor of infancy case (Chapter 5) was supported by *ex vivo* short-term drug screening results alongside clinical considerations. Excitingly, *ex vivo* short-term drug screening, when further validated, may ultimately help predict both treatment response and non-response. Early identification of non-responders is particularly important in pediatric oncology to avoid unnecessary toxicity and long-term side effects in young children.

Effective treatments for the patients of tomorrow

While functional precision medicine aims to identify the most effective treatment strategy for the patient of today, drug (combination) and biomarker discovery is important for treating the patients of tomorrow. Pediatric-specific drug discovery is particularly important, as pediatric cancers differ fundamentally from adult malignancies in their developmental origin, relatively low mutational burden and lower frequency of actionable genomic alterations. Consequently, there is a need

for pediatric-specific drug and biomarker discovery, including the development of novel treatment modalities such as antibody-drug conjugates and targeted protein degraders. International initiatives such as ITCC-P4 and ACCELERATE play an important role in accelerating development and clinical implementation of pediatric-specific therapies.

Scalable and biologically representative model systems are required for drug discovery studies. Compared to *ex vivo* short-term cultured patient samples, organoid models are better suited for large-scale drug and biomarker discovery studies due to their scalability and long-term expandability while retaining key characteristics of the original tumor. The use of organoid models (cultured without animal-derived extracellular matrixes) may contribute to reducing animal-based testing during drug development. Although regulatory agencies increasingly encourage alternatives to animal experiments,[10] PDX models will likely remain important for late-stage preclinical validation due to their physiological complexity.

Current organoid models lack a tumor microenvironment, although these cells influence therapeutic response. The growing clinical importance of immunotherapies emphasizes the need for scalable, reproducible models containing different cell types, like co-cultures of organoids with immune cells. Our microscopy-based cell type-specific viability readout for neuroblastoma-PBMC co-cultures established in Chapter 5 provides ready-to-use protocols for high-throughput drug screening that future studies can use for immunotherapy testing. Integrating other tumor microenvironmental components in co-cultures may further increase physiological relevance.

Concluding remarks

In summary, the studies presented in this thesis investigated how drug screening can expand the pediatric precision oncology landscape by evaluating therapeutic vulnerabilities across multiple model systems of neuroblastoma and other pediatric solid tumors. Across these studies, *ex vivo* short-term drug screens appear feasible within clinically relevant timeframes while largely retaining drug sensitivity patterns observed in organoids that require substantially longer

establishment timelines. Together, these findings demonstrate that different preclinical model systems fulfill complementary roles within functional precision oncology. While *ex vivo* short-term drug screening appears particularly suited for rapid treatment stratification of the patient of today, scalable organoid models, preferably incorporating other cell types of the tumor microenvironment, provide opportunities for drug and biomarker discovery for the patients of tomorrow. Furthermore, low-input culturing strategies, such as MOS, and tumor-specific viability readouts may further enhance the feasibility, scalability and clinical applicability of drug screening. Ultimately, the studies presented in this thesis support the potential integration of *ex vivo* short-term drug screening alongside molecular profiling to improve treatment selection and accelerate development of more effective therapies for children with difficult-to-treat cancers.

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Appendices

Nederlandse samenvatting
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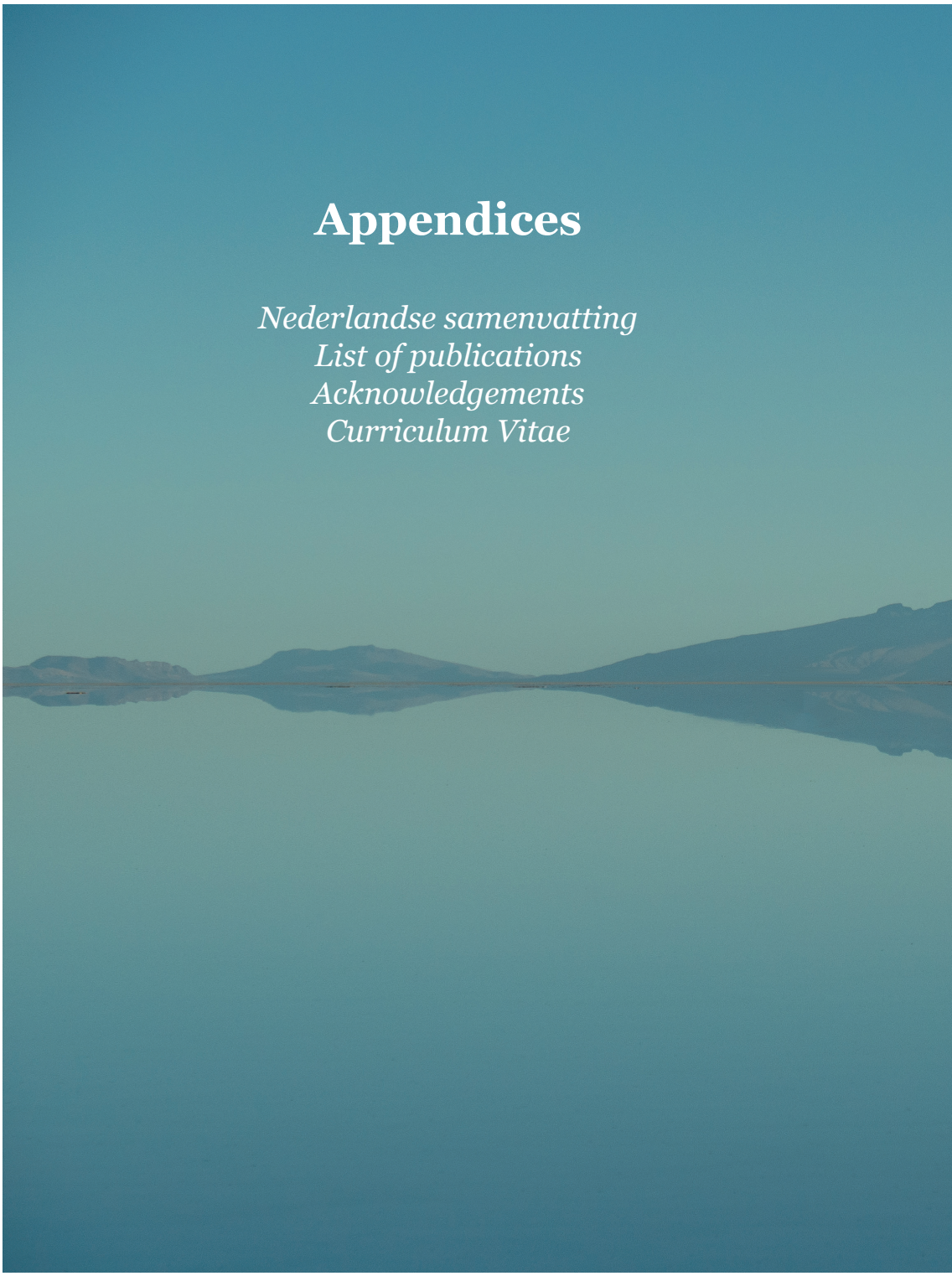

A

Nederlandse samenvatting

Jaarlijks krijgen wereldwijd ongeveer 400.000 kinderen de diagnose kanker. De overlevingskansen voor kinderen met kanker zijn de afgelopen decennia sterk verbeterd, maar voor sommige patiënten blijven de vooruitzichten slecht. Bij hoog-risico neuroblastoom, een vorm van kinderkanker, overleeft bijvoorbeeld slechts de helft van de kinderen de ziekte. Daarnaast kunnen behandelingen ernstige bijwerkingen veroorzaken, die soms ook op latere leeftijd nog merkbaar zijn. Daarom is het belangrijk om behandelingen te identificeren die niet alleen de genezingskansen verbeteren, maar ook minder schadelijk zijn. Dat vraagt om betere manieren om te bepalen welke behandeling bij welk kind past. Er bestaan al methoden om behandelingen beter op de tumor van een patiënt af te stemmen, maar deze leveren niet voor ieder kind een geschikte behandeloptie op. Dit proefschrift richt zich op extracraniale solide tumoren bij kinderen. Dit zijn vaste tumoren die ontstaan in organen of andere weefsels buiten de hersenen. In dit proefschrift onderzoeken we hoe we kunnen testen welke medicijnen effect hebben op de tumor van een individueel kind, zodat we in de toekomst beter kunnen bepalen welke behandeling mogelijk geschikt is. In deze samenvatting bespreken we eerst enkele begrippen en context die nodig zijn om dit proefschrift te begrijpen. Daarna leggen we uit waarom het nodig is om het testen van medicijnen op tumorcellen te verbeteren, en wat we daarvoor in elk hoofdstuk onderzoeken.

Neuroblastoom ontstaat in het zich ontwikkelende zenuwstelsel en komt vooral voor bij jonge kinderen. De tumor ontstaat vaak in de bijnier, maar kan ook op andere plaatsen in het lichaam voorkomen. De ziekte kan zich heel verschillend gedragen: bij sommige kinderen kan de tumor uit zichzelf verdwijnen, terwijl de ziekte bij anderen agressief groeit en uitzaait. Daarom worden kinderen ingedeeld in risicogroepen, waarop de behandeling wordt aangepast. Kinderen met een laag risico kunnen soms met weinig of zelfs zonder behandeling herstellen, terwijl kinderen met hoog-risico neuroblastoom een intensieve behandeling nodig hebben. Deze behandeling bestaat bijvoorbeeld uit chemotherapie, een operatie, stamceltherapie, bestraling en immunotherapie. Ondanks deze intensieve behandeling komt de ziekte bij meer dan de helft van de kinderen terug of

reageert de tumor onvoldoende op de behandeling. Genezing op lange termijn blijft daardoor moeilijk. Er is dus dringend behoefte aan manieren om behandelingen te vinden die beter aansluiten bij de tumor van ieder individueel kind.

Om de behandeling te verbeteren, wordt steeds vaker gebruikgemaakt van precisiegeneeskunde: het afstemmen van de behandeling op de eigenschappen van de tumor van een individuele patiënt. Hiervoor bestaan twee methoden: moleculair onderzoek en functionele profilering. In dit proefschrift ligt de nadruk op functionele profilering. We vergelijken de uitkomsten met die van moleculair onderzoek, dat momenteel standaard wordt ingezet bij kinderkanker, om te onderzoeken of beide methoden elkaar kunnen aanvullen. Bij moleculair onderzoek worden het DNA en RNA van de tumor onderzocht op afwijkingen waarvoor mogelijk al een gericht medicijn bestaat. Deze aanpak heeft de behandeling voor een deel van de patiënten verbeterd. Toch biedt moleculair onderzoek niet voor iedere patiënt een oplossing. Soms wordt er geen geschikte afwijking gevonden, bestaat er nog geen doelgericht medicijn voor, of werkt een veelbelovend doelgericht medicijn minder goed dan verwacht. Daarom zijn aanvullende methoden nodig die kunnen helpen om te bepalen welke medicijnen daadwerkelijk effect hebben op de tumorcellen van een patiënt.

Functionele profilering is een veelbelovende aanvulling waarbij onderzoekers rechtstreeks medicijntesten uitvoeren op tumorcellen. Hiervoor worden tumorcellen buiten het lichaam (*ex vivo*) in het laboratorium gekweekt en blootgesteld aan medicijnen. Zo kunnen onderzoekers meten of een medicijn de tumorcellen kan doden of remmen. Dit kan op drie verschillende manieren, met drie verschillende modellen. De eerste manier is het gebruik van kortdurende *ex vivo* kweken. Hierbij worden tumorcellen tijdelijk in het laboratorium gekweekt, zodat de effecten van medicijnen relatief snel kunnen worden getest. De tweede manier is het gebruik van organoïden. Organoïden zijn kleine, vaak driedimensionale, “mini-organen” die in het laboratorium worden gekweekt. In dit proefschrift gebruiken we organoïden die gemaakt zijn van tumorcellen (“mini-tumoren”). Het duurt enkele maanden voordat deze organoïden stabiel groeien, maar daarna kunnen ze langdurig worden gekweekt, waardoor er veel meer cellen

beschikbaar komen. De derde manier is het gebruik van muismodellen. Hierbij laten onderzoekers tumorweefsel van een patiënt in een muis groeien. Zo kunnen zij onderzoeken hoe een behandeling werkt in een levend organisme. Deze modellen bieden verschillende mogelijkheden voor medicijnonderzoek, maar hebben elk ook beperkingen. Het was bovendien nog niet duidelijk in hoeverre de uitkomsten van deze drie modellen met elkaar overeenkomen.

Voordat functionele profilering kan worden gebruikt om behandelkeuzes voor individuele patiënten te ondersteunen, moet de methode aan drie eisen voldoen. Ten eerste moet de methode haalbaar zijn met de kleine hoeveelheid patiëntmateriaal die beschikbaar is, uit bijvoorbeeld een klein naaldbiopt. Daarnaast is snelheid belangrijk: de methode moet snel genoeg resultaten opleveren om de behandelkeuze te kunnen beïnvloeden. Tot slot moet de methode een betrouwbare voorspelling geven van hoe de patiënt écht op het medicijn zal reageren. De huidige methoden voldoen nog niet optimaal aan deze eisen. Dit proefschrift richt zich daarom op het verbeteren van functionele profilering voor extracraniale solide tumoren bij kinderen. In de volgende hoofdstukken onderzoeken we hoe we veel medicijnen snel kunnen testen (Hoofdstuk 2), hoe de drie modellen zich tot elkaar verhouden (Hoofdstuk 3), hoe we met minder tumorcellen kunnen testen (Hoofdstuk 4) en hoe we tumorcellen kunnen onderscheiden van andere cellen in een tumor (Hoofdstuk 5).

In **Hoofdstuk 2** staat de methodologie centraal. We onderzoeken of kortdurende *ex vivo* kweken kunnen helpen om snel mogelijke behandelopties voor kinderen met solide tumoren te vinden. Hiervoor gebruikten we tumoren die in muismodellen waren gegroeid. We onderzochten negen verschillende tumortypen en testten per tumorsample de werking van ruim tweehonderd medicijnen. De resultaten waren binnen twee weken beschikbaar en bleken betrouwbaar en reproduceerbaar: in twee verschillende instituten vonden we dezelfde medicijngevoeligheden bij samples van dezelfde patiënt. We onderzochten ook of met moleculair onderzoek gevonden genetische afwijkingen samenhingen met de gevoeligheid voor bepaalde medicijnen en vonden dat een genetische afwijking niet altijd leidde tot meer *ex vivo* gevoeligheid. Daarnaast konden we voor bijna alle onderzochte samples één of meerdere medicijnen aanwijzen die uitzonderlijk

effectief waren ten opzichte van de andere samples. Voor de meeste samples zonder genetische afwijkingen konden we met deze methode mogelijke alternatieve behandelopties vinden. Ook bij samples met een bekende genetische afwijking kunnen kortdurende *ex vivo* kweken helpen om verschillende medicijnen te vergelijken die op die afwijking aangrijpen. Kortdurende *ex vivo* kweken zouden moleculair onderzoek dus kunnen aanvullen.

In **Hoofdstuk 3** richten we ons op neuroblastoom en vergelijken we drie modellen om tumorweefsel te onderzoeken: kortdurende *ex vivo* kweken, organoïden en muismodellen. Waar voldoende materiaal beschikbaar was, maakten we alle drie de modellen van tumorweefsel van dezelfde patiënt. Zo konden we de modellen vergelijken op medicijngevoeligheid, snelheid en haalbaarheid. In de vergelijkingen die we konden uitvoeren, kwamen de gemeten medicijngevoeligheden grotendeels overeen, maar de modellen verschilden in haalbaarheid en snelheid. Kortdurende *ex vivo* kweken waren sneller en beter haalbaar en lieten vergelijkbare medicijngevoeligheden zien als organoïden en muismodellen. Daarom lijken kortdurende *ex vivo* kweken veelbelovend om snel een behandeling voor een individuele patiënt te kiezen. Organoïden waren lastiger op te kweken, maar waren, wanneer ze eenmaal goed groeien, juist geschikt voor grootschalig onderzoek naar nieuwe medicijnen. Muismodellen kunnen waardevol zijn om veelbelovende behandelingen verder te onderzoeken in een levend organisme, voordat ze bij patiënten worden onderzocht.

In **Hoofdstuk 4** staat de haalbaarheid centraal. We onderzoeken hoe we medicijntesten haalbaarder kunnen maken met de beperkte hoeveelheid patiëntmateriaal die beschikbaar is, door ze efficiënter uit te voeren. Hiervoor gebruikten we de MicroOrganoSphere™-technologie, waarbij tumorcellen in zeer kleine druppels van een gel groeien en driedimensionale clusters vormen. We lieten zien dat neuroblastoomcellen hierin minstens even goed groeiden als met de gebruikelijke kweekmethode. Daarnaast konden we betrouwbare medicijntesten uitvoeren met 2,5 keer minder tumorcellen, terwijl de resultaten vergelijkbaar bleven. Hierdoor kunnen we met hetzelfde tumorweefsel een hoger aantal medicijnen testen. Dit kan vooral waardevol zijn wanneer slechts een klein stukje tumor beschikbaar is.

In **Hoofdstuk 5** verbeteren we de techniek voor samples waarin meerdere celtypen aanwezig zijn. We ontwikkelen een methode waarbij we een microscoop gebruiken om celtype-specifieke gevoeligheid voor medicijnen te kwantificeren. Een tumor bestaat niet alleen uit tumorcellen, maar ook uit omliggende cellen, waaronder immuuncellen. Deze cellen vormen samen de tumor-micro-omgeving en kunnen invloed hebben op de reactie van de tumor op een behandeling. Daarnaast richten steeds meer nieuwe behandelingen zich op het activeren van het eigen immuunsysteem tegen de tumor. Bij de huidige medicijntesten worden de reacties van verschillende celtypen vaak gezamenlijk gemeten. Daardoor is niet altijd duidelijk of een medicijn vooral de tumorcellen of juist de omliggende cellen beïnvloedt. We ontwikkelden daarom een microscopiemethode die de tumorcellen apart kan herkennen tussen de andere celtypen. We pasten deze methode toe op kweken van neuroblastoom organoïden met immuuncellen en op kortdurende *ex vivo* kweken van vers patiëntmateriaal. Met deze methode kunnen we beter onderzoeken welke medicijnen rechtstreeks effect hebben op tumorcellen en of de aanwezigheid van immuuncellen de werking van een medicijn verandert.

Tot slot worden in **Hoofdstuk 6** de bevindingen van dit proefschrift in perspectief geplaatst. Er bestaat niet één model dat voor alle onderzoeksvragen het beste is. Voor de patiënt van vandaag, waarbij snel een behandelbeslissing nodig is, zijn kortdurende *ex vivo* kweken veelbelovend omdat ze binnen korte tijd resultaten opleveren. Voor de patiënt van morgen, waarvoor we nieuwe medicijnen en combinaties kunnen ontwikkelen, zijn organoïden waardevoller omdat zij op grotere schaal gebruikt kunnen worden. De grootste openstaande uitdaging is het aantonen dat de uitkomsten van medicijntesten in het lab daadwerkelijk voorspellen hoe een patiënt op diezelfde behandeling zal reageren. Wanneer dit lukt, is het testen van medicijnen op tumorcellen in het lab een waardevolle aanvulling op moleculair onderzoek. Door genetische informatie te combineren met medicijntesten op tumorcellen in het lab, kan het landschap van precisiegeneeskunde worden verbreed en kunnen behandelkeuzes in de toekomst mogelijk beter worden afgestemd op ieder individueel kind met kanker.

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Schoonbeek, M. C., Amo-Addae, V., Koomen, M., Swaak, S., Looze, E. J., Langenberg, K., Lampert, J. C., Geoerger, B., Eggert, A., Schleiermacher, G., Schueler, J., Gürgen, D., Vassal, G., van Hooff, S., van den Boogaard, M. L., Molenaar, J. J., Driehuis, E., & Eising, S. (2026). MicroOrganoSphere™ technology to advance drug screening in neuroblastoma precision medicine. *EJC Paediatric Oncology*, *8*, 100534. https://doi.org/10.1016/j.ejcped.2026.100534

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Curriculum Vitae

Marlinde Schoonbeek was born on April 28th, 1998, in Haren, the Netherlands. She obtained her high school degree at Praedinius Gymnasium in Groningen in 2016 and then began a bachelor's in Biomedical Sciences at Utrecht University. In 2019, she started the master's program in Cancer, Stem Cells and Developmental Biology at Utrecht University, graduating *cum laude* in 2021. During her master's, she conducted research on extracellular vesicles and the cell cycle (Regenerative Medicine Centre, Utrecht), paediatric leukaemia (Princess Máxima Centre), and CDK4/6 inhibition in breast cancer (Dana-Farber Cancer Institute).

She began her career in 2021 as an education developer for the bachelor's program in Biomedical Sciences, focusing on data science, career orientation, student well-being and professional skills. In May 2022, Marlinde started her PhD research in the Molenaar group at the Princess Máxima Centre. Her research focused on improving preclinical models for determining drug sensitivity in neuroblastoma and other paediatric solid tumours. She collaborated with the biotechnology company Xilis B.V. and a large public-private consortium with academic institutions, contract research organizations and pharmaceutical companies. She received the Scholar-in-Training Award at the AACR Precision Medicine conference in Boston and the Young Investigator Award at Innovation for Health in Leiden and delivered plenary presentations at both events. She also received a personal travel grant from Villa Joep.

Alongside her scientific work, Marlinde has organized activities on personal development, leadership and inclusion. She is a Junior Ambassador for TOPX, a network for ambitious women in life sciences, where she enjoys organizing and moderating events and panel discussions. At the Princess Máxima Center, Marlinde was part of Ace, a community promoting authenticity for excellence in science through role model lunches, coaching programs and workshops.

Since October 2026, Marlinde works as Application Specialist at the biotechnology start-up Orgonex. Here, she helps drive the growth and development of the company and supports researchers and companies in making organoid culture more efficient.